



dihydrolipoamide dehydrogenase

lipoamide acetyltransferase component (E_2) of a human pyruvate dehydrogenase complex precursor (PDC- E_2): (incomplete sequence) ODP2_HUMAN, 615 amino acids (65.32 kDa).

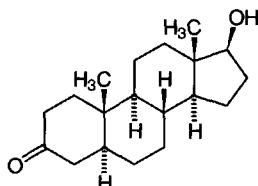
dihydrolipoamide dehydrogenase EC 1.8.1.4; *other names*: lipoamide reductase (NADH); lipoyl dehydrogenase; dihydrolipoyl dehydrogenase; diaphorase; the E_3 component of α -ketoacid dehydrogenase complexes, an FAD-flavoprotein enzyme that catalyses the oxidation by NAD^+ of dihydrolipoamide to lipoamide and NADH. Example (precursor) from human: DLDH_HUMAN, 509 amino acids (54.09 kDa). *See also* **oxoglutarate dehydrogenase complex**; **pyruvate dehydrogenase complex**.

dihydrolipoamide S-succinyltransferase EC 2.3.1.61; an enzyme that catalyses the formation of succinyl-CoA and dihydrolipoamide from S-succinyl-dihydrolipoamide and CoA. It is the E_2 component of the **oxoglutarate dehydrogenase complex**. Example from human (precursor): database code ODO2_HUMAN, 453 amino acids (48.69 kDa); the enzyme is formed from residues 68–453, 1–67 forming the transit peptide.

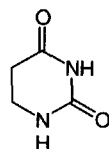
dihydroorotase EC 3.5.2.3; *systematic name*: (S)-dihydroorotate amidohydrolase; *other names*: carbamoylaspartic dehydratase; DHOase. An enzyme that catalyses the formation of N-carbamoyl-L-aspartate from (S)-dihydroorotate. In many eukaryotes this is an activity of **CAD protein**.

dihydropyridine receptor a protein located in skeletal muscle transverse tubules that specifically binds 1,4-dihydropyridine compounds. It forms one of the subunits of the L-type **calcium channel**.

5 α -dihydrotestosterone 4,5-dihydrotestosterone; 5 α -androstan-17 β -ol-3-one; a potent metabolite of testosterone, formed in the cytoplasm of target cells by the action of progesterone 5 α -reductase, EC 1.3.1.30. It mediates many of the differentiating, growth-promoting, and functional actions of testosterone by binding to the **androgen receptor** and forming a complex. *See also* **testicular feminization**.



dihydrouracil 5,6-dihydrouracil, a minor pyrimidine base. It occurs in the **dihydrouridine arm** of **transfer RNA**.

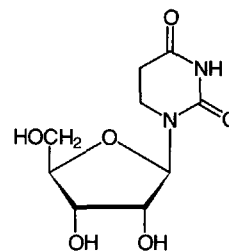


dihydrouridine *symbol*: D or hU; 5,6-dihydrouracil riboside; 1- β -D-ribofuranosyl-5,6-dihydrouracil; a rare nucleoside occurring in the **dihydrouridine arm** of **transfer RNA**.

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dihydrouridine

dihydrouridine arm *abbr.*: DHU arm; the base-paired segment of the **clover leaf structure** of **transfer RNA** that has attached to it the loop containing 5,6-dihydrouracil (see **dihydrouracil**).

dihydroxy+ prefix (*in chemical nomenclature*) indicating the presence of two substituent hydroxyl groups.

dihydroxyacetone *older name for glyceralone*.

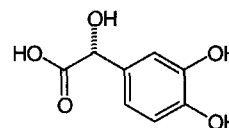
dihydroxyacetone phosphate *older name for glyceralone phosphate*.

dihydroxyacetone synthase *see* **formaldehyde transketolase**.

dihydroxyacetone transferase *see* **transaldolase**.

dihydroxyeicosatetraenoate *abbr.*: diHETE; any of several products resulting from dioxygenation of arachidonate at two sites, normally resulting from **lipoxygenase** action on **hydroxy-eicosatetraenoates**. (6*E*,8*Z*,10*E*,14*Z*)-(5*S*,12*S*)-5,12-dihydroxy-eicosa-6,8,10,14-tetraen-1-oate (*abbr.*: 5(*S*),12(*S*)-diHETE) is synthesized in human leukocytes by two successive lipoxygenase reactions and acts as a leukotriene B_4 antagonist. *See* **leukotriene B**. (6*E*,8*Z*,11*Z*,13*E*)-(5*S*,15*S*)-5,15-dihydroxy-eicosa-6,8,11,13-tetraen-1-oate (*abbr.*: 5(*S*),15(*S*)-diHETE) is synthesized in human neutrophils and enhances neutrophil degranulation. (5*Z*,9*E*,11*Z*,13*E*)-(8*S*,15*S*)-8,15-dihydroxyeicosa-5,9,11,13-tetraen-1-oate (*abbr.*: 8(*S*),15(*S*)-diHETE) is synthesized in human leukocytes, eosinophils, and platelets and is chemotactic for human polymorphonuclear leukocytes.

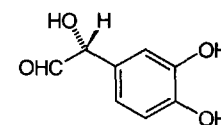
3,4-dihydroxymandelate an intermediate in the formation of 4-hydroxy-3-methoxymandelate from **norepinephrine**.



3,4-dihydroxyphenylalanine *see* **dopa**.

3,4-dihydroxyphenylethylamine *see* **dopamine**.

3,4-dihydroxyphenylglycolaldehyde the product of the action of monoamine oxidase on **norepinephrine**. It is converted by an NAD^+ -dependent aldehyde dehydrogenase to 3,4-dihydroxymandelate.

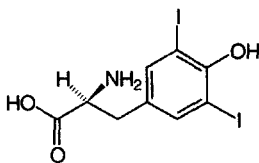


1,25-dihydroxyvitamin D₃ *another name for* **calcitriol**.

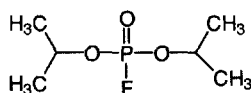
diiodotyrosine *abbr.*: DIT; 3,5-diiodotyrosine; iodogorgoic acid, 3,5-diiodo-4-hydroxyphenylalanine, 2-amino-3-(3,5-diiodo-4-hydroxyphenyl)propanoic acid. Residues of the L enantiomer, together with residues of the monoiodo com-

diisopropylfluorophosphate

pound, 3-iodo-L-tyrosine, are formed by iodination of L-tyrosine residues in thyroglobulin as precursors of thyroid hormones.



diisopropylfluorophosphate diisopropylfluorophosphate; dyflos; *abbr.*: DIPF or DFP; an inhibitor of **cholinesterase** and other **serine proteases**. It phosphorylates the serine in the enzyme's active site, causing long-lasting inhibition. Recovery of enzymic activity requires *de novo* enzyme synthesis.

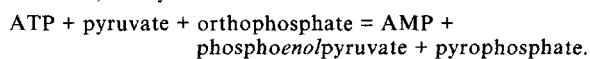


diisopropylphospho *symbol*: Dip; *abbr.*: DIP; the doubly esterified phosphoric acyl group, $[(CH_3)_2CH-O]_2P(O)-$, derived from diisopropyl fluorophosphate by loss of a fluorine atom.

dikaryon or **dicaryon** a living cell containing two nuclei, or a structure (e.g. a mycelium) composed of such cells. It may contain nuclei of the same genetic constitution (i.e. **homokaryons**) or of differing genetic constitutions (i.e. **heterokaryons**). —**dikaryotic** or **dicaryotic** *adj.*

diketose any monosaccharide derivative containing two (potential) ketonic carbonyl groups. *Compare* **ketose**.

dikinase either of the two enzymes belonging to the sub-class EC 2.7.9, comprising phosphotransferases with paired acceptors. For example, pyruvate, orthophosphate dikinase, EC 2.7.9.1, catalyses the reversible reaction:



dil. *abbr.* for dilute (def. 2).

dilate to expand or cause to expand; to make larger or wider; e.g. of the pupil of the eye or a chamber of the heart. —**dilation** or **dilatation** *n.*

dilatometer any instrument for measuring small changes in the volume of liquids. One design consists of a (glass) bulb fitted with a graduated hollow stem in which changes in volume may be observed. In another design, the (glass) bulb is fitted with a stopper through which runs a capillary tube so that any increased volume of the contained liquid may escape; the change in mass of the bulb and its contained liquid may then be determined. —**dilatometry** *n.*

diluent a diluting agent; *see* **dilute** (def. 1).

dilute **1** a to lower the concentration of a solute in a solution (or of particles in a suspension). **b** to mix one fluid with another thereby lowering the concentration of the first. **2** *abbr.*: dil.; describing a solution (dispersion or mixture) having a relatively low concentration (of a specified solute, type of dispersed particle, or component). *Compare* **concentrated** (def. 1).

diluted (of a solution or suspension) lowered in concentration and increased in volume by addition of solvent or other diluent. *Compare* **concentrated** (def. 2).

dilution **1** the act or process of making more **dilute** (def. 2). **2** a diluted solution or suspension. **3** the (specified) extent to which a solution or suspension has been diluted.

dilution end-point analysis the determination of the concentration of a substance in a solution by measuring the amount of a substance that reacts with a known amount of a reagent.

dimethylaminoazobenzene



tration of antibody in a solution by allowing a given amount of antigen to react with increasing dilutions of the antibody. The most dilute solution that produces a detectable effect, e.g. precipitation, is taken as the end point.

dilution law *see* **Ostwald dilution law**.

dilution quenching the lowering of **counting efficiency** of a liquid scintillation counter by the addition of solvent to the contents of the scintillation vial. *See also* **quenching**.

dilution rate *symbol*: D ; the rate, F , at which existing medium is replaced with fresh medium in a **continuous culture**, divided by the volume, V , of the culture: $D = F/V$. Under steady-state conditions the dilution rate numerically equals the specific growth rate of the culture. *See also* **auxostat**, **chemostat**.

dilution value of a buffer *symbol*: $(\Delta pH)_{1/2}$; the change in pH that occurs when a buffer solution is diluted with an equal volume of water.

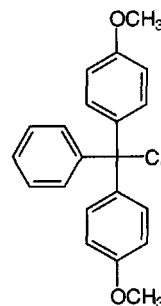
dimer **1** any molecular structure of any size in which two initially identical chemical entities have become covalently combined. **2** any macromolecular structure in which two (either identical or nonidentical) subunits are noncovalently associated. *See also* **heterodimer**, **homodimer**. —**dimeric** *adj.*

dimercaprol or **British anti-lewisite** (*abbr.*: BAL); 2,3-dimercaptopropanol; an antidote to the arsenical war gas, lewisite, that is also useful as a chelating agent and detoxicant for heavy-metal ions and as a protective agent for sulphydryl groups in enzymes, coenzymes, or the like.

dimerize or **dimerise** cause a **dimer** to be formed. —**dimerization** or **dimerisation** *n.*

dimerizer hypothesis or **dimeriser hypothesis** an unsubstantiated model of facilitated diffusion proposed to explain the rates of entry of sugars into human erythrocytes from various mixtures of sugar species. According to this hypothesis, permeation occurs as a result of pairs of sugar molecules interacting with a membrane component, followed by the association of the sugars into dimers, a form in which a reduced number of free hydrogen-bond-forming groups would be available.

dimethoxytrityl *abbr.*: DMT; 4,4'-dimethoxytriphenylmethyl; a group used for protecting hydroxyl groups (especially the 5'-hydroxyl groups of nucleosides) during chemical synthesis or other procedures. It is usually introduced by reaction with dimethoxytrityl chloride, 4,4'-dimethoxytriphenylmethyl chloride. *Compare* **trityl**.



4,4'-dimethoxytriphenylmethyl chloride

dimethylallyl transterase EC 2.5.1.1; *systematic name*: dimethylallyl-diphosphate:isopentenyl-diphosphate dimethylallyltransterase; *other names*: prenyltransferase; geranyl-diphosphate synthase. An enzyme that catalyses the reaction of dimethylallyl diphosphate with isopentenyl diphosphate to form pyrophosphate and geranyl diphosphate. It participates in the pathway for the formation of **polyprenyl** groups, sesquiterpene and higher terpenes, and **cholesterol**.

dimethylaminoazobenzene isothiocyanate *see* **DABITC**.

dimethylaminoazobenzene sulfonyl chloride *see* **dabsyl**.

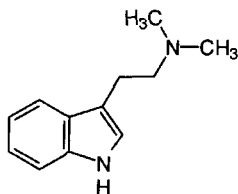


dimethylaminobenzaldehyde *see* Ehrlich's reagent.

dimethylaminonaphthalene sulfonyl chloride *see* dansyl, dansylate.

dimethylformamide *see* DMF.

dimethyltryptamine *N,N*-dimethyltryptamine; 3-[2-(dimethylamino)ethyl]indole; a hallucinogenic agent extracted from several plant species including *Piptadenia peregrina*. It is also a metabolite of 5-hydroxytryptamine. It is a controlled substance under the US Code of Federal Regulations.



dimorphism 1 the existence of a chemical compound in two distinct physical forms, e.g. two different crystalline forms. 2 the existence of individuals belonging to two distinct forms within the same species of organism; e.g. in **sexual dimorphism** the males and females of the same species are markedly different. 3 the occurrence in certain plants of two distinct forms of leaves or other parts in the same individual. —**dimorphic** or **dimorphous** *adj.*

dinactin *see* nonactin.

dinitrofluorobenzene *abbr.*: DNFB; a (less correct) name for fluorodinitrobenzene. *See* Sanger's reagent.

dinitrogen *systematic name* for the diatomic compound N_2 , commonly termed (molecular) nitrogen.

dinitrophenate the strongly yellow-coloured anion, $(NO_2)_2-C_6H_3-O^-$, formed from dinitrophenol.

dinitrophenol *abbr.*: DNP; 2,4-dinitrophenol; a pale-yellow crystalline solid. It is a powerful uncoupler of **oxidative phosphorylation**, and hence toxic.

dinitrophenyl *symbol*: Dnp; the group formed by loss of the phenolic hydroxyl group from 2,4-dinitrophenol. *See* Sanger's method.

dinitrophenylate to introduce a dinitrophenyl group into a compound. —**dinitrophenylation** *n.*

dinoprost a generic name for prostaglandin $F_{2\alpha}$. *See* prostaglandin.

dinor+ *prefix (in chemical nomenclature)* indicating shortening of a carbon chain or contraction of a ring by two methylene groups. *See also* **nor+**.

Dintzis procedure a procedure for investigating the direction and rate of biosynthesis of a polypeptide. It requires the addition to a protein-synthesizing system *in vitro*, at a given time, of a **pulse** of a labelled form of an amino acid, of which there are several residues at different positions in the structure of the polypeptide. After various periods of elapsed time, within the time required for synthesis of a complete chain, completed molecules of the polypeptide are isolated, and the extents of incorporation of the labelled amino acid into them are determined for each position in which a residue occurs naturally. Over all the samples, a gradient of radioactivity (increasing from the N terminus to the C terminus of the chain) is found, indicating that the pulse tends to enter the more C-terminal positions and that the more N-terminal region has already been synthesized. By these means it was clearly shown for the first time that polypeptide chains are assembled from amino acids on a template and that synthesis takes place from its N terminus. [After Howard Marvin Dintzis (1927–), US biochemist, who conceived the procedure and described its use in 1961.]

dinucleotide 1 any **oligonucleotide** consisting of two mononucleotides in 3'–5' phosphodiester linkage. 2 any **nucleotide coen-**

zyme in which either of the mononucleotides adenosine 5'-phosphate or adenosine 2',5'-bisphosphate is in 5'–5' diphosphate (i.e. phosphoric anhydride, pyrophosphate) linkage with nicotinamide mononucleotide or riboflavin 5'-phosphate.

dinucleotide fold an alternative name for **nucleotide binding fold**.

diode 1 a semiconductor device, containing one p–n junction, used in circuits to rectify **alternating current** to a **direct current**: semiconductor diode. 2 a thermionic valve containing an anode and a cathode and functioning in a similar manner: **thermionic diode**.

dioestrus *see* **estrous cycle**.

diol any organic compound (or portion of such a compound) containing two hydroxyl groups.

diol lipid any of various neutral lipids that are fatty-acid esters of dihydroxy alcohols (i.e. are not glycerides). They occur in small amounts in seed oils; e.g. 1-palmitoleoyl-2-*trans*-vacenoyl-erythro-butane-2,3-diol, which is found in seeds of *Coix lacrima*.

dioxin receptor *see* **ARNT**.

dioxygen *systematic name* for the diatomic compound O_2 , commonly termed (molecular) oxygen. *Compare* **trioxygen**.

dioxygenase any oxidoreductase enzyme of the sub-subclass EC 1.13.11. Such enzymes catalyse reactions involving the incorporation of both oxygen atoms of dioxygen into a second substrate; *see* e.g. **ethylene-forming enzyme**, **benzoate 1,2-dioxygenase**.

Dip *symbol* for the diisopropylphospho group.

DIP *abbr.* for the diisopropylphospho group.

dipentene a racemic mixture of the (+) and (–) enantiomers of **limonene**.

dipeptide any **oligopeptide** consisting of two (usually, α -) amino-acid residues.

dipeptide mimic any synthetic dipeptide that inhibits **angiotensin-converting enzyme**. An example is **captopril**.

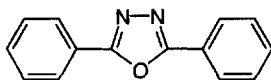
dipeptidyl-peptidase *abbr.*: DPP; any member of a group of enzymes belonging to the sub-subclass EC 3.4.14, dipeptidylpeptide and tripeptidylpeptide hydrolases, that release an N-terminal dipeptide from an oligo- or polypeptide. **Dipeptidyl-peptidase I** (or cathepsin C), EC 3.4.14.1, is a lysosomal cysteine-type peptidase that acts on polypeptides, but not if the terminal amino-acid residue is Arg or Lys, and not if either of the adjacent residues is Pro. Example from *Rattus norvegicus*: database code CATC_RAT, 462 amino acids (52.38 kDa). **Dipeptidyl-peptidase II**, EC 3.4.14.2, is a lysosomal serine-type peptidase that acts preferentially on tripeptides. **Dipeptidyl-peptidase III**, EC 3.4.14.4 (*other names*: red cell angiotensinase; enkephalinase B), is a cytosolic serine-type peptidase that acts on tetra- or longer peptides. **Dipeptidyl-peptidase IV**, EC 3.4.14.5 (*other names*: dipeptidyl aminopeptidase IV; Xaa-Pro-dipeptidyl-aminopeptidase; Gly-Pro naphthylamidase; post-proline dipeptidyl aminopeptidase IV), catalyses the release of an N-terminal dipeptide, Xaa-Xbb-|-Xcc, from a polypeptide, preferentially when Xbb is Pro, provided Xcc is neither Pro nor hydroxyproline. The protein is a serine proteinase, type II transmembrane glycoprotein; it is a homodimer: database code DPP4_HUMAN, 766 amino acids (88.22 kDa); T-cell activation antigen CD26. **Dipeptidyl-dipeptidase**, EC 3.4.14.6, is a thiol-activated peptidase from *Brassica oleracea* that acts preferentially on tetrapeptides.

dip-F *symbol* for diisopropyl fluorophosphate.

DIPF *abbr.* for diisopropylphosphofluoridate. *See* **diisopropyl fluorophosphonate**.

diphenylamine reaction the reaction of an acidic solution of diphenylamine with deoxypentoses to give a blue colour. It is used in the quantitative determination of DNA, and alone or in combination with other reagents as a detection reagent.

2,5-diphenyloxazole *abbr.*: PPO; a compound used as a primary scintillant in scintillation counting. *See* **scintillation counter**. *See also* **scintillation cocktail**.



2,5-diphenyloxazole

diphosphate **1** the tetravalent anion, $P_2O_7^{4-}$, derived from diphosphoric acid (earlier known as pyrophosphoric acid), heptaaxodiphosphoric(v) acid, $(HO)_2PO-O-PO(OH)_2$. **2** any mixture of free diphosphoric acid and its anions. *see also inorganic diphosphate, pyrophosphate*. **3** any (partial or full) salt or ester of diphosphoric acid. **4** any organic compound containing two phosphoric residues linked by an oxygen atom. **5** name of the univalent group $-O-PO(OH)-O-PO(OH)_2$ (irrespective of its state of ionization). **6** *formerly*, any organic compound containing two independent phosphoric residues attached to different parts of the molecule; now incorrect, use **bisphosphate**.

(+)-diphospho+ **1** *prefix to (or infix in)* a chemical name indicating the presence in ester (or diester) linkage of a diphosphoric residue. **2** *formerly*, *prefix to* a chemical name indicating the presence of two independent phosphoric residues in ester linkages; now incorrect, use **bisphospho+**.

diphosphoglycerate *former name for* **1** **2,3-bisphospho-D-glycerate**. **2** **3-phospho-D-glyceroyl phosphate**.

diphosphoglycerate pocket *abbr.*: DPG pocket; *see* **bisphosphoglycerate pocket**.

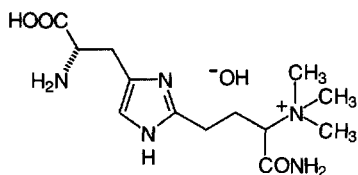
diphosphoinosotide *former name for* **phosphatidylinositol 4-phosphate**.

diphosphopyridine nucleotide (oxidized) *abbr.*: DPN; *obsolete name for* the oxidized form of nicotinamide adenine dinucleotide; *see* **NAD⁺**.

diphosphopyridine nucleotide (reduced) *abbr.*: DPNH; *obsolete name for* the reduced form of nicotinamide adenine dinucleotide; *see* **NADH**.

diphosphoric *symbol: PP or (in the one-letter convention for nucleotides) pp*; *general name for* a residue of diphosphoric acid, $(HO)_2PO-O-PO(OH)_2$, whether singly or multiply attached through oxygen to other groups and whether or not any residual hydroxyl groups are dissociated.

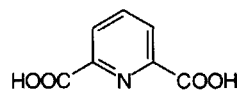
diphthamide a modified histidine residue on EF-2. *See* **elongation factor**.



diphtheria toxin an enzyme toxin produced by strains of *Corynebacterium diphtheriae* which are lysogenic for corynephage, whose genome contains the *tox* structural gene (carried by phage β_{tox}^+). The toxin has the activity of EC 2.4.2.36; *recommended name*: **NAD⁺-diphthamide ADP-ribosyltransferase**; *systematic name*: **NAD⁺:peptide-diphthamide N-(ADP-D-ribosyl)transferase**. It catalyses the reaction between **NAD⁺** and peptide diphthamide to form nicotinamide and peptide **N-(ADP-D-ribosyl)diphthamide**. The toxin molecule is a proenzyme comprising a single polypeptide chain of ≈ 62 kDa; the toxin is lethal for susceptible animals at around 100 ng per kg of body mass. On cleavage of one peptide bond

and one disulfide bond, the toxin splits into an N-terminal fragment, A (21.15 kDa), the enzyme that specifically inhibits protein synthesis by catalysing transfer of the ADP-ribosyl moiety of **NAD⁺** to eukaryotic EF-2 (*see* **elongation factor**) in the cytoplasm, and a C-terminal fragment, B (≈ 40 kDa), that is required for recognition of specific surface receptors on sensitive cell membranes. Fragment B is necessary for the toxicity of fragment A because it enables the latter to cross the plasma membrane. Example (precursor) from phage β : database code DTX_CORBE, 567 amino acids (61.53 kDa); fragment A, amino acids 33–225; fragment B, amino acids 226–567.

dipicolinate 2,6-pyridinedicarboxylate; a substance that is abundant (10%) in bacterial spores. Its production is a striking feature of sporulation and it may play a part in making spores resistant to heat.



dipicolinic acid

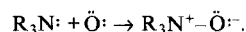
diplochromosome a 'double chromosome', comprising the arms of two daughter chromosomes attached to a single centromere. It arises due to failure of the centromere to divide following duplication of a chromosome.

diploid describing a cell or nucleus having two sets of homologous chromosomes; i.e. containing twice the **haploid** number. — **diploidy** *n*.

diploid state the chromosome state of a cell in which each type of chromosome, except for the sex chromosomes, is always represented twice.

diplotene the fourth phase of prophase I in meiosis.

dipolar bond or **coordinate bond** or **coordinate covalence** or **coordinate link** or **dative bond** or **semipolar double bond** a covalent bond formed (actually or conceptually) between two initially uncharged moieties by **coordination** (def. 2). The consequent generation of opposing charges on the linked structures creates a **dipole**; e.g.



The term is preferred to the alternatives, which are now either obsolescent or obsolete. *Compare* **polar bond**.

dipolar correlation time time taken for an atomic spin system to rotate through one radian from its previous orientation, the rotation being caused by interactions with other, neighbouring spins.

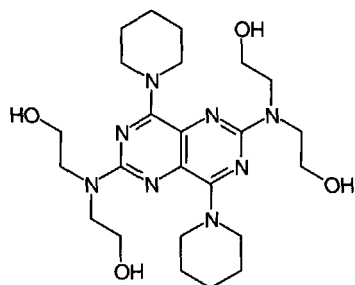
dipolar interaction the interaction between two **magnetic moments** due to the effect of the magnetic field of one on the other.

dipolar ion or **zwitterion** a molecule containing ionic groups of opposite charge in equal numbers. *Compare* **amphion**.

dipole a pair of equal and opposite electric charges or magnetic poles (especially in a molecular entity) separated by a, usually, small distance. — **dipolar** *adj*.

dipole moment or **electric dipole moment** *symbol: p or μ* ; a vector quantity equal to the product of the absolute magnitude of either charge in a dipole (the two are equal in an uncharged molecule) multiplied by the distance separating them. If a molecule possessing a dipole moment is placed in a uniform electric field, it will be subject to torque tending to line it up in the direction of the field. $p = \sum Q_i r_i$ coulomb metres, where Q_i is the charge and r_i is the position vector.

dipyridamole an inhibitor of cyclic AMP phosphodiesterase and 5-lipoxygenase, used as a coronary vasodilator.



dipyridamole

diradical an alternative term for **biradical**.

direct current *abbr.*: d.c. or DC; an electric current that flows in only one direction.

directional selection any genetic **selection** that shifts the population mean in the direction desired by the breeder.

dis+ *prefix* indicating 1 reversal. 2 deprivation, exclusion, or lack. 3 negation. 4 removal or release.

dis2 a gene in fission yeast that is the equivalent of **DIS2 S1** in budding yeast.

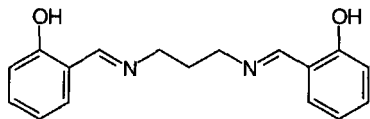
disabled strain any strain of organism, especially a microorganism, that because of some character, e.g. a nutritional requirement, is unable to reproduce in the normal environment, including the normal host, and can only survive in laboratory conditions.

disaccharidase any enzyme within sub-subclass EC 3.2.1 that hydrolyses a disaccharide to its constituent monosaccharides.

disaccharidase deficiency *see* **lactase deficiency**.

disaccharide any **oligosaccharide** consisting of two monosaccharide residues. It may be a **nonreducing disaccharide** or a **reducing disaccharide**, depending on the mode of linkage between the residues.

disalicylidenepropanediamine *abbr.*: DSPD; *N,N'*-disalicylidene-1,3-diaminopropane; an inhibitor of photosynthetic electron flow at the ferredoxin level.



disassociation a variant form of **dissociation** (def. 1).

disc a variant spelling (*esp. Brit.*) of **disk**.

discharge 1 (*in physics*) the release of an electric charge, e.g. from a condenser or a battery, or the equalization of an electric potential difference. 2 (*in physics*) the intermittent or continuous flow of electricity through a gas by the formation of electrons and ions. 2 (*in medicine*) an abnormal secretion from a body cavity, or the secretion from a wound.

discoidin any of a group of endogenous carbohydrate-binding proteins (lectins) produced by cells of the slime mould *Dictyostelium discoideum* during differentiation. Similar to **pallidin**, they may be concerned in cell adhesion. Examples (all from *D. discoideum*): discoidin I, A chain: database code DISA_DICDI, 253 amino acids (28.24 kDa); B and C chains: database code DISC_DICDI, 253 amino acids (28.39 kDa).

discontinuous epitope an immunological determinant (i.e. **epitope**) on an antigen that comprises different sections of a protein molecule brought together by folding of the polypeptide chain. *Compare* **continuous epitope**.

discriminator 1 any (electronic) device that gives an output signal only when the input signal is above a certain preset level

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or has certain preset characteristics. 2 any criterion used in sorting objects, experimental results, signals, etc.

disease any anatomical abnormality or impairment of the normal functioning of an organism or of any of its parts other than one arising directly from physical injury. It may be caused by environmental factors (e.g. malnutrition, toxic agents, etc.), infective agents (bacteria, viruses, etc.), inherent defects in the organism (e.g. genetic disease), or any combination of these factors.

disequilibrium assay any radioimmunoassay in which the reagents are not added at the same time and are not allowed to reach thermodynamic equilibrium. *Compare* **equilibrium assay**.

disinfection the destruction or removal of all organisms capable of giving rise to infection. *Compare* **sterilize** (def. 1).

disintegration or nuclear disintegration any process of nuclear **decay** involving a splitting into more nuclei or the emission of particles (either spontaneously or as a result of a collision).

disintegration constant or decay constant *symbol*: λ ; for a radionuclide, the probability of the nuclear decay of one of its nuclei in unit time. It is given by:

$$\lambda = -(1/t) \ln(A/A_0),$$

where A_0 is the radioactivity at zero time and A is the activity at time t . *See also* **average life**.

disintegrin any of a group of compounds that act as inhibitors of platelet-fibrinogen interaction and other aggregations. They are found in certain snake venoms. Example, disintegrin CTF-I/CTF-II from *Trimeresurus flavoviridis*: database code DISC_TRIFL, 75 amino acids (8.32 kDa); two motifs. Example, from *Agkistrodon rhodostoma* (Malayan pit viper) venom: database code DISI_AGKRH, 68 amino acids (7.32 kDa).

disjunction the separation of chromosomes at **anaphase** during nuclear division.

disk or (esp. Brit.) disc 1 a circular, flattened structure, or any object having or resembling such a shape. *See also* **retinal rod**.

2. (*in computing*) a disk-shaped device on which files are stored electromagnetically.

disk (gel) electrophoresis a technique of separation by electrophoresis in which a tube is loaded first with the sample, applied in a gel support (usually polyacrylamide), followed by a spacer gel and finally by the running gel, in which separation occurs. The gels are so formed as to give a gradient of progressively smaller pore size, and a discontinuous buffer system is used. This arrangement causes the separating components of the applied mixture to stack as a series of very thin bands and permits very high resolution.

dismembrator an ultrasonic device for rapidly disrupting cells in bacteriological and cytological studies. It consists of a probe oscillating at ≈ 20 kHz and an acoustical power of up to 150 W that is dipped into a cell suspension, contained in either a stainless-steel or glass beaker.

dismutase *see* **superoxide dismutase**.

dismutation a chemical reaction in which a single compound serves both as an oxidizing agent and as a reducing agent; e.g. the reaction of two molecules of pyruvate and one of water giving rise to one molecule each of lactate, acetate, and carbon dioxide. *Compare* **disproportionation**.

disodium chromoglycate *see* **chromolyn**.

disodium etidronate *see* **etidronate**.

disome a **polysome** in which two ribosomes are attached to a strand of messenger RNA. *Compare* **monosome**, **nucleodisome**.

Disase *proprietary name* for a highly stable neutral metalloproteinase obtained from *Bacillus polymyxa*. It is so named from its use, alone or with collagenase, for gentle enzymic disaggregation and dispersal of animal tissues into separated cells for primary **cell culture** and for redispersal of cells on subculturing. *See also* **Chaotropase**, **Pronase**.

disperse phase or dispersed phase the phase in a two-phase sys-

dispersion

tem that consists of the particles in a colloid, emulsion, or suspension.

dispersion 1 the selective separation of an inhomogeneous emission according to some criterion (especially frequency, particle mass, speed, or energy); e.g. the separation of heterochromatic light into its components. 2 any system in which particles of any nature (e.g. solid, liquid, gas, or composite) are dispersed in a solid, liquid, or gas of different composition. 3 (in statistics) the extent to which values of a statistical frequency distribution are scattered around a mean or median value.

dispersion force or **London force** a long-range attractive interaction occurring between atoms or molecules whether or not they have permanent charges or dipole moments. Since the electrons in an atom or molecule are moving, an instantaneous dipole moment is induced in the atom or molecule; this averages to zero over a short time. This instantaneous dipole moment will induce an oppositely oriented dipole moment in a neighbouring atom or molecule in a manner that does not time-average to zero. This interaction energy, $V(R)$, is given approximately by:

$$V(R) = -\frac{3}{2} \left(\frac{h\nu_A\nu_B}{\nu_A + \nu_B} \right) \frac{\alpha_A\alpha_B}{(4\pi\epsilon_0)} \frac{1}{R^6}$$

where ν_A and ν_B are the approximate frequencies of the first electron transitions of entities A and B, α_A and α_B are their polarizabilities, ϵ_0 is the permittivity of vacuum, h is the Planck constant, and R the distance between the entities. [After Fritz London (1900–54), German-born US physicist.]

dispersive replication of DNA see **replication of DNA**.

displacement binding a technique in which the displacement of radioligand bound to a population of receptors is measured as the concentration of an unlabelled ligand, which binds to the same receptors, is increased. It is used to distinguish different affinity states of the receptor population or receptor selectivity of the radioligand.

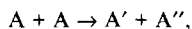
displacement chromatography an analytical chromatographic technique in which a mixture is applied to a chromatographic column and the components of the mixture are then successively displaced by elution with a solution containing another substance of higher affinity for the column material than that of the most firmly held component. The relative proportions of the components of the mixture are determined by **frontal analysis** of the eluate.

displacement electrophoresis an alternative term for **isotachopheresis**.

displacement loop an alternative name for **D-loop**.

displacement reaction a chemical reaction in which an atom or group is displaced from its covalent attachment to a central (frequently a carbon) atom following an attack by either an **electrophile** or a **nucleophile**.

disproportionation any chemical reaction of the type:



where A, A', and A'' are different chemical species. Compare **dismutation**.

DIS2 S1 a gene of budding yeast encoding protein phosphatase I (its equivalent in fission yeast is **dis2**). In the absence of gene function **mitosis** is arrested. The product in *Saccharomyces cerevisiae* is: database code YSCDIS2S1A, 312 amino acids (35.93 kDa).

dissimulation a less common alternative term for **catabolism**. Compare **assimilation**. —**dissimulatory** adj.

dissimilatory nitrate reduction see **nitrate reduction**.

dissociation 1 or **disassociation** a reversible splitting of a chemical substance into simpler entities; compare **association**. In this sense, the term may be applied to: a the **heterolysis** of a covalent bond, yielding, in the case of an initially uncharged molecule, positively and negatively charged ions: **ionic dissociation**; b the disaggregation of a macromolecular homopolymer or heteropolymer into its constituent units (i.e. **protomers**), usually by a stepwise process through components of interme-

diate size (i.e. **oligomers**) and without rupture of covalent bonds; c the breakdown of an enzyme–substrate complex, which may or may not involve cleavage of a covalent bond; or d the separation of a complex of two or more molecules into the constituent molecules. 2 a splitting of a chemical substance by the **homolysis** of a covalent bond, yielding **radicals**. 3 **translocation** between a large chromosome and a small supernumerary one, in effect splitting the large chromosome into two chromosomes. 4 separation (induced or spontaneous) of the nuclear components of a heterokaryotic **dikaryon**. —**dissociated**, **dissociable**, **dissociative** adj.; **dissociate** vb.

dissociation constant symbol: K_{diss} or K_d ; the **equilibrium constant** for the reversible breakdown of a complex or a chemical compound into two or more simpler entities; the reciprocal of the **association constant**, K_{ass} . In some cases, an **apparent dissociation constant** (K'_d) is determined (for the distinction see **equilibrium constant**). It is also known as the **ionization constant** in cases where a substance dissociates into ions. For a dissociation (or ionization) of the general type $AB \rightleftharpoons A + B$, the dissociation (or ionization) constant is given by:

$$K_{\text{diss}} = [A][B]/[AB] = 1/K_{\text{ass}}$$

dissociation energy symbol: D ; the enthalpy (per mole) required to break a given chemical bond in a specified chemical entity by homolysis.

distal situated furthest from an origin, central line, or point of attachment.

distal protein a protein that binds late in the assembly of complex structures, such as ribosomes. Compare **proximal protein**.

distil or (US) **distill** to undergo, to subject to, to purify, or to separate by **distillation**.

distillate the liquid (or solid) product of **distillation**.

distillation the act or process of evaporating a liquid by boiling so that its components, which are vaporized at different temperatures or at different rates, can be separated and condensed back to a liquid (or solid). It is used to purify one liquid from a mixture of liquids or from (dissolved) solids. See **fractional distillation**.

distribution 1 (in chemistry) an alternative term for **partition**.

2 (in biochemistry) the pattern of occurrence of a substance within or between cells or tissues, or within a group of organisms, taxa, etc. 3 (in biology) the geographical range or pattern of occurrence of a particular taxon or genetic variant. 4 (in statistics) the frequency of occurrence of a variable at each of a number of discrete values. See also **normal distribution**.

distribution coefficient 1 an alternative term for **partition coefficient**. 2 in **gel chromatography**, the fraction of the stationary phase that is available for diffusion of a given solute species.

distribution law an alternative name for **partition law**.

distributive 1 of, relating to, or characterized by distribution.

2 (in molecular biology) or **nonprocessive** describing (the action of) any enzyme or catalytic complex that progressively synthesizes or degrades a biopolymer by effecting several or many cycles of the same type of reaction or reaction sequence and dissociates from the template or intermediate product after each catalytic event. Compare **processive**. —**distributively** adv.

disulfide or **disulphide** (in organic chemistry) any compound containing two bivalent sulfur atoms linked by a covalent bond and each attached to a carbon atom.

disulfide bond or **disulfide bridge** or **disulfide link** a covalent bond between two sulfur atoms. Such bonds in biomolecules are usually formed by the oxidation of **sulphydryl groups** in two neighbouring cysteine molecules or residues. Disulfide bonds may link two half-cystine residues in the same polypeptide chain, as in ribonuclease, or in different peptide chains, as in insulin or oxidized glutathione, and they often are important in maintaining the three-dimensional structure of polypeptides and proteins. They may also be reversibly formed in oxidation-reduction reactions as in the interconversion of

disulfide interchange

lipoamide and dihydrolipoamide. Also: **disulphide bond** or **disulphide bridge** or **disulphide link**.

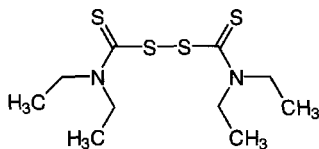
disulfide interchange or **disulphide interchange** any chemical reaction in which there is an interchange in the groups attached to two or more disulfide bonds; e.g.:



where R, X, Y, and Z are different polypeptide chains or other SH-bearing molecules. See also **protein disulfide isomerase**.

disulfide knot or **disulphide knot** a region at the centre of a **fibrinogen** molecule where the six polypeptide chains are linked together by eight disulfide bonds. Compare **cystine-knot**.

disulfiram *trivial name* for tetraethylthiuram disulfide; bis(diethylthiocarbamoyl) disulfide; a drug used to deter alcohol abuse in the treatment of alcoholism. Alcohol ingestion after disulfiram causes vasomotor disturbances, nausea, vomiting, and even unconsciousness and death. It acts by inhibiting the enzyme acetaldehyde dehydrogenase and hence slows the removal of acetaldehyde. It occurs naturally in the otherwise edible fruit body of the agaric mushroom *Coprinus atramentarius*. One proprietary name is Antabuse.



DIT *abbr.* for diiodotyrosine.

dithioerythritol *abbr.*: DTE; *erythro*-1,4-dimercapto-2,3-butanediol; both this and **dithiothreitol** are known as **Cleland's reagent**. It has similar properties to dithiothreitol but is less often used.

dithiothreitol *abbr.*: DTT; *threo*-1,4-dimercapto-2,3-butanediol; both this and **dithioerythritol** are known as **Cleland's reagent**. A water-soluble solid compound which, because of its low redox potential (−0.33 V, pH 7.0), is capable of maintaining monothiol completely in the reduced state and of reducing disulfides quantitatively, so becoming oxidized to a cyclic disulfide; it has little tendency to be oxidized directly by air.

diuresis increased or excessive flow of urine.

diuretic 1 producing an increase in the volume of urine. 2 any agent that increases the volume of urine.

diurnal 1 occurring during the day. 2 repeating every day; having a periodicity of approximately 24 hours. Compare **circadian**.

Diuron see DCMU.

divergent evolution or **divergence** biological evolution that tends to produce an increasing difference in some characteristic(s) between initially similar groups of, e.g., organisms or gene products. Compare **convergent evolution**.

divergent transcription the transcription of adjacent genes from opposite strands of duplex DNA and in opposite directions.

dixenic see **synxenic**.

Dixon plot a graphical method for the presentation of enzyme-kinetic data by which a Michaelis constant, K_m (see **Michaelis kinetics**), or an **inhibitor constant**, K_i , may be determined. The method is primarily used as a means of readily obtaining K_i . The reciprocal of the initial velocity (i.e. $1/V$) is plotted against a series of inhibitor concentrations $[I]$, at constant substrate concentration, $[S]$. When this is done for a number of values of $[S]$, the resulting lines intersect at a point corresponding to K_i . The value of $[I]$ at which this occurs is to the left of the ordinate (for competitive inhibition) and corresponds to $-K_i$. For non-competitive inhibition the lines meet on the abscissa. [After Malcolm Dixon (1899–1985), British enzymologist, who described it in 1972.]

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dizygotic describing twin offspring formed from two separate ova fertilized by two separate sperms. Compare **monozygotic**.

DL prefix (*in chemical nomenclature*) denoting an equimolar mixture of D and L enantiomers. See **D/L convention**.

dl- prefix (*in chemical nomenclature*) formerly used to denote **racemic**; now obsolete, use (±)-, *racemo*-, or *rac*-.

D/L convention a convention of symbols used to designate the **absolute configuration** around a chiral element in α -amino acids, cyclitols, monosaccharides, and derivatives of these classes of compounds. Within these classes it can be applied to any molecule of the type R_1CHXR_2 that can be oriented in a **Fischer projection** with the most highly oxidized atom at the top; then if X is on the right the configuration is designated by the prefix D-, whereas if X is on the left the configuration is designated by the prefix L-, relative to D-(+)- or L-(-)-glyceraldehyde. Racemic (i.e. equimolar) mixtures of D and L enantiomers are designated by the prefix DL-. α -Amino acids are designated D- or L- according to the configuration at the α -carbon atom. Cyclitols are designated D- when the formula is drawn in such a way that the substituent on the lowest-numbered asymmetric carbon atom is above the plane of the ring and the numbering is anticlockwise, and L- when it is clockwise; the locant of the defining centre should precede the prefix D- or L-, e.g. 1D-1-O-methyl-*myo*-inositol. For monosaccharides the highest-numbered chiral centre (i.e. carbon atom) is the reference atom determining whether the monosaccharide is D or L. When a compound contains more than one class of residue the configuration designators are D_s or L_s if they apply to an amino-acid residue, D_c or L_c to a cyclitol residue, and D_g or L_g to a monosaccharide residue (the subscript s denoting the reference α -amino acid serine, the subscript c denoting cyclitol, and the subscript g denoting the reference monosaccharide glyceraldehyde). The absolute configurations of chiral compounds of classes other than those listed are designated by application of the **sequence rule**. Compare **ambo**-, **d**-, **l**-, **dl**-, **isomerism**.

D-loop or **displacement loop** a DNA structure generated when an additional strand of DNA is taken up by a DNA duplex, so that one strand of the original duplex is displaced, forming a D-shaped loop. Formation of a D-loop is favoured in negatively supercoiled DNA and can occur spontaneously. Such D-loops are formed in covalently closed mammalian mitochondrial DNA, and are prominent as intermediates in genetic recombination, in which their formation is catalysed by strand-exchange enzymes, the best characterized of which is the RecA protein of *Escherichia coli*.

DMF or **DMFA** *abbr.* for *N,N*-dimethylformamide; a neutral liquid of low melting point (−61 °C) and a reasonably high boiling point (153 °C), miscible with water and with most common organic solvents, hence sometimes termed the **universal solvent**.

DMSO *abbr.* for dimethylsulfoxide (use discouraged, Me₂SO preferred).

DMT *abbr.* for the dimethoxytrityl group.

dna symbol for any of various bacterial genes that are involved in DNA replication. For example, *dnaG* is the gene for **DNA primase** in *Escherichia coli*.

DNA *abbr.* and *common name* for **deoxyribonucleate** (def. 1), or **deoxyribonucleic acid**; one of the two main types of **nucleic acid**, consisting of a long, unbranched macromolecule formed from one, or more commonly, two, strands of linked deoxyribonucleotides, the 3'-phosphate group of each constituent deoxyribonucleotide being joined in 3',5'-phosphodiester linkage to the 5'-hydroxyl group of the deoxyribose moiety of the next one. Compare **RNA**. The absence of a 2'-hydroxyl group on each deoxyribose moiety renders these phosphodiester linkages resistant to hydrolytic attack by alkali, in contrast to those of RNA. A DNA strand, unless circular, has polarity, with one 5' end and one 3' end, the two strands in duplex DNA having opposed polarities. Two purines, adenine and guanine, and two pyrimidines, cytosine and thymine, are the major bases present. The linear sequence of the purine and pyrimi-



dine bases carries genetic information, whereas the deoxy-sugar and phosphoric residues play a structural role; the base sequence is written in the 5'→3' direction. *See also* **double helix**. Specific forms, functions, molecules, or preparations of DNA may be designated by prefixes or suffixes, thus:

A-DNA or **DNA-A**, **A form of DNA**; **amdDNA**, **anti-messenger DNA**; **B-DNA** or **DNA-B**, **B form of DNA**; **cdNA**, **complementary DNA**; **cccDNA**, **covalently closed circular DNA**; **C-DNA** or **DNA-C**, **C form of DNA**; **ctDNA**, **chloroplast DNA**; **dsDNA**, **double-strand(ed) DNA** (*see* **duplex DNA**); **idNA**, **intercalary DNA**; **msDNA**, **multicopy single-strand(ed) DNA**; **mtDNA**, **mitochondrial DNA**; **nDNA**, **nuclear DNA**; **pDNA**, **plasmid DNA**; **rDNA**, **ribosomal DNA**; **recDNA**, **recombinant DNA**; **rfDNA**, **replicative-form DNA**; **rtDNA**, **recombinant DNA**; **scDNA**, **single-copy DNA**; **ssDNA**, **single-strand(ed) DNA**; **tDNA**, **transfer DNA**; **T-DNA**, **transfer DNA of the Ti plasmid**. *See* full entry for **T-DNA**. **Z-DNA** or **DNA-Z**, **Z form of DNA**.

DNAase or **DNAse** *abbr.* sometimes used for **deoxyribonuclease**; DNase is more usual.

DNA-binding protein F/G *see* **matrin**.

DNA blot transfer *see* **blot-transfer technique**.

DNA chip *see* **SBH**.

DNA cloning *see* **molecular cloning**.

DNA cycle *see* **cell-division cycle**.

DNA-dependent DNA polymerase *see* **DNA polymerase**.

DNA-dependent RNA polymerase *see* **RNA polymerase**.

DNA-directed DNA polymerase *recommended name for DNA polymerase*, EC 2.7.7.7.

DNA-directed RNA polymerase *recommended name for RNA polymerase*, EC 2.7.7.6.

DNA fingerprinting a term sometimes used to refer to **genetic profiling**.

(DNA) glycosylase or **DNA glycosidase** any of a group of glycohydrolases that initiate repair of DNA by hydrolysing N-glycosyl bonds linking deoxyribose residues with modified or incorrect bases to generate apurinic or apyrimidinic sites in the DNA and release the bases. They include **DNA-3-methyladenine glycosidase I** (EC 3.2.2.20), **DNA-3-methyladenine glycosidase II** (EC 3.2.2.21), and **formamidopyrimidine-DNA glycosidase** (EC 3.2.2.23). The enzymes form part of the pathway of base-excision repair, which acts on oxidative or non-oxidative damage, as determined by the specificity of individual enzymes. Sequences tend to be related; examples from *Escherichia coli*: **DNA-3-methyladenine glycosidase I** is constitutive and specific to the named base; database code 3MG1_ECOLI, 187 amino acids (21.08 kDa); **DNA-3-methyladenine glycosidase II** is inducible and releases any of 3-methyladenine, 3-methylguanine, 7-methylguanine, O-2-methylthymine, and O-2-methylcytosine; database code 3MG2_ECOLI, 282 amino acids (31.36 kDa).

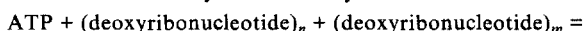
DNA gyrase *see* type II **DNA topoisomerase**.

DNA helicase an enzyme that catalyses unwinding of the DNA double helix during DNA replication and repair. It requires ATP. Example, *rep* gene product from *Escherichia coli*: database code REP_ECOLI, 673 amino acids (76.85 kDa).

(DNA) joinase an alternative name for **DNA ligase**.

DnaK or **heat-shock protein 70** *abbr.*: hsp70; a **heat-shock protein** that plays an essential role in the initiation of DNA replication in **lambda phage** and, in combination with DnaJ protein, acts to release proteins O and P. Example from *Escherichia coli*: database code DNAK_ECOLI, 637 amino acids (68.98 kDa).

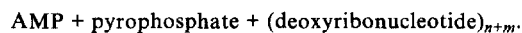
DNA ligase or **(DNA) joinase** or **DNA repair enzyme** either of two enzymes that restore broken phosphodiester bonds in deoxyribonucleic acids. (1) **DNA ligase (ATP)** EC 6.5.1.1; *systematic name*: poly(deoxyribonucleotide):poly(deoxyribonucleotide) ligase (AMP-forming); *other names*: polynucleotide ligase; DNA repair enzyme; (DNA) joinase; *former name*: sealase. An enzyme that catalyses the reaction:



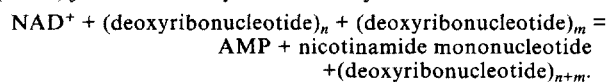
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Example from phage T4: database code DNLI_BPT4, 487 amino acids (55.23 kDa). (2) **DNA ligase (NAD⁺)** EC 6.5.1.2; *systematic name*: poly(deoxyribonucleotide):poly(deoxyribonucleotide) ligase (AMP-forming, NMN-forming); *other names*: polynucleotide ligase (NAD⁺); DNA repair enzyme; (DNA) joinase. An enzyme that catalyses the reaction:

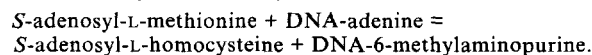


Example from *Escherichia coli*: database code DNLI_ECOLI, 671 amino acids (73.61 kDa).

DNA-like RNA *abbr.*: D-RNA or dRNA; a minor fraction of eukaryotic cellular RNA that has a base composition much closer to that of DNA than the bulk of RNA. It is mainly of high molecular mass, short half-life, and confined to the nucleus, and is probably the same as **giant messenger-like RNA** and **pre-messenger RNA**.

DNA-3-methyladenine glycosidase *see* **(DNA) glycosylase**.

DNA methylase either of two enzymes involved in **modification** and other roles. (1) **Site-specific DNA-methyltransferase (adenine-specific)** EC 2.1.1.72; *other names*: N-6 adenine-specific DNA methylase; modification methylase; restriction-modification system; it catalyses the reaction:



Example from *Escherichia coli*: database code DMA_ECOLI, 278 amino acids (32.06 kDa); it recognizes the base sequence GATC and is part of the system for postreplicative mismatch repair. (2) **Site-specific DNA-methyltransferase (cytosine-specific)** EC 2.1.1.73; *other names*: C-5 cytosine-specific DNA methylase; modification methylase; restriction-modification system; it catalyses the reaction between S-adenosyl-L-methionine and DNA-cytosine to form S-adenosyl-L-homocysteine and DNA-5-methylcytosine. Example from *E. coli*: database code DCM_ECOLI, 472 amino acids (53.40 kDa); it recognizes the base sequence CCWGG and methylates both strands.

(DNA) nicking-closing enzyme or **nicking-and-closing enzyme** alternative names for type I **DNA topoisomerase**.

DNA nucleotidylxotransferase *recommended name for terminal deoxynucleotidyltransferase*.

DNA photolyase EC 4.1.99.3; *recommended name*: deoxyribodipyrimidine photo-lyase; *other name*: photoreactivating enzyme. An enzyme that splits cyclobutadipyrimidine (in DNA) into two pyrimidine residues (in DNA), thereby repairing the DNA after light-induced **pyrimidine dimer** formation. It is light dependent, requiring the cofactors reduced flavin and either (in different enzyme types) a folate coenzyme or a modified flavin. Example, *Escherichia coli* enzyme required for 'light repair': database code PHR_ECOLI, 472 amino acids (53.61 kDa); four motifs.

DNA polymerase *common name* for either of two categories of enzymes that catalyse the synthesis of DNA from deoxyribonucleoside triphosphates in the presence of a nucleic-acid primer. Based on structural similarities, DNA polymerases have been grouped in two families. In family A are *Escherichia coli* and various other bacterial polymerases I, *Thermus aquaticus* Taq polymerase, some bacteriophage polymerases, and yeast mitochondrial polymerase γ . In family B are many polymerases of higher eukaryotes, yeast polymerases I to III, and polymerase REV3, *E. coli* polymerase II, archaeobacterial polymerases, and many viral polymerases.

(1) **DNA-directed DNA polymerase** EC 2.7.7.7; *other name*: DNA nucleotidyltransferase (DNA-directed); it is required for DNA replication and repair and has 3'-exonuclease activity (*see* **proofreading**). Eukaryotic DNA-directed DNA polymerases are of five types:

(a) DNA polymerase- α (also called DNA polymerase I or Kornberg enzyme) is a high-molecular-mass (>100 kDa) en-

DNA primase

zyme that was first detected in *E. coli*; in association with **DNA primase** it is replicative. Example (human): database code DPOA_HUMAN, 1462 amino acids (165.67 kDa) belongs to polymerase family B, with six conserved regions, four of which are involved in substrate binding. Several DNA viruses have type B polymerases; examples include adenoviruses and herpesviruses.

(b) DNA polymerase- β (or DNA polymerase II) is a low-molecular-mass (<50 kDa) enzyme recovered almost entirely in nuclear extracts.

(c) DNA polymerase- γ (or DNA polymerase III) is an enzyme of high molecular mass (>100 kDa) that copies synthetic polyribonucleotides with high efficiency, but does not copy DNA well. Examples from *E. coli*: α chain: database code DP3A_ECOLI, 1160 amino acids (129.76 kDa); β chain: database code DP3B_ECOLI, 366 amino acids (40.54 kDa); ϵ chain: database code DP3E_ECOLI, 243 amino acids (27.07 kDa). The protein at database code DP3X_ECOLI, 643 amino acids (71.06 kDa) represents ϵ and τ ; the production of the two protein products from this region is due to programmed ribosomal frameshifting.

(d) DNA polymerase- δ is a zinc finger protein with both polymerase and 3'-exonuclease activities; it is a heterodimer of 125 kDa and 50 kDa subunits.

(e) DNA polymerase- mt is the mitochondrial DNA polymerase.

E. coli has three DNA polymerases: (a) pol I, database code DPO1_ECOLI, 928 amino acids (103.00 kDa); this has 5'- and 3'-exonuclease activities; the 5'-exonuclease activity is involved in DNA replication; the polymerase activity is employed during DNA repair; 3-D structures are known for parts of the enzyme (database codes NRL1DPI1 and NRL1DPI2). (b) pol II, database code DPO2_ECOLI, 782 amino acids (89.82 kDa); is regulated by **lexA** and belongs to family B (above). (c) pol III, the major replication polymerase, has an α subunit with polymerase activity, database code DP3A_ECOLI, 1160 amino acids (129.76 kDa) and a total subunit composition of $(\alpha, \epsilon, \theta)_2 - \tau_2 - (\gamma, \delta, \delta', \psi, \chi)_2 - \beta_4$. The complex $(\alpha, \epsilon, \theta)$, the 'core', is assembled first.

(2) **RNA-directed DNA polymerase** is the EC recommended name for **reverse transcriptase**.

DNA primase a DNA-directed RNA polymerase (EC 2.7.7.-) of the bacterial **primosome**. It associates transiently with the primosome and synthesizes short RNA primers for the **Okazaki fragments** on both template strands at replication forks. Example from *Escherichia coli*: database code PRIM_ECOLI, 581 amino acids (65.49 kDa).

DNA print the pattern formed by denaturing the DNA in colonies of bacteria grown on a support and fixing the DNA to the support.

DNA provirus *see* provirus.

DNA repair enzyme any of the various enzymes involved in the different stages of DNA repair. Three steps are involved, each catalysed by a different set of enzymes: (1) the altered portion of DNA is removed by repair nucleases; (2) **DNA polymerase** fills the gap with a complementary copy; (3) the break is sealed by **DNA ligase**. There are two other related repair pathways, called (base)-**excision repair** and nucleotide-excision repair. *See also* repair enzyme.

DNA replication *see* replication of DNA.

DNA-RNA hybrid a double helix consisting of one DNA chain and one complementary RNA chain, the two chains being held together by hydrogen bonds between the complementary base pairs.

DNA-RNA hybridization or **DNA-RNA hybridisation** the formation of a **duplex** (def. 2) nucleic-acid structure that occurs when DNA and RNA molecules that have complementary base sequences are brought together. This phenomenon is exploited in a technique used to localize genes in a chromosome that depends on allowing radioactively labelled RNA to hybridize with denatured, single-stranded DNA from which the

RNA was transcribed and then determining the position on the DNA where hybridization has taken place by autoradiography. It also forms the basis of **Northern blotting**.

DNase *usual abbr. for deoxyribonuclease. Compare* **DNAase**.

(DNA) sealase an alternative name for **polynucleotide ligase** (def. 1).

DNA sequencing determination of the order in which bases occur in DNA. Usually the DNA is cut with restriction enzymes and subjected to either the **chemical cleavage method** or the **chain-termination method**.

DNA superhelix *see* supercoil.

(DNA) swivelase a former name for type I **DNA topoisomerase**.

DNA topoisomerase any enzyme that alters superhelix density in supercoiled DNA. Topoisomerases are found in all cell types, from microorganisms to those of humans, and in some viruses e.g. bacteriophage T4 and vaccinia. All topoisomerases isolated so far have the ability to relax negatively supercoiled DNA, converting it to a less supercoiled form. They have been divided into type I, which make a transient single-strand break, and type II, which make a transient double-strand break. **Type I DNA topoisomerase**, EC 5.99.1.2; *other names*: DNA topoisomerase, relaxing enzyme, untwisting enzyme, swivelase, nicking-closing enzyme, ω -protein; it catalyses the ATP-independent breakage of a single strand of DNA, which may be followed by passage (i.e., in the case of duplex DNA, transfer of the unbroken strand through the broken strand) and rejoining. This has the effect of relaxing one negative supercoil. Example from *Escherichia coli*: database code TOP1_ECOLI, 864 amino acids (97.29 kDa). This enzyme can relax only negatively supercoiled DNA, whereas some enzymes, e.g. that from calf thymus, can relax both negatively and positively supercoiled DNA. **Type II DNA topoisomerase**, EC 5.99.1.3; *other name*: DNA topoisomerase (ATP-hydrolysing); it catalyses ATP-dependent breakage, passage (in this case, of a double-strand through a double-strand break) and rejoining of the double-stranded DNA. This enzyme can thus decatenate circular duplex DNA. All type II enzymes can relax both positively and negatively supercoiled DNA. Bacterial **DNA gyrase** is a special example of this class of enzyme. In addition to the properties of the other type II enzymes, it catalyses the ATP-dependent introduction of negative superhelical turns into a closed circular DNA. It is an A_2B_2 tetramer. The energy-free topoisomerase activity of the A subunit, which breaks the DNA, is inhibited by 4-quinolone antibiotics such as nalidixic acid and ciprofloxacin. The energy-transducing activity of the B subunit is inhibited by novobiocin and other coumarin antibiotics. Example, subunit A: database code GYRA_ECOLI, 875 amino acids (96.85 kDa); subunit B: database code GYRB_ECOLI, 803 amino acids (89.72 kDa). The breaking and rejoining of DNA is catalysed by the A subunit; the B subunit is the ATPase. A distinctive property of this particular enzyme is that it can introduce negative superhelical turns into a closed circular DNA molecule with the coupled hydrolysis of ATP. *See also* linkage number, supercoil, winding number, writhing number.

(DNA) untwisting enzyme an alternative name for **(DNA) swivelase**.

DNA virus or **deoxyribovirus** any virus in which the genome consists of DNA, either double-stranded (in class I viruses) or single-stranded (in class II viruses). *See* **Baltimore classification of viruses**.

DNFB *abbr. for* (2,4)-dinitrofluorobenzene. *See* **Sanger's reagent**.

Dnp *symbol for* dinitrophenyl group.

DNP *abbr. for* 1 (2,4)-dinitrophenol. 2 deoxyribonucleoprotein (use deprecated).

Dns *symbol for* 5-(dimethylamino)naphth-1-ylsulfonyl; *see also* dansyl.

dNTP *abbr. for* any deoxynucleoside (5'-)triphosphate (the deoxynucleoside being either unspecified or unknown).

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DOC

DOC *abbr. for* (use discouraged) 1 deoxycholate (see **deoxycholic acid**), 2 deoxycorticosterone.

doca or **DOCA** *abbr. for* deoxycorticosterone acetate (see **deoxycorticosterone**).

Docetaxel a proprietary name for a semisynthetic analogue of taxol.

docking the binding of any macromolecule or part of such a molecule to its specific harbouring site on another molecular structure. See also **signal recognition particle receptor**.

docking protein another name for **signal recognition particle receptor**.

docosa+ or (before a vowel) **docos+** prefix indicating twenty-two or twenty-two times.

docosadienoic acid any straight-chain fatty acid having 22 carbon atoms and two double bonds. The (5Z,13Z)-form has been reported to occur in *Limnanthes douglasii*.

docosahexaenoic acid any straight-chain fatty acid having 22 carbon atoms and six double bonds. Only the (all-Z)-4,7,10,13,16,19-isomer, of the *n*-3 family, occurs naturally, being present in substantial amounts (10–15% of total fatty acids) in fish oils, and in variable amounts (a few percent of total) in animal glycerophospholipids. It can be formed metabolically from α -linolenic acid.

docosanoic acid or **behenic acid** a straight-chain saturated fatty acid having 22 carbon atoms that occurs as a constituent of cerebrosides and some seed oils (especially of *Lophira* spp.); in some of the latter it accounts for 20–30% of total fatty acids.

docosapentaenoic acid any straight-chain fatty acid having 22 carbon atoms and five double bonds. The (all-Z)-4,7,10,13,16-isomer is a member of the *n*-6 family and can be formed metabolically from linoleic acid; it is a constituent of animal glycerophospholipids. The (all-Z)-7,10,13,16,19-isomer, of the *n*-3 family, can be formed metabolically from α -linolenic acid; it is a precursor of 4,7,10,13,16,19-docosahexaenoic acid, and is present in small amounts (a few percent) in animal glycerophospholipids and in larger amounts (up to 10%) in fish oils.

docosatetraenoic acid any straight-chain fatty acid having 22 carbon atoms and four double bonds. Adrenic acid is found in animal glycerophospholipids, especially in brain and heart of animals fed sunflower or corn oil, and is presumed to be the (all-Z)-7,10,13,16-isomer.

docosatrienoic acid any straight-chain fatty acid having 22 carbon atoms and three double bonds. The (all-Z)-7,10,13-isomer occurs in animal lipids, especially during essential fatty acid deficiency.

docosenoic acid any straight-chain fatty acid having 22 carbon atoms and one double bond. The (11Z)-isomer, cetoleic acid, occurs in marine oils and rapeseed oil. The (13Z)-isomer, erucic acid, is an important constituent of rapeseed and other oils of plants belonging to the family Cruciferae. The possibility that it may be harmful has prompted efforts to breed crop varieties that yield oils containing low levels or none of this fatty acid.

Dod *symbol for* the dodecyl group.

dodeca+ or (before a vowel) **dodec+** prefix indicating twelve.

dodecandrin a type I ribosome-inactivating protein.

dodecanoic acid *systematic name for* lauric acid, a straight-chain saturated fatty acid having 12 carbon atoms. See also **laurate**.

dodecanoyl *symbol* Lau; *systematic name for* lauroyl.

dodecenoic acid any straight-chain fatty acid having 12 carbon atoms and one double bond.

dodecenoyl the acyl group derived from any dodecenoic acid.
dodecenoyl-CoA Δ^3 -isomerase EC 5.3.3.8; *systematic name*: dodecenoyl-CoA Δ^3 -cis- Δ^2 -trans-isomerase; *other names*: Δ^3 -cis- Δ^2 -trans-enoyl-CoA isomerase; acetylene-allene isomerase; 3,2 trans-enoyl-CoA isomerase. An enzyme that is involved in **unsaturated fatty acid oxidation**; it catalyses the isomerization of 3-cis-dodecenoyl-CoA to 2-trans-dodecenoyl-

dolichyl-diphosphooligosaccharide-pr

CoA, thereby producing a substrate for **enoyl-CoA hydratase**. Example from rat: database code S21091, 298 amino acids (32.86 kDa). See also **peroxisomal bifunctional enzyme**.

dodecyl *symbol*: Dod; *preferred name for* the alkyl group, $\text{CH}_3\text{--}[\text{CH}_2]_{10}\text{--CH}_2\text{--}$ (also known as **lauryl**), derived from dodecane.

n-dodecyl-N,N-dimethylglycine lauryldimethylbetaine; a zwitterionic betaine detergent, CMC 1.6–2.1 mM. *Proprietary name*: Empigen BB.

n-dodecyl- β -D-glucopyranoside a water-soluble nonionic detergent, CMC 0.13 mM.

n-dodecyl- β -D-maltoside a water-soluble nonionic detergent, aggregation number 98, CMC 0.1–0.6 mM.

dodecylsulfate the $\text{CH}_3\text{--}[\text{CH}_2]_{10}\text{--CH}_2\text{--O--SO}_3$ anion. See **sodium dodecyl sulfate**.

doghouse configuration (of a cyclobutadipyrimidine) see **pyrimidine dimer**.

Doisy, Edward Adelbert (1893–1986), US chemist, biochemist, and endocrinologist remembered for the isolation of the female sex hormones estradiol, estriol, and estrone, and especially for his isolation (from putrified fish meal) of a second antihemorrhagic principle named vitamin K_2 (now termed menaquinone) (see also **Dam**) and for his subsequent structural characterization of vitamins K_1 and K_2 ; Nobel Laureate in Physiology or Medicine (1943) 'for his discovery of the chemical nature of vitamin K' [prize shared with H. C. P. Dam].

dolichol the recommended term for any 2,3-dihydroprenol derived from four or more linked isoprene units, i.e. any prenol derivative of general structure $\text{H--}[\text{CH}_2\text{--C}(\text{CH}_3)=\text{CH--CH}_2]_n\text{--CH}_2\text{--CH}(\text{CH}_3)\text{--CH}_2\text{--OH}$ where *n* is greater than 3; in naturally occurring dolichols *n* is commonly in the range 13–23. (Note: Because the isoprene unit carrying the hydroxyl group is saturated in dolichols, they are not true **prenols** or **polyprenols**, though often classed as such. It is therefore recommended that, unless qualified, these collective terms are not used to include dolichols.) The phosphoric esters of dolichol (dolichol phosphate or phosphodolichol or dolichyl phosphate) and the diphosphoric esters (dolichol diphosphate or diphosphodolichol or dolichyl diphosphate) function in eukaryotes as carriers of mono- and oligosaccharide residues in the glycosylation of lipids and proteins within intracellular membranes. First, glycosyl groups are transferred from soluble nucleoside-diphosphosugars such as GDPmannose, UDP-N-acetylglucosamine, UDPglucose, or UDPxylose to form dolichyl phospho- or diphosphosugars. Then dolichyl diphosphooligosaccharides are built up from these by stepwise transfer of additional sugar residues. Finally the completed oligosaccharide moiety is transferred to a lipid or to a growing polypeptide chain with release of dolichyl diphosphate.

dolichol kinase EC 2.7.1.108; an enzyme that catalyses the formation of dolichyl phosphate from CTP and dolichol with release of CDP. Dolichyl phosphate is a precursor of dolichyl diphosphooligosaccharides, dolichyl phosphoglucose, and dolichyl phosphomannose, intermediates in N-linked glycoprotein oligosaccharide biosynthesis. Example from yeast: database code SC59_YEAST, 519 amino acids (58.84 kDa).

dolichol O-acyltransferase EC 2.3.1.123; an enzyme that catalyses the formation of dolichyl palmitate from palmitoyl-CoA and dolichol with release of CoA. Acylated dolichols are found in some cell membranes, though their function is obscure.

dolichyl the alkyl group derived from a **dolichol** by loss of its hydroxyl group.

dolichyl diphosphooligosaccharide a precursor for the transfer of N-linked oligosaccharide chains to protein in the synthesis of glycoproteins. It consists of dolichol esterified to diphosphate with an oligosaccharide chain linked to the diphosphate by the C-1 of the sugar at the reducing end of the chain.

dolichyl-diphosphooligosaccharide-protein glycosyl-

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dolichyl-phosphate β -D-glucosyltransferase

transferase EC 2.4.1.119; an enzyme that catalyses the reaction of dolichyl diphosphooligosaccharide with protein L-asparagine to form a glycoprotein with the oligosaccharide chain attached by glycosylamine linkage to protein L-asparagine, with release of dolichyl phosphate. Example from chicken: database code GSBP_CHICK, 508 amino acids (56.90 kDa).

dolichyl-phosphate β -D-glucosyltransferase EC 2.4.1.117; an enzyme that catalyses the reaction of UDPglucose with dolichyl phosphate to form dolichyl β -D-glucosyl phosphate (an intermediate in the biosynthesis of N-linked glycoprotein oligosaccharides) and UDP. It is a type II membrane protein of the endoplasmic reticulum involved in the glycosylation pathway. Example from yeast: database code ALG5_YEAST, 334 amino acids (38.30 kDa).

dolichyl-phosphate-mannose-protein mannosyltransferase EC 2.4.1.109; an enzyme that catalyses the reaction of dolichyl phosphate D-mannose with protein to form O-D-mannosylprotein and dolichyl phosphate. Two examples from yeast: database code PMT1_YEAST, 817 amino acids (92.57 kDa); and database code PMT2_YEAST, 758 amino acids (86.75 kDa). See also **dolichyl-phosphate β -D-mannosyltransferase**.

dolichyl-phosphate β -D-mannosyltransferase EC 2.4.1.83; an enzyme that catalyses the formation of dolichyl D-mannosyl phosphate, an intermediate in the biosynthesis of N-linked glycoprotein oligosaccharides, from GDPmannose and dolichol phosphate with release of GDP. Example (precursor of isoenzyme I) from yeast: database code DPM1_YEAST, 267 amino acids (30.33 kDa). See also **dolichyl-phosphate-mannose-protein mannosyltransferase**.

domain 1 any topological region having specific characteristics, contained within certain limits, and/or under individual control. 2 or **structural domain** a compact, globular region in the structure of a single protein molecule, which may consist of several such globular regions held together by more flexible parts of the polypeptide chain. It has been suggested that the word be reserved for large subassemblies that would be stable if the polypeptide chain connecting them to the rest of the protein molecule were to be cleaved and that the term **folding unit** be used to define small assemblies of secondary structure segments in a protein molecule. 3 a region of a protein molecule delimited on the basis of function without knowledge of and relation to the molecular substructure, as, e.g., the part of a protein molecule that binds to a receptor, that binds a substrate and possesses a catalytic function forming a **catalytic domain**, or that, in a membrane protein, passes through the membrane from one face to the other, forming a **transmembrane domain**. Such a region may contain more than one structural domain (see def. 2) 4 a poorly characterized length of chromosomal DNA, of the order of 50–200 kilobase pairs, that comprises all the (coding or noncoding) sequences required for the formation of mRNA for any one (specified) protein. 5 a zone within a cellular membrane consisting of one class of component, e.g. lipid. 6 a diverse set of cellular events, processes, and metabolic reactions controlled or affected by a specific agent, e.g. a hormone or another messenger. See **metabolic code**. 7 a mathematical aggregate to which a variable is confined. 8 (in physics) any region in a ferromagnetic solid in which all the atoms have the spins of their unpaired electrons aligned in the same direction: **magnetic domain**. See also **Appendix E**.

domain of bonding the two linked **binding sets** through which a pair of associated **protomers** are associated.

dominant (in genetics) describing a gene that produces the same character whether present in the homozygous state or in the heterozygous state (i.e. together with a different specified allele). The ineffective heterozygous allele is described as **recessive** to the dominant one. Dominant inherited diseases fall into one of three categories: (1) 'change of function' in which the abnormal protein acquires a novel activity that is deleterious to the cell, (2) 'dominant negative' in which the abnormal

protein forms heterooligomeric complexes with the protein from the normal allele thereby knocking out the function of the entire protein complex, (3) 'haplo-insufficiency' in which a single copy of the gene has lost function and the 50% of the normal amount of protein that is produced from the non-mutated allele is not adequate to preclude clinical symptoms. Examples: type (1), sodium channel disorders; type (2), mutations in collagen genes which lead to osteogenesis imperfecta; type (3), loss of function in one copy of the low-density-lipoprotein receptor gene resulting in familial hypercholesterolemia.

Donnan effect or **Donnan (membrane) equilibrium** the effect of the presence of charged macromolecules upon one side of a semipermeable membrane on the distribution of small, permeable ions across the membrane. At equilibrium, the small counterions will be more concentrated on the macromolecules' side of the membrane and the small ions of the same charge as the macromolecules will be less concentrated there, whereas on the side of the membrane with no macromolecules the concentrations of the small ions of opposite charge will be equal. It can be shown that, if the small ions are A^+ and B^- ,

$$[A^+]_{(1)} / [A^+]_{(2)} = [B^-]_{(2)} / [B^-]_{(1)} = r_D,$$

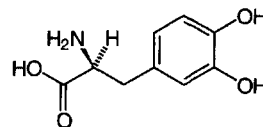
where r_D is the **Donnan ratio** and the subscripts 1 and 2 indicate the two sides of the membrane. [After Frederick George Donnan (1870–1956), British physical chemist.]

donor (in chemistry) any chemical species that is able to donate one or more electrons, atoms, or groups to another species, the acceptor.

donor atom 1 an atom that supplies two electrons in forming a **dipolar bond**. 2 an atom, added as an impurity to a semiconductor, that provides additional electrons in the conduction band. See also **electron donor**.

donor cell any living cell that contributes genetic information to another, **recipient cell**. See **conjugation** (def. 3), **transduction**, **transformation**.

dopa or **DOPA** *abbr. for* 3-hydroxytyrosine; 3,4-dihydroxyphenylalanine; 2-amino-3-(3,4-dihydroxyphenyl)propanoic acid; the naturally occurring enantiomer is 3-hydroxy-L-tyrosine (*abbr.*: L-dopa, also known as levodopa because it is levorotatory; $[\alpha]_{546}^{20} \approx 13.0$). L-Dopa is formed from L-tyrosine by tyrosine 3-monooxygenase, EC 1.14.16.2, and is an immediate metabolic precursor of the neurotransmitter **dopamine** and other **catecholamines**. L-Dopa is used in the treatment of **Parkinson's disease**.



dopac or **DOPAC** *abbr. for* dihydroxyphenylacetate.

dopachrome Δ -isomerase EC 5.3.3.12; *systematic name*: dopachrome Δ^7 - Δ^2 -isomerase; an enzyme that catalyses the conversion of dopachrome (2-carboxy-2,3-dihydroindole-5,6-quinone) to 5,6-dihydroxyindole-2-carboxylate. Dopachrome is formed non-enzymatically from dopaquinone, the product of **monophenol monooxygenase**. These are reactions in the pathway of melanin synthesis. A deficiency in the enzyme results from a mutation in mice known as **slaty**. The enzyme contains zinc. Example from *Mus musculus*: database code TYR2_MOUSE, 517 amino acids (58.57 kDa).

dopa decarboxylase see **aromatic-L-amino-acid decarboxylase**.

dopamine an alternative name for 3,4-dihydroxyphenylethylamine; 3-hydroxytyramine; 3,4-dihydroxyphenylethylamine; 4-(2-aminoethyl)-1,2-benzenediol; a **catecholamine** neurotransmitter formed by aromatic-L-amino-acid decarboxylase, EC 4.1.1.28, from 3,4-dihydroxy-L-phenylalanine. A metabolic

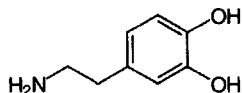
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precursor of **norepinephrine** and **epinephrine**, it is found in **dopaminergic** nerves in the brain and in the adrenal medulla. See also **Parkinson's disease**.



dopamine β -hydroxylase EC 1.14.17.1; *recommended name*: dopamine β -monooxygenase; *systematic name*: 3,4-dihydroxyphenethylamine,ascorbate:oxygen oxidoreductase (β -hydroxylating). An enzyme that catalyses hydroxylation of dopamine to norepinephrine in aminergic neurons releasing norepinephrine and epinephrine; ascorbate is simultaneously oxidized to dehydroascorbate. The enzyme is a copper protein, stimulated by fumarate. It is absent from dopaminergic neurons. Example, homotetramer, human precursor, database code DOPO_HUMAN, 603 amino acids (67.62 kDa).

dopamine β -monooxygenase see **dopamine β -hydroxylase**.

dopamine receptor any of several membrane proteins that bind dopamine and mediate its intracellular effects. Types D₁ to D₅ have been recognized; D₁ and D₅ (also called D_{1A} and D_{1B}) comprise one subfamily; D₂, D₃, and D₄ (also called D_{2A}, D_{2B}, and D_{2C}) comprise a second subfamily. D₃ receptors seem to be insensitive to guanine nucleotides and regulate conductance of Ca²⁺. All are seven-transmembrane-domain receptors that couple to G-proteins. D₁ and D₅ elevate cyclic AMP, but vary in binding characteristics; D₂ has a short and a long form originating from alternative splicing; in humans, the short form (D_{2S}) has 414 amino acids, and the long form (D_{2L}) has 443 amino acids. D₂ lowers cyclic AMP levels and regulates ion channels, opening K⁺ and inhibiting Ca²⁺ channels. D₃ and D₄ have binding characteristics similar to, but distinct from D₂; their action is not mediated by G-protein interaction with adenylate cyclase. Examples, all human; D₁: database code DADR_HUMAN, 446 amino acids (49.29 kDa); D₂: database code D2DR_HUMAN, 443 amino acids (50.62 kDa); D₃: database code D3DR_HUMAN, 400 amino acids (44.22 kDa); D₄: database code D4DR_HUMAN, 387 amino acids (40.89 kDa); D₅: database code DBDR_HUMAN, 477 amino acids (52.95 kDa).

dopaminergic describing a nerve that is activated by dopamine. The term is applied to nerves that act by releasing dopamine at their nerve endings. Compare **adrenergic**, **aminergic**, **cholinergic**, **peptidergic**, **purinergic**, **serotonergic**.

Doppler effect the apparent change in the frequency of acoustic or electromagnetic waves due to relative motion between the source and the observer. The frequency appears to increase when the source and the observer move towards each other, and to decrease when they move away from each other. [After Christian Johann Doppler (1803–53), Austrian mathematician and physicist.]

dorsal protein embryonic developmental morphogenetic nuclear protein; a protein whose concentration in the nucleus of a cell during the blastoderm stage of development determines the cell's lateral or ventral identity. It is similar to the product of the vertebrate proto-oncogene *c-rel*. Example from *Drosophila melanogaster*: database code DORS_DROME, 678 amino acids (75.39 kDa).

dose 1 a measured quantity of a drug, microorganism, radiation, etc. administered to an organism, often at specified intervals. In experimental pharmacology, the units and routes of administration should be specified and the quantity of the drug expressed per unit of animal mass, e.g. mol kg⁻¹, mg kg⁻¹, mg min⁻¹ kg⁻¹. 2 to prescribe or administer a dose or doses. — **dosage** *n*.

dose equivalent *symbol*: H; a measure of the biological effects of **ionizing radiation** upon humans (or other mammals)

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used in radiation dosimetry and health physics. It is usually equal to the **absorbed dose** multiplied by the **quality factor**. In the case of radiation arising from radioactive material within a person's body, the absorbed dose must be multiplied also by a distribution factor (normally unity, but in certain instances given a value of 5) in order to take account of any nonuniform distribution of radioactivity. The CGS unit of dose equivalent is the **rem**; the corresponding SI derived unit is the **sievert** (def. 1).

dose rate in radiation dosimetry, the amount of **ionizing radiation** (expressed in grays, rads, rems, röntgens, or sieverts) received or receivable by an irradiated object or organism in unit time. For personnel, it is usually expressed in sieverts per hour. (There is no separately named unit of dose rate.)

dose ratio an older name for **concentration ratio**.

dose-response curve any graphical representation of the relationship between the size of an administered dose and the extent of the response of the organism, or other test system, to the dose. The response may be some biochemical or physiological change in one or more organisms, tissue samples etc., or the proportion of organisms in a group that show a response.

dosimeter or **dose meter** an instrument for measuring the quantity of ionizing radiation administered to an organism or an object, or the intensity of a source of such radiation.

dosimetry the process of measuring administered quantities of ionizing radiation, drugs, microorganisms, etc.

dot hybridization or **dot hybridisation** or **dot blotting** a development of **blotting** or blot transfer, in which as little as 1 pg of a specific nucleic-acid sequence, bound to nitrocellulose as a small spot, can be detected using a ³²P-labelled probe as a 'dot blot'.

double-antibody method any technique in which a specific antibody, used to detect or measure its corresponding antigen, as in **immunoassay**, is isolated from the reaction mixture by combination with heterologous antibody to the specific antibody. Compare **antiglobulin test**.

double-beam spectrophotometer a **spectrophotometer** in which the incident light beam is split into two halves, one half passing through the sample and the second half passing through the reference so that direct measurement of the difference in absorption can be made. In one type the two emergent beams fall on different detectors and the resulting signal is a direct current; in another type the incident beam passes alternately through the sample and the reference, and the two emergent beams fall on the same detector resulting in an alternating current signal.

double-blind testing a method of evaluating the efficacy of a substance (e.g. a drug) in which neither the subjects nor the experimenter knows the identity of the samples at the time of the evaluation.

double bond a **covalent bond** that consists of two electron pairs shared by two atoms.

double-displacement (enzyme) mechanism or **ping-pong mechanism** or **substituted-enzyme mechanism** any enzymic mechanism in which an enzyme reacts with one substrate to give a covalently modified enzyme together with the release of one product, and the modified enzyme then reacts with the second substrate with the subsequent release of the second product and the return of the enzyme to its initial state.

double helix or **Watson-Crick model of DNA**: the main features of this model are that DNA consists of two antiparallel helical polynucleotide chains coiled around the same axis to form a double helix. Deoxyribose-phosphate backbones are on the outside of the helix and purine and pyrimidine bases lie approximately at right angles to the axis on the inside of the helix. The diameter of the helix is 2.0 nm and there is a residue on each chain every 0.34 nm. The angle between each residue on the same chain is 36°, so that the structure repeats after 10 residues (3.4 nm) on each chain. The two chains are held together by hydrogen bonds between pairs of bases, each mem-

double immunodiffusion

ber of the pair belonging to a different polynucleotide chain. Adenine is always paired with thymine and guanine with cytosine. The two chains are therefore complementary.

double immunodiffusion or **double diffusion** a technique of immunodiffusion in which both the antibody and the antigen diffuse towards each other through a gel to form lines of precipitation. See **Ouchterlony technique**.

double labelling an experimental strategy in which either a single chemical compound is labelled with two different nuclides or two compounds are each labelled with a different nuclide. It is used to follow the simultaneous transformations of two substances or two parts of one substance, or to distinguish identical molecules synthesized at different times. —**doubly labelled** *adj.*

double layer *see* **electric double layer**.

double membrane *see* **unit membrane**.

double-reciprocal plot a plot of $1/x$ versus $1/y$, where x and y are two variables. If the plot of y versus x has the shape of a rectangular hyperbola the corresponding double-reciprocal plot will be a straight line. Note: The x and y axes are not the asymptotes of the rectangular hyperbola. The **Lineweaver-Burk plot** is a double-reciprocal plot. See also **hyperbolic kinetics curve**.

double refraction *see* **birefringence**.

double refraction of flow *see* **flow birefringence**.

double-sector cell a cell (def. 2), used in analytical ultracentrifugation experiments, with two radial sector-shaped cavities, allowing the concentration distribution of a solute in one sector to be related to that of the solvent alone in the other sector; i.e. the second sector provides a reference for the first.

double-sieve hypothesis a hypothesis invoked to account for the unexpectedly low level of error (misreading) observed in protein biosynthesis. According to this hypothesis, the discrete synthetic and hydrolytic sites that exist on certain aminoacyl-tRNA synthetases provide a two-stage stereochemical sorting mechanism – likened in action to a pair of sieves – for the elimination of inappropriate amino acids. The synthetic site first rejects amino acids larger than that genetically specified; the hydrolytic site then preferentially destroys reaction products of amino acids smaller than, or isosteric with, the one specified. The second process has also been termed **editing** or **proofreading**. Such a mechanism would have particular importance in discriminating between a pair of very similar amino acids such as isoleucine and valine. Compare **kinetic proofreading**.

double-stop terminator two consecutive termination codons in a molecule of mRNA.

double-stranded or **double-strand** describing a **duplex** form of a nucleic acid. See **double-stranded RNA**, **duplex DNA**.

double-stranded RNA *abbr.*: dsRNA; a structure that occurs when complementary base sequences in single-stranded RNA form a **duplex** (def. 2). The RNA may fold back on itself to generate an antiparallel duplex structure, known as the stem, with a loop of unpaired bases at the end. The loop and stem together are called a **hairpin**. The two complementary sequences can result from an inverted repeat. See also **palindrome**.

doublet 1 any sequence of two nucleotides in a polynucleotide strand. 2 a closely spaced pair of spectral lines or peaks. Compare **twin**.

doubling dilution a serial dilution effected so that the concentration in each tube of the particular cell, substance, etc. of interest is half that in the preceding tube of the series.

doubling time the average time taken for the number of cells in a population to double. It will equal the **generation time** only if every cell in the population is able to form two daughter cells and there is no cell lysis.

Dounce homogenizer or **Dounce homogeniser** a hand-operated tissue homogenizer, usually of glass, consisting of a ball that is operated as a piston in a tube. The clearance is adjusted to give a fine particle size reduction without damage to cell

nuclei or mitochondria. [After Alexander Latham Dounce, US biochemist (1909–).]

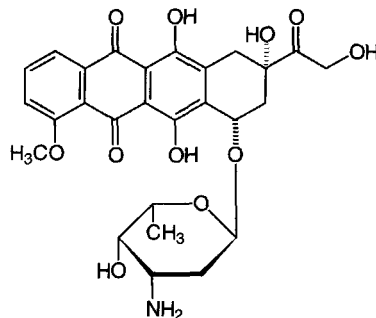
Dowex *proprietary name* for a group of synthetic ion-exchange resins made by Dow Chemical Company. See also **cholestyramine resin**.

down regulation 1 a decrease in response of a cell to a hormone brought about either by a decrease in receptor number or by the uncoupling of an effector component. 2 a decrease in the activity of an enzyme or the amount of a protein resulting from the action of an effector molecule on gene expression or protein synthesis. —**down-regulated** *adj.*

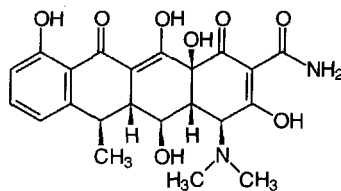
Down's syndrome or (formerly) **mongolism** a type of mental abnormality frequently associated with **trisomy** of chromosome 21. The condition is characterized by a small stature, a rounded head with obliquely slanted ('Mongol-like') eyes and high cheekbones, a fissured tongue with enlarged papillae, and a characteristic palmar print. The degree of mental defect varies considerably. [First described in 1866 by John Langdon Haydon Down (1828–96), British physician.]

downstream 1 (in molecular biology) that part of a strand of DNA lying towards the 3' end from the recognition sequence for a particular restriction enzyme. 2 towards the end of a multistage process, or at a stage nearer to the endpoint of the process than a key stage or one under consideration. Compare **upstream**.

doxorubicin a cytotoxic anthracycline antibiotic from *Streptomyces peucetius* var. *caesius*; its carbohydrate is **daunosamine**. It intercalates with DNA and prevents the actions of type II DNA topoisomerase. It also inhibits reverse transcriptase and acts on cell membranes. It has anti-tumour activity and has been used to treat some forms of cancer.



doxycycline 5-hydroxy-6-deoxytetracycline; an antibiotic having a wide range of activity against both Gram-positive and Gram-negative organisms. It is closely similar to **tetracycline** in structure and mechanism of action. One proprietary name is Vibramycin.



DPG or **2,3-DPG** *abbr.* for (2,3)-diphosphoglycerate (former name for **2,3-bis(phospho)-D-glycerate**).

DPG pocket *abbr.* for diphosphoglycerate pocket. See **bis-phosphoglycerate pocket**.

dpm or **d.p.m.** *abbr.* for disintegrations per minute.

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Dpm

Dpm see **A₂pm**.

DPN *abbr.* for diphosphopyridine nucleotide when its oxidation state is unknown or unspecified (obsolete; NAD now used).

DPN⁺ *abbr.* for diphosphopyridine nucleotide (oxidized) (obsolete name for **NAD⁺**).

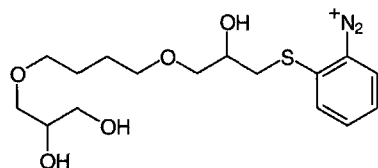
DPNH *abbr.* for reduced diphosphopyridine nucleotide (obsolete; NADH now used).

DPP *abbr.* for dipeptidyl peptidase.

Dpr see **A₂pr**.

dps *abbr.* for disintegrations per second.

DPT *abbr.* for 2-diazophenylthioether, the short name of the group $-O-CH_2-CH(OH)-CH_2-O-[CH_2]_4-O-CH_2-CH(OH)-CH_2-S-C_6H_4-N_2^+$, used to introduce reactive diazonium groups into cellulose or paper, forming **DPT-cellulose** or **DPT-paper**. These are used in **blot transfer** of proteins or nucleic acids. DPT is formed by diazotization of 2-aminophenylthioether (*abbr.*: **APT**) groups previously coupled to the support.



D region the (third) hypervariable region occurring between the V region and the J region in the heavy chains (but not in the light chains) of **immunoglobulin** molecules. It is coded for by the D (for 'diversity') gene segment.

dRib *symbol* for the monosaccharide deoxyribose, 2-deoxy-D-ribose.

Drk *abbr.* for 1 a small protein that bears SH2 domains (see **SH domain**) and acts as a link between **sevenless** protein and **son of sevenless** protein in the *Drosophila* photoreceptor system (downstream of receptor kinases). **2 Drk1** a mammalian integral membrane protein forming a voltage-dependent potassium channel of the delayed rectifier class. The voltage sensor, probably segment S4, is characterized by a series of positively charged amino-acid residues at every third position. Example from rat: database code CIKA_RAT, 853 amino acids (95.17 kDa).

D-RNA or **dRNA** *abbr.* for DNA-like RNA.

dropping-mercury electrode an electrode used in **polarography** in which mercury drops from a reservoir through a capillary tube into the solution under study. It has the advantage of constantly presenting a clean, unpolarized mercury surface to the solution.

Drosophila a genus of small dipterous insects that includes the fruit fly, *D. melanogaster*, also known as vinegar fly or banana fly. This is much used in genetic research because of the giant chromosomes, which occur in the cells of its salivary glands, and because of its short life cycle.

Drude equation an equation describing the variation of optical rotation with wavelength of light. It is useful in the interpretation of measurements of **optical rotatory dispersion**, when it may be expressed in the form:

$$[m']_{\lambda} = \frac{96\pi N_A}{hc} \times \frac{R\lambda_0^2}{\lambda^2 - \lambda_0^2}$$

where N_A is the Avogadro constant, h is the Planck constant, c is the speed of light, R is the rotational strength, λ_0 is the wavelength of the centre of an absorption band, λ is the wavelength of the observation, and $[m']_{\lambda}$ is the reduced mean residue rotation at that wavelength, as defined at **molar optical rotation**. See also **Moffitt-Yang equation**. [After Paul Karl Ludwig Drude (1863–1906), German physicist, noted for, *inter alia*, his work on refraction of light.]

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drug any naturally occurring or synthetic substance, other than a nutrient, that, when administered or applied to an organism, affects the structure or functioning of the organism; in particular, any such substance used in the diagnosis, prevention, or treatment of disease.

drug-metabolizing enzyme or **xenometabolic enzyme** any enzyme that acts on a drug or other foreign compound but is not known to have any action on a normal metabolite. Many enzymes previously included in this category have subsequently been found to act also on normally occurring substrates.

drug resistance the relatively enhanced resistance of an organism to the action of a drug. It may be caused by induction of an enzyme acting on the drug, by mutation, or by the acquisition of a plasmid coding for drug resistance.

dry ice solid carbon dioxide. It is a convenient cooling agent because, at ordinary atmospheric pressures, heat converts it directly to the gaseous form (sublimation point 194.65 K at 101 325 Pa; -78.5°C at 760 Torr).

DsbA a 21 kDa protein that is the bacterial functional equivalent in *Escherichia coli* of **protein disulfide-isomerase** in eukaryotes. There is little similarity in the primary structure of the two proteins. The structure of DsbA has been determined by X-ray crystallography and part of the structure shown to be similar to **thioredoxin**.

DSC *abbr.* for differential scanning calorimetry or differential scanning calorimeter.

dsDNA *abbr.* for double-strand(ed) DNA (see **duplex DNA**).

DSPD *abbr.* for disalicylidenepropanediamine.

dsRNA *abbr.* for double-strand(ed) RNA.

DSS *abbr.* for 2,2-dimethylsilapentane 5-sulfonic acid; a compound used as a water-soluble reference substance in **nuclear magnetic resonance spectroscopy**.

dsx *abbr.* for the *doublesex* gene in *Drosophila melanogaster*. Alternative splicing of the same transcript yields the male-specific protein (database code DSXM_DROME, 549 amino acids (57.35 kDa)); and the female-specific protein (database code DSXF_DROME, 427 amino acids (44.72 kDa)). They are transcription factors that bind to the fat body enhancer.

dT *symbol* for a residue of the deoxynucleoside thymidine (alternative to dThd).

dTDP *abbr.* for deoxyribosylthymine diphosphate or thymidine diphosphate, the common names for 2'-deoxyribosylthymine-5'-diphosphate; thymidine 5'-diphosphate; 5'-thymidylyl phosphate; thymine 2'-deoxyriboside 5'-diphosphate.

DTE *abbr.* for 1,4-dithioerythritol.

dThd *symbol* for a residue of the deoxynucleoside thymidine; 2'-deoxyribosylthymine; thymine 2'-deoxyriboside (alternative to dT).

dThd5'/P *symbol* for thymidine 5'-phosphate.

dThd5'/PP *symbol* for thymidine 5'-diphosphate (alternative to ppdT).

dThd5'/PPP *symbol* for thymidine 5'-triphosphate (alternative to pppdT).

dTMP *abbr.* for deoxyribosylthymine monophosphate or thymidine monophosphate, the common names for 2'-deoxyribosylthymine 5'-phosphate, thymidine 5'-phosphate, 5'-thymidylic acid, thymine 2'-deoxyriboside 5'-phosphate.

DTNB *abbr.* for 5,5'-dithiobis(2-nitrobenzoic acid); see **Ellman's reagent**.

DTT *abbr.* for 1,4-dithiothreitol.

dTTP *abbr.* for deoxyribosylthymine triphosphate or thymidine triphosphate, the common names for 2'-deoxyribosylthymine 5'-triphosphate; thymidine 5'-triphosphate; 5'-thymidylyl diphosphate; thymine 2'-deoxyriboside 5'-triphosphate.

dU *symbol* for a residue of the deoxynucleoside 2'-deoxyuridine (alternative to dUrd).

dual specificity mitogen-activated protein kinase kinase see **MAP kinase kinase**.

Dubnoff shaker a controlled-temperature water bath with a uniformly oscillating load platform and facilities for main-



Duboscq colorimeter

taining a controlled atmosphere. It is used for studies of the metabolism of tissues and microorganisms and for enzyme and blood-coagulation studies. [After Jacob William Dubnoff (1909–), US biochemist.]

Duboscq colorimeter an early, accurate visual colorimeter for determination of small quantities of substances in solution, in which the intensities of colour of two solutions, standard and unknown, were compared by varying the path-lengths of light passing through them from a single source until their apparent colours matched. This was achieved by placing the solutions in flat-bottomed vessels (termed cups) and adjustment of their positions in relation to two identical, fixed, transparent plungers through which the light passed, via prisms, to a divided-field eyepiece. The path-length in each solution was then read from a vernier scale. [After Jules Duboscq (1817–86), French optician.]

Duchenne muscular dystrophy *see* **dystrophin**.

ductless gland *an alternative name for* **endocrine gland**.

dUDP *abbr. for* deoxyuridine diphosphate, the common name for 2'-deoxy-5'-uridylyl phosphate; 2'-deoxyuridine 5'-diphosphate; uracil 2'-deoxyribose 5'-diphosphate.

Duffy system *symbol:* Fy; a blood-group system for which the antigenic determinant depends on the structure of an erythrocyte membrane glycoprotein, of molecular mass 35–55 kDa, the structural details of which are lacking.

Dulbecco, Renato (1914–), Italian-born US virologist notable for demonstrating how certain viruses are able to transform some cells into a cancerous state; Nobel Laureate in Physiology or Medicine (1975) jointly with D. Baltimore and H. M. Temin 'for their discoveries concerning the interaction between tumour viruses and the genetic material of the cell'.

dulcitol *another (older) name for* **galactitol**.

dUMP *abbr. for* deoxyuridine monophosphate, the common name for 2'-deoxyuridine 5'-phosphate; 2'-deoxy-5'-uridylic acid; uracil 2'-deoxyribose 5'-phosphate.

duodenum that part of the small intestine extending from the pylorus of the stomach to the jejunum. In mammals it lies around the head of the pancreas and receives the duct from the exocrine pancreas and the gall bladder. —**duodenal** *adj.*

Duolite *proprietary name for* a group of ion-exchange resins. It includes those formerly known by the names Biodeminrolit, De-Acidite, Zeo-Karb, and Zerolit.

duplex **1** having two parts or elements, especially of similar form or function. **2** describing a molecular structure (or part of one) in which two polynucleotide strands lie side by side in a head-to-tail orientation and are linked together along their length; *see* **duplex DNA**. **3** describing a telegraphic system, computer channel, etc. that permits two messages to be sent simultaneously in opposite directions.

duplex DNA double-stranded **DNA** in which the two polynucleotide chains are linked together by hydrogen bonds between complementary **base pairs** along their lengths, with the 3'→5' phosphodiester bonds of the two chains running in opposite directions. Such a molecule is usually coiled into a double helix and it may or may not be covalently closed into a circular molecule and/or formed into superhelical **DNA** (*see* **supercoil**). *See also* **double helix**.

duplicate (*in biology*) to reproduce by division into two equal parts; *see* **binary fission**.

duplication *see* **gene duplication**.

dUrd *symbol for* a residue of the deoxynucleoside deoxyuridine; 2'-deoxyribosyluracil; uracil 2'-deoxyribose (alternative to dU).

dUrd5'P *symbol for* deoxyuridine 5'-phosphate.

dUrd5'PP *symbol for* deoxyuridine 5'-diphosphate (alternative to ppdU).

dUrd5'PPP *symbol for* deoxyuridine 5'-triphosphate (alternative to pppdU).

dUTP *abbr. for* deoxyuridine triphosphate, the common name for 2'-deoxyuridine 5'-triphosphate; 2'-deoxy-5'-uridylyl diphosphate, uracil 2'-deoxyribose 5'-triphosphate.

dUTPase EC 3.6.1.23; *recommended name:* dUTP pyrophosphatase; *systematic name:* dUTP nucleotidohydrolase; *other name:* deoxyuridine-triphosphatase. An enzyme of nucleotide metabolism in eukaryotes that prevents the incorporation into DNA of deoxyuridylyl residues from dUTP by catalysing the hydrolysis of the latter to dUMP and pyrophosphate. Genes for dUTPase are found in several viruses, e.g. herpes simplex type 1: database code DUT_HSV11, 371 amino acids (39.08 kDa).

Duve *see* **de Duve**.

du Vigneaud, Vincent (1901–78), US biochemist distinguished for the synthesis of the posterior pituitary hormones oxytocin and vasopressin, for establishing the structure of biotin, and for his contributions to the synthesis of penicillin; Nobel Laureate in Chemistry (1955) 'for his work on biochemically important sulphur compounds, especially for the first synthesis of a polypeptide hormone'.

dwarfism underdevelopment of the body due to hyposecretion of **somatotropin**. Affected humans typically attain adult heights of less than 1.25 m.

Dy *symbol for* dysprosium.

dyad **1** two units treated as one. **2** a meiotic chromosome in which the two homologous members of the tetrad have separated. **3** a pair of cells resulting from the first meiotic division. **4** a bivalent atom, molecule, or group. *See also* **dyad symmetry**.

dyad axis a twofold **rotation axis**.

dyad-related describing a phenomenon that occurs at a specific position with respect to a **dyad axis**.

dyad symmetry **1** the type of symmetry occurring in molecular structures in which there is repetition after every 180° rotation around a dyad axis. **2** (*in molecular biology*) the formal description of a palindromic sequence in which the axis of symmetry is at the centre of the palindromic sequence. *See* **palindrome**.

dye a natural or synthetic substance that strongly absorbs certain wavelengths of visible or near ultraviolet light and is used to colour material by becoming firmly attached to it. *See also* **stain**.

dye-exclusion test a test for cell viability based on the phenomenon that living cells exclude some dyes, e.g. Eosin and Trypan Blue, whereas dead cells take them up and become stained.

dye-laser a device for obtaining high-intensity light of chosen wavelength by passing the monochromatic emission beam of a **laser** through a stream of liquid containing a fluorescent dye, and then passing the emitted broad-band fluorescent light through a monochromator. By variation of the dye employed a wide range of wavelengths of high-intensity light can be obtained from the single-wavelength laser emission. The light obtained by this device is necessarily of longer wavelength than that of the laser emission.

dye-ligand chromatography or **pseudo-affinity chromatography** or **pseudo-ligand affinity chromatography** a technique similar to **affinity chromatography** in which the immobilized ligand is any one of certain reactive dyes (coupled usually to cross-linked agarose), the dye commonly being a triazinyl compound, e.g. Reactive Blue 2 or Reactive Red 120. The technique is thus sometimes referred to (inappropriately) as **triazine-dye affinity chromatography**. The fixed dye molecules selectively bind particular enzyme or other protein molecules; the specificity of the interaction varies widely with both protein and dye, although frequently the dye appears to mimic a natural ligand of the protein. Examples of use are the isolation of nucleotide-requiring enzymes (*see also* **nucleotide-binding fold**) and the removal or purification of albumin from serum. The underlying mechanism is not fully understood but appears to involve both hydrophobic (i.e. apolar) and strongly ionic interactions. Many uses may in reality be examples of **affinity-elution chromatography**.

dynactin the largest protein component of the **dynactin complex**, a 20S multiprotein assembly of total mass about 1.2



dynamic

MDa, that activates dynein-based activity *in vivo*. A large structural component of the complex is an actin-like 40 nm filament composed of actin-related protein, to which other components attach. Vertebrate dynactin is a ubiquitously expressed cytoplasmic protein; several isoforms of mass ranging from 117 kDa to 160 kDa are produced by alternative splicing. It has a high degree of homology with the product of the *Drosophila* gene *glued*, essential during embryogenesis, some mutations of which kill homozygous embryos. Example from rat: database code DYNA_RAT, 1325 amino acids (146.05 kDa).

dynamic of, or concerned with, motion or force, or with dynamics.

dynamics 1 the branch of a science, especially mechanics, concerned with the motions of objects and the forces that produce such motions. It includes **kinetics**. 2 those forces that produce change or movement in a system or molecule; the motions (e.g. of residues within a macromolecule) so resulting.

dynamic state of body constituents the phenomenon, discovered in the early 1940s, that body components, especially proteins, are in a state of continuous degradation and resynthesis (i.e. of dynamic equilibrium). It was subsequently found to be applicable to other body constituents, although genomic DNA is a notable exception.

dynammin a microtubule-associated protein able to bind and hydrolyse GTP. The dynammin family includes two types, neuron-specific dynammin 1, and the widely expressed dynammin 2, which exhibit a high degree of sequence similarity in the 300 amino acid N-terminal region containing the GTP-binding site, but show divergence in a 100 amino acid proline-rich C-terminal region that in dynammin 1 but not dynammin 2 is a site for regulation of GTPase activity by phosphorylation–dephosphorylation. Dynammins are activated by protein kinase C and have a **pleckstrin homology domain** which binds acidic (especially inositol-) phospholipids with activation of the GTPase activity. Dynammins are involved in formation of microtubule bundles and in vesicular trafficking; purified dynammin 2 in solution forms helical structures that *in vivo* may facilitate the pinching off of vesicles budding from the plasma membrane and accelerate endocytic recycling. The *Drosophila* protein (a dynammin 2) is a product of *shibire*, a gene known to be involved in endocytosis. Examples, rat dynammin 1: database code DYN1_RAT, 851 amino acids (95.93 kDa); *Drosophila* product of *shibire*: database code DYN_DROME, 883 amino acids (98.54 kDa).

dyne *symbol*: dyn; the CGS unit of force, defined as the force required to give one gram an acceleration of one centimetre

per second per second. A gram weight thus represents a force of about 980 dyn. 1 dyn = 10^{-5} N.

dynein a large, multisubunit protein complex with ATPase activity, that is associated with microtubules and functions as a molecular motor to drive the movement of a diverse range of intracellular structures, such as endosomes, lysosomes, and mitochondria, along microtubules towards the centrosome. It is also attached to the peripheral fibrous components of eukaryotic flagella and cilia and is responsible for their movement; it may also play a role in the separation of chromosomes at cell division. The complex consists of two heavy chains (≈ 500 kDa), three or four intermediate chains (≈ 70 kDa) and four light chains (≈ 50 kDa). Example: cytoplasmic dynein heavy chain (MAP-1C) from rat: database code RAT-DYNEINC, 4644 amino acids, (531 kDa); intermediate chain from *Chlamydomonas reinhardtii*: database code DY06_CHLRE, 567 amino acids (63.52 kDa).

dynorphin an opioid tridecapeptide of high potency found in pig brain, duodenum, and pituitary, having the structure

Tyr-Gly-Gly-Phe-Leu-Arg-Arg-Gln-Phe-Lys-Val-Val-Thr,

the N-terminal pentapeptide sequence of which is identical with [5-leucine]enkephalin. It is synthesized as **preprodynorphin** (also called preproenkephalin B), from which its precursor protein **prodynorphin** (also called β -neoendorphin or proenkephalin B), a 236-residue polypeptide, is formed. See also **endorphin**.

dys+ prefix indicating abnormal, diseased, impaired, painful, or difficult.

dysfunction any abnormality or disturbance in the function of an organ, tissue, cell, or part of a cell. —**dysfunctional** *adj*.

dysgenic genetically deleterious.

dystroglycan a protein precursor yielding a 43 kDa transmembrane glycoprotein and an extracellular 156 kDa dystrophin-associated glycoprotein in skeletal muscle; the 156 kDa component binds **laminin**, acting as a laminin receptor that links the extracellular matrix and **sarcolemma** in skeletal muscle. Example (precursor) from human: database code DSG1_HUMAN, 1049 amino acids (113.59 kDa).

dystrophin a protein present in small amounts in normal muscle but absent or abnormal in patients with **Duchenne muscular dystrophy**. It may play a role in anchoring the cytoskeleton to the plasma membrane. Example from human: database code DMD_HUMAN, 3685 amino acids (426.19 kDa). See also **dystroglycan**, **utrophin**.

dystrophin-related protein (*abbr* DRP) see **utrophin**.

dystrophy any disorder, sometimes genetic, of the structure and functions of an organ or a tissue, as of muscle or bone. —**dystrophic** *adj*.

Ee

● **symbol for 1 electron**; it may be written as e^- to indicate its charge; e^+ denotes a positron. **2** (in mathematics) the transcendental number equal to the limiting value of the series $[1 + (1/n!)]$ for $n = 1$ to infinity; its value is 2.718 281 828 46. It is used as the base of natural (Napierian) logarithms. **3** equatorial; see **conformation**.

● **symbol for pseudo-equatorial**; see **conformation**.

● **symbol for 1 elementary charge** (or proton charge) **2** efficacy (of a drug).

E **symbol for 1 a residue of the α -amino acid L-glutamic acid** (alternative to Glu). **2** elimination reaction. **3** electromeric effect (in electron displacement). **4** exa+ (SI multiplicative prefix denoting 10^{18}).

[E]₀ or **[E]_t** or **[E]_{stoch}** (in enzyme kinetics) **symbol for the total or stoichiometric concentration of active (catalytic) centres**.

E₁ **abbr. for estrone**.

E₂ **abbr. for estradiol**.

E74 a protein, expressed during development of *Drosophila*, that correlates with an ecdysone-induced activity of puff 74EF. It is encoded by *Drosophila* *E1P74Ef*, an ecdysone-inducible gene that encodes two *ETS*-related protein products, E74A and E74B, that differ mainly in their N-terminal regions, and have substantial poly-Ala, poly-Asn, poly-Gln, poly-Gly, and poly-Ser domains.

E **symbol for 1 (in chemical nomenclature) a specific configuration around a double bond**; see **E/Z convention**. **2** electromotive force. **3** electric potential difference (of a galvanic cell). **4** energy; kinetic energy and potential energy may be denoted by the symbols E_k and E_p respectively. **5** extinction coefficient (no longer recommended). **6** (bold italic) electric field strength (see **electric field**).

(E)- prefix (in chemical nomenclature) denoting a geometric isomer in which the highest priority ligands according to the Cahn–Ingold–Prelog rules (see **sequence rule**) are located on opposite sides of a double bond. See also **E/Z convention**.

-E conformational descriptor denoting the envelope conformation of a monosaccharide or monosaccharide derivative.

E⁰ or **E⁺** **symbol for standard electrode potential**.

E^{0'} or **E⁺'** **symbol for standard electrode potential at a specified pH**.

E_a or **E_A** **symbol for (Arrhenius) activation energy**.

E_{ea} **symbol for electron affinity** (def. 2).

Eadie–Hofstee plot a graphical representation of enzyme kinetic data in which the reaction velocity divided by the substrate concentration, $v/[S]$, is plotted as ordinate against the reaction velocity, v , as abscissa. If a straight line is obtained, the intercept on the abscissa gives the maximum velocity, V , and that on the ordinate V/K_m . The slope is $-1/K_m$. [After George Sharpe Eadie (1895–1976) and Barend Hendrik Jan Hofstee (1912–80). US biochemists, who independently described it in 1942 and 1952, respectively.]

Eagle's medium any of various growth or maintenance media used in tissue culture and essentially comprising a balanced salt solution, usually **Earle's balanced salt solution** or **Hanks's balanced salt solution**, supplemented with amino acids, vitamins, serum, and antibiotics.

EAR-3 see **COUP**.

Earle's balanced salt solution a solution used in tissue culture to provide a normal ionic, pH, and osmotic environment for cell growth. It contains, per 100 ml distilled water, 0.68 g NaCl, 0.04 g KCl, 0.02 g CaCl₂, 0.02 g MgSO₄·7H₂O, 0.0125 g NaH₂PO₄·H₂O, 0.22 g NaHCO₃, 0.1 g glucose, and 0.002 g Phenol Red; it has a pH of 7.6–7.8.

early enzyme an enzyme that is transcribed from an **early gene** (def. 2) of a virus. Compare **late enzyme**.

early gene **1** or **immediate gene** any of a number of genes in-

involved in the earliest responses of cells to factors that initiate the transition from quiescence to proliferation. Early genes include those that encode transcription factors, such as the *c-JUN*, *c-FOS*, and *c-EGR* genes, and those coding for structural proteins such as actin. Early genes are further subdivided into immediate-early and delayed-early genes. **2** any viral gene that is transcribed early after the virus infects a host cell. Such genes probably code for proteins that are necessary for viral nucleic acid replication. Compare **late gene**.

early growth response protein any of a family of proteins that are expressed early in response to growth factors and function as transcriptional regulators, binding to the DNA sequence 5'-CGCCCCCGC-3'. Example, early growth response protein 1 from human (other names: krox24, TF ETR103); database code EGR1_HUMAN, 543 amino acids (57.51 kDa).

early protein a protein that is transcribed from an **early gene** (def. 2) of a virus. Compare **late protein**.

early quitter any incomplete polypeptide formed in an *in vitro* translation system.

Easson–Stedman model a model to explain the differences in biological activity of two enantiomeric molecules; it assumes that three of the groups attached to an asymmetric carbon atom in an agonist (or substrate) are involved in its attachment to a specific receptor (or enzyme), one enantiomer will therefore be a better agonist (or substrate) than the other. The concept is similar to, but considerably pre-dates, that developed by Ogston (see **Ogston concept**).

EBV or **EB virus** **abbr. for Epstein–Barr virus**.

EC **abbr. for 1 Enzyme Commission**; it is used as a prefix in the numerical designation of an enzyme (see **enzyme classification**). **2** electron capture. **3** enterochromaffin (as in **EC cell**).

EC₅₀ **abbr. for half effective concentration**; the molarity of an agonist that produces 50% of the maximal possible effect of that agonist. Other percentage values (EC₂₀, EC₄₀, etc.) can be specified. The action of the agonist may be stimulatory or inhibitory.

EC cell **abbr. for enterochromaffin cell**.

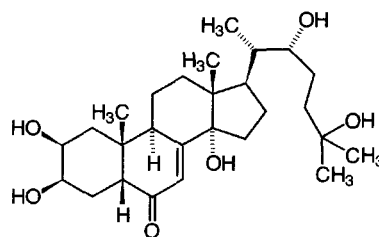
Eccles, (Sir) John Carew (1903–97), Australian neurophysiologist; Nobel Laureate in Physiology or Medicine (1963) jointly with A. L. Hodgkin and A. F. Huxley 'for their discoveries concerning the ionic mechanisms involved in excitation and inhibition in the peripheral and central portions of the nerve cell membrane'.

eccrine **1** describing a secretory cell that discharges its product without loss of cytoplasm. **2** describing a gland made up of such cells. Compare **apocrine**, **holocrine**, **merocrine**.

ECD **abbr. for 1 electron capture detector**. **2** electrochemical detection. See **polarography**.

ecdysis the periodic shedding of the exoskeleton and construction of a new cuticle, as in insects and crustaceans, or the shedding of the outer layer of the skin, as in reptiles. —**ecdysial** *adj.*

ecdysone **1** α -ecdysone (22*R*)-2 β ,3 β ,14,22,25-pentahydroxycholest-7-en-6-one; the **ecdysteroid** that is synthesized in and





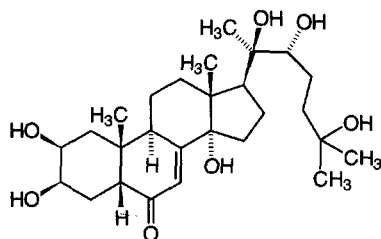
ecdysone receptor

secreted by the prothoracic glands of immature insects and the ovaries of the adult females. First isolated from the silkworm moth (*Bombyx mori*), it is the inactive prohormone of the moulting hormone **ecdysterone**; it may also have intrinsic hormonal activity at other stages of insect development. **2 β -ecdysone** an alternative name for **ecdysterone**. **3** a former name for **ecdysteroid**.

ecdysone receptor a receptor for ecdysone, located in the cell nucleus and related to the steroid/thyroid/retinoic acid family of nuclear hormone receptors. It consists of a heterodimer of a subunit named EcR (ecdysone receptor) and another named USP (ultraspiracle), the insect homologue of vertebrate retinoid X receptor. The heterodimer binds ecdysone with high affinity, and the ternary complex binds to DNA. Example: database code ECR_DROME, 878 amino acids (93.85 kDa).

ecdysteroid or (formerly) **ecdysone** any of a group of polyhydroxylated ketosteroids of which α -ecdysone is the parent substance for the purpose of nomenclature (see **ecdysone**). They are almost invariably C_{27} steroids, differing only in the number and/or steric arrangement of their hydroxyl groups. Ecdysteroids are ubiquitous in insects and other arthropods, in which they initiate postembryonic development, including the metamorphosis of immature forms and the development of the reproductive system and the maturation of oocytes in adult females. They have been found also in annelids, molluscs, and nematodes. Cognate substances, often known as **phytoecdysones**, have been identified in plants. The principal ecdysteroids in insects are α -ecdysone (see **ecdysone** (def. 1)) and **ecdysterone**.

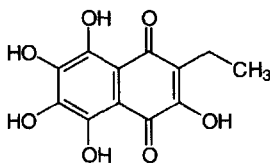
ecdysterone or **β -ecdysone** or **crustecdysone** or (insect) **moulting hormone** 20-hydroxyecdysone; (22*R*)-2 β ,3 β ,14,20,22,25-hexahydroxycholest-7-en-6-one; the **ecdysteroid** isolated both from insects and from crustaceans. In insects, ecdysterone is formed in peripheral tissues from α -ecdysone through the action of ecdysone 20-monooxygenase (EC 1.14.99.22). The hormone causes puffing of polytene chromosomes, and initiates moulting of larval and pupal forms by stimulating the epidermal cells to enlarge, divide, and secrete degradative enzymes that dissolve the protein and chitin components of the cuticle. Moreover, in the adult females of certain insects, e.g. mosquitoes (*Aedes* spp.) and *Drosophila melanogaster*, ecdysterone formed from α -ecdysone in the ovaries stimulates vitellogenin synthesis in the fat body.



ECF *abbr.* for extracellular fluid.

ECG or (*US*) **EKG** *abbr.* for 1 electrocardiograph. **2** electrocardiogram.

echinochrome A 2-ethyl-3,5,6,7,8-pentahydroxy-1,4-naphthalenedione; a red pigment of sea urchin eggs.



echinoderm any of the phylum Echinodermata of marine invertebrates that contains sea urchins, sea cucumbers, starfish, etc. They are more or less radially symmetrical, with calcareous skeletal plates in their skin and a well-developed coelom. Locomotion and gaseous exchange are accomplished by means of retractable tube feet. They are related to the same ancestral stock that gave rise to the vertebrates.

eclipsed conformation *see* **torsion angle**.

eclosion hormone any peptide hormone that programmes the death of certain muscles and neurons during metamorphosis of insects.

ecology the branch of biology dealing with the relations of living organisms to their surroundings. —**ecological** *adj.*; **ecologically** *adv.*; **ecologist** *n.*

ecomone any nontrophic molecule that ensures a flux of information between organisms in an **ecosystem**.

ecosystem a unit of the environment together with the organisms it contains. There is a constant interchange between living organisms and their chemical and physical environment.

ecotin a monomeric periplasmic protein of *Escherichia coli* that inhibits the pancreatic serine proteases chymotrypsin, trypsin, and elastases, enabling the organism to survive in the presence of these enzymes. It also inhibits blood coagulation factors Xa and XIIa and kallikrein, thereby having anticoagulant effects. Example from *E. coli*: database code ECOT_ECOLI, 162 amino acids (18.17 kDa).

ECP *abbr.* for eosinophil cationic protein.

ecstasy *see* **3,4-methylenedioxymethamphetamine**.

ECTEOA cellulose *abbr.* for a weakly basic ($pK \approx 7.5$) anion-exchange material of uncertain structure, prepared by the condensation of epichlorhydrin, triethanolamine (hence e,c + t,e,ola), and cellulose. It is useful for separations of proteins, nucleoproteins, and nucleic acids.

ecto+ *comb. form* indicating external, outer.

ectocrine describing or relating to a metabolite that, when released from the organism in which it was made, differentially affects other organisms of the same or a different species. Such a metabolite might be harmful to some members of a community and beneficial to others. The term includes **allomone**, **ectohormone**, **kairomone**, and **pheromone**. *Compare* **endocrine** (def. 1), **exocrine** (def. 1).

ectoenzyme any enzyme that is attached to the external surface of the plasma membrane of a cell.

ectohormone any **ectocrine** substance whose production and release benefits either the organism producing it or other members of the same species. *Compare* **endohormone**.

ectoparasite any **parasite** that lives on the exterior of its host organism.

ectopic occurring in an unusual place or in an unusual form or manner. For example, an **ectopic protein** is a protein produced by a neoplasm derived from a tissue that does not normally produce that protein. An example is the production of **vasopressin** and **corticotropin** by small cell lung cancer. Ectopic pregnancy occurs when the fertilized ovum implants outside the uterus e.g. in a fallopian tube. *Compare* **entopic**.

ectoplasm the outer, relatively rigid part of the **cytoplasm**. *Compare* **endoplasm**. —**ectoplasmic** *adj.*

ectoplast the part of the plasma membrane of a plant cell that is in contact with the cell wall.

ectoprotein any individual protein found on the exterior of cells. Such proteins may function, e.g., as mediators of cell-cell or cell-surface interactions, or as receptors for substances with regulatory actions on cells.

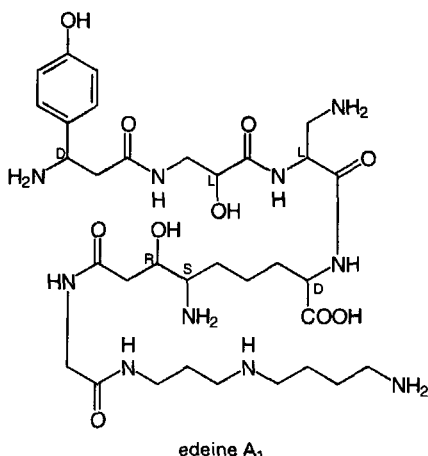
ectosymbiont a partner in a symbiotic relationship that remains outside the tissues and cells of the other partner, often occupying a body cavity; e.g. one of the cellulose-metabolizing microorganisms occurring in the digestive tract of ruminants. *Compare* **endosymbiont**. *See also* **symbiosis**.

ED₅₀ *abbr.* for median effective dose; i.e. the dose of a drug or other agent that produces, on average, a specified all-or-none response in 50% of a test population, or, if the response is

edeine

graded, the dose that produces 50% of the maximal response to that drug or agent.

edeine any of several pentapeptide-amide antibiotics elaborated by a strain of *Bacillus brevis* and effective against Gram-positive and Gram-negative bacteria, some fungi and other eukaryotic cells, and some neoplastic cells. Edeine A₁, N^α[(N²-[N-(β-tyrosyl)isoserine]-2,3-diaminopropionyl]-2,6-diamino-7-hydroxyazela-9-ylglycyl]spermidine, and its N^ω-amidinospermidine analogue, edeine B₁, are highly active; their respective constitutional isomers, edeines A₂ and B₂, have low intrinsic activity. In intact bacteria, low concentrations of edeine reversibly inhibit DNA synthesis and enhance RNA synthesis but do not affect protein synthesis. In cell-free systems, edeine A₁ interacts with both ribosomal subunits (preferentially the smaller one) and strongly inhibits translation.



Edelman, Gerald Maurice (1929–), US biochemist; molecular biologist, and neuroscientist; Nobel Laureate in Physiology or Medicine (1972) jointly with R. R. Porter 'for their discoveries concerning the chemical structure of antibodies'.

edema or (esp. Brit.) **oedema** swelling of an organ or tissue due to the accumulation of fluid. —**edematous** or (esp. Brit.) **oedematous** adj.

edestin a 300 kDa globulin obtained from hemp seed, castor-oil beans, and certain other seeds. It readily forms polymorphic crystals and will support the growth of animals in the absence of other dietary proteins. [From Greek *edestos*, edible.]

edetate see EDTA.

EDF abbr. for erythroid differentiation protein; see **activin**.

editing 1 the process or act of altering or adapting erroneous molecular structures, especially nucleotide sequences in polynucleotides, so as to preserve fidelity in the transfer and expression of genetic information. For example, the 3'→5' exonuclease activity of DNA polymerase I has an editing function in DNA polymerization by removing mismatched residues at the primer terminus of the growing chain. Another editing function is shown by aminoacyl-tRNA synthetases. In the presence of ATP and the 'wrong' amino acid, these enzymes do not catalyse the formation of the incorrect aminoacyl-tRNA but act as ATP pyrophosphatases, hydrolysing ATP to AMP and pyrophosphate. The term is applied also to the processes of conversion of RNA precursors, e.g. hnRNA, to the mature form, mRNA. 2 the insertion, deletion, or substitution of nucleotides within nascent RNA transcripts to produce RNA molecules with sequences that differ from those coded genomically. Most examples have been found in organelle-encoded RNAs, e.g. mitochondrial RNAs, and include

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ApoB, several **glutamate receptor** mRNAs, and **Wilms tumour** susceptible gene mRNA. In the case of ApoB, a cytidine residue is deaminated to a uridine residue so that the codon for glutamine (CAA) becomes a stop codon (UAA); hence, translation results in a shortened form of the protein.

editosome a macromolecular complex involved in the editing of RNA transcripts. Specific ribonucleotide complexes include guide RNA (see **gRNA**), which specifies the edited sequence.

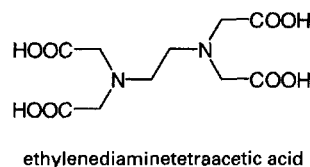
Edman degradation a procedure used in sequencing (poly)peptides in which amino-acid residues are removed stepwise from the N terminus by reaction with phenylisothiocyanate, C₆H₅-N=C=S, to form the phenylthiocarbamyl-peptide (abbr.: PTC-peptide). This is then cleaved in anhydrous acid (which favours attack by the sulfur atom on the carbonyl carbon of the terminal peptide) to give a thiazolinone intermediate and the release of the remainder of the peptide chain. The thiazolinone intermediate is hydrolysed to the phenylthiocarbamyl derivative of the N-terminal amino acid, which then cyclizes to form the phenylthiohydantoin derivative (abbr.: PTH-amino acid). The PTH-amino acid is extracted into organic solvents and identified by chromatography. These derivatives have a strong UV-absorption at about 259 nm, which permits them to be visualized against a fluorescent screen using an ultraviolet light source. As the remainder of the peptide chain is liberated intact, the procedure may be repeated – up to about ten amino-acid residues may be sequenced in this way, or more in favourable cases. The procedure may be automated, e.g. in a **sequenator**. [After Pehr Victor Edman (1916–77), Swedish protein chemist, who described the method in 1956.]

Edman reagent phenylisothiocyanate; C₆H₅-N=C=S. See **Edman degradation**.

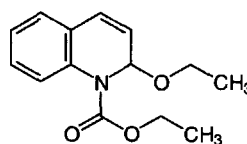
EDR abbr. for equi-effective dose ratio.

EDRF abbr. for endothelium-dependent releasing factor (see **nitric oxide**).

EDTA ethylenediaminetetraacetate; edetate; any anion or salt, esp. the (hydrated) disodium salt, of edetic acid, ethylenediamine-*N,N,N',N'*-tetraacetic acid, (ethylenedinitrilo)tetraacetic acid. Commonly used as the disodium salt, it is a powerful chelating agent for divalent metal ions.



EEDQ abbr. for *N*-ethoxycarbonyl-2-ethoxy-1,2-dihydroquinoline; an agent for peptide condensation that causes little or no racemization; it is useful in coupling ligands to insoluble polymers. It also causes progressive and irreversible inhibition of mitochondrial F₁-ATPase.



eeF-2 a eukaryotic **elongation factor** that is equivalent to the prokaryotic **EF-G**.

EET (formerly) abbr. for epoxyeicosatetraenoate.

EF abbr. for elongation factor.

EF-1 see **EF-T**, **elongation factor**.



EFE *abbr. for ethylene-forming enzyme.*

effective dose *see* ED₅₀.

effective theoretical plate number *symbol:* N ; a number indicative of chromatographic column performance when resolution is taken into account: $N = 16R_s^2/(1 - \alpha)^2$, where R_s is the peak resolution and α the separation factor (i.e. the ratio of the distribution ratios or coefficients for the two substances being resolved when measured under identical conditions); by definition α is greater than unity. *Compare* **height equivalent to an effective theoretical plate**. *See also* **theoretical plate**.

effectomer the part of a two-component agonist that brings about the biological effect but only when bound to a **haptomer**, which enables it to interact with the cell membrane and exert its toxic action on intact cells. For example, the A fragment of diphtheria toxin can only exert its toxic effects on a target cell when bound to the B fragment.

effector 1 or modifier or modulator (*in molecular biology*) any small molecule or ligand that interacts with an enzyme thereby changing its catalytic behaviour but that is not itself changed during the enzyme action. A **positive effector** enhances catalytic activity while a **negative effector** reduces it. *See also* **allosteric effector**. **2** (*in physiology*) a cell or organ that produces a physiological response when stimulated by the nervous system.

effector site or regulatory site any site on an enzyme molecule that binds an **effector**.

efferent 1 conveying outwards from a part, organ, or centre; e.g. of a blood vessel, nerve, or duct. **2** an efferent part, e.g. a blood vessel or nerve. *Compare* **afferent**.

efficacy *symbol:* e ; a quantitative index of drug action related to the magnitude of a tissue response generated at a given level of receptor occupancy with (full) agonists (of high efficacy) generating the maximum possible response. Drugs with lower efficacy that cannot generate a maximal response even when all receptors are occupied are known as **partial agonists**; **antagonists** have an efficacy of zero. The overall response to a drug will be a function of its efficacy and receptor binding affinity, which determines the proportion of receptors occupied at a given concentration of drug. It may be formulated as the combination of an agonist with its receptors that results in a signal or a **stimulus**, S , equal to the product of the efficacy of the agonist, e_A , and the proportion of receptors occupied, P_{AR} ; i.e. $S_A = e_A P_{AR}$. *See also* **intrinsic efficacy**, **maximal agonist effect**.

efficiency the ratio of the useful output of a machine, device, system, etc. to the input, whether in terms of thermal, mechanical, or radiation energy, or of biological or chemical conversions.

efficiency of counting *see* **counting efficiency**.

efficiency of plating *abbr.:* EOP; a quantification of the relative efficiencies with which different cells can be infected by viruses, and support viral replication. It is the ratio of the plaque count to the number of virions in the inoculum.

efflux *see* **flux**.

EF-G or translocase *abbr. for an elongation factor of Escherichia coli* that promotes the GTP-dependent translocation of the nascent protein chain from the A-site to the P-site of the ribosome. Database code EFG_ECOLI, 703 amino acids (77.36 kDa). An equivalent factor in eukaryotes is named eEF-2.

EF-hand a **helix-turn-helix** motif that binds Ca^{2+} ; it is found in many Ca^{2+} -binding membrane proteins and also in certain others, e.g. **myosin** light chain. It was first recognized in parvalbumin, in which it involves helices E and F. To envisage the motif, use the right hand with thumb and forefinger extended at $\approx 90^\circ$ with the remaining three fingers clenched. The thumb will then point towards the C terminus of helix F and the forefinger will point along helix E in the N-terminal direction, the clenched fingers tracing the course of the E-F loop about the bound calcium ion. Two apposed right-hands, representing the E and F and the C and D helices of the molecule, are related by a 2-fold axis of symmetry.

efrapeptin or efrastatin or A23871 a hydrophobic peptide fun-

gal antibiotic that inhibits oxidative phosphorylation by binding to the soluble component, F_1 , of mitochondrial **H⁺-transporting ATP synthase**.

EF-T or transfer factor a cytoplasmic protein of *Escherichia coli* and other prokaryotes which functions in protein synthesis to promote the GTP-dependent binding of aminoacyl-tRNAs to the A-site of the ribosome during protein chain elongation. EF-T can be dissociated into two polypeptides, EF-Tu (EF-1A) and EF-Ts (EF-1B), the complex being the main form in bacterial cells. These are so described since on their initial isolation one was heat-labile, or unstable, (Tu) and one was heat-stable (Ts). Only Tu functions directly in the GTP-dependent binding of the aminoacyl-tRNA to the A-site of the ribosome but Ts is required for regenerating Tu-GTP after the binding reaction which results in the hydrolysis of GTP to GDP. The factors were originally isolated from both *E. coli* and *Pseudomonas fluorescens*. The structures from *E. coli* are known, database code EFTS_ECOLI, 283 amino acids (30.39 kDa), EFTU_ECOLI, 393 amino acids (43.13 kDa). The 3-D structure of Tu shows high alpha-helix content with identified binding sites for GTP and aminoacyl-tRNA. The Tu from *E. coli* is also a subunit of RNA polymerase of phage Q β (Q beta). *See also* **elongation factor**.

EF-Ts now designated EF-1B; *see* EF-T.

EF-Tu now designated EF-1A; *see* EF-T.

egagasin an amphipathic glycoprotein, found in microsomal (i.e. endoplasmic reticulum) membranes of mouse liver, that forms a complex with microsomal (but not lysosomal) β -D-glucuronidase (EC 3.2.1.31) and thereby anchors it to the microsomal membranes. It may similarly anchor other polar proteins. It hydrolyses carboxylic esters to an alcohol and carboxylic anion. It has a C-terminal consensus sequence, HXEL, that retains the proteins in the endoplasmic reticulum. Example from *Mus musculus*: database code A55281, 562 amino acids (61.51 kDa). [From e (endoplasmic) + g (glucuronidase) + Greek *syn*, with.]

EG cell or L cell an **enteroglucagon**-producing endocrine cell. Such cells are found in the basal part of the mucosal glands in the lower intestine; the highest concentrations occur in the terminal ileum and colon. They contain large, dense, secretory granules and are easily distinguishable from other endocrine cells in the same area.

EGF *abbr. for epidermal growth factor*.

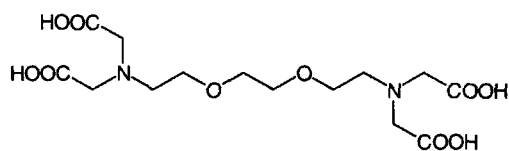
egg the reproductive structure of certain animals, e.g. reptiles, birds, and insects, consisting of an **ovum** together with nutritive and protective tissues, and from which, when fertilized, a young offspring emerges.

egg albumin the common name for **ovalbumin**.

egg cell an alternative name for **ovum**.

egg white the white part of an egg (especially a bird's egg) surrounding the yolk. Its principal organic constituent is egg albumin (*see* **ovalbumin**).

EGTA *abbr. for ethylene glycol-O,O'-bis(2-amino-ethyl)-N,N,N',N'-tetraacetic acid*, [ethylenebis(oxyethylenetrinitrilo)]tetraacetic acid; also for any anion or salt thereof. It is a chelating agent with a high affinity for Ca^{2+} ions.



Ehlers-Danlos syndrome a heterogeneous group of inherited human disorders sharing many phenotypic features. Classically the clinical picture includes hyperelastic, fragile skin, poor wound healing, hyperextensible joints, and easy bruising. Seven distinct genetic forms have been described. Type I,

Ehrlich

which is severe, is inherited as an autosomal dominant; the biochemical lesion is unknown. Types II and III are inherited as autosomal dominants; the biochemical lesions are unknown. Type IV is inherited as an autosomal dominant, but X-linked inheritance is not excluded; patients lack type III collagen and analysis has shown that skin collagen from these patients lacks a peptide of about 100 amino-acid residues that is characteristic of type III collagen. Type V shows X-linked inheritance and fibroblast cultures from a patient show a diminished lysyl oxidase activity, both in the cells and the medium. Type VI is inherited as an autosomal recessive, and collagen from such patients is deficient in hydroxylysine; fibroblast cultures from these patients contain an abnormal lysyl hydroxylase (EC 1.14.11.4). Type VII is presumed to be inherited as an autosomal recessive and the biochemical lesion is a deficiency of procollagen peptidase, with a consequent failure to normally convert procollagen to collagen. [After Edward Ehlers (1863–1937), German dermatologist, and Henri Alexandre Danlos (1844–1932), French dermatologist, whose respective descriptions of the disorders were published in 1901 and 1908.]

Ehrlich, Paul (1854–1915), German medical scientist famous for his work in chemotherapy, especially for his demonstration of the efficacy of synthetic arsenicals in spirochaetal infections, and notable also for his studies of the interaction of dyes with living tissues, for his work on the assay of antisera, and for advancing the side-chain theory of immunity; Nobel Laureate in Physiology or Medicine (1908) jointly with I. I. Mechnikov 'in recognition of their work on immunity'.

Ehrlich's reagent a solution of *p*-dimethylaminobenzaldehyde in conc. HCl. It forms coloured complexes with a number of compounds, such as indoles, aromatic amines, ureides, and hydroxyproline, and is useful in the detection and estimation of these compounds.

EIA *abbr.* for enzyme immunoassay.

eicosa+ or (before a vowel) **eicos+** a variant spelling (*esp.* US) of **icosa+** (*comb. form* denoting twenty or twenty times). This spelling is universally used in naming **eicosanoids**, although not officially recommended (but see **Appendix B**).

eicosadienoic acid any straight-chain fatty acid having twenty carbon atoms and two double bonds per molecule. The *all-Z*-(8,11)-isomer is a normally minor metabolite of oleic acid in mammals, but is important as an intermediate in the pathway synthesizing *all-Z*-eicosa-5,8,11-trienoic acid (**Mead acid**) during essential fatty acid deficiency. The (5,11)-isomer occurs in gymnosperms, and several others have been synthesized. The (11Z,14Z)-isomer occurs as an intermediate in the synthesis of arachidonate from linoleate via dihomog- γ -linolenate, but this is not the major pathway for this biosynthesis.

eicosanoic acid the systematic name for **arachidic acid**.

eicosanoid any of various agents that are C₂₀ polyunsaturated fatty acids, commonly arachidonic acid (see **eicosatetraenoic acid**), or an **eicosapentaenoic acid**, or their skeletal derivatives. They include the **leukotrienes** and the **prostanoids** and are noted for their widespread biological activity, such as contraction or relaxation of smooth muscle, platelet aggregation, and the inflammatory response.

eicosapentaenoic acid any straight-chain fatty acid having twenty carbon atoms and five double bonds per molecule. The most noteworthy is the *all-Z*-(5,8,11,14,17)-isomer, which occurs in animal phospholipids, especially those of many marine species; fish oils are a good source, with eicosapentaenoic acids often representing up to 10% of total fatty acids. Eicosapentaenoic acids act as precursors of the PG₃ series of **prostaglandins**. The suggestion has been made that a higher than normal content of this acid in the diet can protect against heart disease.

eicosatetraenoic acid any straight-chain fatty acid having twenty carbon atoms and four double bonds per molecule. The *all-Z*-(5,8,11,14)-isomer is **arachidonic acid**. The (8,11,14,17)-isomer is an important intermediate in the formation of long-

chain *n*-3 fatty acids from (9,12,15)-linolenic acid, being formed from that acid by desaturation at C-6 and chain elongation. The (5,11,14,17)-isomer is another naturally occurring form.

eicosatetraynoic acid any straight-chain fatty acid having twenty carbon atoms and four triple bonds per molecule. eicosa-5,8,11,14-tetraynoic acid (*abbr.*: ETYA) is a relatively nonspecific inhibitor of reactions involving arachidonate, including phospholipase A₂, **prostaglandin-endoperoxide synthase**, and **lipoygenase**.

eicosatrienoic acid any straight-chain fatty acid having twenty carbon atoms and three double bonds per molecule. The *all-Z*-(5,8,11)-isomer (**Mead acid**) is synthesized in mammals in essential fatty-acid deficiency. The *all-Z*-(8,11,14)-isomer, known also as dihomog-(6,9,12)-linolenic acid (*or* dihomog-linolenic acid), is a precursor of arachidonic acid and of the PG₁ series of **prostaglandins**. The (5,11,14)-isomer has been found in higher plants and algae, cats, marine organisms, and insects.

eicosatriynoic acid any straight-chain fatty acid having twenty carbon atoms and three triple bonds per molecule. 5,8,11-eicosatriynoic acid (*abbr.*: ETI) is an agent that selectively inhibits 5- and 12-lipoygenase without inhibiting **prostaglandin-endoperoxide synthase**.

eicosenoic acid any straight-chain fatty acid having twenty carbon atoms and one double bond per molecule. The 9*E*-isomer is gadolaidic acid (see **gadolaidate**); the 9*Z*-isomer is gadoleic acid (see **gadoleate**), a constituent of fish oils; and the 11*Z*-isomer is gondoic acid, present in seeds and fish oils.

EID *abbr.* for electroimmunodiffusion.

eIF *abbr.* for eukaryotic initiation factor (see **initiation factor**).

Eigen, Manfred (1927–), German physical chemist noted for developing techniques for the study of rapid reactions and applying them to problems of molecular biology; Nobel Laureate in Chemistry (1967), the prize being shared with R. G. W. Norrish and G. Porter 'for their studies of extremely fast chemical reactions, effected by disturbing the equilibrium by means of very short pulses of energy'.

eigenfunction a solution of a differential equation that has solutions only for particular values of some parameters.

eigenvalue any one of the possible values for a parameter of an equation for which the solution will be compatible with the boundary conditions.

Eijkman, Christiaan (1858–1930), Dutch physician, bacteriologist, and nutritionist noted for observing that unpolished rice grains contained an antiberiberi principle (later recognized as a vitamin and named vitamin B₁ or thiamine) and for being the first to produce a dietary deficiency disease experimentally; Nobel Laureate in Physiology or Medicine (1929) 'for his discovery of the antineuritic vitamin' [prized shared with F. G. Hopkins].

einstein a non-SI unit used to express the energy associated with one mole of photons during a photochemical reaction. 1 einstein = $N_A h\nu$, where N_A is the Avogadro constant, h is the Planck constant, and ν is the particular frequency of the electromagnetic radiation effecting the reaction.

Einstein, Albert (1879–1955), German-born Swiss then US patent examiner and mathematical physicist famous above all for his formulation of both the special theory of relativity (in 1903) and the general theory of relativity (in 1916); Nobel Laureate in Physics (1921) 'for his services to Theoretical Physics, and especially for his discovery of the photoelectric effect'.

Einstein-Sutherland relation a relation between the diffusion coefficient at infinite dilution, D_0 , and certain molecular parameters. $D_0 = kT/f$, where T is the thermodynamic temperature, k is the Boltzmann constant, and f is the frictional coefficient of the diffusing particle.

EKG *abbr.* (US) for 1 electrocardiograph. 2 electrocardiogram.

elafin another name for **SKALP**. [From elastase f inhibitor. The



elaidate

'f' was inserted to avoid confusion with elastin and gives an association to the Latin *finire*, to terminate.]

elaidate 1 numerical symbol: 18:1(9); the trivial name for (*E*)-octadec-9-enoate; $\text{CH}_3\text{---}[\text{CH}_2]_7\text{---CH=CH---}[\text{CH}_2]_7\text{---COO}^-$ (*trans* isomer); the anion derived from elaidic acid, (*E*)-octadec-9-enoic acid, a monounsaturated straight-chain higher fatty acid. 2 any mixture of free elaidic acid and its anion. 3 any salt or ester of elaidic acid. *Compare oleate*.

elaidinization or **elaidinisation** the *cis* to *trans* isomerization of mono- and polyunsaturated fatty acids, as in the isomerization of oleic acid to elaidic acid brought about by treatment with nitrous acid or by the action of rumen bacteria.

elaidoyl the trivial name for (*E*)-octadec-9-enoyl; $\text{CH}_3\text{---}[\text{CH}_2]_7\text{---CH=CH---}[\text{CH}_2]_7\text{---CO-}$ (*trans* isomer); the acyl group derived from elaidic acid, (*E*)-octadec-9-enoic acid. It occurs in acylglycerols to an appreciable extent in the body fats of ruminants, possibly having arisen from oleic acid by elaidinization. *Compare oleoyl*.

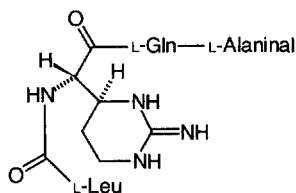
elaioplast any oil-storing leukoplast that occurs in plant cells.

ELAM abbr. for endothelial leukocyte adhesion molecule; i.e. E-selectin (see *selectin*).

elasmobranch 1 any member of the subclass Elasmobranchii, comprising fishes with cartilaginous skeletons such as sharks, skates, and rays. 2 of, pertaining to, or relating to the Elasmobranchii.

elastase historically, any proteinase that hydrolyses elastin. However, the name is now applied to a group of enzymes that vary in specificity. **Pancreatic elastase**, EC 3.4.21.36 (*other names*: pancreatopeptidase E; pancreatic elastase I), catalyses the hydrolysis of proteins, including elastin, with preferential cleavage at Ala-|-Xaa. Example (domestic pig): database code NRL_3EST, 240 amino acids (25.88 kDa); 3-D structure known. **Pancreatic elastase II**, EC 3.4.21.71, preferentially cleaves the carbonyl side of Leu, Met, and Phe. **Leukocyte elastase** EC 3.4.21.37 (*other names*: lysosomal elastase; neutrophil elastase; bone marrow serine protease; medullasin), catalyses the hydrolysis of proteins, including elastin, with preferential cleavage at Val-|-Xaa > Ala-|-Xaa. Pancreatic elastase and leukocyte elastase, in common with most elastases, are serine proteinases, but others, e.g. elastase (EC 3.4.24.26) from *Pseudomonas aeruginosa*, are zinc-containing metalloproteinases. **Elastase III** is pancreatic endopeptidase E (EC 3.4.21.70; *other name*: cholesterol-binding proteinase); it shows preferential cleavage at Ala-|-Xaa, but does not hydrolyse elastin.

elastatinal a peptide produced by actinomycetes that inhibits serine proteinases, particularly elastase.



elastic (of a body or a substance) capable of returning to its original shape after deformation. —**elasticity** *n*.

elasticity coefficient symbol: ϵ_S^v ; for a metabolic system in a steady state, the fractional change in the velocity, v , of an enzyme step caused by a fractional change in substrate, effector, or product concentration, S . It is given by:

$$\epsilon_S^v = \frac{S}{v} \left(\frac{\partial v}{\partial S} \right)_{y,z,\dots} = \frac{\partial \ln v}{\partial \ln S}$$

Since the elasticity coefficient is defined as the partial derivative of a velocity, v , with respect to S (multiplied by the scaling factor, S/v) all other variables that are able to influence the rate through the enzyme are held constant. There are, for any

reaction, as many elasticity coefficients as there are metabolites and effectors that interact with the enzyme in question. *See also* **controllability coefficient**, **response coefficient**, **sensitivity coefficient**.

elastin a major structural protein of mammalian connective tissues, especially elastic fibres, and found at sites such as the aorta, nuchal ligament, and lung. Glycine makes up one third of the amino-acid residues in elastin; glycine, proline, alanine, and valine together account for over 80% of its residues. The polymeric chains are cross-linked together into an extensible 3-D network, lysinonorleucine and desmosine cross-links being formed between allysine and lysine residues. A soluble precursor of elastin, **proelastin**, has been isolated from copper-deficient pigs in which the formation of allysine is blocked. Example (precursor) from human: database code ELS_HUMAN, 730 amino acids (63.19 kDa).

elastonection a protein, M_r 120 000, from human skin fibroblasts and associated with adhesion of mesenchymal cells to elastic fibres.

elective theory of immunity an alternative name for **selective theory of immunity**

electrical coupling the phenomenon of ion transport across a membrane, in the absence of a cation-anion cotransport system, when the movement of anions and cations, following the law of electroneutrality, is governed only by the passive permeabilities of the membrane to the ions in question.

electric birefringence a form of **birefringence**, also known as the **Kerr effect**, caused by the alignment of (macro)molecules in an electric field.

electric capacitance see **capacitance**.

electric cell an apparatus consisting of two reversible electrodes dipping into the same solution, or into separate solutions electrically connected through an electrolyte bridge. It is used to generate an electromotive force, for electrolysis, or to measure the chemical activity of a chemical species in one of the solutions. *See also* **half-cell**.

electric charge or **quantity of electricity** symbol: Q or q ; the attribute of certain elementary particles that is responsible for electric phenomena. Electric charge may be of two kinds, positive or negative, which interact electromagnetically such that like charges repel one another while unlike charges are attracted one to another. The SI derived unit of electric charge is the coulomb. *See also* **charge** (def. 3).

electric conductance or **conductance** symbol: G ; a measure of the readiness with which an electric current can flow through a given component of a circuit. For a direct-current circuit it is equal to the reciprocal of the resistance; for an alternating-current circuit it is equal to the resistance divided by the square of the impedance. The SI derived unit of conductance is the siemens. *See also* **mho**.

electric current or **current** symbol: I or i ; one of the seven SI base physical quantities – flow of electric charge through a conductor. The magnitude of a current is given by the amount of charge flowing per unit time. The SI base unit of electric current is the ampere.

electric dichroism the form of **dichroism** (def. 1) caused by orientation of (macro)molecules in an electric field.

electric dipole moment see **dipole moment**.

electric double layer an alternative name for **electrochemical double layer**.

electric field a field in which a stationary electric charge, Q , experiences a force, F , proportional to the magnitude of the charge. The electric field strength, E , is given by: $E = F/Q$; it is expressed in V m^{-1} . If the field is produced by other stationary charges it is called an **electrostatic field**. *Compare* **Coulomb's law**.

electricity 1 any of the forms of energy associated with moving or stationary electrons, protons, or other charge-bearing entities (whether molecular or otherwise). **Static electricity** is associated with stationary charges, **current electricity** with moving charges. 2 the science dealing with electricity.

electric potential or **potential** *symbol*: V or ϕ ; the work done in bringing unit electric charge from infinity to a specified point in an electric field. The SI derived unit of electric potential is the volt.

electric potential difference or **potential difference** *symbol*: E or U or ΔV or $\Delta\phi$; *abbr.*: pd; the work done when a unit electric charge is moved between two points having different electric potentials. If the points are joined by a conductor an electric current will flow between them. The SI derived unit of electric potential difference is the volt. *Compare* **electromotive force**.

electric resistance or **resistance** *symbol*: R ; the impediment to the flow of an electric current through a component of a circuit, a medium, or a substance. The SI derived unit of resistance is the ohm.

electro+ or (sometimes before a vowel) **electr+** *comb. form* indicating electric, electrically, electrolytic.

electroblot *see* **electrophoretic transfer**.

electroblotting *an alternative name for* **electrophoretic transfer**.

electrochemical analysis or **electrochemical assay** a chemical analysis or assay performed by means of electrolysis or electrodeposition.

electrochemical double layer any double layer of charges formed in a solution by the adsorption of a layer of ions of opposite charge at the surface of another charge-carrying phase, e.g. the walls of the vessel or the surface of a macromolecule.

electrochemical equivalent the mass of an ion deposited or liberated by the passage of 1 coulomb of electricity. In grams it is 1/96487 of the **chemical equivalent**.

electrochemical potential *symbol*: $\tilde{\mu}$; the change in free energy occurring when a charged solute is transported either up or down a concentration gradient. It is the sum of the free energy changes due both to the concentration gradient and to the electric potential. It is expressed in J mol^{-1} .

electrochemical series *an alternative name for* **electromotive series**.

electrochemistry the branch of science concerned with the study of electrolysis, electrolytic cells, and the applications and properties of ions in solution. —**electrochemical adj.**

electrochromatography the form of **chromatography** in which separation of the constituents of a mixture is due to the effect of a constant electric field. *See also* **zone electrophoresis**.

electrochromism any reversible change in optical absorption or emission spectra of a molecule in the condensed phase due to an external electric field. *Compare* **photochromism**, **thermochromism**. —**electrochromic adj.**

electrocortin the former name for **aldosterone** (before establishment of its structure).

electrocrystallization or **electrocrystallisation** **electrodeposition** that results in crystal formation.

electrode any conductor, such as metal, metal covered with one of its salts, carbon, or a thin glass or other membrane, used to establish electric contact with a solution or other non-metallic part of an electric circuit. The potential assumed by the electrode may depend on the (thermodynamic) activity of one of the solutes in the solution with which it is in contact. *See also* **anode**, **calomel electrode**, **cathode**, **enzyme electrode**, **hydrogen electrode**, **ion-selective electrode**, **oxygen electrode**, **quinhydrone electrode**, **reference electrode**.

electrodecentration or **electrophoresis convection** a technique for separating and fractionating proteins and other sols that exploits local differences in concentration and density that may occur during **electrodialysis**. Under the influence of gravity such density differences lead to large-scale separation of sols of high and of low concentrations. This results in a gradual increase in the concentration of a sol in the lower layers of the solution and a decrease in the upper layers.

electrodeposition the deposition of material from a dissolved or suspended state onto an electrode by the application of an electric field; it includes **electrocrystallization**.

electrode potential *symbol*: E ; the electric potential between

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a reversible electrode and the solution in which it is immersed. It may be due to the loss or gain of electrons, if the solution contains a redox couple, or to the formation of ions from uncharged atoms or of atoms from ions, if the solution contains ions of a chemical element present as uncharged atoms in the electrode.

electrodialysis a form of **dialysis** conducted in the presence of an electric field across the membrane(s). It is useful for quickly removing ionic microsolute from a solution containing nondiffusible material.

electroelution a technique in which application of an electric potential causes biomolecules to be transferred from an electrophoresis gel to a membrane during **blotting** procedures. *See also* **transfer membrane**.

electroendosmosis a former term for **electroosmosis**. —**electroendosmotic adj.**

electrofocusing a newer name for **isoelectric focusing**.

electrofuge (in chemistry) a leaving group in a reaction that does not carry away the bonding electron pair.

electrogenesis the generation of an electrochemical potential gradient, esp. across a membrane.

electrogenic pump any biological system that generates an electrochemical potential across a membrane by pumping one ion across the membrane without the concomitant movement of another ion of the same charge in the opposite direction.

electroimmunoassay the original name for **rocket immunoelectrophoresis**.

electroimmunodiffusion *abbr.*: EID; any technique for the analysis of antigenic macromolecules in which a combination of electrophoresis and immunodiffusion is employed. Four types may be distinguished: (1) one-dimensional single, i.e. **rocket immunoelectrophoresis**; (2) one-dimensional double, i.e. **counter immunoelectrophoresis**; (3) two-dimensional single, i.e. **crossed immunoelectrophoresis**; and (4) two-dimensional double, not so far individually named or much used.

electroimmunodiffusogram the immunoprecipitation pattern produced by any type of **electroimmunodiffusion**.

electrokinetic potential or **zeta potential** *symbol*: ζ ; the potential drop across the mobile part of an **electrochemical double layer** that is responsible for electrokinetic phenomena. The electrokinetic potential is positive if the potential increases from the bulk of the liquid phase towards the interface.

electrolectin a lectin of the **galactin** family from *Electrophorus electricus*, and the name of a group of β -D-galactoside-binding lectins found in teleosts, amphibians, and mammals. Example from *E. electricus*: database code LEG_ELEEL, 129 amino acids (14.44 kDa).

electrolysis the chemical change produced by passing a direct electric current through a solution of an electrolyte, or through a fused electrolyte. Positively charged ions (cations) move towards the cathode and negatively charged ions (anions) move towards the anode, thus carrying the electric current. At the electrodes the positively charged ions gain electrons and negatively charged ions give up electrons, forming uncharged atoms, molecules, or radicals, which may either be deposited on the electrode or react with the electrode, the solvent, or each other. Alternatively, atoms of the electrode material may ionize and go into solution.

electrolyte any substance that, when in solution or molten, can undergo (partial or complete) dissociation into ions and hence is able to conduct an electric current by the movement of the ions.

electrolyte balance (in physiology) the sum total of the reactions and processes concerned with the maintenance of a constant internal environment in the body with respect to the distribution of ions between the various fluid compartments and excretions.

electrolytic 1 of, pertaining to, or produced by **electrolysis**.

2 of or pertaining to an **electrolyte**.

electrolytic conductivity or (formerly) **specific conductance** *symbol*: κ ; the ability of an electrolyte to permit the flow of



electromagnetic

electric current, defined by the equation: $\kappa = j/E$, where j is the electric current density and E is the electric field strength. In SI units, κ is in S m^{-1} , j in A m^{-2} , and E in V m^{-1} . Electrolytic conductivity is usually measured in a rigid conductivity cell whose cell constant has been determined by means of a reference potassium chloride solution.

electromagnetic 1 of, relating to, or involving both electricity and magnetism; having both electrical and magnetic properties. 2 of or relating to **electromagnetic radiation**. 3 a of, relating to, containing, or operated by an electromagnet. b of or relating to electromagnetism.

electromagnetic radiation *abbr.*: emr or EMR; radiation consisting of transverse waves of energy with associated electric and magnetic fields that can be represented by vectors at right angles to each other and to the direction of its propagation. The radiation results from the acceleration of electric charges and can be propagated through free space (i.e. it requires no supporting medium). The characteristics of the radiation vary with the frequency (or the wavelength) of the wave motion, the range of possible frequencies constituting the **electromagnetic spectrum**. From the low-frequency end of the spectrum to the high end are ranged: **radiofrequency radiation**, **infrared radiation**, visible radiation or **light**, **ultraviolet radiation**, **X-rays**, and **gamma-radiation**. Radiation of all frequencies travels at the same velocity through space. The speed of light in vacuum, c_0 , is defined as $299\,792\,458\text{ m s}^{-1}$.

electromagnetic unit *abbr.*: emu or EMU; any of the electrical units based on the magnetic properties of electric currents. The force, F , between two magnetic poles of strength m_1 and m_2 placed a distance d apart in a medium of permeability μ is given by $F = m_1 m_2 \mu d^{-2}$. If F , μ , and d each be unity and if $m_1 = m_2 = m$, then m_1 and m_2 are poles of unit strength. Units based on this definition of m are electromagnetic units and are denoted by the prefix 'ab-'.

electromeric effect an obsolescent term for a molecular polarizability effect occurring by intramolecular electron displacement characterized by the substitution of one electron pair for another in the same atomic octet of electrons.

electrometer any instrument of very high input impedance that is used for detecting or measuring (changes in) the magnitude of an electric potential difference or electric charge esp. if of small value. Hence it is useful also for the (indirect) measurement of a very low electric current or a very high electric resistance, and for the detection or measurement of ionizing radiation. —**electrometric** *adj.*

electrometric titration 1 an alternative name for **potentiometric titration**. 2 a name sometimes used generically for **conductometric titration** and **potentiometric titration**.

electromotive force *abbr.*: emf or EMF; *symbol*: E ; the energy supplied by a source of electric current in driving unit electric charge through an electric circuit. The SI derived unit of electromotive force is the volt. *Compare* **electric potential difference**.

electromotive series or **electrochemical series** a series in which the metals are arranged in decreasing order of their tendency to form cations by a reaction of the type: $\text{M} \rightleftharpoons \text{M}^+ + \text{e}^-$.

electromyogram *abbr.*: EMG; the tracing produced by an electromyograph.

electromyograph an instrument for detecting and recording the electrical activity of a muscle or group of muscles.

electron 1 *symbol*: e or e^- or β^- ; a negatively charged elementary particle of rest mass, m_e , $\approx 9.109\,39 \times 10^{-31}\text{ kg}$ and electric charge $1.602\,177\,33(49) \times 10^{-19}\text{ C}$. It is sometimes termed a **negatron**, especially to distinguish it from its antiparticle, the positron. Electrons are present orbiting in all atomic nuclei and are involved in the formation of bonds between atoms. Specific designations of electrons in atoms and molecules include: **σ -electrons**, which participate in single bonds; **π -electrons**, which participate in double or triple bonds; and **p -electrons**, which participate in unshared electron pairs. 2 (sometimes) either a negatron (i.e. **electron** (def. 1)) or a **positron**. *See also* **beta-particle**.

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electron acceptor 1 any substance to which an electron may be transferred. 2 an alternative name for **Lewis acid** (use discouraged). *Compare* **electron donor**.

electron affinity 1 the degree to which any molecular entity attracts one or more additional electrons. 2 *symbol*: E_{ea} ; the minimum energy, expressed in joules, needed to remove an electron from a negatively charged molecular entity, or the reciprocal of the energy required to introduce an electron into a molecular entity.

electron-affinity spectroscopy a sensitive ionization method for the qualitative analysis of small quantities of organic compounds separated by gas chromatography, based on the differences in electron-capture ability that exist between the principal classes of organic compounds. The substances to be detected are passed through an ionized gas; the detector is sensitive to the resulting reduction in current.

electron capture 1 or **K-capture** *abbr.*: EC; a mode of decay of radionuclides in which an electron from an inner orbit of a decaying atom is absorbed into its nucleus, wherein a proton is converted to a neutron. Rearrangement of the orbital electrons then leads to the emission of characteristic X-radiation (see **X-radiation**). Generally the captured electron is derived from the K-shell, hence the alternative name. 2 *see* **electron-affinity spectroscopy**.

electron-capture detector *abbr.*: ECD; the type of detector used in **electron-affinity spectroscopy**.

electron carrier any molecular entity that serves as an electron acceptor and electron donor in an electron transport system.

electron-dense describing any structure, material, or compound that scatters an impinging electron beam, thus becoming apparent in **electron microscopy**, etc.

electron-dense label any atom group or substance (e.g. ferritin) that is **electron-dense** and that has been or may be attached in a specific manner to compounds or structures so as to show them up by electron microscopy, etc.

electron density the electron probability distribution in any molecular entity. If $P(x,y,z)$ is the electron density at the point with coordinates x,y,z , then the probability of finding an electron in the volume element $dx dy dz$ is $P(x,y,z) dx dy dz$. In, e.g., X-ray scattering experiments the scattering at this position is proportional to this probability. The term is often wrongly applied to negative **charge density**.

electron-density map a three-dimensional representation of the structure of a molecule or substance derived from X-ray diffraction data and made by the superposition of the electron-density contours of a series of imaginary parallel planes through the specimen being examined.

electron diffraction an effect observed when a narrow beam of electrons passes through a layer of material, e.g. a crystal, and is deflected by this material. The effect is due to the wave-like nature of electrons.

electron donor 1 any substance that is able to transfer an electron to another chemical species. 2 an alternative name for **Lewis base** (use discouraged). *Compare* **electron acceptor**. *See also* **donor atom**.

electron donor-acceptor chromatography *see* **charge-transfer chromatography**.

electron donor-acceptor complex (sometimes) an alternative name for 1 **charge-transfer complex** (use not recommended). 2 **Lewis adduct** (use not recommended).

electronegative 1 describing an atom or a group of atoms that tends to attract electrons, especially in the formation of a covalent bond. 2 describing any chemical or other entity that carries a negative charge and hence tends to move to the anode in electrophoresis.

electronegativity a measure of the power of an atom or group of atoms to attract electrons to itself from other parts of the same molecular entity.

electron equivalent the **equivalent** of any oxidizing (or reducing) agent that is equivalent to one entity of electrons.



electroneutrality condition

electroneutrality condition the condition that in any solution the sum of the negative electric charges on ions must be equal to the sum of the positive charges on ions.

electron-exchange resin any synthetic resin containing groups able to undergo reversible oxidation-reduction reactions; such resins are useful as insoluble oxidizing or reducing agents.

electron gun a device used in electron microscopes, mass spectrometers, and other instruments to generate a beam of high-velocity electrons. It consists of a white-hot tungsten filament, as the electron source, and an anode kept at a potential of several tens of kilovolts above that of the filament. The electrons are accelerated towards the anode, which contains a small orifice through which some of the fastest electrons pass to form the electron beam.

electron hole the hole in the valency structure of an atom created by the absence of an electron when the energy level of an electron is raised from the valency band to the conduction band. The hole has a great tendency to recapture an electron. *See also* hole.

electronic 1 of or relating to an electron or electrons. 2 of, relating to, or operated by **electronics** (def. 2).

electronics 1 (*functioning as sing.*) the branch of science and technology concerned with electronic devices and circuits and with their development and applications. 2 (*functioning as pl.*) the circuits, devices, and equipment that function by the passage of electrons through semiconductors, transistors, and thermionic valves.

electronic transition the passage of an electron in a chemical entity from one energy level to another.

electron lens any arrangement of coils and/or electrodes producing a magnetic and/or an electric field used to focus an electron beam, e.g. in an electron microscope.

electron micrograph any photograph of an object taken with an electron microscope.

electron microscope *abbr.*: EM or e.m.; an instrument in which a magnified image of a sample is produced using a beam of high-energy electrons. The electrons are accelerated by means of an **electron gun**, using voltages in the range 40–100 kV, and the apparatus is evacuated to minimize electron scatter by air. In the **transmission electron microscope** (*abbr.*: TEM) the electron beam is focused onto a thin sample by a combination of magnetic and electric lenses and, after passing through the sample, the electrons are focused onto a fluorescent screen or photographic plate. The resolving power can be as great as 0.2–0.5 nm. In **cryoelectron microscopy**, lightly fixed frozen sections are employed. If these are used for the detection of antigens with antibodies coupled to colloidal gold, the technique is called **cryoimmunoelectron microscopy**. In the **scanning electron microscope** (*abbr.*: SEM) a narrow beam of electrons is made to scan to and fro across the sample by a varying field, and emitted **secondary electrons** are focused onto a screen to produce an image of the specimen. The specimen may be thicker than in the transmission electron microscope and, though resolution is lower with the scanning instrument, the depth of field is greater and the image gives a three-dimensional effect.

electron multiplier an electronic device for detecting, and counting, electrons, using **secondary emission**. It consists of an evacuated tube containing a series of electrodes, each held at a higher positive electric potential than the preceding one. When an electron hits the first electrode two or more **secondary electrons** are emitted; these are accelerated to the second electrode, where more electrons are produced, and so on.

electron nuclear double resonance *abbr.*: ENDOR; a phenomenon in nuclear magnetic resonance spectroscopy in which saturation of an electron resonance by the simultaneous application of a microwave field causes, under appropriate conditions, an improvement of the resolution of nuclear hyperfine structures.

electron optics the branch of science concerned with the effects of electric and magnetic fields on the movement of electrons.

trons, especially with the focusing and deflection of electron beams in **cathode-ray tubes**, **electron microscopes**, and similar devices.

electron paramagnetic resonance spectroscopy *an alternative name for electron spin resonance spectroscopy.*

electron probe X-ray microanalysis a highly sensitive technique for elemental microanalysis based on the frequencies and/or intensities of the characteristic X-rays emitted by each element present in a sample when it is excited by a beam of accelerated electrons.

electron sink or **electron trap** an **electrophilic** atom or group of atoms that can capture an electron from another part of a molecular system.

electron spectroscopy for chemical analysis *abbr.*: ESCA; *see* photoelectron spectroscopy.

electron spin *see* spin (def. 3).

electron spin-echo spectroscopy a technique giving precise values for electron spin relaxation times, in which intense microwave pulses at a frequency satisfying the resonance condition are applied to a paramagnetic sample in a magnetic field. Phase coherence in the spins is lost through relaxation when the pulse is switched off but can be restored by a second pulse 180° out of phase. The restoration of phase coherence gives rise to a nuclear induction signal (the echo).

electron spin-lattice relaxation time (*in electron spin resonance spectroscopy*) a measure of the time taken for the spin population to return to its equilibrium value by interaction with the fluctuating internal electric fields surrounding it.

electron spin resonance spectroscopy *abbr.*: ESR (spectroscopy); a technique, also called **electron paramagnetic resonance spectroscopy** (*abbr.*: EPR (spectroscopy)) and analogous to **nuclear magnetic resonance spectroscopy**, that is used to investigate paramagnetic centres in the system under study; only electrons whose spin is not paired with the oppositely directed spin of another electron give an ESR signal. Useful information is obtained by this technique about certain transition-metal ions (notably Cu²⁺), (free) radicals, and free electron centres such as may be produced by X-irradiation of (macro) molecules. A **probe** (def. 2) giving an ESR signal may be incorporated into membrane lipids or attached to proteins to enable otherwise inaccessible systems to be studied by this technique.

electron spin-spin relaxation time (*in electron spin resonance spectroscopy*) a measure of the time taken for phase coherence to be lost and equilibrium to be re-established through interaction with neighbouring spins. It is inversely proportional to the line width of the signal.

electron-transfer flavoprotein or **electron-transferring flavoprotein** *abbr.*: ETF; a protein containing flavin adenine dinucleotide (FAD) that together with an acyl-CoA dehydrogenase (EC 1.3.99.3) forms a system that oxidizes an acyl-CoA molecule and reduces ubiquinone and other acceptors in the mitochondrial electron transport system.

electron-transferring flavoprotein dehydrogenase *abbr.*: ETF dehydrogenase EC 1.5.5.1; an enzyme that catalyses the oxidation of reduced ETF by ubiquinone to form ETF and ubiquinol. The ubiquinol is then oxidized by **ubiquinol-cytochrome c reductase**. It is an iron-sulfur flavoprotein involved in coupling the **beta-oxidation system** to the **respiratory chain**. *See also* electron-transferring flavoprotein.

electron-transferring fluoroprotein *see* azurin.

electron transport 1 **respiratory electron transport**; the process by which pairs of electrons derived from intermediates of the **tricarboxylic-acid cycle** and other substrates flow down the **respiratory chain** to dioxygen, the ultimate electron acceptor in respiration. 2 **photosynthetic electron transport**; the process by which pairs of electrons are transported from H₂O to NADP⁺ in the light phase of photosynthesis during noncyclic **photophosphorylation**.

electron-transport chain *an alternative name for respiratory chain.*

electron trap *an alternative name for electron sink.*

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electronvolt

electronvolt or **electron volt** *symbol:* eV; a non-SI unit of energy equal to the kinetic energy acquired by an electron when accelerated through an electric potential difference of 1 volt. $1 \text{ eV} = e \times V \approx 1.602 \times 10^{-19} \text{ J}$, where e is the **elementary charge** and V is the volt.

electroosmosis or (formerly) **electroendosmosis** the motion of a liquid through a membrane (or plug or capillary) consequent upon the application of an electric field across the membrane. A similar phenomenon may occur in electrophoresis, where many of the supporting media used, e.g. paper or agar, acquire negative charges during electrophoresis at alkaline pHs and, since the medium cannot move, H_3O^+ ions move towards the cathode, giving the effect of an osmotic movement of solvent towards the cathode and making electrically neutral molecules appear to be cationic. —**electroosmotic adj.**

electropherogram a variant spelling of **electrophoretogram**.

electrophile or **electrophilic reagent** any chemical species that is preferentially attracted to a region of high electron density in another species during a chemical reaction. Such reagents normally are positively charged or contain electron-deficient chemical groups. They tend to react with electron-rich or negatively charged chemical species. *Compare nucleophile.*

electrophilic 1 of, pertaining to, or being an **electrophile**; having or involving an affinity for regions of high electron density in a chemical reactant. 2 describing a chemical reaction in which an electrophile participates.

electrophilic catalysis catalysis by a Lewis acid, i.e. any chemical species that abstracts an electron pair from the reactant.

electrophilic displacement an alternative term for **electrophilic substitution reaction**.

electrophilicity the relative reactivity of an **electrophile**, measured by the relative rate constants of different electrophiles towards a common reactant.

electrophilic reagent an alternative name for **electrophile**.

electrophilic substitution reaction or **electrophilic displacement** a chemical reaction in which an **electrophile** effects heterolytic substitution in another reactant, both bonding electrons being supplied by that other reactant.

electrophoresis 1 the phenomenon of the movement of ions (including macromolecular ions) or charged particles or ions through a fluid under the influence of an electric field applied to the fluid. A number of different media have been used as the fluid support, including paper, cellulose acetate, starch gel, and polyacrylamide gel. Ions or particles bearing a net positive charge tend to move towards the negative pole of the electric field and vice versa, the rate of movement of a particular variety of ion or particle depending, *inter alia*, on its charge-to-mass ratio. The phenomenon has been widely applied in separating proteins, nucleic acids, and other charged molecular species for analytical or preparative purposes, and also in the analytical or preparative fractionation of heterogeneous populations of dispersed cells or other types of macroscopic particles. 2 the act or process of causing ions or charged particles to migrate; any technique based upon such a phenomenon, e.g. **continuous flow electrophoresis**, **immuno-electrophoresis**, **moving boundary electrophoresis**, **paper electrophoresis**, **polyacrylamide gel electrophoresis**, **zone electrophoresis**. *See also* **electrodecentration**. —**electrophoretic adj.**

electrophoresis convection an alternative term for **electrodecentration**.

electrophoretic effect the phenomenon of decreased **electrophoretic mobility** of a charged macromolecule caused by the movement of counter ions and/or solvent molecules in the opposite direction to that of the macromolecule.

electrophoretic mobility *symbol:* u ; the **electrophoretic velocity**, v , of a charged particle expressed per unit field strength; hence, $u = v/E$, where E is the field strength. The value of u is positive if the particle moves towards the pole of lower potential and negative in the opposite case. The electrophoretic mobility depends only on molecular parameters.

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electrophoretic molecular sieving (sometimes) an alternative term for **polyacrylamide (gel) electrophoresis**.

electrophoretic titration curve the pH-mobility curve of an ampholyte, e.g. a protein, generated by subjecting a zone of it to electrophoresis in a gel slab at right angles to a preformed, stationary pH gradient. *Compare isoelectric focusing.*

electrophoretic transfer or **electroblotting** a development of the technique of blot transfer, in which proteins or nucleic acids are transferred from a separation gel to nitrocellulose or diethylaminoethyl- (DEAE-)cellulose membranes or to diazobenzoyloxymethyl- (DBM)- or diazophenylthioether- (DPT-) paper by electrophoresis, rather than by capillary flow, with a consequent decrease in the time required for the transfer. The membrane or paper bearing a resultant pattern of separated substances has been termed an **electroblot**. *See blotting.*

electrophoretic velocity *symbol:* v ; the velocity of a charged particle during electrophoresis. It is normally proportional to the electric field strength. *Compare electrophoretic mobility.*

electrophoretogram or **electropherogram** the result of a zone-electrophoretic separation, either directly visible or after staining or processing to produce a graph.

electrophysiology the part of physiology concerned with the electrical phenomena associated with bodily processes, such as nervous and muscular activity.

electroporate to create momentary pores in the membranes of living cells, without loss of their viability, by exposing them to a sequence of brief electrical pulses of high field strength. The reversible breakdown of the cell membranes thus caused enables treated cells to take up exogenous material (e.g. drugs or foreign DNA). —**electroporated adj.**; **electroporation n.**

electroporator an apparatus or device for effecting electroporation.

electropositive 1 describing an atom or group of atoms that tends to give up electrons, especially in the formation of a covalent bond. 2 describing any chemical or other entity that carries a positive charge and hence tends to move to the cathode in **electrophoresis**.

electropositivity a measure of the power of an atom or group of atoms to give up electrons to other parts of the same molecular entity.

electrospray a technique used in **mass spectrometry** in which a dilute acidic solution of the macromolecule is sprayed from a metal syringe needle maintained at +5000 V, forming fine highly charged droplets from which the solvent rapidly evaporates.

electrostatic of or pertaining to static electricity or electrostatics.

electrostatic bond any valency linkage between atoms arising from the transfer of one or more outer-shell electrons of one atom to the outer shell of another atom, leading to more complete outer shells in both atoms. The dissociation of an electrostatic bond leads to the production of ions.

electrostatic field any **electric field** produced by stationary charges.

electrostatic interaction any of the attractive or repulsive forces between atoms and/or groups of atoms and/or molecules that are due to the presence of ionized chemical entities and to the electronegative and electropositive properties of these atoms, groups, or molecules. *Compare electric field.*

electrostatic precipitation the removal of small particles suspended in a gas by electrostatic charging followed by precipitation onto a highly charged collector.

electrostatics the branch of physics concerned with static electricity.

electrostatic units *abbr:* esu or ESU; a system of electrical units, used in the cgs system, based upon the **electrostatic unit** of electric charge, i.e. the quantity of electricity that will repel an equal quantity of electricity, 1 cm distant from it in a vacuum, with the force of 1 dyne.

electrostriction the reversible change in dimensions of a dielectric when an electric field is applied to it. For example, the



electrotaxis

electrostatic field associated with a dissolved electrolyte causes a shrinkage in the volume occupied by the solvent.

electrotaxis the directional movement of an organism in an electric field. —**electrotactic** *adj.*

electrotropism any orientation response of a sessile organism to an electrical stimulus.

electrovalency or (*esp. US*) **electrovalence** 1 the type of **valency** characterized by the transfer of one or more electrons from one atom to another atom, with the formation of ions. 2 the number of negative or positive charges acquired by an atom by the respective gain or loss of electrons. —**electrovalent** *adj.*; **electrovalently** *adv.*

electrovalent bond an alternative term for **ionic bond**.

electroviscosity the, usually minor, change in viscosity that occurs when an electric field is applied to some polar liquids.

eledoisin a **tachykinin** with the structure Glp-Pro-Ser-Lys-Asp-Ala-Phe-Ile-Gly-Leu-Met-NH₂; it is obtained from the posterior salivary glands of small octopus (*Eledone* spp.). It is more potent than **phylalaemin** and **substance P** in some assay systems. It is useful as a lachrymal and salivary secretagogue and in the study of receptors for substance P.

element any basic and distinct component of matter that is not resolvable into simpler components with differing chemical properties. Each consists exclusively of atoms with the same unique **proton number** (although such atoms may not necessarily have the same **nucleon number** or **relative atomic mass**). Ninety-three chemical elements are known to occur naturally, either in the free state or in combination with others; to date, a further 16 have been produced artificially. —**elemental** or **elementary** *adj.*

elementary analysis any quantitative chemical analysis of the amounts of different chemical elements present in a sample.

elementary charge or **proton charge** *symbol:* *e*; a fundamental physical constant representing the electric charge on one electron (or proton). Its recommended value is $1.602\,177\,33\,(49) \times 10^{-19}$ C.

elementary particle 1 or **fundamental particle** any of the particles that form the base units of which all matter is composed. The stable elementary particles are protons, electrons, and neutrinos; they combine with neutrons to form stable atoms. Other, short-lived elementary particles are known. 2 or **Fernández-Morán particle** or **oxysome** or **stalked particle** any of the knoblike structures, 8–9 nm in diameter, seen in the electron microscope on the inner surface of the inner membrane of unfixed mitochondria negatively stained with phosphotungstate.

elementary rest mass *symbol:* *m_e*; the mass of an electron, it is a fundamental atomic constant, $9.109\,389\,7(54) \times 10^{-31}$ kg.

eleostearate 1 *numerical symbol:* 18:3(9,11,13); the *trivial name* for octadeca-9,11,13-trienoate; CH₃–[CH₂]₃–[CH=CH]₃–[CH₂]₇–COO[–]; the anion derived from eleostearic acid, octadeca-9,11,13-trienoic acid, a triunsaturated straight-chain higher fatty acid. It is a constitutional isomer of (6,9,12)-linolenate with conjugated double bonds. 2 any mixture of free eleostearic acid and its anion. 3 any salt or ester of eleostearic acid. *See also* **eleostearoyl**.

eleostearoyl *symbol:* eSt_e; the *trivial name* for octadeca-9,11,13-trienoyl; CH₃–[CH₂]₃–[CH=CH]₃–[CH₂]₇–CO–; the acyl group derived from eleostearic acid. The 9*Z*,11*E*,13*E*-isomer occurs naturally as acylglycerols in some seed oils, especially tung oil.

ELF1 a gene encoding transcription factors, related to the protooncogene **ETS** and named from E74-like factor (*see* **E74**), a *Drosophila* protein that binds a regulatory element of the dopa decarboxylase gene. Example from *Drosophila melanogaster*: database code ELF1_DROME, 1063 amino acids (116.15 kDa).

elimination reaction a chemical reaction in which two groups are removed from a molecule of an organic compound without being replaced by other groups. In most such reactions the groups are lost from adjacent carbon atoms; one of the groups

eliminated is commonly a hydron and the other a nucleophile, and the elimination results in the formation of a multiple bond. The reverse process is an **addition reaction**.

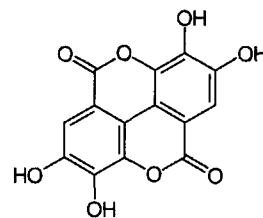
Elion, Gertrude Belle (1918–99), US chemist, biochemist, and pharmacologist distinguished for her introduction, largely in collaboration with her colleague G. H. Hitchings, of a range of widely used synthetic drugs designed as antimetabolites, including the antifolate bactericidal agent **co-trimoxazole**, the immunosuppressants **mercaptopurine** and its derivative **azathioprine**, and the antiviral compound **acycloguanosine**; Nobel Laureate in Physiology or Medicine (1988) jointly with J. W. Black and G. H. Hitchings 'for their discoveries of important principles for drug treatment'.

ELISA *abbr. for* enzyme-linked immunosorbent assay.

Elk a family of cell-surface proteins similar to those encoded by the protooncogene **ETS** and with presumed similar function (named from Eph-like kinase); they are protein tyrosine kinases, restricted to brain and testis, having a single transmembrane domain and two fibronectin-like domains. Example from rat (precursor): database code KELK_RAT, 984 amino acids (109.88 kDa).

Elk-1 a gene-regulating protein, found in lung and testis, that binds to DNA at purine-rich sites. It is a substrate for a **MAP kinase** in the phosphorylation cascade originating with activated protein kinase C or Ras; on phosphorylation it forms with the transcription factor **SRF** a ternary complex, which binds to the **serum response element** of DNA, turning on **FOS** transcription. Example (human): database code ELK1_HUMAN, 428 amino acids (44.86 kDa).

ellagic acid 4,4',5,5',6,6'-hexahydroxydiphenic acid 2,6,2',6'-dilactone; a commonly occurring plant polyphenol that inhibits glutathione S-transferase.



ellipse any closed plane figure whose perimeter is the locus of a point that moves so that the sum of its distances from two fixed points, the foci, is constant. An ellipse has two axes of symmetry, the major axis at the furthest distance between opposing points on its circumference and the minor axis at the nearest distance.

ellipsoid any surface or solid figure whose plane sections are all either ellipses or circles. —**ellipsoidal** *adj.*

ellipsoid of rotation or **ellipsoid of revolution** a solid figure generated by the rotation of an ellipse about one of its axes. If the rotation is about the major axis the ellipsoid of revolution is said to be **prolate**, whereas if the rotation is about the minor axis it is said to be **oblate**. *See also* **equivalent ellipsoid of rotation**, **spheroid** (def. 2).

ellipsosome a compartment in the retinal cones of fish that contains cytochrome-like pigment.

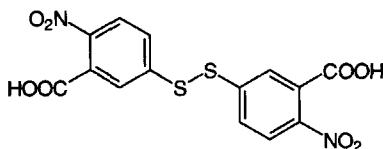
elliptical of, relating to, or having the shape of an **ellipse** or **ellipsoid**. —**elliptically** *adv.*

elliptically polarized or **elliptically polarised** describing light (or other electromagnetic radiation), whose electric vector appears to follow an elliptical path when viewed along the light beam. *See also* **polarized light**.

ellipticity 1 the ratio of the lengths of the major to the minor axis of an **ellipse** or **ellipsoid**. 2 the arc tangent of the ratio of the lengths of the minor to the major axis of an ellipse. *See also* **circular dichroism**, **Cotton effect**, **molecular ellipticity**.



Ellman's reagent 5,5'-dithiobis(2-nitrobenzoic acid) (*abbr.*: DTNB); a reagent used for labelling thiol (SH) groups in protein side-chains. The reagent reacts quantitatively with thiol groups, forming mixed disulfides and releasing the anion 5-sulfido-2-nitrobenzoic acid; this absorbs light at 412 nm, thus permitting the determination of the number of thiol groups in the protein sample. [After George Leon Ellman (1923–).]



elongase any of the long-chain fatty-acyl-coenzyme A elongating enzymes, found in mammalian microsomes, that act on saturated and unsaturated fatty-acyl derivatives having 16 or more carbon atoms per molecule of fatty acid.

elongation factor *abbr.*: EF; any of a group of soluble proteins required for chain elongation during polypeptide synthesis at the ribosome. In *Escherichia coli* there are three elongation factors: EF-Tu, EF-Ts (see **EF-T**), and **EF-G**. EF-Tu helps to bind aminoacyl-tRNA, via a tertiary complex, aminoacyl-tRNA-EF-Tu-GTP, to the ribosomal A-site. After EF-Tu-dependent GTP hydrolysis, the EF-Tu-GDP complex dissociates from the ribosome and is recycled to EF-Tu-GTP, probably via an EF-Tu-EF-Ts intermediate. *E. coli* EF-Tu and GTP form ternary complexes with all aminoacyl-tRNAs except the initiator, tRNA^{Met}. EF-G catalyses the GTP-dependent movement of the ribosome and the codon-anticodon-linked mRNA-peptidyl-tRNA complex relative to each other, resulting in the removal of a deacylated tRNA and the location of peptidyl-tRNA from the ribosomal A-site to the P-site. In eukaryotes three proteins, EF-1 α , EF-1 β , and EF-2, are required for the elongation events. EF-1 α , which occurs in different tissues as aggregates of varying size of a polypeptide chain of M_r 53 000, functions during chain elongation, like EF-Tu, to bring the aminoacyl-tRNA to the ribosomal A-site in a reaction in which GTP is hydrolysed. EF-1 β , of M_r 30 000, functions as EF-Ts. EF-2, like EF-G, catalyses the translocation of the peptidyl-tRNA from the ribosomal A-site to the P-site in a GTP-dependent reaction. Rat liver EF-2 has an M_r of 110 000, while wheatgerm EF-2 has an M_r of 70 000. *Compare* initiation factor, release factor.

Elson-Morgan reaction a colorimetric reaction for the estimation of combined and free hexosamines. The specimen is heated in alkaline solution with acetylacetone to form a pyrole derivative, which then gives a red colour with *p*-dimethylaminobenzaldehyde on acidification. [After Leslie Alderman Elson and Walter Thomas James Morgan (1900–), British biochemist, who together described the reaction in 1933.]

eluant a variant spelling of **eluent**.

eluate **1** (*in chromatography*) the solution flowing from a chromatographic column; the desired material in such a solution. **2** (*in immunology*) a solution of material derived from an insoluble or particulate antigen-antibody complex, e.g. by heating or by treatment with an appropriate buffer solution.

eluent or **eluant**, a liquid used in **elution**, particularly in **chromatography**.

eluent strength function (*in chromatography*) the order of an **eluotropic series** defined as the adsorption energy per unit area of standard activity of the solvent:

$$\log K = \log V_s + \alpha(S^0 - A_s \epsilon^0),$$

where K is the distribution coefficient in mass/volume of adsorbent and is independent of the solute concentration, V_s is the adsorbent surface volume, α is an adsorbent function proportional to the average surface energy of the adsorbent, S^0 is a solute parameter equivalent to the adsorption energy of the

solute from solution, A_s the area of adsorbent occupied by solvent, and ϵ^0 is a solvent strength parameter; the larger ϵ^0 , the stronger the solvent.

eluotropic series a series of solvents arranged in the order of their relative abilities to effect **elution**.

elute to wash out or extract retained material, particularly from a chromatographic column.

eluting **1** of or relating to **elution**, e.g. an eluting solvent. **2** the process of carrying out an **elution**.

elution the process of washing out or extracting adsorbed material from a chromatographic column.

elution volume (*in chromatography*) the volume of mobile phase that must pass through a chromatographic column, after sample application, to produce a peak in the concentration of a particular solute in the effluent or eluate.

elutriation the process of separating and purifying particles in a powder or suspension according to their rates of sedimentation. It may be effected by repeatedly washing, decanting, and settling, or by allowing a liquid or gas containing the particles to flow upwards (against gravity), when particles whose sedimentation rates are less than the rate of upward flow are washed away from the larger, heavier particles. *Compare* centrifugal elutriation, electrodecantation.

elutriator a device for grading finely divided material according to the sizes and weights of the particles by means of a stream of liquid or gas.

elymoclavine an **ergot alkaloid** produced by *Claviceps purpurea* and other fungi.

em+ prefix: a variant of **en+** (sometimes before *b*, *m*, and *p*).

EM or **e.m.** *abbr.* for electron microscope (or electron microscopy).

Embden-Meyerhof pathway or **Embden-Meyerhof-Parnas pathway** alternative names for **glycolytic pathway**. [After Gustav Embden (1874–1933) and Otto Meyerhof (1884–1951), German biochemists, and Jacob Karol Parnas (1884–1949), Polish biochemist.]

EMBL *abbr.* for European Molecular Biology Laboratory; an international research facility based at Heidelberg in Germany. The initials are also used as the name of a major database of nucleic acid and protein sequences maintained by EMBL.

EMBO *abbr.* for European Molecular Biology Organization.

embolus (*pl.* emboli) a thrombus fragment that is transported in the circulation to distant tissues, where it can cause ischemic tissue damage.

embryo the structure that develops from a **zygote**, up to the time of birth or hatching in eutherian animals, or of germination in plants. In mammals the term is restricted to the structure present in the early part of gestation that develops into a **fetus**. —*embryonic adj.*

embryogenesis or **embryogeny** the formation of an embryo from an ovum and the processes of its development. —*embryogenic adj.*

embryology the branch of biology concerned with the study of embryos and their development. —*embryological or embryologic adj.*; *embryologically adv.*; *embryologist n.*

embryonated differentiated into, or having an embryo, e.g. an embryonated hen's egg.

embryonic stem cell *abbr.*: ES cell; a **totipotent** cell that can be cultured from an early embryo.

embryonin another name for bovine α_2 -macroglobulin.

emerimicin see **peptaibohols**.

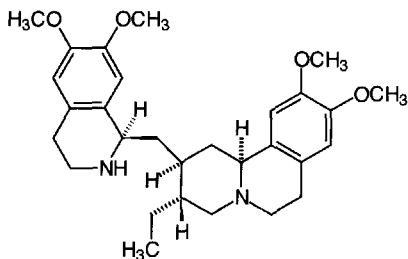
Emerson enhancement the phenomenon in photosynthesis in which the **quantum yield** of light in the far-red region of the spectrum is enhanced by weak background illumination with shorter wavelength light. *Compare* **enhancement spectrum**. [After Robert Emerson (1903–59).]

emeti **1** having the power to cause vomiting. **2** any compound with emetic properties.

emetine 6',7',10,11-tetramethoxyemetan; an alkaloid from *Cephaelis ipecacuanha* that is a gastrointestinal irritant and



potent emetic and used as an expectorant and antiamebic agent. It inhibits protein synthesis by preventing translocation of peptidyl-tRNA from the A- to the P-site on the ribosome.



emf or **EMF** *abbr.* for electromotive force.

EMG *abbr.* for electromyogram.

+emia or (*esp. Brit.*) **+aemia** *suffix* denoting blood, especially some abnormal condition of the blood.

emilin *abbr.* for elastin microfibril interface located protein; an extracellular matrix glycoprotein, gp115. It is a component of elastin fibres, preferentially located at the elastin-microfibril interface.

emiocytosis a process whereby a secretory substance is released from a cell; a form of **exocytosis**. A secretory granule moves to the cell surface, where the membrane sac enclosing the granule fuses with the plasma membrane, ruptures, and liberates the granule into the extracellular space.

emission (*in physics*) the release of energy in the form of radiation or particles.

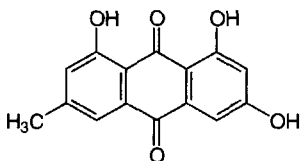
emission spectrophotometry or **emission photometry** the processes of measuring an **emission spectrum**, either photometrically or photographically.

emission spectroscopy the **spectroscopy** of an **emission spectrum**. This differs from emission spectrophotometry in being a generic term. Thus measurement may be photometric or photographic, for example.

emission spectrum any **spectrum** of emitted electromagnetic radiation produced by a sample after its atoms or molecules have been put into excited states by the absorption of energy, each line in the emission spectrum being produced by the decay of a particular species of excited atom or molecule to some lower energy level. The exciting energy may be supplied as heat, bombardment by particles, or irradiation with other electromagnetic radiation. *Compare* **absorption spectrum**, **excitation spectrum**.

EMIT *abbr. and trademark* for enzyme-multiplied immunoassay technique; *see* **homogeneous enzyme immunoassay**.

emodin 1,3,8-trihydroxy-6-methylantraquinone; a compound found (as its rhamnoside) in rhubarb root and other plants.



EMPIGEN BB *proprietary name* for dodecyl-*N,N*-dimethylglycine.

EMR *abbr.* for 1 equi-effective molarity concentration ratio. **2** or **emr** electromagnetic radiation.

emu or **EMU** *abbr.* for electromagnetic unit.

emulsifier **1** an alternative name for **emulsifying agent**. **2** a machine for making emulsions.

emulsify to form or to convert into an **emulsion**. —**emulsifiable** or **emulsible** *adj.*; **emulsification** *n.*

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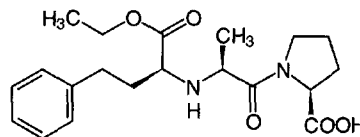
emulsifying agent or **emulsifier** any amphiphile or surfactant, such as a phospholipid or a soap, small quantities of which promote the formation of, and stabilize, an **emulsion**.

emulsin a ferment (i.e. an enzyme preparation) obtained from sweet or bitter almonds and originally studied in 1837 by the German chemists Justus von Liebig (1803–73) and Friedrich Wöhler (1800–82) for its ability to degrade **amygdalin**. Together with invertin, it played an important role in **Fischer's** work on enzyme specificity leading to his **lock and key model**. In modern terms, it was an enzyme mixture containing mainly β -glucosidase activity (EC 3.2.1.21).

emulsion **1** a temporary or permanent dispersion of an oil or other hydrophobic material in an aqueous solution, or vice versa, forming an **oil-in-water emulsion** or a **water-in-oil emulsion** respectively. **2** *short for* **photographic emulsion**.

en+ or (*sometimes before b, m, and p*) **em+** *prefix* **1** (*forming verbs*) put in or into, e.g. encapsulate, encode; surround or cover with, e.g. enmesh; to make into a certain condition, e.g. enrich. **2** (*forming nouns and adjectives*) in, into, e.g. endemic.

enalapril (S)-1-[N-[1-(ethoxycarbonyl)-3-phenylpropyl]-L-alanyl]-L-proline; an inhibitor of **angiotensin converting enzyme** (ACE) and a valuable antihypertensive drug. It has the advantage over **captopril** that it is more easily absorbed and has fewer side effects, and also that enalaprilate, to which it is converted after absorption by removal of the ethyl ester group, is a much more potent ACE inhibitor than captopril. ACE inhibitors were developed from compounds such as succinylproline, an early inhibitor. Conversion of the free succinyl carboxyl group to a sulfhydryl, and addition of a methyl group, yielded captopril with greatly enhanced binding properties. In enalaprilate there was addition of further binding affinity by a phenylethyl group, restoration of carboxyl in place of sulfhydryl and introduction of a secondary amino group. These drugs lower angiotensin II levels which lowers blood pressure, both because of a reduction in direct action of angiotensin as a vasoconstrictor and also by reducing its actions in stimulating aldosterone and norepinephrine release, both of these latter substances having hypertensive effects.



enamel the hard, white, calcified material that forms the outer covering of the crown of a tooth.

enamelin *see* **tufelin**.

enamine or **alkenylamine** any organic molecule having the general structure $R^1R^2NCR^3=CR^4R^5$ (where the R groups may be any hydrocarbyl group or H, and the same or different).

enanthate *see* **heptanoate**.

enantiomer either of a pair of **stereoisomers** whose molecules as a whole display **chirality**, i.e. are mirror images of each other and thus not superposable. They are sometimes referred to as optical isomers or optical antipodes; these terms are not recommended. *See also* **enantiomorph** (def. 2), **racemate** (def. 1). *Compare* **diastereoisomer**.

enantiomeric of, or pertaining to, an **enantiomer**; pertaining to the phenomenon of, or displaying **enantiomerism**. Chiral groups that are mirror images of one another are termed **enantiomeric groups**. *See also* **enantiomorph**, **racemic**.

enantiomeric purity a measure of the proportion of one enantiomer in an enantiomeric mixture expressed as either a fraction or a percentage. *Distinguish from* **optical purity**.

enantiomerism the phenomenon of the existence of **enantiomers** in general, or the existence of enantiomeric molecules in a particular instance; it includes **optical isomerism**.

enantiomorph **1** either of two objects, especially crystals, that



enantiomorph

are mirror images of each other and thus are not superposable. **2** either of the two crystalline forms exhibited by a pair of enantiomers. Use of the term to mean **enantiomer** is deprecated.

enantiomorph or **enantiomorphous** of, or pertaining to, an **enantiomorph**; pertaining to the phenomenon of, or displaying **enantiomorphism**. The term is often used synonymously with **enantiomeric** (enantiomeric molecules frequently form enantiomorphous crystals).

enantiomorphism the phenomenon of being related as between an object and its nonsuperposable mirror image. The term is used especially in relation to enantiomorphous crystals.

enantiomorphous *see* **enantiomorph**.

enantiotopic **1** when chemically-like ligands in constitutionally equivalent locations (generally the two *a* ligands in *Caabc*) are related by a centre or plane of symmetry, or by an alternating axis of symmetry (but not by a simple axis of symmetry), they are enantiotopic. The two ligands are in a stereochemically different, mirror-image environment. If each *a* ligand of *Caabc* is replaced separately by a different, achiral ligand, *d*, the products are the two enantiomers of *Cabcd*. Example: the methylene hydrogens of ethanol are enantiotopic; if ethanol is written as a Fischer projection structure with OH at the top, H-C-H in the middle, and CH₃ at the bottom, the left-hand hydrogen of the central methylene is H_S, while that at the right is H_R (*see pro-R/pro-S convention*). Replacement of ¹H_R by ²H yields (+)-(R)-[1-²H₁]ethanol and the same replacement of ¹H_S yields the enantiomer, (-)-(S)-[1-²H₁]ethanol. In another important compound, citric acid, the two CH₂COOH groups are also enantiotopic. **2** the two faces of a double bond or of a planar cyclic ring system that are related by a symmetry plane but not by a C₂ axis (i.e., a two-fold axis of symmetry) are enantiotopic; the two faces show stereochemically different, mirror-image related environments. Separate addition of the same achiral reagent to the two faces (*see Re/Si convention*) gives enantiomeric products. Example: the simple addition of HCN to CH₃-CHO yields a racemic mixture of the (*R*) and (*S*) cyanohydrins, CH₃-CH(OH)-CN, with both faces of C=O being involved. The reduction of the C=O bond of CH₃-CHO to form ethanol by alcohol dehydrogenase requires addition of a hydride ion from NADH at the C atom and a hydron at the O atom. Thus, reduction of CH₃-CHO with NAD²H at its *A* face (*see diastereotopic* (def. 2)) yields (*R*)-[1-²H₁]ethanol and reduction of CH₃-C²HO with *A*-NADH yields (*S*)-[1-²H₁]ethanol. The enzymatic reduction is stereospecific and only one of the enantiotopic faces of C=O is attacked; it is the same one (the *Re* face) in both of these situations. *Compare* **diastereotopic**.

encapsidate to surround (a particle of viral nucleic acid) with a **capsid**. —**encapsidation** *n*.

encapsis the association of myofibrils into bundles and the further association of these bundles into larger bundles, etc.

encephalin a variant spelling of **enkephalin**.

encephalitis inflammation of the brain.

3' end the end of a linear polynucleotide strand at which the 3'-hydroxyl group of the terminal nucleoside residue is normally not phosphorylated.

5' end the end of a linear polynucleotide strand at which the 5'-hydroxyl group of the terminal nucleoside residue is normally phosphorylated.

endo+ a variant form of **endo+** (sometimes before a vowel).

end capping (in chromatography) the blocking of residual silanol groups on the surface of silica where these remain exposed after the bonding of C₁₈ or other alkyl chains to the silica in the formation of reversed-phase stationary phases for column chromatography. For this purpose hydrocarbyl silanes (*see silane* (def. 3)) having small alkyl (usually methyl) groups are used so that they can penetrate between the main bonded-phase groups.

endemic present in or peculiar to a more or less localized area, e.g. an endemic disease. *Compare* **enzootic**.

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endergonic describing a process or reaction on which work must be done, i.e. one requiring an energy input, for it to take place. At constant pressure and temperature the free energy content of such a system increases. *Compare* **exergonic**. [From **endo+** plus Greek *ergon*, work.]

end group any residue at an extremity of a branched or linear macromolecule.

end-group analysis determination of both the nature and the number of terminal groups in a macromolecule, e.g. in proteins, the N- and C-terminal amino-acid residues; in polynucleotides, the 3'- and 5'-terminal nucleotide residues.

endo+ or (sometimes before a vowel) **endo+** *comb. form* meaning within, inner, absorbing, containing. *Compare* **exo+**. *See also* **intra+**.

endo- prefix (in chemical nomenclature) denoting insertion (of the additional constituent(s) specified) into the structure of (a named compound); e.g. endo-4a-glycine-[5-leucine]enkephalin; endo-Gly^{4a}-[Leu⁵]enkephalin; Tyr-Gly-Gly-Phe-Gly-Leu; a synthetic polypeptide in which a glycine residue has been inserted between residues 4 and 5 of [5-leucine]enkephalin.

endo- prefix (in stereochemistry). *See* **conformation**.

endoamylase any **amylase** that hydrolyses nonterminal glycosidic linkages; it is a subcategory of **endoglycosidase**.

endocrine **1** describing or relating to any gland or other group of cells that synthesizes **hormones** and secretes them directly into the blood, lymph, or other intercellular fluid. **2** describing or relating to a secretion of endocrine tissue. **3** a secretory product of endocrine tissue; a hormone. Originally known as **internal secretion**. *Compare* **exocrine**.

endocrine gland or **ductless gland** any of the ductless glandular structures that secrete (one or more) hormones directly into the bloodstream.

endocrinology the science concerned with the endocrine organs, their products, and the effects of these products. — **endocrinological** *adj*.

endocytic **1** situated within a living cell but not belonging to the cell itself; intracellular. **2** an alternative term for endocytotic (*see* **endocytosis**).

endocytosis the uptake of external materials by cells through the mechanism of **phagocytosis** or **pinocytosis**. The term is often used interchangeably with pinocytosis. *Compare* **exocytosis**, **transcytosis**. *See also* **internalize**, **viropexis**. — **endocytic** or **endocytotic** *adj*; — **endocytose** *vb*.

endocytotic vesicle *see* **pinocytotic vesicle**.

endodeoxyribonuclease *see* **deoxyribonuclease**.

endoenzyme **1** any intracellular enzyme. *Compare* **ectoenzyme**, **exoenzyme** (def. 1). **2** any enzyme that catalyses **endohydrolysis**. It may be an **endoglycosidase**, an **endonuclease**, or an **endopeptidase**. *Compare* **exoenzyme** (def. 2).

endogenous arising or developing within an organism, tissue, or cell, and excluding any consequences of externally added agents or materials. — **endogenously** *adv*.

endoglin a major glycoprotein of vascular endothelium that may be important in the binding of endothelial cells to integrins. It forms a heteromeric complex with the signalling receptors for transforming growth factor β (TGF-β). It has an **RGD** integrin-recognition motif and is a homodimer of disulfide-linked subunits. Example (precursor) from *Sus scrofa*: database code EGLN_PIG, 653 amino acids (70.20 kDa).

endoglycosidase any enzyme within subclass EC 3.2, glycosidases, that hydrolyses nonterminal glycosidic linkages in oligo- or polysaccharides. Many activities of this type are known, e.g. from *Flavobacterium meningosepticum*.

endohormone any hormone acting within the individual organism that produces it. *Compare* **ectohormone**.

endohydrolysis the hydrolysis, esp. by an **endoenzyme**, of any linkage between residues in a biopolymer. For example, **endopeptidases** attack neither the C-terminal nor the N-terminal peptide linkages of an oligo- or polypeptide, and **endoglycosidases** attack the terminal glycosidic linkages at either the reducing or nonreducing end of an oligo- or polysaccharide.

endolyn-78

endolyn-78 a glycoprotein, M_r 78 000, present in substantial amounts in membranes of endosomes and lysosomes, but occurring only at low levels in plasma membrane and the peripheral tubular endosomal compartment.

endomembrane system a hypothetical integrated membrane system of eukaryotic cells, proposed by Morré and Mollenhauer, that represents a developmental and functional continuum. It comprises the endoplasmic reticulum, nucleus, Golgi apparatus and vesicles, plasmalemma, tonoplast, and outer membranes, but not the inner membranes of the mitochondria and chloroplasts.

endometrium the mucous membrane that lines the uterus. It becomes progressively thicker and more glandular in the later stages of the estrous (menstrual) cycle, which prepares it for embryo implantation. If pregnancy occurs the endometrium becomes the decidua, which is shed after birth. If pregnancy does not occur the endometrium returns to its original state; in primates, including humans, much of the endometrium breaks down and is lost in menstruation. —*endometrial adj.*

endomitosis the replication of chromosomes without cellular or nuclear division. It is a form of polyploidization that is fairly common in differentiated or differentiating tissues. It is characterized by an increase in nuclear DNA content.

endonuclease any enzyme of a large group of phosphoric diester hydrolases, forming sub-subclasses EC 3.1.21–31, that catalyses the hydrolysis of nonterminal diester linkages in polynucleotides to yield oligonucleotides. Examples include certain of the **deoxyribonucleases** and **ribonucleases**. Compare **exonuclease**. See also **restriction endonuclease**.

endopeptidase or **endoproteinase** or (formerly) **proteinase** any enzyme within subclass EC 3.4, peptide hydrolases, that hydrolyses nonterminal peptide linkages in oligopeptides or polypeptides, and comprising any enzyme of sub-subclasses EC 3.4.21–99. They are classified according to the presence of essential catalytic residues or ions at their active sites. Four distinct sub-subclasses are: (1) serine proteinases (EC 3.4.21); (2) cysteine proteinases (EC 3.4.22); (3) aspartic proteinases (EC 3.4.23), and (4) metalloproteinases (EC 3.4.24). There are two major families of serine proteinases, the **chymotrypsins** and the **subtilisins**. Aspartic proteinases contain two aspartic residues at their active site. The metalloproteinases contain metal ions, usually zinc. The proteinases can be distinguished by inhibitors, which are usually specific for a particular class or type.

endopeptidase La the enzyme EC 3.4.21.53; *other names*: ATP-dependent serine proteinase; ATP-dependent protease La; a serine proteinase that catalyses the hydrolysis of large proteins such as globin, casein, and denatured serum albumin in the presence of ATP. It is seemingly the sole member of its own superfamily. Example from *Escherichia coli*: database code JN0303, 784 amino acids (87.35 kDa).

endoperoxide a cyclic peroxide formed from a long-chain polyunsaturated fatty acid, especially arachidonic acid, through the action of **prostaglandin-endoperoxide synthase**.

endoplasm the inner, relatively fluid, part of the **cytoplasm**. Compare **ectoplasm**. —*endoplasmic adj.*

endoplasmic reticulum *abbr.*: ER; the irregular network of unit membranes, visible only by electron microscopy, that occurs in the cytoplasm of many eukaryotic cells. The membranes form a complex meshwork of tubular channels, which are often expanded into slitlike cavities called **cisternae**. The ER takes two forms, rough (or granular), with ribosomes adhering to the outer surface, and smooth (with no ribosomes attached). When cells are disrupted by homogenization, the cisternae of rough ER break up into small closed vesicles called rough **microsomes**. Similarly the smooth ER may give smooth microsomes. The functions of the two forms of the ER differ. The ribosomes attached to the rough ER are the site of translation of the mRNA for those proteins which are either to be retained within the cisternae (**ER-resident proteins**), the proteins of the **lysosomes**, or the proteins destined for export from

the cell. Glycoproteins also undergo their initial glycosylation within the cisternae. The smooth ER has several distinct roles. It is the recipient of the proteins synthesized in the rough ER. Those proteins to be exported are passed to the Golgi complex (see **Golgi apparatus**), the resident proteins are returned to the rough ER and the lysosomal proteins after phosphorylation of their **mannose** residues are passed to the lysosomes. Glycosylation of the glycoproteins also continues. The smooth ER is the site of synthesis of lipids, including the phospholipids. The major membrane phospholipids are phosphatidylcholines, phosphatidylethanolamines, phosphatidylserines, and sphingomyelins. The ER also produces cholesterol, triacylglycerols, and ceramides. The membranes of the smooth ER also contain enzymes that catalyse a series of reactions to detoxify both lipid-soluble drugs and harmful products of metabolism. Large quantities of certain compounds such as phenobarbital cause an increase in the amount of the smooth ER.

endoproteinase or (formerly) **proteinase** an alternative name for **endopeptidase**.

ENDOR *abbr.* for electron nuclear double resonance.

endorphin any endogenous peptide with morphine-like activity. The term was originally applied to a general class of substances of morphine-like activity postulated to occur in the brain; 'endogenous opioid' is now used to denote any peptide, found in higher organisms, that has morphine-like activity. The group includes the pentapeptide **enkephalins**, but the term is most often applied to the endogenous opioid peptides of the pituitary gland, including α -endorphin, which has the structure Tyr-Gly-Gly-Phe-Met-Thr-Ser-Glu-Lys-Ser-Gln-Thr-Pro-Leu-Val-Thr; γ -endorphin has the same structure but with an additional Leu at the C terminus; β -endorphin has the same structure as α -endorphin with additionally -Leu-Phe-Lys-Asn-Ala-Ile-Ile-Lys-Asn-Ala-His-Lys-Lys-Gly-Gln-OH at the C terminus. Endorphins have the same N-terminal tetrapeptide sequence as the enkephalins. Tissue endorphins are larger polypeptides (about 7 kDa) found in pancreas, placenta, and adrenal medulla. The members of this group of peptides originate from large precursor molecules: preproopiomelanocortin is the precursor of β -endorphin, both in brain and other tissues, and [Met⁵]enkephalin; preprodynorphin (preproenkephalin B) is the precursor of dynorphin; [Leu⁵]enkephalin is formed from preproopiomelanocortin and from preproenkephalin A. Compare **exorphin**. [From endo(ogenous) + morphine.]

endo- α -sialidase EC 3.2.1.129; *other names*: endo-N-acetylneuraminidase; endoneuraminidase; poly(α -2,8-sialosyl) endo-N-acetylneuraminidase; an enzyme that catalyses the endohydrolysis of α -2,8-sialosyl linkages in oligo- or poly(sialic) acids. Example from bacteriophage k1f: database code ENAN_BPK1F, 919 amino acids (101.77 kDa).

endoskeleton any skeleton lying within an organism, such as the axial skeleton of vertebrates. Compare **exoskeleton**.

endosmosis the osmotic flow of water or of an aqueous solution into a cell, organism, or vessel from a surrounding aqueous medium. Compare **exosmosis**. —*endosmotic adj.*

endosome a membrane-bound organelle in animal cells that carries materials newly ingested by **endocytosis**. It passes many of the materials to lysosomes for degradation.

endosperm the nutritive tissue, found in the seeds of most angiosperms, that surrounds and nourishes the embryo.

endosperm albumin the proteins occurring in the endosperm of, e.g. barley; see **protein Z**.

endospore 1 an asexual spore formed within a cell, especially one produced by some bacteria and algae. 2 or **endosporium** the innermost coat of the wall of a spore or pollen grain.

endosymbiont a partner in a symbiotic relationship that penetrates the tissues or cells of the other partner; e.g. any of the nitrogen-fixing species of bacteria that occur in the root nodules of legumes. See **symbiosis**. Compare **ectosymbiont**.

endosymbiotic infection an infection of cells by viruses that replicate without cytopathic effect.

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endothelial leukocyte adhesion molecule *abbr.*: ELAM; *another name for selectin*.

endothelin *abbr.*: ET; any one of three 21-residue peptides, called endothelins 1, 2, and 3 (*abbr.*: ET-1, ET-2, ET-3). Endothelins 2 and 3 are so named because of homology with ET-1, which is known to be made by the endothelial cell. ET-1 is the most potent vasopressor compound yet discovered, being ten times more potent than angiotensin II. All three compounds have two disulfide bridges, between cysteine residues 1 and 15, and 3 and 11; these hold the structure in a conical spiral shape. The amino-acid sequence of ET-1 is {bicyclic(1→15, 3→11)}[Cys¹-Ser-Cys³-Ser-Ser-Leu-Met-Asp-Lys-Glu-Cys¹¹-Val-Tyr-Phe-Cys¹⁵-His-Leu-Asp-Ile-Ile-Trp]; ET-2 differs in having -Trp-Leu- at positions 6,7; ET-3 has the substitutions (position in parentheses) Thr(2), Phe(4), Thr(5), Tyr(6), Lys(7), and Tyr(14). The endothelins are products of three distinct genes, and are synthesized as part of a large precursor molecule, **preproendothelin** (prepro-ET), which is then proteolytically processed. The 2.2 kb mRNA for human prepro-ET-1 is encoded in five exons, the 5'-flanking region containing response elements for phorbol ester (Fos/Jun-inducible) and nuclear factor 1 (TGF-β-induced expression; *see transforming growth factor*), and its expression is stimulated by vasopressor hormones such as epinephrine, angiotensin II, and arginine vasopressin, as well as TGF-β (from aggregating platelets), thrombin, and interleukin-1. Prepro-ET-1 is a 203-residue peptide, which is cleaved to the 92-residue pro-ET (**big endothelin**), from which ET-1 is formed by **endothelin converting enzyme**. Related peptides are **vasoactive intestinal contractor** and **sarafotoxin**, a peptide produced by *Atractaspis engaddensis*. Only ET-1 is detected in vascular endothelial cells; it is also expressed in nonvascular cells in other tissues, including brain, kidney, and lung. ET-2 and ET-3 are expressed in tissues such as brain, kidney, adrenal, and intestine.

endothelin receptor any of several membrane proteins that bind **endothelins** and mediate their intracellular effects. They are G-protein-coupled seven-transmembrane-domain receptors. The structures of two types have been predicted from human, rat, and bovine cDNA. ET_A (human, 427 amino acids) has affinity ET-1 ≥ ET-2 ≥ ET-3, while ET_B (human, 442 amino acids) is equipotent for all three.

endothelium the single layer of thin, flattened cells of mesoblastic origin that lines the blood vessels and some body cavities, e.g. those of the heart. *Compare epithelium, mesothelium.* —**endothelial** *adj.*

endothelium-derived relaxing factor *abbr.*: EDRF; *see nitric oxide*.

endothermic describing a process or reaction that absorbs heat, i.e. a process or reaction for which the change in enthalpy, Δ*H*, is positive at constant pressure and temperature. *Compare exothermic*.

endothiapepsin EC 3.4.23.22; *other name*: Endothia aspartyl proteinase. An enzyme that catalyses the hydrolysis of proteins with broad specificity similar to that of **pepsin A**; it prefers hydrophobic residues at P1 and P1', but does not cleave the Ala-Leu linkage in the B chain of insulin or the protected dipeptide Cbz-Glu-Tyr; it clots milk. Example from *Cryptonectria parasitica* (*Endothia parasitica*; chestnut blight fungus); database code NRL_4APE, 330 amino acids (33.76 kDa); 3-D structure known; it is similar to other aspartic proteases.

endotoxin any microbial toxin that cannot easily be separated from the structure of the cell. *Compare exotoxin*.

end point or **endpoint** the point in a titration that should (but may not always) correspond to the theoretical **equivalence point**.

end product the final chemical substance formed in a sequence of metabolic (or chemical) reactions.

end-product inhibition the inhibition of a sequence of metabolic reactions by the end product of the sequence, usually by action on a reaction which is often at the beginning of the sequence.

end-window counter a **Geiger counter** or **proportional counter** so constructed that its base is relatively transparent to the radiation to be detected.

+ene suffix (*in chemical nomenclature*) indicating the presence of one or more carbon-carbon double bonds in an organic compound.

enediol any acyclic organic compound in which there is a hydroxyl group attached to each of two carbon atoms that are linked by a double bond. *Compare enol*.

energy *symbol*: *E*; the capacity of a system for doing work. There are various forms of energy – potential, kinetic, electrical, chemical, nuclear, and radiant – which can be interconverted by suitable means. The SI unit of energy is the joule.

energy balance the algebraic balance of the various energy inputs versus energy outputs of a system; it is positive if energy is released and negative if it is absorbed. In physiology it is the relation of the amount of energy taken into the organism to the amount used for internal work, external work, and for the growth and repair of tissues.

energy barrier *see potential energy barrier*.

energy charge *see adenylate energy charge*.

energy coupling the coupling of ATP synthesis to electron transport in the **respiratory chain**.

energy diagram a diagram representing the energy contents of various states of the reactants, activated complexes, and products in a chemical reaction; or of the nuclear energy levels of an atom; or of the electronic energy levels of an atom or molecule.

energy flow (*in ecology*) the transfer of energy between organisms in an ecosystem.

energy level any of the stable energy states that a molecular entity can take up. In **quantum mechanics** only certain discrete energy levels are possible and continuous variation of the energy level of a molecular entity is excluded.

energy metabolism the metabolic reactions of a cell or organism concerned with energy transformations.

energy of activation *see activation energy*.

energy-poor *term sometimes used to describe 1* a compound whose hydrolysis under standard conditions manifests a small negative free energy change. *See also low-energy compound. 2* a chemical group with a low **group potential. 3** the linkage of such a chemical group to some other group.

energy requirement the amount of energy needed to maintain a cell or organism.

energy sink a molecule or a group in a molecule that is able easily to accept energy transferred to it from another component of the system.

energy transduction *see transduction* (def. 2).

energy transfer the transfer of excitation energy from one chromophore or one molecular entity to another by a process not involving radiation; the energy may then be dissipated in a variety of ways, e.g. by fluorescence. Such energy transfer is very dependent on the distances involved and is useful in studies of structural relationships between groups on a macromolecule.

energy trapping the processes by which energy released in a catabolic reaction is coupled to the synthesis of another compound, often a nucleoside triphosphate such as ATP.

engineered 1 (of a cell or organism, or a strain of cells or organisms) *colloquial term for genetically engineered. 2* (of a nucleotide sequence) *jargon term, usually followed by 'into'* artificially inserted (into a longer sequence, such as a vector). **3** (of a protein) *jargon term* describing an altered protein synthesized by a cell or organism (or a strain of cells or organisms) whose genetic material has been manipulated to that end.

engrailed any of various protein products of the **homeotic gene engrailed**; in *Drosophila* this is a segment-polarity gene. Examples: engrailed-1 and engrailed-2 (human) respectively; database codes HME1_HUMAN and HME2_HUMAN, 391 amino acids (40.00 kDa) and 332 amino acids (34.12 kDa); en-

enhancement

grailed chain C, *Drosophila melanogaster*, database code NRL_1HDDC, 57 amino acids (6.94 kDa); 3-D structure of complex with DNA known; one motif.

enhancement 1 the improvement of the effects of ionizing radiation on tissue by dioxygen or other chemical agents. **2** the increased yield of one virus from cells infected by another virus; the term is often used when the mechanism of the process is not known. *Compare* **complementation** (def. 2), **interference**. **3** *see* **Emerson enhancement**. **4** *see* **fluorescence enhancement**.

enhancement of fluorescence an alternative term for **fluorescence enhancement**.

enhancement spectrum a spectrum showing the enhancement of photosynthetic **quantum yield** from that produced at a fixed-wavelength background illumination to that produced by illumination with a variable-wavelength beam. Such spectra show peaks that can sometimes be reasonably well identified with specific pigments in the system. *Compare* **Emerson enhancement**.

enhancer (in molecular biology) a eukaryotic control element that can increase expression of a gene. Enhancers may be located some distance from the gene and may be either upstream or downstream.

enhancesome a dislike structure, formed by a 180 bp loop of DNA and dimers of the *Xenopus* transcription factor xUBF, that is involved in rRNA transcription.

enkephalin any pentapeptide **endorphin** with the sequence Tyr-Gly-Gly-Phe-Xaa. There are two similar naturally occurring enkephalins, present in brain, spinal cord, and gut; their recommended trivial names are: [5-leucine]enkephalin (*abbr.*: [Leu⁵]enkephalin or [Leu]enkephalin), Tyr-Gly-Gly-Phe-Leu (I); and [5-methionine]enkephalin (*abbr.*: [Met⁵]enkephalin or [Met]enkephalin), Tyr-Gly-Gly-Phe-Met (II). (II has the same sequence as residues 61–65 of β -**lipotropin**.) The common precursor of I and II, proenkephalin – itself formed from preproenkephalin by removal of the signal peptide – contains four copies of II and one each of I, the heptapeptide [Met⁵]enkephalinyll-Arg-Phe, and the octapeptide [Met⁵]enkephalinyll-Arg-Phe-Leu. (Note: Designations such as Leu-, leucyl-, or leucine-enkephalin have commonly been given to I, with corresponding terms for II, but all of these incorrectly imply N-terminal extension with a residue of leucine or methionine, respectively.)

enkephalinase *see* **carboxypeptidase H**.

enkephalin convertase a specific carboxypeptidase that converts enkephalin precursors into enkephalin in adrenal chromaffin granules. It is now known as **carboxypeptidase H**.

enniatiin any of various antibiotics from *Fusarium* spp. that function as **ionophores**. They are cyclodepsipeptides (*see* **depsipeptide**).

enol any acyclic organic compound with a hydroxyl group attached to either of two carbon atoms that are linked by a double bond. *Compare* **enediol**. *See also* **keto-enol tautomerism**.

enolase EC 4.2.1.11; *recommended name*: phosphopyruvate hydratase; *other name*: 2-phosphoglycerate dehydratase. An enzyme that catalyses the conversion of 2-phospho-D-glycerate to phosphoenolpyruvate and H₂O; magnesium is a cofactor. Example from *Saccharomyces cerevisiae*: database code NRL_7ENL, 436 amino acids (46.58 kDa); 3-D structure known; six motifs.

enology or (*esp. Brit.*) **oenology** the study of wines.

5-enolpyruvylshikimate-3-phosphate phospholylase *see* **chorismate synthase**.

eNOS *see* **nitric-oxide synthase**.

enoyl an acyl group derived from any alkenoic acid.

enoyl-[acyl-carrier protein] reductase (NADPH, A-specific) EC 1.3.1.39; *other name*: acyl-ACP dehydrogenase; an enzyme of the **fatty acid synthase complex** in liver. It catalyses the reduction by NADPH of *trans*-2,3-dehydroacyl-[acyl-carrier protein] to form acyl-[acyl-carrier protein] and NADP⁺.

enoyl-CoA hydratase 1 EC 4.2.1.17; *systematic name*: (3S)-3-Ban đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

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Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

hydroxyacyl-CoA hydro-lyase; *other names*: enoyl hydratase, unsaturated acyl-CoA hydratase; an enzyme of the **beta-oxidation system**, present in mitochondria and peroxisomes. It catalyses the hydration of (3S)-3-hydroxyacyl-CoA to *trans*-2-(or 3)-enoyl-CoA. It is a mitochondrial matrix protein, a homohexamer; a characteristic transit peptide sequence is found in the precursor. Example (precursor) from rat: database code ECHM_RAT, 290 amino acids (31.48 kDa). **2** EC 4.2.1.74; *recommended name*: long-chain-enoyl-CoA hydratase; an enzyme that catalyses a similar reaction to the above; it does not act on crotonoyl-CoA. *See also* **beta-oxidation system**.

enoyl-CoA isomerase *see* **dodecenoyl-CoA Δ-isomerase**.

3,2-trans-enoyl-CoA isomerase *see* **dodecenoyl-CoA Δ-isomerase**.

enrich to increase in content or abundance; e.g. to increase the abundance of one (usually stable) nuclide of an element above its naturally occurring level; to increase the content in a food-stuff of one or more specific nutrients (such as vitamins or minerals) above the natural level. —**enrichment n.**

enrichment medium a selective culture medium that favours the growth of a desired microorganism.

entactic describing an enzyme poised for catalytic action in the absence of substrate.

entactin or **nidogen** a sulfated Ca²⁺-binding protein occurring in basement membrane and involved in cell adhesion; it binds laminin and type IV collagen. The protein has two globular domains, an **EF-hand** domain, EGF-like domains and a thyroglobulin type I domain. Example (precursor) from human: database code NIDO_HUMAN, 1247 amino acids (136.34 kDa).

enteric or **enteral** of or pertaining to the intestine or gut.

enteric-coated describing a drug or medicament that is prepared in a form enabling it to pass through the stomach unaltered and be released in the intestine.

entero+ or (*before a vowel*) **enter+** *comb. form* denoting the intestine.

enterobactin or **enterochelin** N',N',N''-tris(2,3-dihydroxybenzoyl)-2,6,10-trioxo-1,5,9-trioxacyclododecane-3,7,11-triamine; a **siderochrome** of the catechol-derivative variety produced by certain members of the Enterobacteriaceae, e.g. *Escherichia coli*, *Salmonella* spp. It is the cyclic self-triester (trilactide) of N-(2,3-dihydroxybenzoyl)-L-serine, and a product of the **shikimate pathway**.

enterochelin an alternative name for **enterobactin**.

enterochromaffin cell *abbr.*: EC cell; any gut endocrine cell containing biogenic monoamines that gives a positive **chromaffin** reaction and displays a characteristic yellow formaldehyde-induced fluorescence. Such cells may be divided immunocytochemically and ultrastructurally into functionally distinct types.

enterocrinin a putative peptide hormone, crystallizable from extracts of intestine of certain mammals, that is held to stimulate duodenal and jejunal secretion of digestive enzymes.

enterocyte any of the columnar epithelial cells on the luminal surface of the villi of the small intestine. Enterocytes are immature in the crypts, mature in the middle of the villi, and senile at the exfoliation area at the villous tips. They have numerous microvilli on their luminal surface (the brush border) and are responsible for the synthesis of digestive enzymes and for the absorption of materials from the gut.

enterogastrone the putative hormone from the duodenum that inhibits gastric activity. Some or all of the actions attributed to enterogastrone are probably due to **glucose-dependent insulinotropic peptide**.

enterogastrone effect *see* **gastric inhibitory peptide**.

enteroglucagon a collective term for a small family of peptides derived from **proglucagon** by post-translational processing in the L cells (or **EG cells**) of the distal small intestine and colon. One component is GLP-1⁷⁻³⁷ (*see* **GLIP**).

enteroglucagon or **glucagon-37** or **oxyntomodulin** a peptide isolated from porcine jejuno-ileum that interacts with glucagon receptors and activates adenylate cyclase in hepatocytes.



enterohepatic recirculation

cytes. It has a specific action on the acid-secreting area of gastric mucosa. It has 37 residues, of which the sequence of the N-terminal 29 is identical with that of pancreatic glucagon.

enterohepatic recirculation the cycle of secretion from the gall bladder into the upper intestinal tract with reabsorption in the lower tract and transport to the liver followed by resecretion. *See* **bile acids**.

entero-insular axis the interrelationship of endocrine function between the gastrointestinal tract and the pancreatic islets in which signals arising in the gut after ingestion of nutrients effect endocrine responses by the islets.

enterokinase a former name for **enteropeptidase**.

enteropeptidase or (formerly) **enterokinase** the enzyme EC 3.4.21.9; a serine proteinase enzyme of the brush border of the upper small intestine that activates **trypsinogen** by the selective cleavage of the Lys⁶-Ile⁷ bond in the latter. Example (precursor) from *Bos taurus*: database code ENTK_BOVIN, 1035 amino acids (114.89 kDa); residues 1–800 form the non-catalytic heavy chain, 801–1035 the catalytic light chain.

enterotoxin any bacterial **exotoxin**, whether ingested or produced within the intestine, that has an action upon the intestinal mucosa, usually giving rise to diarrhoea and other unpleasant symptoms. **Cholera toxin** is an example. Other enterotoxins are produced by *Escherichia coli* and some Staphylococci. Of the *E. coli* enterotoxins, LT (from labile, i.e. heat-labile, toxin) is similar in structure to cholera toxin and has a similar mode of action. ST (from heat-stable toxin) has an effect similar to atrial **natriuretic peptide**, activating guanylate cyclase. It has 13 amino acids. Other enterotoxins produced by *E. coli* include **hemolysin** and vero toxin, a Shiga-like toxin similar to that of *Shigella dysenteriae* that has *N*-glycosidase activity, and cleaves an adenine residue from the 28S rRNA of the 60S ribosome subunit.

enthalpimetry the measurement of the **enthalpy** of a body or system.

enthalpy or **heat content** *symbol*: *H*; a physical quantity and a thermodynamic property defined as the sum of the internal energy of a system, *U* and the product of the volume, *V*, multiplied by the pressure, *p*, i.e. $H = U + pV$. Enthalpy is related to the **Gibbs energy**, *G*, by the equation $G = H - TS$, where *T* denotes the thermodynamic temperature and *S* denotes **entropy**.

Entner–Doudoroff enzyme the 'catabolic' D-glucose-6-phosphate dehydrogenase (EC 1.1.1.49) from *Pseudomonas fluorescens*. It is a 220 kDa protein composed of four, apparently identical, polypeptide chains. *See also* **Entner–Doudoroff pathway**.

Entner–Doudoroff pathway a metabolic pathway occurring in *Pseudomonas* spp. in which D-glucose 6-phosphate is first oxidized to 6-phospho-D-gluconate, which is then acted on by 6-phosphogluconate dehydratase (EC 4.2.1.12) to form 2-dehydro-3-deoxy-6-phospho-D-gluconate. This is then split by 2-dehydro-3-deoxyphosphogluconate aldolase (EC 4.1.2.14) to give pyruvate and D-glyceraldehyde 3-phosphate. [After Nathan Entner (1920–) and Michael Doudoroff (1911–).]

entomology the science of the study of insects.

entopic occurring in the usual place. *Compare* **ectopic**.

ENTREZ a suite of computer programs accessible via the Internet that links **MEDLINE** with genome and sequence databases.

entropic union a polymerization, or other reaction, resulting from an energetically unfavourable interaction of a surface of the molecule in question with water molecules; the polymerization causes the removal of water from the molecules' surfaces and a hence a decrease in the free energy content of the system.

entropy *symbol*: *S*; a physical quantity and a thermodynamic property indicating the amount of disorder in a system; i.e. the amount of energy in a system that is unavailable for doing work. In any irreversible process the total entropy of all systems concerned is increased. In a reversible process the total increase in entropy in all systems concerned is zero, while the increase in entropy, *dS*, of any individual system, or part of a

system, is equal to the heat that it absorbs, *dq_{rev}*, divided by the thermodynamic temperature, *T*; i.e. $dS = dq_{rev}/T$. Entropy is related to **Gibbs energy**, *G*, and **enthalpy**, *H*, by the equation $G = H - TS$. The SI unit for entropy is J K⁻¹ —**entropic adj.**

entry exclusion the phenomenon in which a resident plasmid interferes with the entry of genetic material to a cell by (bacterial) **conjugation** (def. 4).

enucleate 1 to remove the nucleus from (a cell). 2 (of a cell) deprived of its nucleus. —**enucleation** *n*.

enuresis the involuntary passing of urine. —**enuretic** *adj.*

env a viral gene for **envelope protein**.

envelope 1 any covering or enclosing structure, e.g. a membrane, capsule, shell, or skin. 2 the cell membrane of a bacterium, together with all the structures external to it including the cell wall and, sometimes, the capsule. 3 or **peplos** an outer lipoprotein coat of a virion that occurs in an **enveloped virus**. 4 *see* **nuclear envelope**.

envelope conformation any **conformation** of a nonplanar five-membered saturated ring compound when four of its ring atoms lie in one plane and the remaining atom lies outside that plane. For a monosaccharide or monosaccharide derivative the conformational descriptor *-E* may be added to its name.

enveloped virus any virus in which a nucleoprotein core is surrounded by a lipoprotein envelope consisting of a closed bilayer of lipid derived from that of the host cell's membrane(s), with glycoprotein on the outside and matrix protein or nucleocapsid protein on the inside. Enveloped viruses include herpesviruses, negative-strand RNA viruses, retroviruses, and togaviruses.

envelope protein any protein (usually a glycoprotein) of the envelope of a virus. Example: envelope polyprotein gp160 precursor of human immunodeficiency virus (HIV) type 1; database code ENV_HV1A2, 855 residues (97.33 kDa). Residues 1–29 are the signal peptide; 30–509 form exterior membrane glycoprotein gp120; 510–855 form transmembrane glycoprotein gp41. The variable (very rapidly evolving) antigenic regions of HIV are in the gp120 sequence.

envelysin EC 3.4.24.12; *other names*: sea-urchin-hatching proteinase; hatching enzyme; an enzyme so named because it dissolves (lyses) the fertilization envelope of the sea-urchin embryo. It is a metalloendopeptidase that catalyses the preferential cleavage of proteins on the amino side of bulky hydrophobic residues, Leu, Ile, Phe, and Tyr. The enzyme requires zinc and calcium. Example (precursor) from the sea urchin *Paracentrotus lividus*: database code HE_PARLI, 587 amino acids (65.14 kDa).

Enzacryl the proprietary name for a group of synthetic carriers for the immobilization of enzymes and other organic biomolecules that are copolymers of acrylamide and various derivatives of *N,N'*-methylene diacrylamide.

enzootic present in or peculiar to animals in a more or less localized area; e.g. an enzootic disease. *Compare* **endemic**.

enzyme any naturally occurring or synthetic macromolecular substance composed wholly or largely of protein, that catalyses, more or less specifically, one or more (bio)chemical reactions at relatively low temperatures. The action of RNA that has catalytic activity (*see* **ribozyme**) is often also regarded as enzymic. Nevertheless, enzymes are mainly proteinaceous and are often easily inactivated by heating or by protein-denaturing agents. The substances upon which they act are known as **substrates**, for which the enzyme possesses a specific binding or **active site**. *See also* **enzyme classification**, **isoenzyme**. [From German *enzym*, possibly from modern Greek *enzumos*, leavened, or more probably from Greek *en*, in, + *zumē*, leaven, yeast.] —**enzymic** or **enzymatic** *adj.*

enzyme I EC 2.7.3.9; enzyme I of the **phosphotransferase** system; phosphoenolpyruvate–protein phosphotransferase; *systematic name*: phosphoenolpyruvate:protein-L-histidine *N*^γ-phosphotransferase. A soluble bacterial enzyme, part of a system for the transport of hexoses across the cell membrane, that cata-



enzyme II

lyses the phosphorylation by phosphoenolpyruvate of a low M_r , heat-stable protein, **HPr**. *Compare enzyme II* (def. 1).

enzyme II 1 EC 2.7.1.69; enzyme II of the **phosphotransferase** system; protein- N^{π} -phosphohistidine-sugar phosphotransferase; *systematic name*: protein- N^{π} -phosphohistidine:sugar N^{π} -phosphotransferase. Any of a group of related membrane-bound bacterial enzymes, part of the system for the transport of hexoses across the cell membrane, that catalyse the phosphorylation of hexoses by phospho-HPr (*see enzyme I*). Enzyme II is responsible for the specificity of the transport process with respect to the sugar. **2** *a former name for acyl-carrier protein*.

enzyme-activated irreversible inhibitor *an alternative name for suicide inhibitor*.

enzyme activation the generation of a catalytically active enzyme from an inactive or poorly active form or from a biosynthetic precursor (proenzyme). Activation may be by enzymic or chemical covalent modification or by addition of a specific activator.

enzyme adaptation *see inducible enzyme*.

enzyme cascade *see cascade sequence*.

enzyme classification the systematic arrangement and naming of enzymes by the **Enzyme Commission**. Each enzyme is denoted by the abbreviation EC followed by a set of four numbers separated by stops. The first number denotes one of the six main divisions: EC 1, oxidoreductases; EC 2, transferases; EC 3, hydrolases; EC 4, lyases; EC 5, isomerases; and EC 6, ligases. The second number denotes the subclass, the third number denotes the sub-subclass; the fourth number is the serial number of the particular enzyme. The most recent edition of the classification, *Enzyme Nomenclature 1992* (IUB/Academic Press; San Diego, London), lists 3196 enzymes.

enzyme cluster or (*sometimes*) **multienzyme cluster** **1** any physiologically significant system of two or more enzymes in physical (i.e. noncovalent) association. The term embraces any **enzyme complex**, **multienzyme complex**, or membrane-bound enzyme array. Clustered enzymes usually display different kinetic and/or regulatory features from their unassociated counterparts; they may be encoded in a cluster-gene. *See also multienzyme system*. **2** *an alternative term for cluster* (def. 2).

Enzyme Commission or the **International Commission on Enzymes**; a body established in 1956 by the International Union of Biochemistry (IUB) to consider the classification and nomenclature of enzymes and coenzymes, their units of activity and standard methods of assay, together with the symbols used in the description of enzyme kinetics. The Commission, which worked closely with the Biological Chemistry Nomenclature Commission of the International Union of Pure and Applied Chemistry (IUPAC), was dissolved in 1961 and its work has been carried on in turn by the Standing Committee on Enzymes of IUB, by the IUPAC/IUB Joint Commission on Biochemical Nomenclature (JCBN), by an Expert Committee on Enzymes, and most recently by the Nomenclature Committees of IUB (NC-IUB) or the International Union of Biochemistry and Molecular Biology (NC-IUBMB). *See also enzyme classification*.

enzyme complex an operational term for any structural and functional entity composed of a number of dissociable enzymes that catalyse a sequence of closely related chemical reactions. *Compare multienzyme*.

enzyme detergent any detergent preparation incorporating an enzyme to assist its cleansing action. The enzymes used in such detergents are usually proteinases of high thermal and alkaline stability, e.g. **Alcalase**.

enzyme differentiation the process whereby, during the development of an organism, each tissue acquires its own characteristic quantitative pattern of enzymes, which underlies the physiological functions and morphological features of the tissue. From a fairly uniform enzymic make-up in the cells of the early embryo, the enzyme patterns of different tissues become progressively more differentiated as development into the mature organism proceeds.

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enzyme electrode any electrode, incorporating an enzyme into its structure, that responds to the concentration of one of the substrates or products of the reaction catalysed by the enzyme. The enzyme is trapped within a gel matrix surrounding the electrode or is kept in contact with the electrode by a semi-permeable membrane.

enzyme engineering or **enzyme technology** the branch of biomolecular engineering concerned with processes designed to produce, isolate, purify, and immobilize enzymes and to use them for the catalysis of specific chemical reactions.

enzyme immunoassay or **enzymimmunoassay** *abbr.*: EIA; any immunoassay in which an enzyme-catalysed reaction is used as the indicator. *See also heterogeneous immunoassay, homogeneous immunoassay*.

enzyme induction the synthesis of an enzyme in a cell or organism at a markedly increased rate in response to the presence of an **inducer**. The inducer is thought to combine with a **repressor** thereby preventing the latter from blocking an **operator**, which controls the translation of the structural gene for the enzyme.

enzyme-inhibition immunoassay a variation of **enzyme immunoassay** in which the inhibition of an enzyme-catalysed reaction is used as the indicator.

enzyme labelling a method used to detect or locate (and sometimes estimate) an antigen in, e.g., a tissue section. The section is exposed to a complementary antibody that has been covalently linked to an enzyme; the antibody binds to the antigen, and its location (and its amount) is determined by an assay dependent on the catalytic activity of the linked enzyme. *See also enzyme immunoassay*.

enzyme-linked immunosorbent assay *abbr.*: ELISA; a form of quantitative immunoassay based on the use of antibodies (or antigens) that are linked to an insoluble carrier surface, which is then used to 'capture' the relevant antigen (or antibody) in the test solution. The antigen-antibody complex is then detected by measuring the activity of an appropriate enzyme that had previously been covalently attached to the antigen (or antibody).

enzyme membrane any (semipermeable) membrane to which an enzyme has been covalently bound. Such membranes are useful in constructing **enzyme electrodes**.

enzyme-multiplied immunoassay technique *abbr. and proprietary name*: EMIT; *an alternative name for homogeneous enzyme immunoassay*.

enzyme-paper graft an enzyme immobilized on (filter) paper, which is frequently impregnated with indicators. It is useful for making analytical devices.

enzyme reactor a device for using immobilized enzymes or enzyme systems for synthetic or other processing reactions, especially on an industrial scale.

enzyme recruitment the exploitation of substrate-ambiguous enzymes or transport proteins in the evolution of new biochemical pathways.

enzyme repression inhibition of the formation of an enzyme by a compound formed in or taken in by a cell or organism.

enzyme specificity *see specificity*.

enzyme-substrate complex *abbr.*: ES; the stoichiometric complex of an enzyme molecule and a substrate molecule bound at the enzyme's active site.

enzyme technology *an alternative term for enzyme engineering*.

enzyme unit *symbol*: U; *abbr.*: EU; an obsolete unit of activity of enzymes, defined as the amount of enzyme that will catalyse the transformation of one micromole of the substrate per minute under standard conditions. It has now been superseded by the **katal**.

enzymic or **enzymatic** by, of, involving, or relating to an enzyme or enzymes; catalysed by an enzyme or by enzymes. — **enzymically** or **enzymatically** *adv.*

enzymic activity the rate of reaction of substrate that may be attributed to catalysis by an enzyme. The concept is now obsolete, having been superseded by **catalytic activity**. The unit of



enzymic cycling

enzymic activity was initially the **enzyme unit** and subsequently the **katal** (as originally defined). The derived quantity concentration of enzymic activity was defined as activity divided by volume and was expressed as katals per litre. *See also catalytic activity concentration, molar (enzymic) activity, specific (enzymic) activity.*

enzymic cycling a form of **catalytic assay** in which there is no coupled indicator reaction; instead, the regenerating system is allowed to cycle for a set time, after which the reactions are stopped by destruction of all enzymes and the most easily determined product is measured in a separate assay. The coupled system thus acts as a chemical amplifier in the determination of the intermediate catalyst.

enzymo+ *comb. form* denoting an **enzyme** or enzymic activity.

enzymoimmunoassay an alternative name for **enzyme immunoassay**.

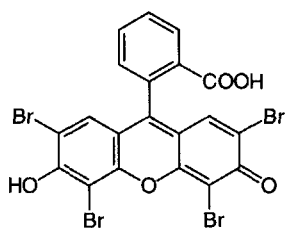
enzymology the science of the study of enzymes and enzyme-catalysed reactions. —**enzymological** *adj.*; **enzymologist** *n.*

enzymolysis the enzyme-catalysed splitting of a chemical compound into smaller ones. The term commonly refers to enzymic hydrolysis. —**enzymolytic** *adj.*

obiogenesis the first occurrence of the formation of living matter from nonliving material.

obiont any system showing some characteristics of living systems but not enough to be generally accepted as living.

eosin any of a number of similar red acidic dyes, derivatives of fluorescein, especially Eosin Y (yellowish), 2',4',5',7'-tetrabromofluorescein disodium salt, Acid Red 87 (CI 45380), or Eosin B (bluish), 4',5'-dibromo-2',7'-dinitrofluorescein disodium salt, Acid Red 91 (CI 45400). They are widely used as stains in histology and hematology.



2',4',5',7'-tetrabromofluorescein

eosinophil(e) 1 or **eosinophil leukocyte** or **eosinophilic leukocyte** a polymorphonuclear leukocyte, present in blood and other connective tissues, that has numerous large cytoplasmic granules that stain readily with eosin. The granules contain numerous hydrolytic enzymes including acid phosphatase, arylsulfatase, cathepsins, glucuronidase, peroxidase, and ribonuclease. The relative number of eosinophils in the blood, normally low, rises markedly in certain allergic conditions and parasitic infections but their function is poorly understood. 2 any cell whose cytoplasm stains readily with eosin, especially an eosinophilic cell of the anterior pituitary that secretes prolactin and somatotropin.

eosinophil cationic protein *abbr.*: ECP; a highly basic zinc-containing **ribonuclease** that binds avidly to negatively charged surfaces and is particularly effective at damaging the tegument of schistosomes.

eosinophilic having an affinity for **eosin**.

eosinotactic having an attractive or repulsive action on eosinophils. [A contraction of eosinophil + chemotactic.]

eosome the suggested primary particulate biogenetic precursor, via the **neosome**, of the **ribosome**. Eosomes of *Escherichia coli* have an average sedimentation coefficient of 14S.

EPA *abbr. for* 1 eicosapentaenoic acid. Normally indicating the (all-Z)-5,8,11,14,17 isomer. 2 erythroid potentiating activity (*see* **TIMP**).

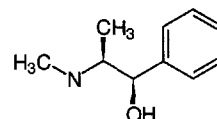
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EpETRE *abbr. for* epoxyeicosatrienoate.

EPH a gene encoding a subfamily of protein tyrosine kinases. It is named from erythropoietin-producing hepatocellular carcinoma cell line, from which the probe used to isolate the gene was obtained. Related proteins are Elk (Eph-like kinase), Eek (Eph and Elk-related kinase), Eck (epithelial cell kinase), and Erk (Elk-related kinase). Overexpressed **EPH** has tumorigenic potential; the ligand of Eph is unknown; database code for the expressed protein: KEPH_HUMAN, 984 amino acids (108.80 kDa); residues 1–23 are the signal 24–984 Eph.

ephedrine (1R,2S)-1-phenyl-1-hydroxy-2-methylamino-propane; an alkaloid obtained from several species of gymnosperm shrubs belonging to the genus *Ephedra*. A structural analogue of epinephrine, it mimics central and peripheral effects of noradrenergic and adrenergic neurons through direct and indirect actions, hence its CNS-stimulant effects and its use as a decongestant in upper respiratory tract infections.



epi+ or (*before a vowel*) **ep+** *prefix* denoting 1 on, upon, above, over, beside, near, close to, in addition to. 2 a chemical compound or group related in some way to a specified chemical compound or group, e.g. an **epimer**. 3 a chemical compound or group distinguished from a specified chemical compound or group by having a bridge connection in the molecule, e.g. an **epoxide**.

epiarginase *see* **epiprotein**.

epichlorohydrin 3-chloropropylene oxide; chloromethyloxirane; 1-chloro-2,3-epoxypropane; a colourless liquid widely used in industry as a solvent and an intermediate in the manufacture of glycerol, glycerol ethers and epoxy resins. In the presence of potassium hydroxide it reacts with compounds containing alcoholic hydroxyl groups to form diethers of glycerol and hence finds application in the manufacture of cross-linked derivatives of polysaccharides such as the proprietary materials Sephacel, Sephadex, and Sepharose.

epidemic 1 describing a disease that affects many persons simultaneously in a more or less restricted area. 2 the widespread occurrence of a disease in a human population. *Compare* **endemic**.

epidemiology the study of the nature and spread of a disease in a human population.

epidermal growth factor *abbr.*: EGF; any member of a group of heat-stable, hormonal, ≈6 kDa proteins that consist of a single polypeptide chain of 49–53 residues with three intrachain disulfide bonds; they are classified as **cytokines**. One form (mEGF), originally known as tooth-lid factor, can be isolated from submaxillary glands of the male mouse (*see* **nerve growth factor**). A similar protein (hEGF), extracted from human pregnancy urine and named **urogastrone**, is now known to be highly homologous to EGF (37 of 53 residues identical, disulfide bonds preserved). The main effects of EGF *in vivo* are to stimulate the growth of epidermal and epithelial cells, and to inhibit the secretion of gastric acid. It also has a potent mitogenic action on many types of cultured mammalian cells and displays a number of insulin-like properties towards them (*see* **insulin-related growth factors**). It is synthesized as a large preprotein, EGF membrane glycoprotein precursor, which contains, as well as the mature EGF protein, nine EGF-like repeats and urogastrone. Example (precursor) from human: database code EGF_HUMAN, 1207 residues (133.79 kDa). The action of EGF is mediated by the **epidermal growth factor receptor**.

epidermal growth factor receptor or **EGF receptor** the mediator of the biological signal of **epidermal growth factor** (EGF)



epidermis

and also of **transforming growth factor- α** (TGF- α). It is a membrane protein with a single transmembrane domain, and a **tyrosine kinase** activity in the cytoplasmic domain; the extracellular domain, which binds EGF, is highly glycosylated. The *v-erbB* oncogene codes for a protein highly homologous to a truncated EGF receptor, and the gene that codes for the receptor is now regarded as the protooncogene *c-erbB* (see **erbB**). Binding of EGF to the receptor leads to the induction of tyrosine kinase activity, and formation of dimers, as a result of which autophosphorylation occurs; stimulation of cell DNA synthesis and cell proliferation follows. Internalization of the EGF-receptor complex occurs subsequent to binding EGF. Example (precursor) from human: database code EGFR_HUMAN, 1210 amino acids (134.14 kDa).

epidermis the outermost layer of cells of an animal or plant. — **epidermal** *adj.*

epienzymatic control a form of control of arginine metabolism, occurring in some *Saccharomyces* spp., in which arginases bind stoichiometrically to ornithine carbamoyltransferase (EC 2.1.3.3), inhibiting the latter's activity but without substantially modifying the arginase activity. See also **epiprotein**.

epigenesis the concept that an organism develops from an originally undifferentiated mass of living material through the appearance of structures and functions not originally present.

epigenetic describing any of the mechanisms regulating the expression and interaction of genes, particularly during the development process. These include changes that influence the phenotype but have arisen as a result of mechanisms such as inherited patterns of DNA methylation rather than differences in gene sequence: **imprinting** is an example of this.

epigenetics the branch of biology dealing with the analysis of the mechanisms causing phenotypic effects to be produced by the genes of a genotype.

epiligrin an extracellular matrix protein, secreted by cultured epidermal keratinocytes. It is found in epithelial basement membranes but not membranes of endothelia, nor in muscle or nerve cells. It is a ligand for cell adhesion via **integrins**, with three chains linked by disulfide bonds, and is also an adhesive ligand for T lymphocytes. Example from the gene sequence of human epiligrin $\alpha 3$: database code HUMLAMAB, 145 amino acids (15.77 kDa). See also **laminin**.

Epilim a proprietary name for sodium valproate (see **valproic acid**).

epimer (in stereochemistry) either of two **diastereoisomers** that differ in configuration at only one chiral centre. The term includes **anomer**.

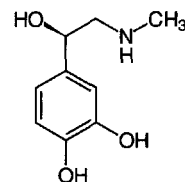
epimerase any enzyme within subclass EC 5.1, that catalyses **epimerization** (note that subclass EC 5.1 also includes **racemases**). Some epimerases, e.g. UDParabinose 4-epimerase (EC 5.1.3.5), contain tightly bound NAD, which undergoes oxidation and reduction during the reaction.

epimeric of or relating to an epimer or the chiral centre that specifies an epimer.

epimerization or **epimerisation** the process of converting an epimer into its diastereoisomer by altering the configuration at the epimeric chiral centre. — **epimerize** or **epimerise** *vb.*

epimino+ an alternative term for **imino+** (def. 2).

epinephrine or **adrenaline** (–)-1-(3,4-dihydroxyphenyl)-2-(methylamino)ethanol; a hormone secreted by the adrenal medulla and a neurotransmitter secreted by neurons in the brainstem. It is synthesized by the methylation of **norepinephrine**. It is an agonist for **adrenoceptors**, through which it has powerful glycogenolytic and lipolytic effects (both of which are mediated by cyclic AMP) and also affects the activity of smooth muscle (notably of the cardiovascular system and bronchi) and glandular tissue. Its mode of action critically depends on whether it acts through α or β adrenoceptors. The BP official name for this substance is Adrenaline; the USP official name for this substance is Epinephrine; Adrenalin is a USA trademark name. It was named adrenalin by Takamine in 1901.



epinephrine

epiphase the upper, less dense layer of a two-phase system. Compare **hypophase**. — **epiphasic** *adj.*

epiplasm a membrane skeleton of certain protists, that contributes to cell shape and patterning of the species-specific cortical architecture. This cytoskeleton contains proteins known as **articulos**.

epiprotein a specific protein, produced in certain circumstances in some *Saccharomyces cerevisiae* strains, that binds stoichiometrically to ornithine carbamoyltransferase (EC 2.1.3.3), thereby inhibiting its activity. It has subsequently been shown to have arginase activity and so is sometimes known as **epiarginase**. See also **epienzymatic control**.

episemantic molecule any molecule that is synthesized under the control of a tertiary **semantide**. All molecules built by enzymes in the absence of a template are included because, although they do not express extensively the information contained in the semantide, they are a product of this information. See also **asemantic molecule**.

episome 1 a genetic element that is sometimes found in cells, especially those of bacteria, and that can replicate either when integrated into or independently of the host chromosome. 2 any fragment of DNA that exists in a cell as an extrachromosomal element. This DNA may or may not be replicated and passed to daughter cells. The **plasmid** is an example of an episome. DNA elements that use transposases involved in phage lambda integration are also examples of episomes.

epithelium 1 (in animal anatomy) a sheet of closely packed cells, arranged in one or more layers, that covers the outer surfaces of the body or lines any internal cavity or tube (except the blood vessels, heart, and serous cavities). Compare **endothelium**, **mesothelium**. 2 (in plant anatomy) any of certain layers of parenchymal cells that line an internal cavity or tube. — **epithelial** *adj.*

epitope any immunological determinant group of an antigen. — **epitopic** *adj.*

epitype any class or group of related epitopes.

epoprostenol international nonproprietary name for prostaglandin I₂; see **prostacyclin**.

epothilone see **taxol**.

epoxidation the formation of an **epoxide** by addition of an oxygen atom as a bridge across the double bond of an alkene.

epoxide any compound where molecules contain an epoxy group attached to adjacent carbon atoms to form a saturated three-membered ring system.

epoxy+ prefix (in chemical nomenclature) indicating the presence in a molecule of an oxygen atom directly attached (by single covalent bonds) to two (adjacent or nonadjacent) carbon atoms of a carbon chain or ring system. Epoxy compounds are cyclic ethers.

epoxyeicosatrienoate abbr.: EpETRe or (formerly) EET; any of various metabolites of arachidonate that contain an epoxy group. They are formed by the action of cytochrome P450 – this occurs in numerous tissues and has a high degree of stereospecificity. The products are predominantly present esterified to membrane-bound phosphatidylcholine. They have a wide spectrum of biological activities, including stimulation of peptide hormone release, inhibition of Na⁺K⁺-ATP-

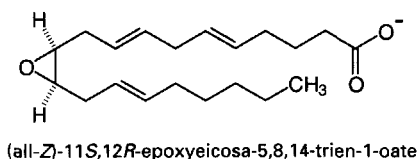
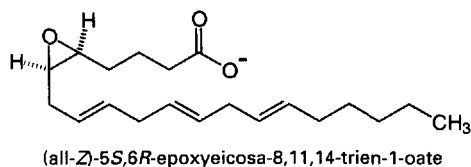
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epoxy resin

ase, mobilization of microsomal Ca^{2+} , and inhibition of prostaglandin-endoperoxide synthase. Examples are (all-*Z*)-5*S*,6*R*-epoxyeicosa-8,11,14-trien-1-oate and (all-*Z*)-11*S*,12*R*-epoxyeicosa-5,8,14-trien-1-oate. EpETREs can be further metabolized to *vic*-diols, diepoxides, and other products.



epoxy resin or **epoxide resin** any of a class of synthetic resins, containing ether linkages and epoxy groups, formed by the copolymerization of an epoxide (e.g. 1-chloro-2,3-epoxypropane) and a polyphenol (e.g. 2,2-bis(4-hydroxyphenyl)propane). When mixed with certain accelerators, e.g. an amine or an anhydride, such resins produce hard, clear, resistant, thermosetting polymers that are used as adhesives and coatings and for embedding materials for microscopy, especially electron microscopy.

Eppendorf pipette any of a range of push-button, plunger-operated, precalibrated, dispensing micropipettes with disposable polypropylene tips. [After the name of the manufacturer.]

EPPS an alternative name for **Hepps**.

EPR or **epr** *abbr.* for electron paramagnetic resonance spectroscopy (see **electron spin resonance spectroscopy**).

epsilon *symbol:* ϵ (lower case) or E (upper case); the fifth letter of the Greek alphabet. For uses see **Appendix A**.

epsilon chain or **ϵ chain** the **heavy chain** of IgE immunoglobulin molecules.

EPSP *abbr.* for 5-enolpyruvylshikimate phosphate; an intermediate in aromatic amino-acid biosynthesis by the shikimate pathway (see **shikimate**).

EPSP synthase an alternative name for **3-phosphoshikimate 1-carboxyvinyltransferase**. See also **shikimate**.

Epstein-Barr virus human (gamma) herpesvirus 4; a DNA virus, first isolated from specimens of tumour tissue obtained from African children affected by Burkitt's lymphoma, that causes infectious mononucleosis (glandular fever) in young adult humans. [After (Sir) Michael Anthony Epstein (1921–), British pathologist and immunologist, and Yvonne M. Barr.]

eq. or **Eq.** *abbr.* for equivalent (def. 4).

equation of state any equation connecting the pressure, volume, and temperature of a substance or a mixture.

equatorial 1 of, relating to, or being an equator; located at or near an equator. *Compare* **axial** (def. 1). 2 (in stereochemistry) *symbol:* *e*; see **conformation**.

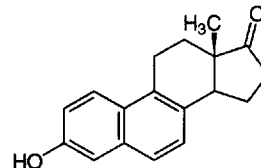
equi+ *comb. form* denoting equal or equality.

equi-effective dose ratio *abbr.:* EDR; the ratio of the doses of test and reference substances that produce the same biological effect (whether activation or inhibition).

equi-effective molarity ratio *abbr.:* EMR; the ratio of the molarity of test and reference substances that produce the same biological effect (whether activation or inhibition).

equilenin the trivial name for 3-hydroxyestra-1,3,5,7,9-pen-

taen-17-one; a weakly estrogenic steroid hormone isolated from the urine of pregnant mares. *Compare* **equilin**.

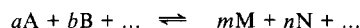


equilibrate to bring to equilibrium, or cause (something) to be in equilibrium (with some other thing), especially with the environment or with a (specified) system. —**equilibration** *n*.

equilibrium 1 a state of balance between or among opposing forces or processes that results in the absence of net change. 2 (in chemistry) a state of dynamic balance in a reversible chemical reaction when the reaction velocities in both directions are equal.

equilibrium assay a type of **radioimmunoassay** in which the reagents used are allowed to reach thermodynamic equilibrium. *Compare* **disequilibrium assay**.

equilibrium constant *symbol:* K ; an expression of the position of the **equilibrium** (def. 2) of a reversible chemical reaction under specified physical conditions (e.g. temperature, pressure, nature of solvent, ionic strength, etc.). For a reaction of the generalized form:



when the thermodynamic **activity** of each of the components A, B, M, N, etc. is written as (A), (B), (M), (N), etc., and a , b , m , n , etc. are the respective numbers of molecular entities participating in the reaction, the equilibrium constant is given by:

$$K = \frac{[(M)^m(N)^n \dots]}{[(A)^a(B)^b \dots]}$$

Frequently in biochemical systems it is not possible to evaluate the activities of all the components; the **concentration equilibrium constant** (*symbol:* K_A) rather than the true or **thermodynamic equilibrium constant** is then calculated, K being expressed in terms of the molarities of each of the components in place of their activities. When the specified concentration of a component includes more than one chemical species (e.g. ionized plus un-ionized forms, or chelated plus unchelated metal ion) in unknown proportions, an **apparent (concentration) equilibrium constant** (*symbol:* K') constrained with respect to certain variables (e.g. pH, or total concentration of metal ion), is written. See also **affinity constant**, **association constant**, **dissociation constant**, **pK**.

equilibrium density-gradient centrifugation a method for the separation of cells, cell organelles, macromolecules, or other particles of different densities by centrifugation in a solution that increases in solute concentration, and hence in density, from the top to the bottom of the centrifuge tube. At equilibrium, particles of the same density are found as a band or zone in the density gradient. *Compare* **differential centrifugation**.

equilibrium dialysis a technique used to measure the binding of a **microsolute** or ions to a **macrosolute**. A solution of the macrosolute is placed inside a dialysis bag (through which the macrosolute will not pass); this represents phase α . The bag is suspended in a solution containing the microsolute (phase β). At equilibrium, any excess concentration of the microsolute in phase α is taken as evidence for binding. From a measurement of the excess concentration of the microsolute and knowledge of the concentration of the macrosolute the extent of the binding may be determined.

equilin the trivial name for 3-hydroxyestra-1,3,5(10),7-tetraen-17-one; a weakly estrogenic steroid hormone isolated from the urine of pregnant mares. *Compare* **equilenin**.

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equimolar

equimolar of equal molarity.

equimolecular containing equal numbers of molecular entities.

equine of, relating to, or being a member of the family Equidae (horses, zebras, and asses); of, belonging to, characteristic of, obtained from, or relating to, a horse or horses; resembling a horse.

equiv. *abbr.* for equivalent (def. 3).

equivalence 1 the state of being equal, equivalent, or interchangeable. 2 **equivalency**. 3 the point or zone in a precipitin test at which antigen and antibody are present in optimal proportions for combination and precipitation.

equivalence factor *symbol* (for a chemical species X): $f_{eq}(X)$; a number pertaining to a given reacting component of a specified titrimetric reaction derived from consideration of the overall stoichiometry of the reaction. For a reaction of the following type: $\nu A + \nu B \rightarrow \text{products}$, where νA and νB are the respective numbers of reacting entities, then the equivalence factor of species B is given by: $f_{eq}(B) = \nu B / \nu A$. When $\nu A > \nu B$ and $f_{eq}(A)$ is taken as unity, then $f_{eq}(B)$ is unity or less than unity. In the case of a reaction that can be clearly identified as acid–base or oxidation–reduction, the equivalence factor must be related to one entity of titratable hydrons or of transferable electrons, respectively.

equivalence point or **stoichiometric point** or **theoretical end point** the point in a titration at which the amount of titrant added is chemically equivalent to the amount of substance titrated.

equivalency or **equivalence** 1 the state of having equal valencies. 2 the state of having equal **equivalence factors**.

equivalent 1 equal or interchangeable in amount, importance, meaning, or value. 2 (in chemistry) having equal valencies. 3 something that is equivalent (to something else). 4 (in chemistry) *a* *abbr.*: equiv. or eq. or Eq.; a unit of amount of substance, defined as the entity of a chemical species that in a specified reaction would combine with, displace, or in any other appropriate way be equivalent to, one entity of titratable hydrons (in an acid–base reaction) or one entity of electrons (in an oxidation–reduction reaction). It is the arithmetic product of the **molar mass** of the chemical species and its **equivalence factor**. *b* (formerly) **equivalent weight**.

equivalent activity a measure of the radiation output of a real radiation source containing a particular radionuclide, defined as being equal to the **activity** (def. 1b) of a hypothetical point source of the same radionuclide that would give the same exposure rate at the same distance from the centre of the source. It is customarily used to express the strengths of most sources emitting high-energy gamma radiation.

equivalent ellipsoid of rotation the **ellipsoid of rotation** having the same volume as an actual hydrodynamic unit of a macromolecule in solution (which consists of the macromolecule and tightly bound solvent).

equivalent weight *abbr.*: equiv. wt.; an obsolete term defined as the weight in grams of an element, compound, or group that, in a specified reaction, will combine with or displace 8 grams of oxygen, or the equivalent weight of another chemical species. *See also* **equivalent** (def. 4).

ER *abbr.* for endoplasmic reticulum.

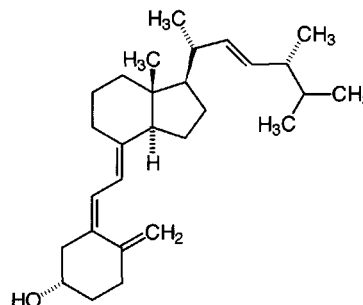
erbA an oncogene (v-*erbA*) originally found in avian erythroblastosis virus; the corresponding protooncogene (c-*erbA*) encodes thyroid hormone receptor. Example, product of v-*erbA* from avian erythroblastosis virus: database code ERBA_AVI, 385 amino acids (43.98 kDa); it requires the activity of other oncogenic agents for tumorigenesis. *See also* **erbB**.

erbB an oncogene (v-*erbB*) from avian erythroblastosis virus; the corresponding protooncogene (c-*erbB*) encodes the **epidermal growth factor receptor**. In tumours the extracellular ligand-binding domain and the C terminus are deleted, contributing to a system with a constitutively active receptor protein. Example of protein product of v-*erbB* from avian erythroblastosis virus: database code ACBERBBH, 31 amino acids (3.50 kDa). The *erb* oncogene system is complicated; the transform-

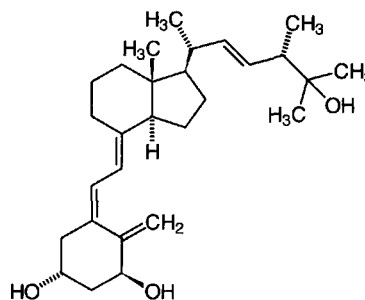
ing protein from the virus is a polypeptide product from *gag*, *erbA*, and *erbB*.

ercalcidiol or (formerly) **25-hydroxyergocalciferol** the recommended trivial name for (5Z,7E,22E)-(3S)-9,10(19),22-ergostatetraene-3,25-diol; a secosteroid with **vitamin D** activity. It is formed in the liver from **ercalcinol** and converted to **ercalcitriol** in the kidney.

ercalcinol or **ergocalciferol** the recommended trivial name for calciferol, vitamin D₂, (5Z,7E,22E)-(3S)-9,10-seco-5,7,10(19),22-ergostatetraen-3-ol; a secosteroid with **vitamin D** activity, obtained synthetically by UV-irradiation of **ergosterol**. It is the form of vitamin D most commonly used as a dietary supplement and for treatment of vitamin D deficiency. *International nonproprietary name*: ergocalciferol.



ercalcitriol or (formerly) **1 α ,25-dihydroxyergocalciferol** the recommended trivial name for (5Z,7E,22E)-(1S,3R)-9,10-seco-5,7,10(19),22-ergostatetraene-1,3,25-triol; a secosteroid with **vitamin D** activity. It is the biologically active metabolite of **ercalcidiol** and **ercalcinol**.



ERE *abbr.* for estrogen response element (*see* **response element**).

erepsin a once-supposed single enzyme secreted by the mucosa of the small intestine and thought to be responsible for completing the breakdown in the gut of partially digested proteins into amino acids. It is now known to be a mixture of **peptidases**.

erg the cgs unit of work, defined as the work done by a force of 1 dyne acting over a distance of 1 cm; equivalent to 10⁻⁷ J.

ERG a gene encoding a family of transcription factors and related to the protooncogene *ETS*; it is named from 'ETS-related gene'. It is expressed in some tumour-derived cell lines. Database code for expressed protein: ERG_HUMAN, 462 amino acids (52.03 kDa).

ergastoplasm a term formerly applied to the basophilic, fibrillar structures seen especially in pancreatic secretory cells; now termed rough **endoplasmic reticulum**.

+ergic suffix (in pharmacology) indicating agonist activity on a receptor in a neuron. *See, e.g.,* **adrenergic**.

ergo+ (comb. form) 1 or (before a vowel) **erg+** denoting work. 2 or (before a vowel) **ergot+** denoting obtained from **ergot** or derived from a substance occurring naturally in ergot.

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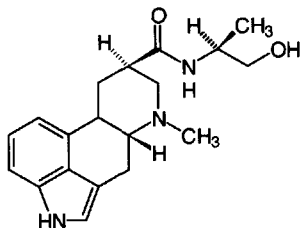
ergocalciferol

ergocalciferol the alternative recommended trivial name and the international nonproprietary name for **ercalcio**.

ergoflavin the major pigment from **ergot**.

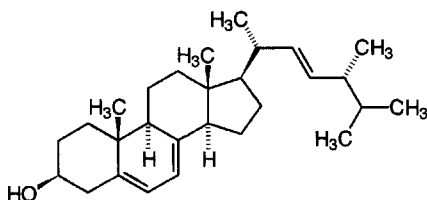
ergometrine see **ergonovine**.

ergonovine D-lysergic acid L-2-propanolamide; **ergometrine**; an **ergot alkaloid** noted for inducing sustained uterine contractions and used in the treatment of postpartum hemorrhage.



ergosome (sometimes) a **polyribosome**.

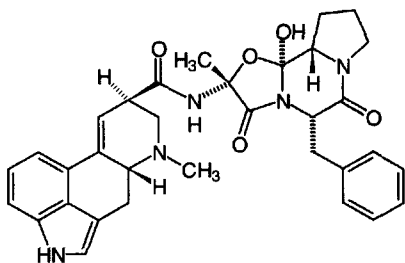
ergosterol trivial name for (22*E*)-ergosta-5,7,22-trien-3 β -ol; (22*E*)-(24*R*)-24-methylcholesta-5,7,22-trien-3 β -ol; a sterol found in ergot, yeast, and moulds. It is usually obtained industrially from yeast, which synthesizes it from simple sugars. It is the most important of the provitamins D. On irradiation with ultraviolet light it is converted to **ercalcio** (vitamin D₂).



ergot the common name for the sclerotial phase of the fungus *Claviceps purpurea*, which parasitizes the ovaries of cereals such as rye, and other grasses, leading to ergot-contaminated seeds. Ergot yields various medically useful alkaloids, including ergotamine and ergometrine, and is a source of lysergic acid. However, ingestion of contaminated seeds by livestock or humans can result in poisoning (**ergotism**).

ergot alkaloid any of a group of about 30 indole alkaloids obtainable from **ergot**. All are derivatives of **lysergamide** and are biosynthesized from L-tryptophan.

ergotamine 12'-hydroxy-2'-methyl-5' α -(phenylmethyl)ergotaman-3',6',18-trione; an **ergot alkaloid** with vasoconstrictor activity, used in the treatment of migraine.



ERIC abbr. for enterobacterial repetitive intergenic consensus sequence; any of a number of highly conserved partly palindromic sequences. They are reminiscent of **REPs** but are longer (120 bp) and are in extragenic sites.

ERK abbr. for from extracellular signal-regulated kinase; any of a sub-subclass (EC 2.7.-), protein kinases, also (more frequently) known as **MAP kinases**. Examples (all human): ERK1 (or insulin-stimulated MAP2 kinase) phosphorylates MAP2 and myelin basic protein; database code ERK1_HUMAN, 379 amino acids (43.09 kDa); ERK2 (or mitogen-activated protein kinase 2 or MAP kinase 2) is similar; database code ERK2_HUMAN, 360 amino acids (41.34 kDa); ERK3 (or MAP kinase isoform p63) has similar activity; database code ERK3_HUMAN, 557 amino acids (62.55 kDa).

ER mannosidase EC 3.2.1.-; other name: Man(9)- α -mannosidase; a glycoprotein enzyme of the *N*-glycosylation pathway of the endoplasmic reticulum that catalyses the removal of just one mannose residue from the core oligosaccharide of glycoproteins (Man(9)) to produce a specific isomer of Man(8)GlcNAc. Example (precursor) from human: database code MAN9_HUMAN, 625 amino acids (70.74 kDa). See **mannosidase**.

Ernst, Richard Robert (1933–) Swiss physical chemist noted for the application of pulsed radiofrequency radiation in nuclear magnetic resonance spectroscopy and the resultant extension of the technique to biological macromolecules; Nobel Laureate in Chemistry (1991) 'for his contributions to the development of the methodology of high resolution nuclear magnetic resonance (NMR) spectroscopy'.

erp60 see **calreticulin**.

ER-resident protein any protein that is retained by the endoplasmic reticulum. Their retention is regulated by the heterotetramer of a phosphoprotein (**calnexin**), a phosphoglycoprotein (see **Ssr α**), and two glycoproteins (see **Ssr β**). See also **KDEL**.

'error catastrophe' hypothesis a hypothesis stating that since the protein synthetic machinery of the cell is itself made of protein, any imprecision in protein synthesis will lead to the formation of protein synthetic machinery of lowered accuracy, which will lead in turn to the production of even less precise machinery. If this positive feedback of error into the protein synthetic machinery is sufficiently high it will cause instability and an accelerating deterioration of the cell, resulting in cell death or loss of function.

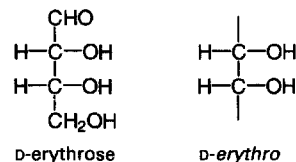
error-prone repair the processes of DNA repair, following damage, that give rise to 'fixed' mutations. Whether or not a particular damaged site will eventually lead to a mutation is dependent both on the genetic constitution of the cell and on environmental factors. It has been suggested that such mutations are due to errors in filling gaps in daughter DNA strands.

erucic acid the trivial name for (Z)-docos-13-enoic acid (22:1 n-9).

ERYF1 see **GATA-1**.

erythritol i-erythritol; meso-erythritol; ms-1,2,3,4-tetrahydroxybutane; erythro-1,2,3,4-butanetetrol; a polyol configurationally isomeric with (active) **threitol** but optically inactive through internal compensation. It occurs in certain fungi (where it appears to act as a storage carbohydrate), and is found also (as an ester of orsellinic acid) in some green algae and lichens. It is about twice as sweet as sucrose.

erythro- the configurational prefix to the systematic name of a polyhydric alcohol, especially of a monosaccharide, used to indicate the particular stereochemical configuration of a set of two contiguous CHOH groups that occurs in D- or L-erythrose; e.g. D-erythro-2-pentulose for D-ribulose. Compare **threo-**.



erythro+ or (before a vowel) **erythr+** or (sometimes) **eryth+** comb. form 1 indicating red-coloured. 2 pertaining to red blood cells.

erythroblast

erythroblast any of the nucleated red blood cells that are intermediates in the formation of a nonnucleated erythrocyte from a hemocytoblast. —**erythroblastic** *adj.*

erythrocrucorin any of a group of respiratory pigments of invertebrates in the range of 0.4–6.7 MDa and containing 30–400 heme groups per molecule.

erythrocuprein an alternative name for **superoxide dismutase**.

erythrocyte a mature red blood cell; in mammals it is non-nucleated and lacks mitochondria. Erythrocytes contain, but are no longer capable of synthesizing, **hemoglobin** and they function in the transport of oxygen. Mammalian erythrocytes obtain energy from anaerobic glycolysis and also metabolize glucose via the phosphogluconate pathway. —**erythrocytic** *adj.*

erythrocyte sedimentation rate *abbr.*: ESR; the rate at which erythrocytes sediment as measured under defined conditions. An increased ESR is almost always indicative of organic disease and is associated with an increased concentration of fibrinogen, γ -globulin, α_2 -macroglobulin, or γ_1 -macroglobulin in the plasma.

erythrodextrin any of the larger **dextrins** that give a red colour with diiodine.

erythrogenesis an alternative term for **erythropoiesis**.

erythro- β -hydroxyaspartic acid an amino acid formed by post-translational modification of an aspartyl residue at position 41 of the light chain of **protein C**.

erythroid 1 red or reddish in colour. 2 of or relating to **erythrocytes** or their precursors.

erythroid differentiation factor *see* **inhibin**.

erythrolabe a pigment that is sensitive to red light (570 nm) and is found in some retinal cones of the mammalian eye. It is lacking in subjects with red-blindness but is present in those with green-blindness. *Compare* **chlorolabe**, **cyanolabe**.

erythromycin any of several wide-spectrum macrolide antibiotics obtained from *Streptomyces erythreus*. Three components, A, B, and C, are produced during fermentation, the major component being erythromycin A. They inhibit protein synthesis in prokaryotes by binding to the 50S ribosomal subunit and inhibiting translocation.

erythromycin resistance factor any of several proteins that confer resistance to the antibiotic **erythromycin**. There are three mechanisms of resistance: (1) erythromycin esterase (EC 3.1.1.–), which is itself of two types (I and II); (2) a transmembrane export system (e.g. *erm* genes of *Arthrobacter* and certain Gram-positive bacteria); (3) rRNA (adenine- N^6 -methyltransferase (EC 2.1.1.48), an enzyme that catalyses the reaction between S-adenosyl-L-methionine and rRNA to form S-adenosyl-L-homocysteine and rRNA containing N^6 -methyladenine; example from *Staphylococcus aureus*: database code MLS2_STAAU, 244 amino acids (28.83 kDa); this protein produces a dimethylation of the adenine residue at position 2058 in 23S rRNA, resulting in reduced affinity between ribosomes and all macrolide, lincosamide, and streptogramin B-type antibiotics.

erythron a collective term for the circulating **erythrocytes** and their precursors.

erythrophore a type of chromatophore, coloured red and found especially in some fishes and crustaceans. The pigment, a carotenoid, is contained in bright red granules, which normally are dispersed throughout the cytoplasm, but which aggregate rapidly in response to epinephrine leading to near discoloration of the cell.

erythropoiesis or **erythrogenesis** the formation of red blood cells, in bone marrow or elsewhere. —**erythropoietic** or **erythrogenetic** *adj.*

erythropoietin a 46 kDa mammalian glycoprotein **cytokine**, formed in the kidneys and liver of mammals, that stimulates cellular differentiation of bone-marrow stem cells at an early stage of erythropoiesis, accelerates the proliferation and maturation of terminally differentiating cells into erythrocytes, and maintains a physiological level of circulating erythrocyte mass. The protein is now prepared from cDNA expressed in

CHO cells and other cells. It is now established as the accepted treatment of chronic renal failure. Its receptor is a type I membrane protein. Examples (precursors, human): erythropoietin, database code EPO_HUMAN, 193 amino acids (21.28 kDa); erythropoietin receptor, database code EPOR_HUMAN, 508 amino acids (55.00 kDa).

erythrose the trivial name for the aldotetrose *erythro*-tetrose; it has D and L enantiomers, which are respectively diastereoisomeric with those of threose. *See* **erythro**- for structure.

erythrosome an artificial membrane preparation in which detergent-extracted, glutaraldehyde-fixed erythrocyte ghosts are coated with phospholipid.

erythrulose the trivial name for the ketotetrose *glycero*-tetrulose; it has D and L enantiomers.

ES *abbr.* for enzyme–substrate complex.

esc a regulatory gene in *Drosophila* that controls expression of **BX-C** genes in embryos. It is named from extra sex combs.

ESCA *abbr.* for electron spectroscopy for chemical analysis. *See* **photoelectron spectroscopy**.

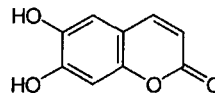
ES cell *abbr.* for embryonic stem cell.

Escherichia a genus of Gram-negative, rod-shaped, usually motile, chemoorganotrophic bacteria belonging to the family Enterobacteriaceae. It contains a single species, the colon bacillus, *Escherichia coli*. [After Theodor Escherich (1857–1911), German physician, who (in 1885) isolated the colon bacillus and gave it its original name of *Bacterium coli commune*.]

Escherichia coli or **colon bacillus** *abbr.*: *E. coli*; the sole member of the bacterial genus *Escherichia* and arguably the most widely used experimental cell system in biochemistry and molecular biology. The cells are straight, round-ended rods, commonly 0.5–1 $\mu\text{m} \times 1\text{--}4 \mu\text{m}$, and usually occurring singly or in pairs. The organism is present in the intestinal tract of humans and other animals and is common in soil and water. The numerous strains, some of which are enteropathogenic, are commonly distinguished serologically by their O, K, and H antigens. Strain O157 is a life-threatening enteropathogen with genes seemingly derived (at least in part) from *Shigella* spp. It produces both hemolysin and enterotoxin. *E. coli* takes part in bacterial conjugation and other forms of genetic transfer, and can be infected with some bacteriophages and plasmids. *See also* **toxin**.

Eschweiler-Clarke reaction *see* **reductive amination**.

esculetin 6,7-dihydroxycoumarin; an agent that selectively inhibits 5- and 12-lipoxygenase, without inhibiting prostaglandin-endoperoxide synthase.



E-selectin *see* **selectin**.

eserine an alternative name for **physostigmine**.

E-site the deacylated tRNA binding site in ribosomes; in bacteria this involves **ribosomal proteins** of the large subunit and 23S rRNA.

ESQL *abbr.* for embedded **SQL**; a computer language that allows SQL to interface with other programming languages, e.g. FORTRAN and C.

ESR or **esr** *abbr.* for 1 electron spin resonance. 2 erythrocyte sedimentation rate.

essential 1 of or being the essence of something; inherent. 2 absolutely necessary, basic, fundamental, indispensable, vitally important. 3 necessary for the normal growth of an organism but either not synthesized by that organism or synthesized at an inadequate rate. 4 (*in pathology*) describing a clinical condition or disorder that is not attributable to any discernible cause; **idiopathic** (def. 1). 5 relating to or derived from an extract of a plant, especially as an essence.

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essential amino acid

essential amino acid any amino acid that cannot be synthesized by a cell or an organism in an amount corresponding to need. The extent to which a particular amino acid is essential in any one species of organism depends on the stage of development of the subject and its physiological state. The following L-amino acids are nutritionally essential for the maintenance of nitrogen equilibrium in an adult human: isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine. In addition, arginine and histidine are essential in growing children. The initial letters of the L-amino acids needed for optimal growth of infants or rats may be remembered using the mnemonic 'Many Very Hairly Little Pigs Live In The Torrid Argentine'. *Compare nonessential amino acid.*

essential fatty acid any polyunsaturated fatty acid that cannot be synthesized by a cell or an organism in an amount corresponding to need or from any dietary precursor. For mammals, linoleic acid (*see* **linoleate**) and (9,12,15)-linolenic acid (*formerly collectively known as vitamin F*) are nutritionally essential. *Compare nonessential fatty acid.*

essential fatty acid deficiency a condition that arises from the absence of **essential fatty acids** in the diet. A deficiency of fatty acids of the *n*-6 family (e.g. linoleic acid) in the diet of young animals results during growth in decreased body weight and, after some months, in increased skin permeability and capillary fragility, and dermatitis. If prolonged into adulthood, poor litter sizes (e.g. in rats) and infertility can result. Deficiency of fatty acids of the *n*-3 family, in the presence of *n*-6 fatty acids, does not result in such obvious manifestations, partly because these fatty acids are tenaciously retained in tissues, and partly because *n*-6 fatty acids (e.g. 22:5 *n*-6) not normally present in significant amounts are synthesized and appear to compensate for the deficiency of *n*-3 fatty acids of similar length and degree of unsaturation. That the effects of deficiency are not solely due to lack of substrates for prostanoid synthesis is indicated by the ability of **columbinic acid** to relieve some of the effects.

essential fructosuria a relatively rare, hereditary, asymptomatic metabolic disorder characterized by the presence of fructose in the urine. In humans it is probably inherited as an autosomal recessive trait. Fructosuria occurs only after ingestion of fructose, which can also cause vomiting and aversion to sweet foodstuffs; long-term consequences can include failure to thrive, renal tubular dysfunction, and cirrhosis. Fructose metabolism involves formation of fructose 1-phosphate by ketohexokinase (EC 2.7.1.3); fructose 1-phosphate is then an (alternative) substrate for fructose-bisphosphate aldolase (EC 4.1.2.13) (I). Of the three isoenzymes of I (A, B, and C), B is much more active than A and C with fructose 1-phosphate as substrate. In essential fructosuria, the B isoenzyme of I is deficient and as a result there is a failure to metabolize fructose.

essential oil any of a group of volatile, odorless, natural oils obtained from plants. Such oils are usually benzene derivatives or **terpenes**, and are useful as flavouring agents or perfumes.

essential pentosuria *see* **pentosuria**.

EST *abbr.* for expressed sequence tag.

EST1 a gene involved in telomere maintenance, possibly coding for **telomerase**. It is named from 'even shorter telomeres'. It encodes a DNA-binding protein with RNA-directed DNA polymerase activity. Yeast gene product: database code EST1_YEAST, 699 amino acids (81.62 kDa).

established cell line cultured cells of a single origin that have the potential to be subcultured indefinitely whilst maintaining stable characteristics.

eSte *symbol* for the eleostearoyl group, $\text{CH}_3\text{--}[\text{CH}_2]_3\text{--}[\text{CH}=\text{CH}]_3\text{--}[\text{CH}_2]_7\text{--CO--}$.

ester any organic compound formed, either actually or formally, by the elimination of the elements of water between a hydroxyl group of an **oxoacid** (def. 1) and a hydroxyl group of

either an alcohol or a phenol. Esters formed by condensation of one, two, or three molecular proportions of alcohol or phenol with one of a tribasic oxoacid may be termed monoesters, diesters, or triesters, respectively.

esterase any enzyme that catalyses the hydrolysis of an ester. Esterases form subclass EC 3.1 of the class EC 3, hydrolases, and are divided according to the nature of their substrates into the following sub-subclasses: EC 3.1.1, carboxylic ester hydrolases; EC 3.1.2, thioester hydrolases; EC 3.1.3, phosphatases, phosphoric monoester hydrolases; EC 3.1.4, phosphodiesterases, phosphodiester hydrolases, phosphoric diester hydrolases (other than nucleases; *see* EC 3.1.11–3.1.31 below); EC 3.1.5, triphosphoric monoester hydrolases; EC 3.1.6, sulfatases, sulfuric ester hydrolases; EC 3.1.7, pyrophosphatases, diphosphoric monoester hydrolases; EC 3.1.8, phosphoric triester hydrolases; EC 3.1.11, exodeoxyribonucleases producing 5'-phosphomonoesters; EC 3.1.13, exoribonucleases producing 5'-phosphomonoesters; EC 3.1.14, exoribonucleases producing other than 5'-phosphomonoesters; EC 3.1.15, exonucleases active with either ribo- or deoxyribonucleic acids and producing 5'-phosphomonoesters; EC 3.1.16, exonucleases active with either ribo- or deoxyribonucleic acids and producing other than 5'-phosphomonoesters; EC 3.1.21, endodeoxyribonucleases producing 5'-phosphomonoesters; EC 3.1.22, endodeoxyribonucleases producing other than 5'-phosphomonoesters; EC 3.1.25, endodeoxyribonucleases specific for altered bases; EC 3.1.26, endoribonucleases producing 5'-phosphomonoesters; EC 3.1.27, endoribonucleases producing other than 5'-phosphomonoesters; EC 3.1.30, endonucleases active with either ribo- or deoxyribonucleic acids and producing 5'-phosphomonoesters; EC 3.1.31, endonucleases active with either ribo- or deoxyribonucleic acids and producing other than 5'-phosphomonoesters.

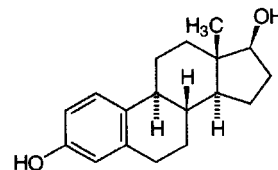
esteratic of, by means of, or having the activity of an esterase.

esterification the act or process of forming an ester from an oxoacid by condensing it with an alcohol or a phenol, or vice versa.

esterify to effect or to undergo esterification.

ester value *an alternative term for saponification number.*

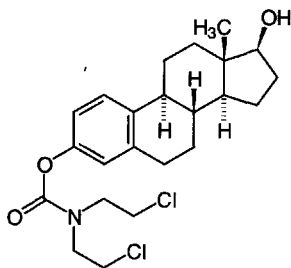
17 β -estradiol or (*esp. Brit.*) **17 β -oestradiol** 1,3,5-estratriene-3,17 β -diol; the most potent of the major **estrogens**. It is synthesized by the ovary, from which it is secreted directly into the circulation. Its formation depends on the presence of both **lutinizing hormone** and FSHRH; it is synthesized from testosterone by the aromatase complex, which is localized in ovarian granulosa cells, and is stimulated by follitropin; lutropin, on the other hand, stimulates production of testosterone, the aromatase substrate, by ovarian thecal cells. Apart from being important during development in inducing female characteristics, 17 β -estradiol is important during the first half of the menstrual cycle (i.e. before ovulation) in stimulating proliferation of the epithelial and stromal layers of the endometrium in preparation for ovulation; it also stimulates production by the liver of proteins that bind steroids and thyroxine. During pregnancy, synthesis of estrogens is greatly increased, most of this increase being due to synthesis by the fetal adrenal gland of the precursors dehydroepiandrosterone and 16-hydroxydehydroepiandrosterone.



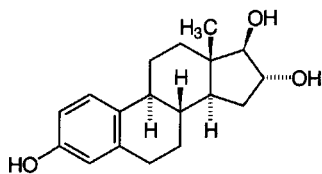
estramustine or (*esp. Brit.*) **oestramustine** estradiol 3-bis(2-chloroethyl)carbamate; an ester formed between estradiol and



the carbonic-acid analogue of nitrogen mustard (mustine); it has antineoplastic activity.

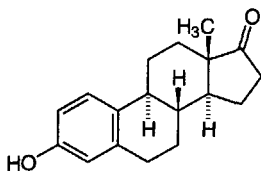


estriol or (*esp. Brit.*) **oestriol** 16 α -estradiol; 1,3,5-estratriene-3,16 α ,17 β -triol; a relatively weak estrogen produced mainly during pregnancy. It is synthesized by the placenta from 16 α -hydroxydehydroepiandrosterone sulfate, which is formed in fetal liver from dehydroepiandrosterone sulfate secreted by the fetal adrenal gland. Urinary estriol is measured to monitor fetal well-being in high-risk pregnancies. Low levels are also associated with the inherited X-linked disorder, placental sulfatase deficiency.



estrogen or (*esp. Brit.*) **oestrogen** any substance, natural or synthetic, that produces changes in the female sexual organs similar to those produced by 17 β -estradiol, the principal hormone of the vertebrate ovary. Naturally occurring estrogens are C₁₈-steroid hormones; they are produced also by the placenta, adrenal cortex, and testis, as well as by many species of plants. *Compare* **androgen**. —**estrogenic** or **oestrogenic** *adj.*

estrone or (*esp. Brit.*) **oestrone** 1,3,5-estratrien-3-ol-17-one; a major estrogen, also called folliculin, synthesized and secreted by the ovary, in which it is formed from androstenedione by the aromatase complex. However, estrone is mainly synthesized in peripheral tissues, from androstenedione taken up from the circulation. Its actions are similar to those of 17 β -estradiol.



estrone sulfotransferase EC 2.8.2.4; an enzyme involved in the inactivation of estrogens. It catalyses the reaction of 3'-phosphoadenylylsulfate with estrone to form the inactive estrone 3-sulfate with release of adenosine 3',5'-bisphosphate. Example from *Bos taurus*: SUOE_BOVIN, 295 amino acids (34.60 kDa).

estrophile (*sometimes*) estrogen receptor.

estrous cycle or (*esp. Brit.*) **oestrous cycle** the hormonally controlled cycle of activity of the reproductive organs of non-

pregnant, sexually mature females of many species of mammal. It may be continuous or seasonal, depending on species. There are four principal phases. (1) **Follicular phase** or **estrus** – ovarian Graafian follicles mature, estrogen secretion by the ovary is at its highest, and the endometrium proliferates; the phase culminates in ovulation; in seasonal animals it is the time of sexual receptivity. (2) **Luteal phase** or **metestrus** – the corpus luteum develops in the ovary, estrogen secretion diminishes, and progesterone secretion reaches its maximum. (3) **Diestrus** – the corpus luteum and endometrium regress, new follicular growth begins, and progesterone secretion diminishes. (4) **Proestrus** – the corpora lutea involute, Graafian follicles re-emerge, and gonadal hormone secretion is at its lowest. The cycle then recommences.

estrus or (*esp. Brit.*) **oestrus** the first phase of the **estrous cycle**. —**estrous** or (*esp. Brit.*) **oestrous** *adj.*

esu or **ESU** *abbr.* for electrostatic unit.

Et *symbol* for the ethyl group, CH₃–CH₂–.

ET *abbr.* for endothelin.

eta 1 *symbol*: η (lower case) or H (upper case); the seventh letter of the Greek alphabet. For uses see **Appendix A**. 2 (*in chemical nomenclature*) a less common alternative name for **hapto**.

ETF *abbr.* for electron-transfer flavoprotein or electron-transferring flavoprotein.

ethambutol (+)-N,N'-bis(1-hydroxymethylpropyl)ethylenediamine; an antitubercular drug.

ethanol or (*formerly*) **ethyl alcohol** CH₃–CH₂–OH; a colourless, water-miscible, flammable liquid. It is produced by **alcoholic fermentation** and is thereby probably the single most important product of bioindustry in economic terms. However, most ethanol (not destined for human consumption) is now manufactured from ethylene as a by-product in the petroleum industry. Commonly known as alcohol.

ethanolamine-phosphate cytidylyltransferase EC 2.7.7.14; *other name*: phosphorylethanolamine transferase; an enzyme that catalyses the reaction of CTP and ethanolamine phosphate to form **CDPethanolamine** with release of pyrophosphate.

ethanolaminephosphotransferase EC 2.7.8.1; an enzyme of the pathway for *de novo* synthesis of phosphatidylethanolamines. It catalyses the reaction of CDPethanolamine with 1,2-diacylglycerol to form phosphatidylethanolamine with release of CMP. Example from yeast: database code EPT1_YEAST, 391 amino acids (44.51 kDa). *Compare* **phosphatidylserine decarboxylase**.

ethanol fermentation an alternative term for **alcohol fermentation**.

ethanolic of, containing, or derived from ethanol; dissolved in pure or aqueous ethanol.

ethanolysed cellulose a powdered and highly purified form of cellulose, formerly in use as a supporting medium in preparative zone electrophoresis of proteins, e.g. serum proteins. It is prepared by heating fibrous cotton (cotton wool) under reflux for approximately 24 h with acid-ethanol (absolute ethanol made 1 molar with respect to HCl either by introduction of dry hydrogen chloride gas or by reaction with acetyl chloride). By this means, partial alcoholysis of the cellulose occurs and acidic groups are neutralized by esterification, the cotton fibres being partially degraded into nonclumping particles; at the same time, light-absorbing and other interfering impurities are extracted from the cellulose.

ether 1 any anhydride, of general formula R¹–O–R², formed between two (identical or nonidentical) organic hydroxy compounds. 2 *the common name* for **diethyl ether** (or ethyl ether or ethoxyethane), C₂H₅–O–C₂H₅.

ethereal (*in chemistry*) of, containing, or dissolved in an ether, especially diethyl ether.

ethereal sulfate an obsolete term for any biogenic half-ester of sulfuric acid with an aromatic hydroxy compound. Ethereal sulfates are found in the urine as end products of sulfur metabolism and metabolism of some aromatic compounds.

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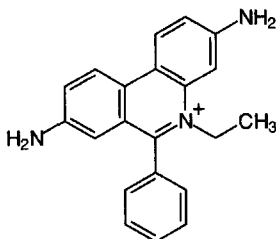
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ether lipid

ether lipid any lipid that contains (normally) one fatty alcohol in ether linkage to one of the carbon atoms (normally C-1) of glycerol. They are found in low concentrations in glycerophospholipid fractions, the constituent at C-2 of the glycerol moiety then being an acyl group. **Platelet-activating factor** is an important physiologically active ether glycerophospholipid. Ether linkages in neutral acylglycerols are known, and contain one fatty alcohol linked to glycerol by an ether linkage (usually at C-1), with fatty acids in ester linkage at the other two positions. The ether linkage resists hydrolysis, and typical hydrolysis products are the glyceryl ethers such as **batyl alcohol**, **chimyl alcohol**, and **selachyl alcohol**.

ethidium the 3,8-diamino-5-ethyl-6-phenylphenanthridinium cation, usually used as the bromide salt. It binds by **intercalation** to double-stranded regions of DNA and RNA molecules with marked enhancement of fluorescence. It is used as a trypanocide and to reveal double-stranded DNA and RNA in gel electrophoresis.



ethynylestradiol the international nonproprietary name for ethynylestradiol; 17 α -ethynyl-1,3,5(10)-estratriene-3,17 β -diol; (17 α)-19-norpregna-1,3,5(10)-trien-20-yne-3,17 β -diol; a synthetic steroid with potent estrogenic activity, commonly used in oral contraceptive preparations.

ethionine S-ethyl-L-homocysteine; a synthetic, carcinogenic analogue of methionine. Its toxicity is due to the fact that it competes with methionine for the enzyme **methionine adenosyltransferase**, with the result that S-adenosylmethionine is formed. This leads to a deficiency of ATP.

ethoxy symbol: OEt or Eto; the alkoxy group, CH₃-CH₂-O-, derived from ethanol by loss of a hydrogen atom.

ethyl symbol: Et; the CH₃-CH₂- alkyl group, derived from ethane.

ethyl alcohol the former name for **ethanol**.

ethylene CH₂=CH₂; a colourless gas and the first member of the olefin (or alkene) series of hydrocarbons. It occurs naturally as a phytohormone, with a number of physiological effects, the most important being stimulation of ripening of fleshy fruit and stimulation of abscission of leaves. *See also* **ethylene-forming enzyme**.

ethylenediaminetetraacetate *see* **EDTA**.

ethylene-forming enzyme abbr.: EFE; other name: 1-aminocyclopropane-1-carboxylate (ACC) oxidase. An enzyme component of the biosynthetic pathway for the formation of the plant hormone **ethylene**; it catalyses a reaction between 1-aminocyclopropane-1-carboxylate and O₂ to form ethylene, HCN, CO₂, and two molecules of 2H₂O; iron and ascorbate are cofactors. The enzyme is a **dioxygenase** but structurally unrelated to other examples of these. Example from tomato: database code ACC1_LYCES, 316 amino acids (35.93 kDa).

ethylenic containing one or more aliphatic carbon-carbon double bonds, as in **ethylene**.

N-ethylmaleimide abbr.: NEM; an agent that reacts irreversibly with thiol groups in proteins; it is useful as an inhibitor in transport studies and of thiol enzymes.

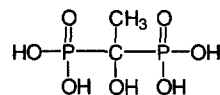
N-ethylmaleimide-sensitive fusion protein abbr.: NSF protein; a homotetrameric cytoplasmic protein required for

vesicle-mediated transport and transport from the endoplasmic reticulum to the Golgi stack. Example from Chinese hamster: database code SC18_CRIGR, 744 amino acids (82.44 kDa). *See also* **SNAP** (def. 3).

ethynylestradiol *see* **ethynylestradiol**.

ETI *see* **eicosatriynoic acid**.

etidronate any anion of etidronic acid, (1-hydroxyethylidene)bisphosphonic acid, an analogue of diphosphoric acid in which the latter's bridging oxy group is replaced by a 1-hydroxyethylidene group. Disodium etidronate is useful clinically for reducing the turnover rate of bone, e.g. in osteoporosis or in Paget's disease of bone.



etidronic acid

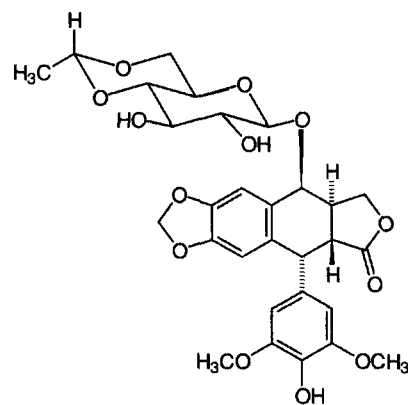
etiology or (*esp. Brit.*) **aetiology** the cause or causes of disease and their study.

Etn symbol for a residue of ethanolamine, H₂N-CH₂-CH₂-O-.

EtO or **OEt** symbol for the ethoxy group, CH₃-CH₂-O-.

ES cell abbr. for embryo-derived stem cell.

Etoposide a semisynthetic lignan derivative prepared from **podophyllotoxin**. It is useful as an antitumour agent in multidrug therapies, exerting its effect probably through interaction with topoisomerase II, which it inhibits; it also inhibits nucleoside transport in mammalian cells.



ETS abbr. for expression tagged site; a jargon phrase from genome mapping projects, etc. meaning a DNA database code (usually short) corresponding to an actively transcribed gene (i.e. cDNA from mRNA) of known sequence but, in general, of unknown function. Large ETS libraries are used in the Human Genome Project and similar projects on other organisms.

ETS a protooncogene related to *v-ets*, one of the oncogenes of the acutely transforming avian erythroblastosis virus E26; is named from 'E twenty-six specific'. It encodes a family of transcription factors: Ets1 activates stromelysin and collagenase genes and binds to the enhancer sequence of the TCR α gene; it is expressed in tumours of the peripheral nervous system and in Ewing's sarcoma; Ets1 and Ets2 are kinase substrates. Example of expressed protein: database code ETS1_HUMAN, 441 amino acids (50.41 kDa).

ETYA 5,8,11,14-eicosatetraynoic acid (*see* **eicosatetraynoic acid**).

eu+ prefix 1 indicating well, pleasant, good. 2 indicating normal, true, typical. In medicine, sometimes used interchangeably with **normo+**.

e.u.

e.u. symbol for entropy unit.**Eu** symbol for europium.**EU** abbr. for enzyme unit (obsolete).

Eubacteria a name suggested for one of three major lineages of cellular organisms, the others then being **Archaeobacteria** and **eukaryotes**, defined when organisms are classified on the basis of sequence homologies of their ribosomal RNA. It comprised all typical bacteria, the major subdivisions being the cyanobacteria, the Gram-positive bacteria, and the Gram-negative bacteria. This classification has now been superseded; see **Archaea**, **Bacteria**, and **Eukarya**.

eucaryon a variant spelling of **eukaryon**.**eucell** any cell of a **eukaryote**.

euchromatin the dispersed less dense form of **chromatin** in the interphase nucleus. It exists in at least two forms, about 10 per cent being in the form of transcriptionally active chromatin which is the least condensed, while the rest is inactive euchromatin which is more condensed than active chromatin but less condensed than **heterochromatin** which represents about 10 per cent of the chromatin in the interphase nucleus.

ecollagen **1** a highly modified form of **collagen** that can be easily and completely transformed into gelatin. **2** a hypothetical limiting structure for collagen in which Gly occurs at intervals of three residues and Hyp at intervals of ten residues; it is considered to be responsible for the 28.6 repeat distance in the X-ray diffraction pattern.

eugenics (*functioning as sing.*) the study of methods of improving the hereditary characters of the human race, especially by selective breeding. —**eugenic** *adj.*

Euglena a genus of fusiform, photosynthetic protozoa (sometimes classified as algae) of the division Euglenophyta. They are useful in biochemical research as unicellular photosynthetic organisms.

euglobulin any 'true' globulin of blood plasma that is soluble in isotonic salt solutions, insoluble at low ionic strengths (or in pure water), and precipitated by addition of saturated ammonium sulfate solution to a final concentration of 33%. Compare **pseudoglobulin**.

euglycemia or **normoglycemia** the condition or state in which the blood glucose level is within the normal range. See also **glycemia**. —**euglycemic** *adj.*

Eukarya one of the three major kingdoms of cellular organisms, the others being **Archaea** and **Bacteria**. The **eukarya** are characterized by the presence of a **eukaryon**.

eukaryon or (*sometimes*) **eucaryon** (*pl.* **eukarya** or **eucarya**) the type of cell nucleus bounded by a nuclear membrane and containing true chromosomes. It is characteristic of all multicellular and unicellular organisms except bacteria, actinomycetes, and cyanobacteria. Compare **prokaryon**. —**eukaryous** or **eucaryous** *adj.*

eukaryosis or (*sometimes*) **eucaryosis** the condition of possessing a **eukaryon**. —**eukaryotic** or **eucaryotic** *adj.*

eukaryote or (*sometimes*) **eucaryote** any organism whose cells contain a **eukaryon** (or **eukarya**) and undergo meiosis. Compare **prokaryote**.

eukaryotic porin see **porin**.

Euler **1** Hans Karl August Simon von Euler-Chelpin (1873–1964), father of U.S. von Euler (see **von Euler** (def. 2)) and generally known as Hans von Euler or Euler. German-born Swedish chemist and biochemist distinguished for his work on the enzymic mechanism of fermentation and in particular for his isolation, naming, and study of the chemical nature and functions of cozymase (now known as NAD); Nobel Laureate in Chemistry (1929), the prize being shared with A. Harden 'for their investigations on the fermentation of sugar and fermentative enzymes'. **2** Ulf Svante von Euler see **von Euler** (def. 2).

Euler-Chelpin see **Euler** (def. 1).**eumelanins** see **melanin**.

eupeptide a name suggested for any physiologically active gastrointestinal polypeptide.

eupeptide bond a name suggested for a peptide bond formed

specifically between C-1 of one amino-acid residue and N-2 of another residue. Compare **isopeptide bond**.

euploid having a chromosome number that is an exact multiple of the monoploid number; i.e. each chromosome of the set is present in the same number. Compare **aneuploid**. —**euploidy** *n.*
euthyroid having a normally functioning thyroid gland and normal levels of thyroid hormones.

eutopic binding (*sometimes*) an alternative term for **productive binding**.

eutrophic describing a habitat, esp. a lake or other mass of water, that contains abundant nutrients, both organic and inorganic, and hence favours excessive growth of aerobic plants and microorganisms. Such growth results in depletion of dissolved dioxygen and, ultimately, extinction of aerobic life. Compare **dystrophic** (def. 2), **oligotrophic**.

eutrophication the process of becoming or rendering eutrophic.

eV symbol for electronvolt.

event marker a feature of some chart recorders by means of which a mark is made on the chart to record the occurrence of a specifiable event or series of events, e.g. the stepwise operation of a fraction-collecting device.

evolution **1** (*in biology*) the process of cumulative change occurring in the form and mode of existence of a population of organisms in the course of successive generations related by descent. **2** (*in chemistry*) the liberation of a gas, or of heat, in the course of a chemical reaction. —**evolutionary** or **evolutional** *adj.*

evolve **1** to develop gradually; to cause to develop. **2** (*in biology*) to undergo **evolution** (def. 1). **3** (*in chemistry*) to give off a gas or heat.

ex+ **1** prefix indicating out of, away from, outside of; without, lacking; former. **2** *comb. form* a variant of **exo+** (*sometimes before a vowel*).

exa symbol: E; the SI prefix denoting 10^{18} times.

EXAFS abbr. for extended X-ray absorption fine structure (spectroscopy).

excelsin a crystallizable globulin of ≈ 290 kDa obtained from the seed of the brazil-nut tree, *Bertholletia excelsa*. It structurally resembles arechin, edestin, and glycinin.

exchange **1** to cause two different objects to change places with each other. **2** the act or process of exchanging. **3** (*in genetics*) the reciprocal exchange of chromatid segments between chromosomes at meiosis or mitosis that results in genetic recombination.

exchange diffusion the exchange of ions or molecules across a (biological) membrane; it can be by simple diffusion or be carrier-mediated.

exchange labelling the catalysed exchange of one nuclide for another radioactive or stable nuclide of the same element in a chemical compound in order to produce an isotopically labelled form. The procedure is useful esp. in the preparation of tritium-labelled compounds.

exchanger a substance upon which, or a device by which, an **exchange** (def. 2) can be effected; e.g. an ion exchanger, or a heat exchanger.

exchange reaction a chemical reaction that results in the production of a substance that is chemically identical with the starting material. It is not detectable by traditional chemical analysis but (often) is demonstrable with the help of (radio)isotopes.

excimer an adduct formed between a molecular entity that has been excited by a photon and an identical unexcited molecular entity; the adduct exists until it fluoresces by emission of a photon. An excimer is recognized by the production of a new fluorescent band at a longer wavelength than that of the usual emission spectrum. Its formation is distinguishable from resonance energy transfer in that the excitation spectrum is identical with that of the monomer.

excitation: $A + \text{photon} \rightarrow A^*$ excimer formation: $A^* + A \rightarrow D^*$ excimer fluorescence: $D^* \rightarrow A + A + \text{photon}$

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excinuclease

excinuclease a DNA-repair endonuclease. An example is the ABC excision nuclease of *Escherichia coli*; this comprises three subunits, named from the corresponding *uvr* genes. UvrA is an ATPase and DNA-binding protein that preferentially binds single-stranded or UV-irradiated double-stranded DNA; it belongs to the **ABC transporter** family; database code UVRA_ECOLI, 940 amino acids (103.75 kDa). UvrB stimulates the activities of UvrA; database code UVRB_ECOLI, 673 amino acids (76.14 kDa). UvrC is a nuclease component; database code UVRC_ECOLI, 588 amino acids (65.64 kDa).

excisase an enzyme involved in lambda phage induction. See **integrase**.

excise to cut out or remove, as of a tumour, organ, or part of a linear polymer.

excision **1** the act or process of excising, especially the enzymic removal of an oligonucleotide segment from a nucleic-acid molecule. **2** (*in genetics*) the separation of one or more replicons from a cointegrate.

excisionase see **Xis protein**.

excision repair one of the intracellular mechanisms for the repair of DNA lesions (single-strand breaks, damaged bases, etc.). It occurs in the following stages: (1) recognition of the damaged region; (2) removal of the damaged oligonucleotide by two enzymic nucleolytic reactions (excision); (3) synthesis by DNA polymerase of the excised oligonucleotide using the second (intact) DNA strand as template; and (4) covalent joining by DNA ligase of the newly synthesized segment to the existing ends of the originally damaged DNA strand. The process is light-independent. Compare **photoreactivation**, **post-replication repair**. See also **short-patch repair**.

excitability proteins a collective name for intrinsic membrane proteins that are ion channels, receptors, and ion pumps.

excitable capable of responding to a stimulus; used especially of a biological membrane, neuron, or other living matter. —**excitability** *n*.

excitable membrane any biological membrane capable of responding to a specific chemical or physical stimulus. Such membranes are commonly characterized by: (1) the presence of a highly specific, integral protein receptor; (2) the occurrence of a conformational change in the receptor in response to the specific stimulus with a consequential change in membrane permeability or in the activity of a membrane-bound enzyme; and (3) reversibility of the alterations in conformation of the receptor and functional activity of the membrane.

excitation-contraction coupling a general term often used to refer to the coupling of an excitatory stimulus to the contraction of muscle; the process by which the fibres of a muscle are caused to contract by the stimulation of a neuron.

excitation spectrum any **spectrum** of electromagnetic radiation that, when applied to a sample, causes the atoms and molecules of the sample to become excited by the absorption of energy and, when reverting to their normal energy levels, to emit radiation of different frequencies to that of the absorbed radiation. Compare **absorption spectrum**, **emission spectrum**.

excitatory able to excite; stimulatory.

excitatory amino-acid receptor a receptor for any excitatory amino-acid in the central nervous system. These comprise two main groups: ionotropic, regulating an intrinsic ion channel, or metabotropic, coupled to either the cAMP or inositol phospholipid second messenger systems. The three main types of ionotropic receptors all have four transmembrane domains and are characterized pharmacologically by selective agonists: *N*-methyl-D-aspartate (NMDA), α -amino-3-hydroxy-5-methylisoxazole (AMPA), and kainate. NMDA receptors have a further binding site for glycine, which modulates receptor activation. All such ionotropic receptors regulate Na⁺ and K⁺ conductance; NMDA receptors additionally regulate Ca²⁺ conductance. Seven subtypes of **metabotropic** receptor have been cloned, all with seven transmembrane domains (mGlu1 to mGlu7): mGlu1 and mGlu5 modulate inositol phospholipid

turnover while mGlu2, mGlu3, mGlu4, mGlu6, and mGlu7 are all negatively coupled to adenylate cyclase.

excite **1** to raise a molecular entity from its ground state to an excited state by the input of energy. **2** to cause an increase in activity or other response in an organism or part of an organism. —**excitation** *n*.

excited state any state of a molecular entity that has a higher energy level than its ground state.

exciter any device that excites, or raises a substance to an excited state.

exciton any nonconducting nonlocalized neutral entity that occurs in semicrystalline semiconductors, consisting of an excited electron bound to a positive hole, the combination forming a concentration of energy with certain properties characteristic of a particle.

exciton splitting the splitting of electronic absorption bands of a population of identical chromophores that are arranged in space such that sets of two or more are in close proximity. The splitting arises from electronic interactions between excited and nonexcited chromophores within the sets.

exclusion chromatography an alternative name for **permeation chromatography**.

exclusion limit an attribute of a specific gel used in **gel-permeation chromatography**. It is the relative molecular mass of the largest molecule of a particular shape that can be effectively fractionated by its use.

exclusion principle see **Pauli exclusion principle**.

exclusion reaction the reaction of a phage-infected bacterium that prevents the entry of additional phages. It is brought about by the strengthening of the bacterial cell envelope.

exclusion volume the volume of solvent retained by the column bed in **gel-permeation chromatography** that is outside the gel particles. Compare **void volume**.

excrete to eliminate waste materials (from a body or organism). —**excretion** *n*.; **excretive** or **excretory** *adj*.

exendin any of several peptides belonging to the glucagon family that have been isolated from *Heloderma* spp., exendin-1 (or **helospectin I**), exendin-2 (or **helodermin**) and exendin-4 from *Heloderma suspectus* and exendin-3 from *Heloderma horridus*. They have biological activity similar to vasoactive intestinal peptide and secretin. Exendin-4 shares 53 per cent similarity in its primary structure with glucagon-like peptide 1 (GLP-1) (7–36) amide. The latter binds to exendin receptors on pancreatic acinar cells, and exendin (9–39) (from exendin-3 or -4) is a GLP-1 receptor antagonist which blocks the inhibitory effect of GLP-1 on food intake of fasted rats. Exendin-4 is

HGEGTFTSDLSKQMEEEEAVRLFI-
EWLKNGGPSSGAPPPS

(database code EXE4_HELISU, 4.188 kDa); exendin-3 is similar except for residues 1–3 which are HSD.

exergonic describing any process or reaction that can produce work; i.e. one that, at constant pressure and temperature, results in a negative change in free energy content. Compare **endergonic**.

exhaustive methylation the maximal methylation of all groups in a substance that are capable of being methylated; it is useful in structural studies of carbohydrates, alkaloids, etc.

exo+ or (*sometimes before a vowel*) **ex+** *comb.* form indicating outside, external, outer. Compare **endo+**. See also **exo-**.

exo- prefix (*in stereochemistry*) see **conformation**.

exocellular describing cellular components that are external to the cell membrane but attached to its outer surface. Compare **extracellular**.

exocrine **1** describing or relating to a gland or other group of cells that discharges its secretion through a duct. **2** relating to a secretion of exocrine tissue. **3** a product of exocrine tissue. Compare **endocrine**.

exocytic **1** situated outside a living cell and not belonging to



exocytose

the cell itself; extracellular. **2** an alternative term for exocytotic (see **exocytosis**).

exocytose to effect exocytosis. —**exocytosed** *adj.*

exocytosis the discharge by a cell of intracellular materials to the exterior by **emiocytosis** or reverse **pinocytosis**. Compare **endocytosis**, **transcytosis**. See also **externalize**. —**exocytic** or **exocytotic** *adj.*

exodeoxyribonuclease see **deoxyribonuclease**.

exodeoxyribonuclease I EC 3.1.11.1; *other names*: exonuclease I; *Escherichia coli* exonuclease I. An enzyme that catalyses exonucleolytic cleavage of DNA in the 3'- to 5'- direction, releasing 5'-phosphomononucleotides. It shows a preference for single-stranded DNA. Database code EX1_ECOLI, 475 amino acids (54.44 kDa).

exodeoxyribonuclease V EC 3.1.11.5; *other names*: exonuclease V; *Escherichia coli* exonuclease V; RecBC. An enzyme that catalyses exonucleolytic cleavage of DNA (in the presence of ATP), in either the 5'- to 3'- or the 3'- to 5'-direction, to yield 5'-phosphooligonucleotides. Database codes of the subunits (products of genes *recA*, *recB*, and *recC*): RecA: EX5A_ECOLI, 608 amino acids (66.89 kDa); RecB: EX5B_ECOLI, 1180 amino acids (133.81 kDa); RecC: EX5C_ECOLI, 1122 amino acids (128.70 kDa).

exodeoxyribonuclease VII EC 3.1.11.6; *other names*: exonuclease VII; *Escherichia coli* exonuclease VII. An enzyme that catalyses exonucleolytic cleavage of DNA, in either the 5'- to 3'- or the 3'- to 5'-direction, to yield 5'-phosphomononucleotides.

exodeoxyribonuclease (phage SP3-induced) EC 3.1.11.4; *other names*: phage SP3 DNase; DNA 5'-dinucleotidohydrolase. An enzyme that catalyses exonucleolytic cleavage of DNA in the 5'- to 3'-direction to yield 5'-phosphodinucleotides.

exoenzyme **1** any enzyme that occurs attached to the outer surface of a cell (i.e. an **ectoenzyme**) or in the periplasmic space, or one that is secreted by a cell into the medium. Compare **endoenzyme** (def. 1). **2** any enzyme that cleaves a linkage to the terminal residue of a biopolymer. Compare **endoenzyme** (def. 2).

exogenous originating outside an organism, tissue, or cell, e.g. nutrients; of or pertaining to external factors that affect an organism, e.g. light. —**exogenously** *adv.*

exomorphin see **casomorphin**.

exon **1** any coding or messenger sequence of deoxynucleotides; i.e. any intragenic region of DNA in eukaryotes that will be ultimately expressed in (mature) mRNA or rRNA residues. An arrangement of exons alternating with **introns** constitutes a **transcriptional unit**. **2** ordinary title of any of the four most junior officers of the Yeomen of the Guard (Beefeaters, Warders of the Tower of London, appointed by the monarch) who in turn act as resident commanders in the absence of superior officers. —**exonic** *adj.*

exon shuffling the recombination of different coding regions of eukaryotic structural genes (i.e. **exons**) through crossing-over, a mechanism proposed to explain the mosaic structure of eukaryotic genes. Such a mechanism is also responsible for the generation of, e.g., antibody diversity.

exonuclease any enzyme of a group of phosphoric diester hydrolases, forming sub-subclasses EC 3.1.11–16, that catalyses the hydrolysis of terminal diester linkages in polynucleotides to yield mononucleotides, the degradation of the chain occurring sequentially from one end. Compare **endonuclease**. See also **deoxyribonuclease**, **nuclease**, **ribonuclease**.

exopeptidase any enzyme of a group of peptide hydrolases within subclass EC 3.4 that catalyses the hydrolysis of peptide bonds adjacent to the terminal amino or carboxyl group of an oligo- or polypeptide. They include: EC 3.4.11, aminopeptidases; EC 3.4.13, dipeptidases; EC 3.4.14, di- and tripeptidyl peptidases; EC 3.4.15, peptidyl dipeptidases; EC 3.4.16, serine-type carboxypeptidases; EC 3.4.17, metallo-carboxypeptidases; EC 3.4.18, cysteine-type carboxypeptidases; and EC 3.4.19, omega peptidases.

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exophthalmic goitre or **Graves' disease** a form of hyperthyroidism characterized by enlargement of the thyroid gland and protrusion of the eyeballs.

exopolysaccharide any of a range of extracellular homo- or heteropolysaccharides produced by microorganisms, either as insoluble capsules or as soluble slimes. They serve for protection and possibly also in recognition by host organisms.

exoribonuclease see **ribonuclease**.

exorphin any peptide with morphine-like activity (see **opioid peptide**) that originates from outside the body. Such substances occur, e.g., in proteolytic digests of certain dietary proteins. Compare **endorphin**.

exo- α -sialidase EC 3.2.1.18; *other names*: sialidase; neuraminidase; *N*-acylneuraminate glycohydrolase; α -neuraminidase; an enzyme that catalyses the hydrolysis of α -2,3-, α -2,6-, and α -2,8-glycosidic linkages joining terminal nonreducing *N*- or *O*-acylneuraminy residues to galactose, oligosaccharides, glycoproteins or glycolipids. Examples: hemagglutinin-neuraminidase from measles virus (strain aik-c) (subacute sclerosing panencephalitis virus): database code HEMA_MEASA, 617 amino acids (69.24 kDa); sialidase (neuraminidase) from *Clostridium perfringens*: database code NANH_CLOPE, 382 amino acids (42.76 kDa).

exoskeleton any skeleton covering the outside of an organism or lying in the skin, such as the hard chitinous cuticle of an arthropod or the shell of a mollusc. Compare **endoskeleton**.

exosmosis the osmotic flow of water or of an aqueous solution from a cell, vessel, or organism into the surrounding aqueous medium. Compare **endosmosis**. —**exosmotic** *adj.*

exosome a fragment of exogenous DNA that, when taken up by a cell, is not readily integrated into the chromosome but can replicate, and be expressed.

exothermic describing a process or reaction that is accompanied by the evolution of heat; i.e. a process or reaction for which the change in enthalpy, ΔH , is negative at constant pressure and temperature. Compare **endothermic**.

exotic describing an introduced (i.e. foreign), nonendemic, nonacclimatized organism.

exotoxin any toxin formed by a microorganism and secreted into the surrounding medium. Compare **endotoxin**.

exp (in mathematics) symbol for exponential; see **exponential function**.

expansin any of a group of proteins, located within the cell walls of plants (both dicotyledons and grasses), that play an essential role in loosening cell walls during cell growth. They are hydrophobic, nonglycosylated proteins of $M_r \approx 30$ kDa.

experiment *abbr.*: expt.; **1** any procedure or series of procedures, carried out in defined conditions and designed to obtain new information about substances, materials, organisms, or natural phenomena, or to test or confirm a hypothesis. **2** an alternative term for **experimentation**. **3** to carry out an experiment or experiments. —**experimenter** *n.*

experimental *abbr.*: exptl.; **1** of, relating to, or based on the results of, an experiment or experimentation. **2** describing a disease produced deliberately in laboratory plants or animals for the purposes of study or as a model for the natural disease condition. **3** any particular object or system in an experiment that includes, or is subjected to, one of the variables. Compare **control** (def. 1). —**experimentally** *adv.*

experimentation the act, process, or practice of carrying out an experiment or experiments.

explant **1** to transfer a piece of living tissue from its normal situation to a culture medium. **2** any piece of tissue treated in this way. —**explanted** *adj.*

exponent or **index** (in mathematics) any number (or expression, or symbol for a number) indicating the power to which another number (or expression or symbol) is raised; e.g., in the expression a^n , n is the exponent, indicating that a is multiplied by itself n times.

exponential (in mathematics) **1** of, containing, or involving one or more exponents, e.g. an exponential equation. **2** de-

exponential function

cribing something approximately expressible by an exponential equation.

exponential function (*in mathematics*) any function of the type e^x or Ae^{ax} , where A and a are constants, e is an irrational number that is the base of natural logarithms, and x is a variable; e^x is sometimes written $\exp x$.

exponential growth *an alternative term for logarithmic growth.*

exponential phase *an alternative term for logarithmic phase.*

export *informal term* describing a substance synthesized in, and then secreted from, a cell or type of cell in a multicellular organism; e.g. export protein.

exposure **1** (*in photography or radiography*) the act or process of exposing sensitized photographic material, e.g. a film or plate, to light or to ionizing radiation. **2** (*in photography*) a measure of the amount of light admitted into a photographic device expressed in terms of the lens aperture (f number) and time. **b** or **light exposure symbol:** H ; a physical quantity equal to the product of the illuminance and the time, Δt , for which the area is exposed. The SI unit is the lux second. **3** (*in radiation dosimetry*) a measure of the amount of X- or gamma radiation to which a subject or object is exposed, expressed in terms of the quantity of electric charge of the ions of one sign produced when all the electrons of both signs liberated in a volume of air of unit mass are completely stopped. The SI unit is $C\ kg^{-1}$, which has replaced the **röntgen**. *Compare absorbed dose.*

express **1** (*in genetics*) **a** to allow or cause the information in a gene to become manifest. **b** to activate the cellular functions involved in **gene expression**. **c** (*in cDNA technology*) to effect the synthesis of protein from the corresponding cloned gene using an **expression vector** in an **expression system**. **2** (*in mathematics*) to indicate by means of a symbol, formula, or equation. **3** to force or squeeze out a liquid from an object.

expressed sequence tag *abbr.:* EST; a partial sequence of a clone picked at random from a cDNA library and used in the identification of genes being expressed in a particular tissue. The technique exploits recent advances in automated DNA sequencing and sequence data handling, and a remarkably high number of ESTs turn out to represent previously unknown genes. These are identified by the predicted primary structure of the proteins that would be expressed and their relation to proteins of known structure. ESTs have proved extremely valuable in mapping the human genome, and there has been controversy surrounding attempts to patent them.

expression **1** the act or process of allowing information to become manifest. *See gene expression, expression vector, expression system.* **2** (*in mathematics*) a variable function or some combination of constants, variables, and/or functions. **3** the act or process of forcing or squeezing fluid from or out of something.

expression system a cell system into which an **expression vector** has been introduced and that contains all the enzyme systems needed for translation of messenger RNA. Ideally, it also contains the enzymes needed for post-translational modification of the synthesized protein. Expression in *Escherichia coli* or other bacterial cells may be inadequate for eukaryotic protein expression, but expression in mammalian cell lines may be inconvenient. Use of a baculovirus expression vector in insect cells is one technique that has been used successfully for eukaryotic protein expression.

expression tagged site *see ETS.*

expression vector (*in cDNA technology*) a **vector** that promotes transcription of a cloned gene, leading to synthesis of protein in a suitable **expression system**. An expression vector carries a **promoter**, e.g. *EcoRI*(UV5), which includes the regulatory region of the *lac* operon and the UV5 mutation, so that the system is insensitive to catabolite repression.

expressor a chemically undefined positive regulator of eukaryotic gene expression.

expt. *abbr. for experiment* (def. 1).

exptl. *abbr. for experimental* (def. 1, 2).

extein a peptide counterpart of an **axon**. *See intein, splicing*

extracellular :

extended X-ray absorption fine structure (spectroscopy)

abbr.: EXAFS (spectroscopy); a method for determining short-range order and local structure around specific atoms in noncrystalline materials. In this technique the X-ray absorption spectrum of the material is measured over an extended energy range near the absorption edge of one particular element in the material and structural details may be obtained from the observed modulation of the X-ray absorption on the high-energy side of the absorption edge. It is useful for elucidating the environment of metal ions in biological macromolecules.

extensin a glycoprotein found in plant cell walls in association with pectin. It resembles collagen of animal cells in being rich in *trans*-4-hydroxy-L-proline residues, and it contains L-arabinofuranose and D-galactopyranose residues attached in oligosaccharide units to the hydroxyl groups of the protein.

extension sequence a terminal sequence of amino-acid residues that is present in a nascent polypeptide as generated by translation of mRNA but is absent from any mature polypeptide or protein formed from it. An N-terminal extension sequence may be a signal sequence.

extensive property any of the properties of a substance or object that depend on the amount of the substance or the size of the object being considered, e.g. internal energy, mass, volume. *Compare intensive property.*

externalize or **externalise** to render external, especially to secrete a substance from a cell or organism. —**externalization** or **externalisation** *n.*

external standard any standard that is added to a sample at some point in an assay and serves as a reference by which the unknown being investigated can be quantified or identified.

external-standard method of quench correction or **external standardization** a method for correcting for quenching in **liquid scintillation counting** in which a source of gamma radiation external to the sample is brought into proximity with it after the sample has been counted and the sample is then counted again. The observed increase in count rate is correlated with that obtained using a set of differently quenched standards and the appropriate correction to the sample-count rate obtained. The method is simple and can be automated but on its own is less accurate than other methods. *Compare internal-standard method of quench correction.*

extinction *an older term (no longer recommended) for absorbance.*

extinction angle in **flow birefringence** between concentric cylinders, with polarizer and analyser at right angles, the angle between the **cross of isocline** and the cross formed by the polarizer and the analyser. It ranges from 45° (no orientation) to zero (complete orientation).

extinction coefficient *symbol* E ; *an alternative term (no longer recommended) for molar (decadic) absorption coefficient. See absorption coefficient.*

extra+ *prefix* signifying outside or beyond.

extra arm or **variable arm** the base-paired segment of variable length in the cloverleaf model of some tRNA molecules. *See transfer RNA.*

extracellular present outside a cell, expelled from a cell, or happening outside a cell. *Compare exocellular, intracellular.*

extracellular fluid *abbr.:* ECF; the portion of the total liquid content that occurs outside the cells, esp. as distinct from **intracellular fluid**. In higher animals it comprises the **interstitial fluid**, the **lymph**, and the **blood plasma**. In a 70 kg man the volume of ECF is approximately 15.5 L and it represents approximately 22% of body mass. *Compare intracellular fluid.*

extracellular matrix a layer consisting mainly of proteins (especially collagen) and glycosaminoglycans (mostly as proteoglycans) that forms a sheet underlying cells such as endothelial and epithelial cells. The constituent substances are secreted by cells in the vicinity, especially fibroblasts.

extracellular space that part of a multicellular organism outside the cells proper — usually taken to be outside the plasma membranes — and occupied by fluid.

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extrachromosomal

extrachromosomal describing structures, including DNA molecules, that are not part of a chromosome, or processes that proceed outside the chromosomes.

extrachromosomal inheritance inheritance controlled by extrachromosomal genes, such as those in mitochondrial or chloroplast DNA or in plasmids.

extract **1** to remove from or separate; to obtain a substance from a material, mixture, organism, or part of an organism by some chemical and/or physical process. **2** something extracted; a solution containing an active principle that has been extracted from plant or animal material.

extraction **1** the act or process of extracting. **2** something that has been extracted. *Compare* **extract** (def. 2).

extra piece *an alternative name for signal peptide.*

extrapolate (*in mathematics*) to estimate a value or the value of a function beyond the data values already obtained by either extending a graph or by calculation. *Compare* **interpolate**. —**extrapolation** *n.*

extremophile an organism that requires extreme physicochemical conditions for its optimum growth and proliferation. Such organisms include **thermophiles** or **psychrophiles**, **halophiles**, **alkalophiles** or **acidophiles**, **osmophiles**, and **barophiles**, based on their growth at extremes of temperature, salt concentration, pH, osmolarity, or pressure, respectively.

extrinsic originating or acting from the outside. —**extrinsically** *adv.*

extrinsic factor *a former name for vitamin B₁₂.*

extrinsic fluorescence fluorescence caused by a molecule of a fluor of small molar mass attached by either chemical coupling or simple binding to a (macro)molecule under study.

extrinsic pathway (of blood coagulation) *see* **blood coagulation**.

extrinsic protein *an alternative name for peripheral (cell membrane) protein.*

extrude to squeeze or force out; used especially of a solid or semisolid material. —**extrusion** *n.*

extrusome an organelle excreted as an anterior vacuole by a parasitic protist. It contains substances used for penetration of host cells.

ex vivo describing any biological process, reaction, or experiment in which the nature or magnitude of a change occurring

in living tissues in intact organisms is subsequently measured **in vitro** following the excision of affected tissues. *Compare* **in vivo** [*Latin*, meaning from life.]

eye (*in molecular biology*) a region of DNA undergoing replication, in which the separated strands give the appearance of an eye.

E/Z convention a convention of stereochemical descriptors for an unsaturated compound showing *cis-trans* isomerism that indicate unequivocally the steric relations around a (given) double bond. The **sequence rule** is applied in turn to the pair of atoms or groups attached to one of the doubly bonded atoms and to the pair attached to the other doubly bonded atom, and the order of preference in each case is established. If the two preferred atoms or groups are on opposite sides of the reference plane of the molecule the arrangement is called *E* [from German *entgegen*, opposite], and if they are on the same side of the plane it is called *Z* [from German *zusammen*, together]. For use as a prefix (in chemical nomenclature) the descriptor is placed in parentheses and connected with a hyphen, e.g. (*E*)-butenedioate and its stereoisomer (*Z*)-butenedioate (the respective systematic names for *trans*-1,2-ethylenedicarboxylate (= fumarate) and for *cis*-1,2-ethylenedicarboxylate (= maleate)). When a molecule of an unsaturated compound contains more than one double bond the prefix is composed of multiple symbols, each preceded if appropriate by the lower-numbered locant of the relevant double bond; e.g. (2*E*,4*Z*)-2,4-hexadienoate. The convention is now generally preferable to that using the structural prefixes *cis*- and *trans*-, especially for olefinic compounds having more than two substituents at a double bond (where it avoids possible ambiguity as to which pair of substituents is being designated). *See also* **pro-E**, **pro-Z convention**.

ezrin or **cytovillin** or **villin-2** or **P81** a protein involved in connections of major cytoskeletal structures to the plasma membrane, having domains in common with **talin**. It undergoes tyrosine phosphorylation after activation of specific growth factor receptors, and functions as a membrane-to-cytoskeleton linker that is involved in assembly of apical microvilli in parietal cells. It participates in the regulation of acid secretion. Example from human: database code EZR1_HUMAN, 585 amino acids (69.19 kDa).

Ff

f symbol for femto+ (SI prefix denoting 10^{-15} times).

f symbol for 1 frequency (alternative to ν). **2** activity coefficient referenced to Raoult's law. **3** fugacity (alternative to $\tilde{p}$). **4** frictional coefficient. **5** furanose form (added after the symbol for a monosaccharide, e.g. Araf for arabinofuranose). **6** (*in mathematics*) function.

F symbol for 1 a residue of the α -amino acid L-phenylalanine (alternative to Phe). **2** fluorine. **3** the fluoro group in an organic compound. **4** (*in genetics*) filial generation. **5** farad. **6** Fick unit. **7** *abbr.* for fertility; see **F plasmid**. **8** *abbr.* sometimes used (not recommended) to designate **folate** or folic acid or their reduced derivatives.

F1, F2a1, F2a2, F2b, F2c, F3 older nomenclature for histones. **F** signifies fraction; in early work, the histones were extracted from the nuclei with sulfuric acid solutions, and separated according to their solubility.

F₄₂₀ see **coenzyme F₄₂₀**.

F₄₃₀ see **coenzyme F₄₃₀**.

°F symbol for degree Fahrenheit. See **Fahrenheit temperature scale**.

F⁺ symbol for a donor bacterial cell containing an **F plasmid**; **F⁻** denotes a recipient bacterial cell lacking an **F plasmid**.

F symbol for 1 Faraday constant. **2** force (bold italic in vector equations). **3** fluence (alternative to *H*).

FA *abbr.* for 1 fatty acid. **2** folic acid or folate (not recommended). **3** filtrable agent.

Fab fragment a ≈ 45 kDa protein fragment obtained (together with **Fc fragment** and **Fc' fragment**) by papain hydrolysis of an immunoglobulin molecule. It consists of one intact light chain linked by a disulfide bond to the N-terminal part of the contiguous heavy chain (the **Fd fragment**). Two Fab fragments are obtained from each IgG antibody molecule; each fragment contains one antigen-binding site. The term Fab is derived from ['fragment antigen-binding']. Compare **F(ab)₂ fragment**. See also **Fv fragment**.

F(ab')₂ fragment a ≈ 90 kDa protein fragment obtained (together with **pFc' fragment**) by pepsin hydrolysis of an immunoglobulin molecule. It consists of that part of the immunoglobulin molecule N-terminal to the site of pepsin attack and contains both **Fab fragments** held together by disulfide bonds in a short section of the **Fc fragment** (the hinge region). One **F(ab')₂ fragment** is obtained from each IgG antibody molecule; it contains two antigen-binding sites but not the site for complement fixation. Compare **Fab fragment**.

FABP *abbr.* for fatty acid binding protein (e.g. **albumin**, etc.); four motifs are characteristic of this function.

Fabry's disease an X-linked inborn error of human metabolism due to defective lysosomal α -D-galactosidase A activity. It is a **sphingolipidosis** or sphingolipid lysosomal storage disease. The enzyme defect leads to the progressive deposition in most visceral tissues of neutral glycosphingolipids with terminal α -D-galactosyl residues, principally globotriaosylceramide, Gal(α 1-4)Gal(β 1-4)Glc(β 1-1)Cer. [After Johannes Fabry (1860–1930), German physician.]

facilitated diffusion (def. 2) across a biological membrane through the participation of specific transporting agents (transporters) or carriers. The equilibrium distribution reached is the same as that achieved by **simple diffusion** but at any one site facilitated diffusion is mediated by a transport protein that exhibits specificity for the transported species. The existence of the transporter mediates passage across the membrane of molecules to which it would otherwise be impermeable. Sometimes termed **mediated transport** to distinguish it from **active transport**.

FACIT *abbr.* for fibril-associated collagens with interrupted triple helices; a group of collagens (types IX, XII, and XIV).

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composed of triple-helical regions interrupted by non-triple-helical regions of 8–42 amino acids.

FACS *abbr.* for fluorescence-activated cell sorter.

F-actin the form of **actin** found in filaments.

factor 1 any component or cause that contributes to an effect or result; a term often used to denote an uncharacterized (or incompletely characterized) component of a biological system. **2** (*in mathematics*) any integer or polynomial that can be divided exactly into another integer or polynomial.

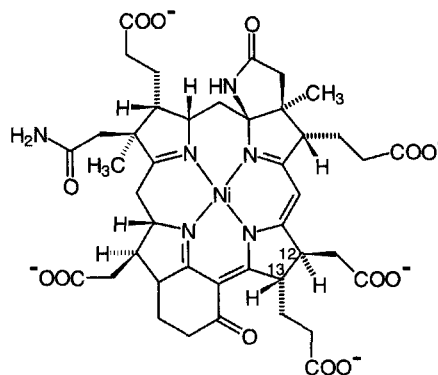
factor I to factor XIII, factor Va, etc. names of the **blood coagulation factors**.

factor B a component of **complement**.

factor D a component of **complement**.

factor F an alternative name for **initiation factor**.

factor F₄₃₀ a nickel-containing tetrapyrrole compound involved in the terminal reductive step of methanogenesis in **methanogenic** bacteria, in which methane is formed from methyl-coenzyme M (see **coenzyme M**). It exists both as the protein-bound and free forms. The free form has an absorption maximum at 430 nm, which shifts to 421 nm in the bound form.



factor G an alternative name for **EF-G**; see also **elongation factor**.

factor H a component of **complement**.

factor I a component of **complement**.

factorial symbol: !; the product of all positive integers from 1 up to and including the integer in question, *n*; hence

$$n! = n(n-1)(n-2)(n-3)\dots 3 \times 2 \times 1.$$

factor IF an alternative name for **initiation factor**.

factor R an alternative name for **release factor**.

factor S a tetrapeptide, of unknown structure, isolated from cerebrospinal fluid of goats, that slowly induces delta (slow wave) sleep in rats on infusion into the cerebral ventricles.

factor T an alternative name for **EF-T**; see also **elongation factor**.

facultative able to live under more than one set of environmental conditions.

facultative aerobe see **aerobe**.

facultative anaerobe see **anaerobe**.

FAD *abbr.* for flavin-adenine dinucleotide.

fade the waning of a response in the continued presence of an agonist. Compare **desensitization**.

FADH₂ *abbr.* for flavin-adenine dinucleotide (reduced).

F agent see **sex factor**.

Fahrenheit temperature scale a temperature scale in which 32 °F is set equal to 273.15 K and 212 °F is set equal to 373.15 K. Thus 1 Fahrenheit degree = $5/9 \times K$, and Fahrenheit tem-



fall curve

perature ($^{\circ}\text{F}$) = $9/5 \times (\text{Celsius temperature}) + 32$. [After G. D. Fahrenheit (1686–1736), German scientist.]

fall curve the curve describing the decrease in the colour intensity of a sample with time, obtained in a colorimetric analysis using an autoanalyser.

falling-drop method a method for the accurate determination of relative densities of liquids in which a drop of the liquid is allowed to fall to its equilibrium position in a column containing a defined density gradient of an immiscible liquid. It was formerly used for determination of the deuterium oxide content of samples of water.

falling-mercury electrode an alternative term for **dropping-mercury electrode**.

falling-sphere viscometer an instrument for measuring the viscosity of a liquid from the time taken for a solid sphere to fall through a given distance in a column of the liquid compared with the time taken for it to fall through the same distance in a liquid of known viscosity.

Fallopian tube an oviduct of a mammal.

familial 1 of or pertaining to a given family. 2 describing something transmitted between members of a family, especially a familial character, condition, or disease.

family 1 (in taxonomy) a taxonomic category of related organisms ranking below the **order** and above the **genus**. 2 (in chemistry and biochemistry) an imprecise term for a group of substances closely related structurally. See also **superfamily**. 3 (in molecular biology) see **Appendix E**.

Fanconi syndrome an inherited human disorder characterized by glycosuria, amino aciduria, phosphaturia, and acidosis. It is due either to functional deficiency of the proximal renal tubules or to abnormalities in amino-acid metabolism. See also **renal rickets**. [After Guido Fanconi (1892–1979), Swiss pediatrician.]

farad symbol: F ; the SI derived unit of electric capacitance. A capacitor has a capacitance of one farad when a charge of one coulomb raises the potential between its plates to one volt: i.e. $1 \text{ F} = 1 \text{ C V}^{-1}$. [After Michael Faraday (1791–1867), British physicist.]

faradaic current an electric current flowing through an electrode that corresponds to the electrolytic oxidation or reduction of one or more chemical species.

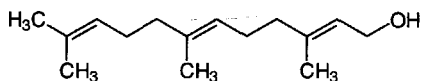
Faraday constant symbol: F ; a fundamental constant, defined as the quantity of electricity required to deposit one mole of a univalent ion from a solution of an electrolyte. It is the arithmetic product of the Avogadro constant, L (or N_A), and the elementary charge, e ; hence

$$F = Le \text{ (or } F = N_A e) = 9.648\,530\,9(29) \times 10^4 \text{ C mol}^{-1}.$$

Faraday effect the rotation of the plane of vibration of polarized light, or of other polarized electromagnetic radiation, when passing through an isotropic, transparent medium in a magnetic field that has a component in the direction of the radiation.

Farber's disease or **Farber's lipogranulomatosis** an alternative name for **ceramidase** deficiency.

farnesol the trivial name for any of four possible stereoisomers of 3,7,11-trimethyl-2,6,10-dodecatrien-1-ol; a sesquiterpene alcohol. The 2*E*,6*E*-isomer is found in many essential oils. The 2*Z*,6*E*-isomer is a minor constituent of some essential oils.



2*E*,6*E*-isomer

farnesyl the (2*E*,6*E*)-3,7,11-trimethyl-2,6,10-dodecatrien-1-yl group; the alkenyl group derived from the sesquiterpene alcohol **farnesol**. 2*E*,6*E*-Farnesyl diphosphate (= *trans,trans*-farnesyl diphosphate).

is an intermediate in carotenoid, sesquiterpene, squalene, and sterol biosynthesis and is also a substrate in the addition of the farnesyl group to proteins (see **prenylation**).

farnesyltransferase EC 2.5.1.21; recommended name: farnesyl-diphosphate farnesyltransferase; systematic name: farnesyl-diphosphate:farnesyl-diphosphate farnesyltransferase; other name: presqualene-diphosphate synthase. An enzyme that catalyses the formation of presqualene diphosphate from two molecules of farnesyl diphosphate with release of pyrophosphate. It occurs in the pathway for the synthesis of **squalene** and derivatives (cholesterol, sesquiterpenes). In eukaryotes it is a monomeric microsomal enzyme. Example from yeast: database code ERG9_YEAST, 444 amino acids (51.61 kDa).

farnesyltranstransferase EC 2.5.1.29; systematic name: *trans,trans*-farnesyl-diphosphate:isopentenyl-diphosphate farnesyltranstransferase; other name: geranylgeranyl-diphosphate synthase. An enzyme that catalyses the formation of geranylgeranyl diphosphate from *trans,trans*-farnesyl diphosphate and isopentenyl diphosphate with the release of pyrophosphate. It occurs in the pathway for sesquiterpene and cholesterol synthesis and forms the **prenyl** derivative geranylgeranyl diphosphate. In some fungi and plants this activity is part of a multifunctional protein that also has the activities of geranylgeranyl-diphosphate synthase (see **dimethylallyltranstransferase**) and **geranyltranstransferase**. Example from *Neurospora crassa*: database code GPPP_NEUCR, 428 amino acids (47.83 kDa).

farnoquinone an alternative name for menaquinone-6; see **menaquinone**.

Farr test a radioimmunoassay technique for measuring the absolute antigen-binding capacity of an antiserum. The antibody is allowed to react with radiolabelled antigen and the antigen-antibody complex is then precipitated by addition of saturated ammonium sulfate solution to a final concentration of 40% w/v. The antigen that has reacted can be determined in the precipitate.

far ultraviolet see **ultraviolet**.

fascicle or **fibre tract** a tight parallel bundle of nerve fibres that are either axons or dendrites. It is formed during the growth of nerves, and is mediated in part by cell **adhesion molecules**.

fasciclin any of the proteins of a related group that are involved in nerve **fascicles** (bundles). They include fasciclin I, a neuronal cell **adhesion molecule**; example (precursor) from *Drosophila melanogaster*: database code FAS1_DROME, 652 amino acids (72.52 kDa); and fasciclin II, which is a neuronal recognition molecule related to NCAM (see **adhesion molecules**): database code FS21_DROME, 873 amino acids (96.82 kDa).

fascin a protein involved in the formation of actin bundles and actin polymerization. Example from *Strongylocentrotus purpuratus*: database code SUSFASCIN, 496 amino acids (54.88 kDa).

fast 1 (of a component of a mixture or specified substance) one that migrates further or furthest from the origin in chromatography or electrophoresis. 2 uninfluenced by a specified agent, e.g. by light, acid, etc. See also **acid-fast bacilli**. 3 to abstain from, or to be prevented from, taking nourishment for at least 24 hours (in the case of laboratory animals). In clinical chemistry, the requirement is often for an overnight fast, normally at least 10 hours. —**fasted** *adj.*; —**fasting** *n*.

FASTA or **FASIN** a computer program used for searching and analysing protein and nucleic acid sequences.

fast component an unusual component of a mixture, e.g. a hemoglobin variant, that moves in a specified buffer system more rapidly in chromatography or electrophoresis than does the normal component.

fast reaction any reaction, or step in a reaction sequence, that has a large rate constant and proceeds rapidly. In a reaction sequence, a fast reaction is not the rate-limiting step.

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fat

fat 1 any triacylglycerol or mixture of triacylglycerols that is solid below 20 °C; those that are liquid at such temperatures are usually referred to as oils. 2 an alternative name for lipid. 3 an alternative name for adipose tissue; see brown adipose tissue, white adipose tissue.

fatal familial insomnia a rare inherited neurological disease, first described in 1986 among inhabitants of mountainous regions of Italy. It is characterized by intractable insomnia and other neurological abnormalities. It is one of three familial forms of human prion disease. Neuropathologically, there is neuronal loss with associated gliosis. It is associated with a pathogenic mutation of the prion gene at codon Asn¹⁷⁸ (shared with Creutzfeldt–Jakob disease, but for unknown reasons of different phenotype) linked to codon Met¹²⁹, a site that exhibits Met/Val polymorphism.

fat body any of the fat-containing cellular structures that serve as energy reserves in amphibians, insects, and reptiles.

fat cell or **fat-cell** or **lipocyte** or (in animals) **adipocyte** any living cell containing noticeable amounts of lipid, primarily as fat or oil.

fat index the mass of diethylether-extractable fat per unit mass of nonfat material (on a dry-weight basis) in, e.g., a foodstuff or tissue.

F₁-ATPase see **H⁺-transporting ATP synthase**.

fat-soluble vitamin any of a diverse group of vitamins that are soluble in organic solvents and relatively insoluble in water. The group includes vitamins A, D, E, and K.

fat solvent any **lipid solvent** e.g. chloroform or diethylether, that will extract triacylglycerols from biological or other materials.

fatty derived from or containing fat.

fatty acid any of the aliphatic monocarboxylic acids that can be liberated by hydrolysis from naturally occurring fats and oils. Fatty acids are predominantly straight-chain acids of 4 to 24 carbon atoms, which may be saturated or unsaturated; branched fatty acids and hydroxy fatty acids also occur, and very long chain acids of over 30 carbons are found in waxes. See also **fatty-acid nomenclature**, **polyunsaturated fatty acid**.

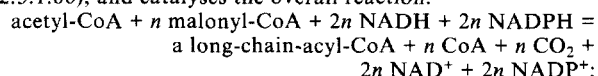
fatty-acid activation the conversion of a fatty acid molecule to its fatty acyl-coenzyme A thioester, the first step in the reactions of **beta oxidation**. Fatty acyl-coenzyme A can be formed in a reaction catalysed by the acid-thiol ligases EC 6.2.1.1, 6.2.1.2, and 6.2.1.3, or by a CoA-transferase of sub-subclass EC 2.8.3.

fatty-acid nomenclature a system of symbols for describing fatty acids. The basic symbolism comprises the number of carbon atoms in the molecule, followed by the number of double bonds; the two numbers are separated by a colon. Thus, 16:0 represents palmitic acid, 16:1 palmitoleic acid, etc. To avoid ambiguity, double bond positions in unsaturated fatty acids should be indicated; 18:3 (9,12,15) represents α -linolenic acid, 18:3 (9,11,13) represents eleostearic acid. These symbols may also include the *E/Z* configurational descriptors (see ***E/Z* convention**). Although not now recommended, a still used system numbers the first double bond from the carboxyl group using the Greek letter Δ . Thus, oleic acid is 18:1 Δ^9 or 18:1 Δ^9 . This system persists in some enzyme names; e.g., the systematic name for linoleate isomerase (EC 5.2.1.5) is linoleate Δ^{12} -*cis*- Δ^{11} -*trans*-isomerase. See also **fatty acid oxidation complex**. Double bonds may also be designated from the end of the chain remote from the carboxyl group. If *n* is the total number of carbons, *n*-3 represents CH₃–CH₂–CH=CH–, etc. Since in many unsaturated fatty acids the double bonds are successively separated by methylene groups, only the position of the first double bond may be given. Thus linoleic acid [18:2 (9,12)] is also represented as 18:2 *n*-6. The Greek letter ω has also been used in the same way; e.g. for linoleic acid, 18:2 ω -6. However, it is not recommended that a double bond position, e.g. that of oleic acid, should be designated as ω 9. See also **fish oil**, **linoleic family**, **linolenic family**, **oleic family**, **palmitoleic family**.

fatty-acid oxidation complex a bacterial multifunction pro-

tein complex that catalyses the **beta oxidation** of fatty acids. The example from *Escherichia coli* comprises α and β subunits. The α subunit contains **decanoyl-CoA Δ -isomerase**, **enoyl-CoA hydratase**, **3-hydroxyacyl-CoA dehydrogenase**, and 3-hydroxybutyryl-CoA epimerase; database code FADB_ECOLI, 729 amino acids (79.50 kDa). The β subunit contains 3-ketoacyl-CoA thiolase; database code THIK_ECOLI, 387 amino acids (40.84 kDa). For a mammalian system see **beta-oxidation system**.

fatty acid synthase complex a large, well-defined multienzyme complex found in eukaryotes. These enzyme complexes, which are cytoplasmic, catalyse a cyclic set of reactions whereby fatty acids are synthesized from one molecule of acetyl coenzyme A and successive molecules of malonyl coenzyme A. In *Escherichia coli* and most bacteria the activities are in separate proteins. In animals, the enzyme complex (EC 2.3.1.85) produces a long-chain fatty acid (typically palmitate), whereas in yeast the product is a long-chain-fatty-acyl-CoA. The synthase complex of yeast has a mass of 2300 kDa and appears in the electron microscope as an ellipsoid, 25 μ m long and 21 μ m in cross section. The yeast enzyme comprises α and β subunits. It is termed fatty-acyl-CoA synthase (EC 2.3.1.86), and catalyses the overall reaction:



To synthesize palmitoyl-S-[acyl-carrier protein] requires seven repetitions of the following set of reactions, where X = S-[acyl carrier protein]:

- (1) $\text{R-CH}_2\text{-CO-X} + ^-\text{OOC-CH}_2\text{-CO-X} \rightarrow \text{R-CH}_2\text{-CO-CH}_2\text{-CO-X} + \text{CO}_2 + \text{H-X}$
- (2) $\text{R-CH}_2\text{-CO-CH}_2\text{-CO-X} + \text{NADPH} \rightarrow \text{R-CH}_2\text{-CH(OH)-CH}_2\text{-CO-X} + \text{NADP}^+$
- (3) $\text{R-CH}_2\text{-CH(OH)-CH}_2\text{-CO-X} \rightarrow \text{R-CH}_2\text{-CH=CH-CO-X} + \text{H}_2\text{O}$
- (4) $\text{R-CH}_2\text{-CH=CH-CO-X} + \text{NADH} \rightarrow \text{R-CH}_2\text{-CH}_2\text{-CH}_2\text{-CO-X} + \text{NAD}^+$

The hydroxacyl product of reaction (2) has the 3*R* configuration and the enoylacyl product of reaction (3) has *E* configuration. Before the assembly begins, acetyl and malonyl units are transferred from their CoA derivatives to acylcarrier protein (ACP) by the activities of two transferases, EC 2.3.1.8 and EC 2.3.1.9. Representative enzyme activities for the four reactions are named as follows: reaction (1), EC 2.3.1.41, 3-oxoacyl-[ACP] synthase; reaction (2), EC 1.1.1.100, 3-oxoacyl-[ACP] reductase; reaction (3), EC 4.2.1.61, 3-hydroxypalmitoyl-[ACP] dehydratase; reaction (4), EC 1.3.1.9, enoyl-[ACP] reductase (NADH).

Similar enzymes with different chain length specificities may also be involved. For the first round, R = H, the two synthase substrates thus being acetyl-[ACP] and malonyl-[ACP]; the product is 3-oxobutanoyl-[ACP]. After seven repetitions of the four reactions, R = C₁₂H₂₅, and the product is palmitoyl-[ACP]. The yeast α subunit catalyses reactions (1) and (2); database code FAS2_YEAST, 1894 amino acids (207.86 kDa). The yeast β subunit catalyses the two acyl transfer reactions and reactions (3) and (4); database code FAS1_YEAST, 2051 amino acids (228.43 kDa).

The mammalian complex is a dimer of identical subunits (260 kDa). Since the product is palmitic acid itself a further enzyme is present: EC 3.1.2.14 (oleoyl-[ACP] hydrolase). Each polypeptide chain is folded into three domains: domain 1 contains the acyltransferase activities and catalyses reaction (1); domain 2 catalyses reactions (2), (3), and (4); domain 3 catalyses the hydrolysis of palmitoyl-[ACP] to palmitate. In liver, reaction (4) is catalysed by EC 1.3.1.39, **enoyl-[acyl-carrier-protein] reductase (NADPH, A-specific)**.

fatty-acid thiokinase an alternative name for 1 butyryl-CoA



fatty-acyl-CoA dehydrogenase

ligase (EC 6.2.1.2). **2** long-chain-fatty-acid-CoA ligase; see **fatty-acyl-CoA ligase**.

fatty-acyl-CoA dehydrogenase any of the acyl-CoA dehydrogenase components of the **beta-oxidation system**.

fatty-acyl-CoA desaturase see **stearoyl-CoA desaturase**.

fatty-acyl-CoA ligase either of two enzymes involved in fatty-acid activation.

(1) Long-chain-fatty-acid-CoA ligase (EC 6.2.1.3); *other names*: acyl-activating enzyme; acyl-CoA synthetase; fatty acid thio kinase (long-chain); lignoceroyl-CoA synthase. It catalyses a reaction between ATP, a long-chain carboxylic acid, and CoA to form an acyl-CoA and pyrophosphate. In mammals there are tissue isoforms. Example from rat liver: database code LCFA_RAT, 699 amino acids (78.09 kDa).

(2) Medium-chain-fatty-acid-CoA ligase (EC 6.2.1.-); *other name*: medium-chain-fatty-acid-CoA synthase. It catalyses the same reaction as above but with medium-chain fatty acids. Example from *Pseudomonas oleovorans*: database code ALKK_PSEOL, 546 amino acids (59.19 kDa).

fatty-acyl-CoA synthase **1** the multifunctional protein responsible for the synthesis of long-chain fatty acyl moieties in some eukaryotes, e.g. yeast. See **fatty acid synthase complex**. **2** an enzyme of the endoplasmic reticulum, EC 6.2.1.3, long-chain-fatty-acid-CoA ligase, that carries out a reaction between ATP, CoA, and a long-chain fatty acid to form the corresponding fatty-acyl CoA, AMP, and pyrophosphate.

fatty-acyl group any acyl group derived from a fatty acid.

fatty alcohol an alternative term for **long-chain alcohol**.

fatty degeneration or **fatty infiltration** the deterioration of a tissue due to the deposition of abnormally large amounts of fat, usually as globules, in its cells.

fatty liver a pathological condition of liver tissue that has undergone **fatty degeneration**. It results from administration of various poisons, especially chlorinated hydrocarbons, or as a result of a dietary deficiency of choline or threonine.

favism a disease of humans in which hemolysis is caused by eating fava beans (*Vicia faba*) or by certain drugs, e.g. sulfonamides, primaquine, and nitrofurantoin antibiotics. The basic defect is an inherited deficiency of glucose-6-phosphate dehydrogenase in the red blood cells; about 70 variants are known. See also **glucose-6-phosphate dehydrogenase deficiency**.

FBP *abbr.* for fructose 1,6-bisphosphate.

FCCP *abbr.* for carbonyl cyanide *p*-trifluoromethoxyphenylhydrazide; an uncoupling agent.

F cell a former name for **PP cell**.

F₁-CF₀ the ϵ subunit of the **H⁺-transporting ATP synthase complex** in the chloroplasts of *Chlamydomonas reinhardtii*, in which the subunit arrangement is different from that in higher plants.

Fc fragment a crystallizable protein fragment obtained (together with **Fab fragment**) by papain hydrolysis of an **immunoglobulin** molecule. That obtained from human IgG is a 50 kDa protein consisting of the C-terminal halves of two heavy chains linked by two disulfide bonds. The Fc fragment has no antigen-binding activity but carries sites for complement fixation. [Fc is from 'fragment-crystallizable'.] Compare **Fc' fragment**, **Fd fragment**.

Fc' fragment a protein fragment obtained in small amounts by papain hydrolysis of an **immunoglobulin** molecule, in addition to **Fab fragments** and **Fc fragment**. That obtained from human IgG is a 24 kDa protein; it forms noncovalently bonded dimers consisting of the C-terminal moieties of two Fc fragments minus their C-terminal tridecapeptide segments. It normally occurs in small quantities in human urine. See also **pFc' fragment**.

Fc receptor any of a number of proteins that bind the Fc region of immunoglobulins A, E or G, and facilitate their recognition and uptake, especially by phagocytic cells (see **phagocyte**). Several types have been described; all of the proteins involved have a single transmembrane helical region. Fcγ receptor I (*abbr.*: FcγRI) or CD64, is a high affinity receptor for IgG, capable of binding monomeric antibody. It is a char-

acteristic marker of mononuclear phagocytes, but may also be expressed on activated neutrophils; example: database code FCγG_HUMAN, 347 amino acids (42.6 kDa). FcγRII or CD32 is a low affinity receptor for IgG present on monocytes, neutrophils, eosinophils, B cells and platelets. Crosslinking of the surface antibody and the receptor leads to downregulation of the B cell. Binding of immune complexes to this receptor leads to degranulation of platelets and release of inflammatory mediators; example: database code FCGA_HUMAN, 317 amino acids (34.99 kDa). Another low affinity IgG receptor, FcγRIII or CD16 is found on macrophages, polymorphs, eosinophils and NK cells. On large granular lymphocytes it is a transmembrane glycoprotein that can link the cells to antibody-sensitized target cells, while on macrophages and neutrophils it has a **GPI anchor**. Two types of IgE receptor are known: (1) a high affinity receptor, FcεRI, found on basophils and mast cells, which on activation leads to the release of histamine and the manifestation of allergy. It is a tetramer of one α chain, one β chain and two disulfide-linked γ chains, examples: α chain, database code FCEA_HUMAN, 257 amino acids (29.6 kDa); β chain, database code FCEB_HUMAN, 244 amino acids (26.53 kDa); γ chain, database code FCEG_HUMAN, 86 amino acids (9.67 kDa). (2) a low affinity receptor, FcεRII, found on lymphocytes and monocytes and involved in IgE production and differentiation of B cells. An IgA receptor is found on myeloid cells; example: database code FCAR_HUMAN, 287 amino acids (32.26 kDa).

FCS *abbr.* for fetal calf serum.

Fd *symbol* for ferredoxin.

Fd₅₄₀ *symbol* for a membrane-bound **ferredoxin** with a redox potential of -0.54 V.

Fd₅₉₀ *symbol* for a membrane-bound **ferredoxin** with a redox potential of -0.59 V.

FDA *abbr.* for Food and Drug Administration (of the USA).

Fd fragment a protein fragment obtained by papain hydrolysis of an **immunoglobulin** molecule followed by reduction of the disulfide bonds. It consists of the part of the **heavy chain** N-terminal to the papain hydrolysis site. The N-terminal moiety is variable and forms part of the original antigen-combining site; the C-terminal moiety is constant. Compare **Fc fragment**.

FDNB *abbr.* for fluorodinitrobenzene; **Sanger's reagent**.

FDP *abbr.* for fructose 1,6-diphosphate (now named **fructose 1,6-bisphosphate**).

FdU *abbr.* for fluorodeoxyuridine (the symbol FdUrd is recommended).

FdUrd *symbol* for (a residue of) the deoxyribonucleotide fluorodeoxyuridine (see **flurouracil**).

Fe *symbol* for iron.

Fe(II) *symbol* for ferrous (divalent) iron.

Fe(III) *symbol* for ferric (trivalent) iron.

FEBS *abbr.* for Federation of European Biochemical Societies.

fecal fat fat in the feces; the output of fecal fat is normally below about 18 mmol 24 h⁻¹ in the adult human. Tests that measure the fat content of feces may assist in diagnosis of generalized malabsorption syndrome; an alternative method is the [¹⁴C]triolein breath-test.

fed batch process a process during which one or more nutrients are supplied to a bioreactor, usually without removing biomass and products. See **batch process**.

Federation of European Biochemical Societies *abbr.*: FEBS; an international scientific organization founded in 1964 and now representing some 32 countries, including all those of mainland Europe plus Israel. FEBS is governed by a Council composed of one delegate from each of the adhering national biochemical societies plus the members of the Executive Committee. FEBS organizes an annual meeting, advanced courses, and supports fellowships and three annual prizes. It publishes two journals, *FEBS Letters* and *European Journal of Biochemistry*.

feedback **1** any feature of a system, e.g. of a metabolic pathway or of an electronic circuit, whereby information about the



feedback control

output of the system is used to influence the input to that system. Such feedback may increase the input to the system – **positive feedback** – or reduce the input to the system – **negative feedback**. A cyclic system of components participating in feedback forms a **feedback loop**, and the system is known as a **closed-loop system**. *Compare* **open-loop system**. **2** that part of the output of an electronic, mechanical, or other system that is returned to the input, so modifying its characteristics. *Compare* **feedforward**. — **feed back** *vb*.

feedback control control of the activity of any system by a mechanism involving **feedback**.

feedback inhibition or **retroinhibition** inhibition of the activity of an enzyme by the accumulation of a substance produced further along a metabolic pathway of which the enzyme is a constituent. *Compare* **enzyme repression**, **feedback repression**.

feedback repression an inhibitory feedback system in which the accumulation of a substance produced in a metabolic pathway represses the synthesis of an enzyme that is required at an earlier stage of the pathway. *Compare* **enzyme repression**, **feedback inhibition**.

feeder cell an irradiated cell that is capable of metabolizing but not of dividing. Such cells are sometimes added to cultures of unirradiated cells to help in the provision of nutrients.

feeder layer a layer of **feeder cells** in a cell culture.

feeder pathway any (minor) metabolic pathway that supplies metabolites to another (major) metabolic pathway.

feedforward **1** any feature of a system, e.g. a metabolic pathway, whereby information about the input to the system is used to influence the output of the system. *Compare* **feedback** (def. 1). **2** that part of the input of a system that influences the output of the system.

Fehling's solution a test reagent for glucose and other reducing substances. It is prepared as two solutions: **Fehling's solution A** containing 0.278 M copper (II) sulfate, and **Fehling's solution B** containing approximately 1.65 M potassium sodium tartrate and 4.3 M potassium hydroxide; equal volumes of each solution are mixed before a test. When boiled with a reducing substance a yellow or red precipitate of copper (I) oxide is formed. *Compare* **Benedict's solution**. [After Hermann Christian von Fehling (1812–85), German organic chemist.]

feline of, pertaining to, or resembling cats.

felix a protein designed for theoretical modelling studies and expressed from a synthetic gene in *Escherichia coli*. Database code NRL_1FLX, 79 amino acids (8.77 kDa), which also gives the predicted 3-D structure. [From four-helix bundle.]

female *symbol*: ♀; **1** of, pertaining to, or designating the sex that only produces gametes that can be fertilized by male gametes. **2** describing a flower that lacks stamens or has non-functional stamens.

female sex hormones any of the **estrogens** (estradiol-17β, estrone, and estriol) that are concerned with the normal growth and development of the mammalian female reproductive tract and, together with **progesterone**, control the estrous and reproductive cycles.

FeMo-co *abbr.* for iron–molybdenum cofactor (of component I of nitrogenase).

FeMo protein any iron–molybdenum-containing protein.

FEMS *abbr.* for Federation of European Microbiological Societies.

femto+ *symbol*: f; SI prefix denoting 10⁻¹⁵ times.

femtomolar (*informal*) describing a solution containing 10⁻¹⁵ mol dm⁻³ (of a specified solute).

femtomole *symbol*: fmol; one 10⁻¹⁵th part of a mole (of a specified substance).

F episome an alternative name for **F plasmid**.

Fe protein any iron-containing protein.

feral wild; having escaped from cultivation or domestication and reverted to the wild state.

ferment **1** any agent or substance that causes fermentation.

2 a former name for **enzyme**. **3** an alternative name for **fermentation**.

4 to undergo or to effect **fermentation**. — **fermentable** *adj.*; **fermentability** *n*.

fermentation **1** (or **ferment**) the decomposition of chemical substances, brought about by **ferments** (def. 1), resulting in the production of simpler substances and, often, of energy; an instance of this, especially the anaerobic breakdown of glucose to lactate or ethanol. **2** (*in biotechnology*) the use of microorganisms or cultured cells to produce useful materials, such as antibiotics, beverages, enzymes, and some commodity chemicals.

fermenter **1** or **fermentor** or **biofermenter** an apparatus or a vessel in which to conduct **fermentation** under more or less controlled conditions. **2** a person or other agent that causes or is able to cause fermentation. **3** a genus, species, or strain of microorganism that ferments (or does not ferment) glucose or a specified sugar; e.g. **lactose fermenter**, **non-lactose fermenter**.

fermentor a variant spelling of **fermenter** (def. 1).

Fernández-Morán particle another name for **elementary particle** (def. 2), attached to the cristae of the mitochondrion.

ferralaterin a soluble nonheme, iron–sulfur-containing chloroplast protein, mediating the light activation of chloroplast fructose 1,6-bisphosphatase.

ferredoxin *symbol*: Fd; any simple, nonenzymic iron–sulfur protein that is characterized by having equal numbers of atoms of iron and labile sulfur (releasable as hydrogen sulfide by acidification). The iron and sulfur atoms are present in one or two clusters of two or four atoms of each. Ferredoxins are of *M*_r 6000–24 000; many have a particularly low redox potential (*E*_{o'} = –0.2 V to –0.6 V). They act as electron carriers in a variety of oxidation–reduction systems and have been found in a wide range of microorganisms, in chloroplasts, and in mitochondria from adrenal cortex (*see* **adrenodoxin**) and heart muscle. Example: 2Fe-2S type from *Spirulina platensis*: database code NRL_3FXC, 98 amino acids (10.49 kDa); three conserved motifs; 3-D structure known. *Compare* **rubredoxin**. *See also* **high-potential iron–sulfur protein**, **nitrogenase**.

ferredoxin-NADP⁺ reductase another name for **NADPH:adrenodoxin oxidoreductase**.

ferredoxin–nitrite reductase *see* **nitrite reductase**.

ferri+ *comb. form* denoting containing ferric iron.

ferric of or containing trivalent iron, Fe³⁺ or Fe(III).

ferrichrome any of a group of growth-promoting Fe(III) chelates formed by various genera of microfungi, e.g. species of *Aspergillus*, *Neurospora*, and *Ustilago*. They are homodetic cyclic hexapeptides made up of a tripeptide of glycine (or other small neutral amino acids) and a tripeptide of an *N*⁴-acyl-*N*⁴-hydroxy-L-ornithine (e.g. the *N*⁴-acetyl compound). They are thus **siderochromes** of the hydroxamic-acid-derivative category.

ferricytochrome any **cytochrome** in which the iron atom is in the ferric form.

ferriheme or (*esp. UK*) **ferrihaem** any Fe(III)–porphyrin coordination complex.

ferriheme chloride another name for **hemin**.

ferriheme hydroxide another name for **hematin**.

ferrihemochrome any **hemochrome** containing Fe(III).

ferrihemoglobin or **methemoglobin** (*abbr.*: MetHb) an oxidation product of hemoglobin in which all its four iron atoms are in the ferric, Fe(III), state, the sixth coordination valencies of which are occupied, in the acid form, by water molecules, or, in the alkaline form, by hydroxide ions; these two forms have characteristic absorption spectra. Unlike hemoglobin, it cannot bind and transport dioxygen. The 3-D structure is known (Brookhaven file 1HGB.PDB). *Compare* **ferrimyoglobin**.

ferrimyoglobin or **metmyoglobin** (*abbr.*: MetMb) an oxidation product of myoglobin in which its iron atom is in the ferric, Fe(III), state, the sixth coordination valency of which is occupied, in the acid form, by a water molecule, or, in the alkaline form, by a hydroxide ion. Unlike myoglobin, it cannot bind and store dioxygen. The 3D structure is known (Brookhaven file 1MBC.PDB). *Compare* **ferrihemoglobin**.

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ferrioxamine

ferrioxamine *see* **desferrioxamine**.

ferritin any of a group of electron-dense, major iron-storage proteins that are widely distributed in animal and plant tissues; the iron is stored in a soluble, readily available form. Ferritins consist of a hollow protein shell; the functional molecule, called **apoferritin**, comprises 24 chains and is roughly spherical, with an external diameter of 12.4–13 nm and an internal diameter of 7–8 nm. The central cavity is able to accommodate a core of up to 4500 Fe(III) atoms, mainly as ferric hydroxide-phosphate, $(\text{FeO}(\text{OH}))_8(\text{FeO}_4\text{PO}_4\text{H}_2)_2$. There are two types of ferritin subunits: L (light) and H (heavy); the major chain can be L or H, depending on species and tissue type. In clinical chemistry, low plasma ferritin is an indication of a decrease in the body's iron stores; a concentration of less than $12 \mu\text{g L}^{-1}$ indicates a complete absence of stored iron. Examples from human: H chain, database code FRIH_HUMAN, 182 amino acids (21.07 kDa) (3D structure known: database code NRL_1FHA); L chain, database code FRIL_HUMAN, 174 amino acids (19.87 kDa).

ferritin-labelling the use of **ferritin** as an electron-dense label by covalently linking ferritin molecules with the aim of making their position in microscopic specimens apparent by electron microscopy.

ferro+ *comb. form* denoting containing **ferrous** iron.

ferrochelatase EC 4.99.1.1; *systematic name*: protoheme ferro-lyase; *other names*: protoheme ferro-lyase; heme synthetase; an enzyme of the pathway for heme biosynthesis. It catalyses the formation of protoheme from protoporphyrin and Fe^{2+} with release of 2H^+ . Hereditary ferrochelatase deficiency occurs in humans and leads to **porphyria** with accumulation of protopyroxin IX. In bacteria the enzyme is part of **siroheme synthase**. Example (precursor) from human: database code HEMZ_HUMAN.

ferrocytochrome any **cytochrome** in which the iron atom is in the Fe(II) state.

ferroheme or (*esp. Brit.*) **ferrohaem** **1** heme. **2** any Fe(II)–porphyrin coordination complex.

ferrohemochrome any **hemochrome** containing Fe(II).

ferromagnetism the property displayed by certain substances, typically metallic iron, that have a high magnetic susceptibility and that show increasing magnetization with increasing applied magnetic field strength. Some ferromagnetic materials retain their magnetization in the absence of an applied magnetic field, i.e. they become permanent magnets. Ferromagnetism is due to alignment, temporary or permanent, of the spins of the unpaired electrons in domains in the material. —**ferromagnetic adj.** Compare **diamagnetism**, **paramagnetism**.

ferroprotoporphyrin *see* **heme**.

ferrous of or containing divalent iron, Fe^{2+} or Fe(II).

ferroxamine *see* **desferrioxamine**.

ferroxidase *see* **ceruloplasmin**.

fertile **1** capable of producing offspring; capable of producing gametes. **2** capable of undergoing growth and development, e.g. fertile seeds or fertile eggs.

fertility the state of being **fertile**.

fertility factor an alternative name for **F plasmid**.

fertility vitamin a former name for **vitamin E**.

fertilization or **fertilisation** **1** the fusion of two gametes of opposite sex to form a zygote. **2** the act or process of fertilizing.

fertilize or **fertilise** **1** to provide a female gamete with a male gamete to effect fertilization. **2** to supply soil or water with mineral and/or organic nutrients to assist the growth of plants.

fertilizer or **fertiliser** **1** any substance or mixture of substances that is added to soil or water to assist the growth of plants. **2** any object or organism, e.g. an insect, that fertilizes an animal or plant.

fertilizin a mucopolysaccharide from the jelly-coat of the eggs of some species that plays a part in sperm recognition and attracts sperms of the same species.

ferulic acid 4-hydroxy-3-methoxycinnamic acid; 3-(4-hydroxy-3-methoxyphenyl)-2-propenoic acid; an aromatic acid widely distributed in plants. It is a component of **suberin**.

FES/FPS a family of genes encoding nonreceptor tyrosine kinases; *v-fps* is an oncogene of Fujinami-PRCII sarcoma, while *v-fes* is a cognate gene of feline sarcoma virus. FES/FPS proteins contain an SH2 domain (*see* **SH domain**) adjacent to the kinase domain. Overexpression of *c-fes* transforms fibroblasts. Example (human): database code KFES_HUMAN, 822 amino acids (93.47 kDa).

Fe-S protein *abbr. for* iron–sulfur protein.

fet+ a variant form of **feto+** (before a vowel).

fetal calf serum *abbr.*: FCS; a serum prepared from the blood of a fetal calf, widely used in tissue culture media.

fetal hemoglobin an alternative name for hemoglobin F (*see* **hemoglobin**).

fetal-lung maturity test a test for maturity of the fetal lung based on its content of dipalmitoylphosphatidylcholine (dipalmitoyllecithin). Dipalmitoyllecithin appears in lung surfactant in embryonic life only when the lung reaches maturity. Its concentration in amniotic fluid reflects its production in the lung and can be measured to determine lung maturity as a prenatal test to aid the decision when to induce labour. Formerly, the lecithin/sphingomyelin ratio was measured.

feto+ or (before a vowel) **fet+** *comb. form* denoting fetus.

fetomodulin *see* **β -thromboglobulin**.

α -fetoprotein *abbr.*: **afp** or **AFP**; a protein formed in fetal liver and found in amniotic fluid. In humans, increased levels in the amniotic fluid between the 14th and 17th weeks of gestation reliably predict most cases of anencephaly and spina bifida cystica, and **afp** determination is performed as an antenatal screening test for such defects. Increased serum levels may be found in adults with hepatoma and teratoma; **afp** is a valuable marker for liver tumours, blood levels being elevated in about 70% of cases. The normal level in adult human plasma is $<10 \text{ kU L}^{-1}$, but rises in normal pregnancy. It is similar to **albumin** (serum) and **vitamin D binding protein**. Example (precursor) from human: database code FETA_HUMAN, 609 amino acids (68.60 kDa).

fetuin *see* **α_2 -HS-glycoprotein**.

fetus or **foetus** the embryo of a mammal in the later stages of development, from the time when the main characteristics of the mature animal can be recognized. In humans the products of conception are termed the fetus from the end of the eighth week of pregnancy until birth. —**fetal** or **foetal adj.**

Feulgen reaction a specific cytochemical reaction to indicate the presence of DNA. It is based on converting DNA, but not RNA, to apurinic acid by acid hydrolysis with release of the aldehyde functions; the aldehyde functions then give a magenta colour on treatment of the specimen with **Schiff reagent**. Tissue giving such a reaction is termed **Feulgen positive**, whereas the absence of such a reaction characterizes **Feulgen-negative** tissues. The reaction is not suitable for living cells. [After Robert Joachim Feulgen (1884–1955), German chemist.]

FFA *abbr. for* free fatty acid(s). Compare **NEFA**.

F factor an alternative name for **F plasmid**.

F' factor an alternative name for **F' plasmid**.

F₀F₁ complex *see* **H⁺-transporting ATP synthase**.

(F₁ + F₂) fragment *see* **heavy meromyosin**.

F₁ fragment the separated globular head portion of a **myosin** molecule.

F₂ fragment the separated fibrous part of **heavy meromyosin**.

F₃ fragment *see* **light meromyosin**.

FGAM synthase *see* **phosphoribosylformylglycinamide synthase**.

FGAR *abbr. for* formylglycinamide ribonucleotide; an intermediate in **purine biosynthesis**.

FGF *abbr. for* fibroblast growth factor.

FGR a gene encoding a nonreceptor tyrosine kinase and belonging to the **src** family; *v-fgr* is the oncogene of the feline sarcoma virus. Gardner-Rasheed. **FGR** transforms most mammalian



fibroblasts, but not epithelial cells. Example of expressed protein from human: database code KFGR_HUMAN.

FH₂ *abbr. (not recommended) for dihydrofolate or dihydrofolic acid.*

FH₄ *abbr. (not recommended) for tetrahydrofolate or tetrahydrofolic acid.*

FIA *abbr. for fluorescence immunoassay.*

fiber *the US spelling of fibre.*

fibre or (US) **fiber** any natural filamentous structure, typically pliable and often strong; by extension the term is applied to filamentous glass or filamentous synthetic polymeric materials. —*fibrous adj.*

fibre diagram the X-ray diffraction pattern obtained from a stretched, hence orientated, sample of an amorphous, solid, naturally occurring or synthetic fibrous linear polymer when analysed by the rotating crystal method.

fibre optics **1** a flexible glass or plastic fibre, or a bundle of such fibres, used as a light guide for the transmission of light, especially as images or digitized information. **2** the study of the optical properties of transparent fibres.

fibriform having the form of a fibre or fibres.

fibril any small thread or fibre, bundles of which may constitute a fibre. —*fibrillar, fibrilliform adj.*

fibrillar collagen *see collagen.*

fibrillarin a component of a nucleolar small nuclear ribonucleoprotein particle (snRNP) thought to participate in the first step in processing pre-rRNA and associated with U3, U8, and U13 snRNP RNAs. It is the nucleolar scleroderma antigen, rich in N^G,N^G-dimethylarginine and common to the major family of nucleolar snRNPs. Example from human: database code FBRL_HUMAN, 321 amino acids (33.78 kDa).

fibrillin a large glycoprotein isolated from fibroblast cell cultures. It is a structural Ca²⁺-binding component of connective tissue microfibrils, and contains 34 six-cysteine (EGF-like) repeats and five eight-cysteine (TGF-β1 binding protein-like) repeats. Defects are associated with the autosomal dominant disorder, Marfan's syndrome. Example (fibrillin 1 precursor) from human: 2871 amino acids (311.96 kDa).

β-fibrillosis *see amyloidosis.*

fibrin the product(s) formed from **fibrinogen** when blood plasma clots. The initial stage of this process is the formation of a **fibrin monomer** (≈340 kDa) through the action of **thrombin**, which cleaves negatively charged **fibrinopeptides** A and B from fibrinogen. In intact fibrinogen, fibrinopeptides A and B prevent self-association of fibrinogen. Their removal enables the resultant molecule to self-associate readily in a staggered side-by-side arrangement, forming the fibrous **soluble fibrin polymer**. This in turn is converted into **insoluble fibrin polymer** (*see desmofibrin*) through the action of plasma transglutaminase (factor XIIIa, EC 2.3.2.13) which, in the presence of calcium ions, creates covalent cross-links between the monomers by transamidating glutamine residues with the amino groups of the side chains of lysine residues in other monomers. *See also blood coagulation.*

fibrinase *an alternative name for plasmin.*

fibrin monomer *see fibrin.*

fibrinogen or **factor I** a highly soluble, 340 kDa, elongated protein of blood plasma concerned in clot formation. It is converted into **fibrin** monomer by the action of **thrombin**. Fibrinogen is a hexamer, 46 nm long and 9 nm maximal diameter, containing two sets of three nonidentical chains (α, β, and γ), linked to each other by disulfide bonds. The N termini of all chains are contained in a central nodule, diverging from which are two three-chain coiled coils; these connect the central nodule to the distal nodules containing the distal domains; there are long C termini of the α chains. Thrombin cleaves four Arg–Gly bonds, releasing two each of **fibrinopeptides** A and B from the α and β chains, and forming fibrin monomer (the intact chains in fibrinogen being referred to as the Aα and Bβ chains). This exposes the N-terminal polymerization sites responsible for the formation of the soft clot, which is converted

into the hard clot by **factor XIIIa**. This catalyses the ε-(γ-glutamyl)lysine cross-linking between γ chains (stronger) and between α chains (weaker) of different monomers. Fibrinogen also acts as a cofactor in platelet aggregation (a binding site on the γ chain participates). Example sequences (precursors, human): α, database code FIBA_HUMAN, 644 amino acids (69.68 kDa); β, database code FIBB_HUMAN, 491 amino acids (55.87 kDa); γ-A, database code FIBG_HUMAN, 437 amino acids (49.43 kDa); an alternative (γ-B or γ') is produced by differential splicing; database code FIBH_HUMAN, 453 amino acids (51.44 kDa). *See also blood coagulation.*

fibrinogenase *an alternative name for thrombin.*

fibrinolysin *an alternative name for plasmin.*

fibrinolysis the process of the dissolution of fibrin in blood clots resulting from the proteolytic action of **plasmin** or other enzymes, e.g. **streptokinase** or the proteases in snake venom. — *fibrinolytic adj.*

fibrinopeptide either of the types of negatively charged peptide cleaved off **fibrinogen** when it is converted to **fibrin** by the action of **thrombin**. Human **fibrinopeptide A** is a 16-residue peptide cleaved from each of the fibrinogen Aα chains; structure (human): ADSGEGDFLAEGGGVR; **fibrinopeptide B** is a 14-residue peptide cleaved from each of the Bβ chains.

fibrin polymer *see fibrin.*

fibrin stabilizing factor or **fibrin stabilising factor** *abbr.:* FSF; *an alternative name for factor XIIIa; see blood coagulation.*

fibroblast or **fibrocyte** a flattened, irregular, branched, motile cell found distributed throughout vertebrate connective tissue. Such cells form, secrete, and maintain the extracellular collagen and mucopolysaccharide of this tissue. Similar cells occur in many invertebrates.

fibroblast growth factor *abbr.:* FGF; any of various growth factors with strong mitogenic potential, characterized by their high affinity for heparin; they are mitogenic for mesodermal and neuroectodermal cells but not for endodermally derived cells. Two major subtypes are recognized, based on their ease of elution from heparin columns: an acidic group (aFGFs), and a less-easily eluted basic group (bFGFs). FGFs have angiogenic properties, may function as inductive agents during embryonic development, and are useful as a partial or total substitute for serum in tissue culture. Those of the acidic group have been isolated mainly from brain or retina; they include acidic brain fibroblast growth factor (brain aFGF), astroglial growth factor-1 (AGF-1), retina-derived growth factor (RDGF), eye-derived growth factor-2 (EDGF-2), endothelial cell growth factor (ECGF), and heparin-binding growth factor (HGFα and HGFβ). Those of the basic group have been isolated from brain, retina, and cartilage; they include basic brain fibroblast growth factor (brain bFGF), eye-derived growth factor-1 (EDGF-1), astroglial growth factor-2 (AGF-2), and cartilage-derived growth factor (CDGF). Example, basic fibroblast growth factor (human): database code NRL_2FGF, 126 amino acids (14.41 kDa); 3-D structure known (beta barrel). FGF-3, so named because aFGF and bFGF had already been identified, a monomer of 239 amino acids, is expressed during embryonic development and by mammary tumour epithelial cells; it is the product of the *int-2* oncogene, and has properties similar to those of basic FGFs. FGF-4 is the product of the *hst/KS3* gene, comprises 205 amino acids, is expressed during development and by some tumours, is mitogenic for fibroblasts and endothelial cells, and is angiogenic *in vivo*. FGF-5 (identified by DNA probes in human bladder carcinoma, hepatoma, and endometrial carcinoma) is a monomer of 267 amino acids; it is expressed by fibroblasts, epithelium, some tumour cells, and during embryonic development, and is mitogenic for fibroblasts and endothelial cells. FGF-6 is the product of *hst-2* (identified by a screen for *hst*-related genes) and is a monomer of 208 amino acids; it is expressed during embryonic development and in adult testis, heart, and skeletal muscle, and is mitogenic for fibroblasts but only weakly for endothelial cells. FGF-7 (ker-

fibroblast growth factor receptor

atinocyte growth factor) is a monomer of 194 amino acids; it is expressed by epithelial tissue and is mitogenic for epithelial cells but not for fibroblasts or endothelial cells. *See also* **KS/hst**.

fibroblast growth factor receptor *abbr.*: FGFR; any of the plasma membrane **tyrosine kinase** proteins that bind **fibroblast growth factors** and mediate their intracellular effects. All have a single transmembrane helical domain of 21 amino acids, an extracellular domain of about 350 amino acids with three immunoglobulin domains (but variants with fewer are known), and a cytoplasmic domain of about 410–425 amino acids within which is a tyrosine kinase domain which is activated by ligand binding. Six to nine potential N-linked glycosylation sites are present in the extracellular domain. FGFRs are classified into five types. FGFR1 was isolated as a partial cDNA clone encoding an *fms*-like gene (*flg*) and is also known as *flg*. Common variants have only two immunoglobulin (Ig) domains. Example from *Rattus norvegicus* (precursor): database code FGR1_RAT, 822 amino acids (91.82 kDa). FGFR2 was isolated as a tyrosine kinase gene encoding a bacterial expressed kinase (*bek*). Common variants of this also have deletion of the first Ig domain. Example from *Xenopus laevis* (precursor): database code FGR2_XENLA, 813 amino acids (91.34 kDa). FGFR3 is the product of *cek2* (*see cek*); example, human (precursor): FGR3_HUMAN, 806 amino acids (87.71 kDa). FGFR4 was isolated as a clone from human erythroleukemia cells; example, human (precursor): database code FGR4_HUMAN, 802 amino acids (87.94 kDa). A fifth type of FGFR is known as *flg-2*. FGFR1 and 2 are expressed by a variety of species, tissues, and cell lines, but FGFR3, FGFR4 and *flg-2* have been isolated from a limited number of sources. Heparan sulfate proteoglycans promote binding of FGFs to FGFRs. Binding of ligand to the extracellular domain of FGFRs induces oligomerization of receptors, autophosphorylation, and phosphorylation of intracellular proteins that associate with the activated receptor through their SH2 domains (*see SH domain*). Prominent among these is phospholipase Cγ1 (*see phospholipase*).

fibroblast surface antigen a variant form of surface fibroblast antigen; *see fibronectin*.

fibrocyte an alternative name for fibroblast.

fibroglycan *see syndecan-2*.

fibroin or silk **fibroin** the protein of the silk fibres that are secreted by the posterior gland of the larva of the silk moth, *Bombyx mori*. It comprises two chains, heavy and light, linked by disulfide bond(s). Both heavy and light chains assembled as a mature protein are needed for efficient intracellular transport and secretion of fibroin; the secondary structure is largely beta sheet. Example (light chain) from *B. mori*: database code FBOL_BOMMO, 262 amino acids (27.64 kDa).

fibronectin any of a group of related adhesive glycoproteins of high molecular mass found on the surface of animal cells, in connective tissue matrices, and in extracellular fluids. They bind to cell surfaces, collagen, fibrin, heparin, DNA, actin, etc. and are involved in cell adhesion, cell motility, opsonization, wound healing, and maintenance of cell shape. Fibronectin is a disulfide-bonded heterodimer. Fibronectins can be classified into two immunologically cross-reacting groups: **plasma fibronectins** (*also called* cold-insoluble globulin or cell attachment protein or cell adhesion factor or cell spreading factor or anti-gelatin factor or opsonic α₂-SB-glycoprotein); and **cellular fibronectins** (*also called* cell surface protein or large external transformation sensitive (LETS) protein or surface fibroblast antigen or galactoprotein A or zeta protein). Cellular fibronectin (which is the major cell surface protein of cultured fibroblasts) is greatly reduced or even lost when a variety of cell types are transformed by oncogenic viruses, chemical carcinogens, or temperature-sensitive viruses at the permissive temperature and several of the morphological changes associated with transformation can be partially reversed by addition of appropriate amounts of purified cellular

fibronectin. There are three types of fibronectin domain (I, II, III) with several versions in the same protein; extra domains and a connecting strand 3 are present in some forms of fibronectin and absent in others; these differences are due to alternative splicing. Human (precursor): database code FINC_HUMAN, 2386 amino acids (262.39 kDa). Fibronectin binds to **integrins** α₄β₁, α_vβ₁, α_vβ₃, α_vβ₅, α_vβ₆, and α₄β₇. [From Latin *fibra*, fibre and *necto*, to attach.]

fibrosome an artificial macromolecular complex of fibronectin layered on an agar-coated substrate.

fibrous protein any protein able to aggregate into fibres or filaments and serving mainly as a structural protein. The polypeptide chains in such proteins are either extended or coiled in one dimension and their structure is maintained largely by interchain hydrogen bonds.

fibulin a Ca²⁺-binding glycoprotein of the extracellular matrix with nine EGF-like repeats. Four forms (A–D) differ only in their C-terminal region and arise by alternative splicing. Example, isoform A precursor from human: database code FBLA_HUMAN, 566 amino acids (61.53 kDa).

ficain or **ficin** a cysteine endopeptidase, EC 3.4.22.3, from *Ficus* spp. with preferential cleavage at Lys-, Ala-, Tyr-, Gly-, Asn-, Leu-, and Val-. The name is also used for the crude dried latex from *Ficus* spp., but extracts of only a small proportion of these (in one study only 13 out of 43 species) contain proteolytic activity. Its action is similar to that of **papain**.

ficin another name for **ficain**.

Fick's first diffusion law a mathematical description of the physical process of solute diffusion; the flux (mass transferred) of particles, *J*, diffusing across a unit plane in unit time is given by:

$$J = -D(\partial c / \partial x),$$

where the mass concentration gradient at *x* is $\partial c / \partial x$ and *D* is the **diffusion coefficient** for the solute. [After Adolph Eugen Fick (1829–1901), German physiologist.]

Fick's second diffusion law a mathematical description of the rate of change in mass concentration, *c*, at point *x* of the diffusing particles with time, ($\partial c / \partial t$), during solute diffusion, for unsteady conditions ($\partial c / \partial x \neq 0$); it is given by:

$$(\partial c / \partial t) = D(\partial^2 c / \partial x^2),$$

where *D* is the **diffusion coefficient**, assumed to be independent of the concentration.

Fick unit symbol: F; a non-SI unit sometimes used to express a **diffusion coefficient**; it is equal to 10^{−7} cm² s^{−1}.

ficolin a multimeric protein so named because of its fibrinogen- and collagen-like domains; it was originally isolated as a TGF-β1 binding protein. Recombinant ficolin-α and ficolin-β expressed from cDNA do not bind TGF-β1. Ficolin-α is expressed preferentially in uterus, lung, and placenta whereas ficolin-β RNA is most abundant in skeletal muscle. The function is at present unknown. Example, ficolin-α from pig: database code A47172, 323 amino acids (34.64 kDa).

Ficoll or **Ficoll 400** a synthetic copolymer of sucrose and epichlorohydrin. It has a branched structure, a high content of hydroxyl groups, which renders it very water-soluble (up to 50% w/v), and a mass average *M_r* of 400 000 ± 100 000. Solutions of Ficoll have low osmotic pressures and are useful in preparing density gradients; Ficoll is widely used in preparing continuous or discontinuous gradients for separating cell types. Ficoll 70, with an *M_r* of 70 000, is useful in perfusion studies.

Ficoll-Paque the proprietary name for an aqueous solution of **Ficoll** and diatrizoate sodium with a density of 1.077 ± 0.001 g cm^{−3}. It is useful for the one-step separation of lymphocytes from a sample of anticoagulant-treated blood and for the preparation of other cell types by density-gradient centrifugation.

fidelity the degree to which the output of any system accurately describes or reflects the input to the system. The term is used



field

especially of the processes of replication, translation, and transcription, and of electronic amplifiers.

field 1 a region of space in which a force is exerted on an object because of its charge (**electric field**), magnetic dipole (**magnetic field**), mass (**gravitational field**), or other attribute. **2** a region of space through which (ionizing) radiation is passing. **3** or **field of view** the area within which an object is observable with a microscope or other optical instrument.

Fieser's solution a solution for removing dioxygen from any other nonacidic, nonoxidizing gas. It contains potassium hydroxide, anthraquinone β -sulfonate, and sodium thiosulfate. Red when fresh, it becomes colourless when exhausted. [After Louis Frederick Fieser (1899–1977), US chemist.]

FIGLU *abbr. for* formiminoglutamate.

figure of merit the square of the efficiency divided by the background count for a given liquid scintillation system (*see liquid-scintillation counting*). It is useful for optimizing the scintillator composition and the discriminator settings for the counting channel(s).

filaggrin a protein that aggregates keratin intermediate filaments and promotes disulfide-bond formation amongst them during terminal differentiation of mammalian epidermis. It is initially synthesized as a large, insoluble, highly phosphorylated precursor containing many tandem copies of 324 amino acids, which are not separated by 'large linker' amino-acid sequences. The precursor is deposited as keratohyalin granules; during terminal differentiation it is dephosphorylated and proteolytically cleaved. Example from human (partial sequence): database code FILA_HUMAN, 416 amino acids (44.06 kDa).

filament any very thin, threadlike structure. *See also* **actin**, **microfilament**, **thick filament**, **thin filament**. —**filamentary** or **filamentous** *adj.*

filamin a protein, extractable from smooth muscle and other tissues, that induces a calcium-ion-insensitive gelation of actin. It promotes orthogonal branching of actin filaments and links actin filaments to membrane glycoproteins. It exists as a homodimer and contains an **actin-binding domain**. Example, nonmuscle filamin (human): database code ABP2_HUMAN, 2647 amino acids (280.45 kDa).

filensin a membrane- and cytoskeleton-associated protein. It is an intermediate filament type III unique to lens fibre cells. It has primary and secondary structural similarity to other intermediate filament proteins and serves as an anchorage site for **vimentin** intermediate filaments. Example (bovine): database code S32103, 755 amino acids (82.86 kDa).

filial generation *symbol:* F; any of the generations of living organisms following a particular parental generation. The generation resulting from a mating of the parents is the **first filial generation** *symbol:* (F₁), while that resulting from crossing of F₁ individuals is the **second filial generation** *symbol:* (F₂), and so on.

filipin a neutral polyene antibiotic from *Streptomyces filipinensis*. It is similar in action to **amphotericin B**.

film badge a masked photographic film worn by workers exposed to ionizing radiation to indicate the extent of their exposure to radiation.

filter 1 any piece or layer of material used or useful for (completely or partially) removing selected components, especially in suspension, from liquid or gaseous mixtures, e.g. a sheet of paper, a stratum of sand (or similar material), or a piece of porous ceramic or plastic. *See also* **membrane filter**, **ultrafilter**. **2** a device incorporating such a piece or layer of material. **3** a block or sheet of material for reducing the intensity of particulate radiation. **4** a screen for selectively absorbing or attenuating electromagnetic radiation or sound waves of some particular or all frequencies. **5** a passive electronic circuit or device that selectively attenuates or allows the passage of certain frequencies of an electrical signal. **6** to remove or separate (suspended particles, macromolecular solutes, selected frequencies of electrical signal or electromagnetic radiation, etc.)

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from (a heterogeneous or composite fluid, beam, signal, etc.) by means of a filter; to pass (something) through a filter. **7** (*as modifier*) containing or using a filter or filters, or used for filtering, as in a filter fluorometer or filter photometer.

filterable or **filtrable** capable of being filtered; able to pass through a (specified) filter.

filterable agent a former name for bacteriophage, virus, or other submicroscopic noncellular microorganism.

filterable virus a former name for a virus the particles of which are capable of passing through a filter that will retain bacteria and other cellular microorganisms.

filter membrane an alternative term for **membrane filter**.

filter paper porous, unsized paper used for filtration; a disk, piece, or sheet of such paper. Such paper may be chemically treated for special applications, e.g. for use in quantitative analysis; it is also used in **paper chromatography**.

filter press a device for effecting filtration, consisting of a series of (metal, plastic, or wooden) frames the two sides of which are covered with filter cloth. The frames are clamped together and the liquid to be filtered is pumped into them in such a way that the solid residue forms a cake between the cloths and the liquid filtrate is conducted away separately.

filter pump a type of vacuum pump used to assist filtration, in which a jet of water forced through a narrow orifice entraps air from the system being evacuated.

filtrable a variant spelling of **filterable**.

filtrate the liquid or gas that has passed through a **filter** (def. 1).

filtration the act or process of separating or (completely or partially) removing selected components (of a mixture) by means of a **filter** (def. 1), to produce a **filtrate**. The term includes **gel filtration**, **membrane filtration**, and **ultrafiltration**.

filtration enrichment a method for isolating fungal auxotrophs. Mutagenized spores are placed in a minimal medium in which only normal spores germinate and produce a mycelial network. This mycelial network is then removed by filtration and the remaining auxotrophic spores are provided with an enriched medium to allow their germination, growth, and propagation.

filtration fraction (*in physiology*) the fraction of the plasma that is filtered through the renal glomeruli. It is equal to the **glomerular filtration rate** divided by the renal plasma flow and is frequently taken to be equal to the ratio of the inulin clearance to *p*-aminohippurate clearance.

fimbria (*pl.* **fimbriae**) **1** (*in botany and zoology*) a fringe or fringed edge or margin, e.g. especially at the opening of a fallopian tube. **2** (*in bacteriology*) an alternative term for **pilus** (especially a common pilus). —**fimbrial** *adj.*

fimbriate or **fimbriated** characterized by or possessing fimbriae; fringed at the edge or margin.

fimbrin an **actin-binding protein** involved in the development and maintenance of cell polarity. Example from *Saccharomyces cerevisiae*: database code FIMB_YEAST, 642 amino acids (71.69 kDa).

fine structure or **microstructure** any structural detail observable in genetic material, microscopic specimens, spectra, etc. when examined at high magnification or resolution. *See also* **hyperfine structure**.

fingerprinting any pattern obtained by a variety of means that is characteristic of a biological entity, including **1** the electrophoretic or chromatographic pattern obtained after a process in which a protein is digested, usually enzymically, in defined conditions to produce a mixture of peptides, the nature of the mixture being dependent on the particular protein and the conditions of digestion. The distribution pattern of spots is rarely, if ever, duplicated if a different protein is used. **2** a set of one or more amino-acid **motifs** (def. 2) obtained by aligning conserved regions of proteins found in protein sequence databases to provide a sequence or set of sequences characteristic of a particular function e.g. the rat metabotropic glutamate receptor **2** precursor fingerprint has seven motifs, the first of which is the 22-residue

firefly lantern

VGPVTIACLGALATLFVTVIFI.

The technique involves alignment followed by maximization of sequence information through iterative scanning, with multiple motifs, of a large composite database. The results are held in the **PRINTS** database. See also **Appendix E. 3** the result of **genetic profiling**.

firefly lantern or **lantern** the luminescent abdominal organ, or photophore, of the American firefly, *Photinus pyralis*, a common source of the enzyme **luciferase** and its luminogenic substrate **luciferin**.

firefly luciferase see **luciferase**.

firefly luciferin see **luciferin**.

first law of photochemistry an alternative name for **Grothuss and Draper law**.

first law of thermodynamics see **thermodynamics**.

first messenger an alternative term sometimes used for **hormone** to distinguish it from **second messenger**. See also **chemical messenger**, **pheromone**.

first-order kinetics the kinetics of a first-order reaction; see **order of reaction**.

first-order reaction see **order of reaction**.

first-pass clearance or **first-pass elimination** or **first-pass metabolism** the effect of passage of a drug administered by mouth through the liver before it reaches the systemic circulation. Some drugs, in spite of their rapid uptake through the gut mucosal membrane, appear in low concentration in the systemic circulation and hence more has to be administered by mouth than by intravenous injection. As little as 10–20% of the drug may reach the systemic circulation unchanged as a result of their metabolism, mainly by the liver. Some authors extend the term to include metabolism also by the cells of the gut mucosal membrane.

Fischer 1 **Edmond Henri** (1920–), Swiss-born biochemist; Nobel Laureate in Physiology or Medicine (1992) jointly with E. G. Krebs 'for their discoveries concerning reversible protein phosphorylation as a biological regulatory mechanism'.

2 **Emil Hermann** (1852–1919), father of H. O. L. Fischer (see def. 5), German organic chemist, stereochemist, enzymologist, and one of the most famous chemists of the nineteenth century, renowned in particular for his synthetic work on dyes, for elucidating the structures of and synthesizing a wide range of natural products (including sugars, purines, amino acids, and polypeptides), for establishing the steric configurations of all the aldohexoses and aldopentoses (see also **Fischer convention**), and for recognizing the stereochemical specificity of enzyme action (see **lock-and-key model**); Nobel Laureate in Chemistry (1902) 'in recognition of the extraordinary services he has rendered by his work on sugar and purine syntheses'.

3 **Ernst Otto** (1918–), German inorganic chemist; Nobel Laureate in Chemistry (1973), the prize being shared with G. Wilkinson 'for their pioneering work, performed independently, on the chemistry of the organometallic, so called sandwich compounds'. 4 **Hans** (1881–1945), German organic chemist noted for his structural and synthetic work on various linear or macrocyclic tetrapyrroles; Nobel Laureate in Chemistry (1930) 'for his researches into the constitution of haemin and chlorophyll and especially for his synthesis of haemin'.

5 **Hermann Otto Laurenz** (1888–1960), son of Emil H. Fischer (see def. 2), German organic chemist and biochemist noted particularly for his unequivocal chemical syntheses (with E. Baer) of glyceraldehyde 3-phosphate and glycerone phosphate.

Fischer conventions conventions used in the construction of plane projection formulas and first formulated by E. H. Fischer (see **Fischer** def. 2). The first convention may be stated as 'All of the tetrahedral apices that connect asymmetric carbon atoms lie in a straight line in the plane of the paper. This line shall contain the lower edges of all these tetrahedra, their other apices being above the plane and on opposite sides of the line' (see **Fischer projection**). The second convention is the

assumption of one specified configuration for (+)saccharic acid and hence for glucose. At the time of Fischer's work, absolute configurations were not known. Fortunately, he guessed the correct configuration for (+)glucose (now designated D-glucose).

Fischer ester former name for glyceraldehyde 3-phosphate (see **D-glyceraldehyde 3-phosphate**). [After H. O. L. Fischer (see **Fischer** def. 5).]

Fischer projection a diagrammatic representation in two dimensions (as on paper) of the three-dimensional structure of an open-chain chiral molecule of the type R^1CHXR^2 , useful particularly for monosaccharides ($X = OH$) and their derivatives and for amino acids ($X = NH_2$) and their derivatives. According to the first **Fischer convention**, the carbon chain is written vertically with the most highly oxidized carbon atom (number 1) at the top, the bonds between the carbon atoms being either in or below the plane of the paper; the hydrogen atoms and the X groups, written to the left and right of the carbon atoms to which they are attached, are considered as being in front of the plane of the paper. If X is on the right, the configuration at that carbon atom is designated by the prefix D- (from Latin *dexter*, turned to the right), whereas if X is on the left, the configuration is designated L- (from Latin *laevus*, turned to the left). See also **D/L convention**.

FISH abbr. for fluorescence *in situ* hybridization.

fish-odour syndrome a condition of humans characterized by an offensive odour of rotting fish, due to abnormal excretion of trimethylamine in the breath, urine, sweat, saliva, and vaginal secretions, especially after eating foods containing trimethylamine (e.g. sea fish) or choline. It results from a reduced ability to oxidize trimethylamine N-oxide, normally effected by liver microsomal monooxygenases.

fish oil oil obtained from the liver or flesh of fish; it is comparatively rich in fatty acids of the n-3 family (see **fatty-acid nomenclature**). Generally, the flesh of so-called oily fish, such as herring, mackerel, salmon, and trout, has a higher content of C_{20} and C_{22} n-3 acids than that of white fish such as cod or plaice. Recently, increasing the dietary content of long-chain n-3 fatty acids has been promoted as a prophylactic measure in the prevention of heart disease, though the mechanisms suggested are conjectural. See also **cod-liver oil**.

Fiske and SubbaRow method a colorimetric method for determination of inorganic phosphate, based on the production of a blue colour when the sample is treated with ammonium molybdate and 1-amino-2-naphthol-4-sulfonic acid. [After Cyrus Hartwell Fiske (1890–1978) and Yellapragada SubbaRow (1896–1948).]

FISP-12 another name for mouse **connective tissue growth factor**. **fission** tending to undergo **fission**.

fission 1 or **nuclear fission** (in physics) a nuclear reaction in which a heavy atomic nucleus splits into two (or, rarely, three or four) approximately equally sized nuclides, with release of a considerable amount of energy. 2 (in biology) cleavage of cells or division of unicellular organisms into two (or more) parts.

fission product any of the stable or unstable nuclides resulting from **fission** (def. 1).

FITC abbr. for 1 fluorescein isothiocyanate. 2 the fluorescein isothiocarbamoyl group.

fitness (in biology) the survival value and the reproductive capability of a given genotype as compared with the average of the population or of other genotypes in the population.

fix 1 to render, or to become, stable, firm, or insoluble; to attach (to something). 2 (in chemistry) to convert a free element into one of its compounds; especially to convert atmospheric dinitrogen into nitrogen-containing substances (see **nitrogen fixation**); to convert a volatile or fluid substance into a non-volatile or solid form; to combine an inorganic substance with an organic substance. 3 (in biology) to preserve (and harden) cells, tissues, or organisms for subsequent (microscopic) study. 4 (in photography) to treat a film or plate with fixer so as to render permanent the developed image. —**fixation** n.

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fixative

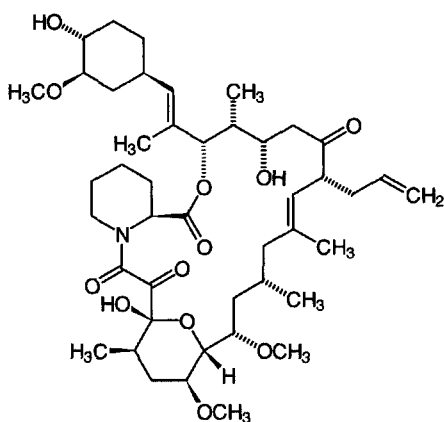
fixative a protein-denaturing fluid, e.g. formaldehyde, that is used to **fix** (def. 3) biological material for (microscopic) study.

fixed antibody see **cell-bound antibody**. Compare **humoral antibody**.

fixed nitrogen any product of **nitrogen fixation**.

fixer (in *photography*) a chemical solution used to remove unexposed, hence unreduced, silver halide granules from a photographic emulsion after the image has been developed.

FK506 an immunosuppressant, isolated from *Streptomyces tsukubaensis*. It is a member of the macrolide antibiotic family, has an M_r of 803, and binds to the **FKBP**-binding proteins. It is a potent inhibitor of T-cell activation and prevents allograft rejection. The *in vivo* function of this **immunophilin** is not clear but in a complex with **FKBP** it blocks the action of **calcineurin** in the liver. It is also a powerful neuroprotective agent in an *in vivo* model of focal cerebral ischemia and may have clinical potential for the treatment of stroke. FK506 also reduces **nitric oxide** production by inhibiting the calcineurin-mediated dephosphorylation of **nitric-oxide synthase** (EC 1.14.13.39). One proprietary name is Tacrolimus.



FKBP *abbr.* for FK506-binding protein; any of an extensive family of small proteins to which the immunosuppressant **FK506** binds. They are found in many species from bacteria to humans. Like **cyclophilin** they possess **peptidyl-prolyl isomerase** activity (EC 5.2.1.8), but have no similarity in structure. Both types belong to the class of proteins termed **immunophilins**. Example from *Drosophila melanogaster*: database code FKBP_DROME, 108 amino acids (11.6 kDa).

flagella the plural of **flagellum**.

flagellate 1 describing unicellular organisms that have a **flagellum** or flagella. 2 resembling a flagellum. 3 an organism that propels itself by means of a flagellum or flagella, especially any member of the protozoan superclass Mastigophora.

flagellin any member of a group of similar monomeric soluble globular proteins that constitute the subunits of bacterial flagella. They also comprise the major antigens of several bacteria, including *Salmonella* spp., which alternate between the production of two antigenic forms of flagella, termed phase 1 and phase 2. Example, phase-1 flagellin from *S. typhimurium* SL877: database code STYFLGH11, 490 amino acids (51.28 kDa); there are three flagellin motifs.

flagellum (*pl.* **flagella** or **flagellums**) 1 the specialized locomotory appendage of bacteria of certain taxonomic groups, consisting of a noncontractile, filiform extension through the cell surface, borne singly, in groups, or as a covering layer. It is commonly 3–20 μm long and 12–25 nm in diameter, is built up of several (often three) longitudinally arranged chains of **flagellin** subunits (often in a spiral), and is anchored to the cell envelope by a **basal body** (def. 1), which imparts rotatory motion to it. Bacterial flagella carry the H (or flagellar) antigens, which are useful in some instances, e.g. in *Salmonella* spp., for classification. 2 the specialized locomotory appendage of the motile cells of unicellular eukaryotes of certain taxonomic groups (e.g. some algae and protozoa) and also of the male gametes (spermatozoa) of most groups of animals. Usually borne singly or in small groups, it is a whiplike structure, similar to, but generally longer than a **cilium**, commonly about 40 μm long and 200–300 nm in diameter. It is built up of an outer membranous sheath, which is a continuation of the cell membrane, and an inner **axoneme**, which terminates within the cell at a **basal body** (def. 2). Eukaryotic flagella often exhibit undulatory motion, which in multiflagellate organisms can occur as coordinated beating. —**flagellar** *adj.*

flame emission spectrophotometry a type of **atomic emission spectrophotometry** in which excitation is produced by a flame.

flame ionization detector or **flame ionisation detector** a sensitive detector used in **gas chromatography** that depends on changes in the electrical conductivity of a flame (burning a mixture of dihydrogen and dinitrogen in air or dihydrogen and dioxygen), brought about by ionization of the vapours of organic substances emerging from the column.

flame photometer an instrument for **flame photometry**.

flame photometry an earlier and simpler, often direct-reading, version of **atomic emission spectrophotometry** in which isolation of the required emission line(s) is achieved by the use of an appropriate filter.

flame spectrophotometry any form of **atomic absorption spectrophotometry** or **atomic emission spectrophotometry** in which excitation is produced by a flame.

flanking marker (DNA) a DNA marker on either side of a locus; a DNA sequence on either side of a gene.

flash evaporation a technique for concentrating a solution at a relatively low temperature thus preventing denaturation of nonvolatile solutes. The solution is placed in a round-bottomed flask or other vessel, which is then rapidly rotated and evacuated; the vapour removed from the solution is reliquified in a condenser and retained in a cold trap. The rotation of the vessel causes the liquid to form a comparatively thin film, which increases its surface-to-volume ratio, thereby assisting evaporation.

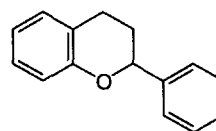
flash evaporator an apparatus for **flash evaporation**.

flash photolysis a technique for investigating short-lived primary or subsequent intermediates, e.g. radicals, in a **photochemical reaction**. The sample is subjected to a brief but powerful flash of light of appropriate wavelength and the **photolysis** products are identified by their absorption or fluorescence spectra, using an independent source of radiation. The kinetics of very fast reactions can be investigated by this technique.

flash point the lowest temperature at which the vapour above a volatile combustible substance will ignite to produce a momentary flash when a small flame is applied under specified conditions.

flat-bed indicating that a technique is carried out in a horizontal plane, e.g. flat-bed chromatography.

flavan 2-phenylchroman; 2,3-dihydro-2-phenylbenzopyran; the parent ring structure on which the structures of flavanols, flavanones, flavones, flavonols, and other classes of aglycons of flavonoids are based.



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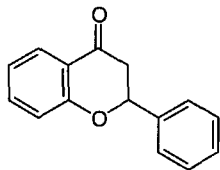
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flavanol

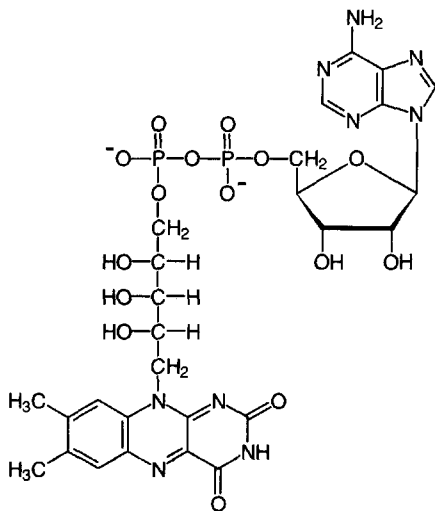
flavanol any hydroxylated derivative of **flavan**. Glycosides of flavan-3-ol and of flavan-3,4-diol respectively form two of the classes of **flavonoid** (def. 1).

flavanone 1 2,3-dihydroflavone; 2-phenylchroman-4-one; 2,3-dihydro-2-phenylbenzopyran-4-one; it is the 4-oxo derivative of **flavan** and the 2,3-dihydro derivative of **flavone**. 2 any of a group of hydroxylated derivatives of flavanone. Flavanone glycosides are colourless; they are found in flowering plants and form one of the classes of **flavonoid** (def. 1).



flavin or (formerly) **flavine** the stem name and trivial name for any member of a group of yellow-coloured, 10-substituted derivatives of 7,8-dimethylisoalloxazine, as in **riboflavin**, **flavin-adenine dinucleotide**, etc.

flavin-adenine dinucleotide *abbr.*: FAD; adenosine diphosphate riboflavin; adenosine(5')diphospho(5')riboflavin; riboflavin 5'-(adenosine 5'-diphosphate); 7,8-dimethyl-10-[5-(adenosine(5')diphospho)-D-ribityl]isoalloxazine; an atypical dinucleotide, not only in that it contains, as does **nicotinamide-adenine dinucleotide**, a phosphoric anhydride linkage between the component mononucleotide units, which in this instance are **adenosine 5'-phosphate** and **flavin mononucleotide**, but also because the latter is not strictly a nucleotide in any case, although considered as one (*see* **nucleotide** (def. 3)). It forms the coenzyme or the prosthetic group of various flavoprotein oxidoreductase enzymes, in which it functions as an electron acceptor by being reversibly converted to **flavin-adenine dinucleotide (reduced)**. *See also* **riboflavin**.

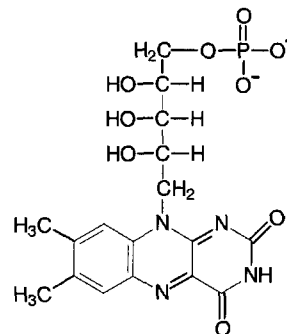


flavin-adenine dinucleotide (reduced) *abbr.*: FADH₂; adenosine diphosphate dihydriboflavin; 1,5-dihydro-7,8-dimethyl-10-[5-(adenosine(5')diphospho)-D-ribityl]isoalloxazine. The reduction occurs at the two unprotonated nitrogen atoms at positions 1 and 5 of FAD. *See* **flavin-adenine dinucleotide**.

flavin coenzyme a general name sometimes used for a **flavin nucleotide** when not covalently linked to an apoenzyme.

flavin enzyme or **flavoenzyme** any **flavoprotein** that is an enzyme.

flavin mononucleotide *abbr.*: FMN; riboflavin 5'-phosphate, 7,8-dimethyl-10-(5-phospho-D-ribityl)isoalloxazine. Although not strictly a mononucleotide, since it contains an alditol moiety rather than a glycoside moiety, it nevertheless is considered as one (*see* **nucleotide** (def. 3)), and as such is a component of **flavin-adenine dinucleotide**. It forms the coenzyme (or occasionally the prosthetic group) of various flavoprotein oxidoreductase enzymes, where it functions as an electron acceptor by being reversibly converted to **flavin mononucleotide (reduced)**. *See also* **riboflavin**.



flavin mononucleotide (reduced) *abbr.* FMNH₂; dihydro-riboflavin 5'-phosphate, 1,5-dihydro-7,8-dimethyl-10-(5-phospho-D-ribityl)isoalloxazine. The reduction occurs at the two unprotonated nitrogen atoms at positions 1 and 5 of FMN. *See* **flavin mononucleotide**.

flavin nucleotide a general name sometimes used for **flavin-adenine dinucleotide** and **flavin mononucleotide**, and their reduced forms.

flavin reductase (in luminescent bacteria) the product of the *luxG* gene in several marine luminescent bacteria. It is presumed from sequence similarity to be an FAD/NAD(P) flavoprotein oxidoreductase. Example from *Vibrio fischeri*: database code LUXG_VIBF1, 236 amino acids (26.47 kDa).

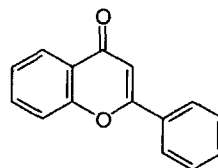
flavo+ *comb. form* 1 yellow. 2 containing a **flavin nucleotide**.

flavocytochrome any protein that is both a **flavoprotein** and a **cytochrome**, e.g. yeast L-lactate dehydrogenase (cytochrome), (EC 1.1.2.3), also known as **cytochrome b₂**.

flavodoxin any of a group of small protein electron carriers, widely distributed in anaerobic bacteria, photosynthetic bacteria, and cyanobacteria, that contain covalently linked flavin mononucleotide. They readily form stable blue semiquinones, and can substitute functionally for **ferredoxin**. Example from *Escherichia coli*: database code FLAV_ECOLI, 175 amino acids (19.58 kDa).

flavolans *see* **tannin**.

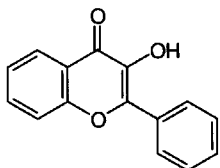
flavone 1 2-phenylchromen-4-one; 2-phenyl-4H-1-benzopyran-4-one; a potent inhibitor of **prostaglandin-endoperoxide synthase**. 2 any of a group of hydroxylated derivatives of flavone. Flavone glycosides occur as pigments, usually yellow, in flowering plants and form one of the classes of **flavonoid** (def. 1). *Distinguish from* **flavanone**.





flavonoid 1 any of a very large and widespread group of water-soluble phenolic derivatives, often brightly coloured and usually found in the vacuoles of plant cells. They are glycosides with the structures of their aglycon moieties based on the **flavan** skeleton, and are classified according to the oxidation state of its pyran ring. Some 10–12 classes are recognized. 2 any flavone, isoflavone, or neoflavone, or any of their derivatives, including bioflavonoids. *See also* **apigenin**, **myricetin**, **quercetin**. 3 of, relating to, or resembling flavone, isoflavone, or neoflavone.

flavanol 1 3-hydroxyflavone. 2 any of a group of hydroxylated derivatives of flavanol. Flavanol glycosides occur as pigments, usually yellow, in flowering plants, and form one of the classes of **flavonoid** (def. 1). *Distinguish from flavanol.*



flavoprotein *abbr.*: FP; any enzyme or other protein of which a flavin nucleotide is a coenzyme or prosthetic group. They occur in virtually all cells, where they act as oxidoreductases or as electron carriers in the terminal electron transport chains.

Fleming, (Sir) Alexander (1881–1955), British bacteriologist famous for his discovery of the bacteriolytic agents lysozyme and penicillin; Nobel Laureate in Physiology or Medicine (1945) jointly with E. B. Chain and H. W. Florey 'for the discovery of penicillin and its curative effect in various infectious diseases'.

Fletcher factor *see* **plasma kallikrein**.

flg *see* **fibroblast growth factor receptor**.

flickering cluster model a model for the structure of liquid water in which small, tetrahedrally coordinated, icelike regions, held together by hydrogen bonds, are thought to exist for short periods, continually breaking up and re-forming.

flightin the enzyme **glycerol 3-phosphate dehydrogenase (NAD⁺)** obtained from certain insects. It is so called because this and mitochondrial **glycerol 3-phosphate oxidase** play a major role in the transfer of energy needed for flight in insects.

flip-flop 1 (*in electronics*) a circuit or device having two stable states and two inputs corresponding to these states. The device remains in either state until caused to change to the other state by application of an appropriate signal to the inputs. 2 (*in enzymology*) a mechanism in which a functional interrelationship between distinct active sites on identical monomers, which are parts of an oligomeric enzyme showing Michaelian kinetics, is mediated by a structural modification, the **flip-flop transition**. *Compare* **half-of-the-sites reactivity**. 3 (*in genetics*) the informal name for a reversible, reciprocal exchange between recombining segments of DNA in site-specific recombination systems in prokaryotes or eukaryotes. *See also* **transposable element**.

flipping a jargon term used in **nuclear magnetic resonance spectroscopy** for the change in orientation of an atomic nucleus from an angle θ° to $(180 - \theta)^\circ$, with respect to the direction of the magnetic field, caused by absorption of radiation energy of a specific radiofrequency.

flocculate to aggregate or cause to aggregate into 'woolly' clusters or masses.

flocculation 1 the act or process of flocculating (*see* **flocculate**). 2 the product of such an act or process.

flocculation reaction or **flocculation test** a **precipitin test** in which the precipitate appears as floccules.

floccule a small aggregated mass of flocculent material.

flocculent 1 having the appearance of wool or down. 2 (*in*

chemistry) consisting of or occurring as small woolly or downy aggregates. 3 (*in biology*) covered with waxy flakes or tufts resembling wool.

flora 1 *collective term* for all the plants, or plant life, characteristic of a given place or period. 2 **microbial flora**; the microbial population of part of the body, e.g. of the gut, skin, etc. 3 a comprehensive treatise or list of all the plants or plant life of a particular region or period.

Florey, (Sir) Howard Walter, Lord Florey (1898–1968), Australian physiologist and experimental pathologist distinguished particularly for his role in the development of penicillin (with E. B. Chain) and cephalosporin C as clinically useful antibacterial agents; Nobel Laureate in Physiology or Medicine (1945) jointly with E. B. Chain and A. Fleming 'for the discovery of penicillin and its curative effect in various infectious diseases'.

Florisil the proprietary name for a type of activated magnesium silicate. It is an adsorbent and is useful for column chromatographic separation of lipids.

flotation coefficient a name sometimes given to a negative **sedimentation coefficient**.

flow birefringence or **double refraction of flow** the **birefringence** induced in a fluid by shear, which causes orientation of any asymmetric molecules in the fluid.

flow cabinet another name for **safety cabinet**.

flow cell any cell (def. 2) so constructed that appropriate measurements (as of absorbance) may be made on a flowing fluid.

flow cytofluorimeter a **flow cytometer** in which fluorescence of the objects being counted is monitored.

flow cytometer an apparatus for **flow cytometry** in which cells, or subcellular components (e.g. isolated chromosomes), in aqueous suspension are made to flow at high speed, in single file, through a sensing region where optical or electric signals indicative of some important physical or chemical properties of the cells, or components, are generated. These signals are analysed and accumulated in a computer for quantitative evaluation. Typical flow rates that can be achieved are in the region of 1000 cells per second. *See also* **flow sorter**, **fluorescence-activated cell sorter**.

flow cytometry the technique for counting or measuring some property of cells, or subcellular components, using a **flow cytometer**, often after labelling with a fluorescent marker.

flow dichroism **dichroism** (def. 1) induced in a fluid by shear, which causes orientation of any asymmetric molecules in the fluid.

flower the reproductive structure of angiosperm plants. It consists generally of sepals, petals, and stamens and/or carpels. It is basically a highly modified leafy shoot.

flow-signal an earlier term for **rheosome**.

flow sorter (for cells) a **flow cytometer** to which has been added a device for physically separating cells, or subcellular components, of desired characteristics in order to provide subpopulations of high homogeneity. Typically the cell suspension is forced out of an orifice, 50–100 μm in diameter, forming a high-speed liquid jet in air. Optical sensing of the jet takes place near the orifice and the jet is then broken up into uniform droplets by applied ultrasonic vibration. The droplets then traverse a region of high-intensity constant electric field where the decision-making and charging circuits, activated by the sensors, cause only droplets containing the wanted cells to be charged and deflected by the electric field into a separate collector.

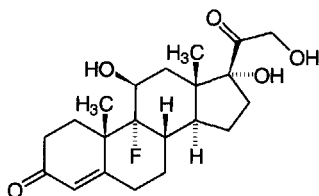
fluctuation test a method of statistical analysis used to determine whether selected variants, e.g. phage- or drug-resistant bacterial mutants, arose spontaneously prior to exposure to the selective agent or were caused by the agent.

fludrocortisone 9 α -fluorohydrocortisone (11 β)-9-fluoro-11,17,21-trihydroxypregn-4-ene-3,20-dione; a **corticosteroid** with about 10 times the glucocorticoid action of cortisol, but with 300–600 times greater mineralocorticoid action. It is used topically in ointments.

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fludrocortisone

fluorescence symbol: F or H ; (in photochemistry) the energy delivered in a given time interval, dt ; $F = \int I dt$ where I is the radiant intensity and t is time. It is expressed in $J m^{-2}$.

fluorescence rate per unit time; it is commonly referred to (strictly incorrectly) as **flux**.

fluid 1 any substance that can flow; any portion of matter in which the components (molecules or particles) continuously change their relative positions. In physics, the term comprehends any liquid or gas; in biology (and in common usage) it generally denotes only a liquid, especially one present in, or secreted by, a living organism. 2 able to flow or to change shape easily; not solid or rigid; liquid. 3 of, concerned with, or using fluid, a fluid, or fluids.

fluidity (in physics) the reciprocal of the **viscosity** of a fluid.

fluid mosaic model a model of cell-membrane structure in which the proteins that are integral to the membrane are considered as a heterogeneous set of globular molecules, each arranged in an amphipathic structure, i.e. with their ionic and highly polar groups protruding from the membrane into the aqueous phase and their nonpolar groups largely buried in the hydrophobic interior of the membrane. These globular molecules are partially embedded in a matrix of phospholipid, the bulk of which is organized as a discontinuous, fluid bilayer, although a small fraction of the lipid may interact specifically with the membrane proteins. Thus the fluid mosaic structure is formally analogous to a two-dimensional oriented solution of integral proteins (or lipoproteins) in a viscous phospholipid bilayer solvent.

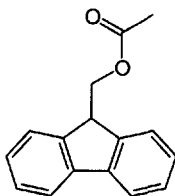
Fluon a proprietary name for **polytetrafluoroethylene**.

fluor or **lumiphor** or **scintillator** any substance that displays **fluorescence** when excited by electromagnetic or particulate radiation. Fluors are useful especially in **scintillation counters**.

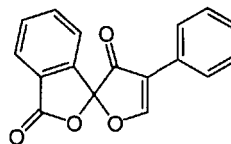
fluor+ a variant form of **fluoro+** (sometimes before a vowel).

fluoroacetate see **fluoroacetate**.

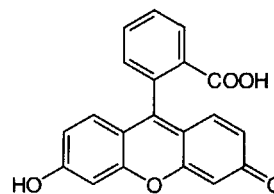
fluorenylmethoxycarbonyl symbol: $Fmoc$; abbr.: $Fmoc$; fluorenyl-9-ylmethoxycarbonyl; a group used for linkage to amino groups for the purpose either of forming fluorescent amino-acid derivatives that can readily be detected after column chromatography, or to protect the amino groups of amino acids or nucleotides while other functional groups are undergoing reaction. Reagents useful for introducing the group are 9-fluorenylmethyl chloroformate and 9-fluorenylmethyl succinimidyl carbonate.



escent structures on reaction with primary amines. *Proprietary name:* **Fluoram**.



fluorescein 9-(2-carboxyphenyl)-6-hydroxy-3H-xanthen-3-one; Acid Yellow 73; CI 45350; an orange-red dye, the sodium salt of which is freely soluble in water to give a solution with yellowish-green fluorescence (to a dilution of 0.02 ppm) under UV light. Certain derivatives, e.g. fluorescein isothiocyanate, are useful for **fluorescein labelling** and others, e.g. the nonfluorescent fluorescein diacetate, are useful as substrates for the detection and assay of esterases.



fluorescein labelling the introduction of fluoresceinyl groups (see **fluorescein**) into biopolymers for fluorescence polarization studies and fluorescence immunoassay techniques.

fluorescence 1 a type of luminescence that consists of the emission by a substance of electromagnetic radiation, especially visible light, as a result of and immediately (10–100 ns) after the absorption of energy derived from exciting radiation of another, usually shorter, wavelength or from incident subatomic particles (especially electrons or alpha particles); the property of emitting such radiation. 2 the radiation emitted during **fluorescence** (def. 1). Compare **phosphorescence**. —**fluorescent**, **fluorescing** adj.; **fluoresce** vb.

fluorescence-activated cell sorter or **fluorescence-assisted cell sorter** abbr.: **FACS**; a type of **flow sorter** in which cells are characterized and sorted by the intensity of the fluorescence they emit when passing through an exciting laser beam.

fluorescence depolarization or **depolarization of fluorescence** the phenomenon occurring when fluorescence is excited by plane polarized radiation; the emitted radiation is always partially depolarized (see **fluorescence polarization**). The extent of such depolarization is due, *inter alia*, to the motion of the absorber (i.e. Brownian motion) and to energy transfer between like chromophores. Fluorescence depolarization can be used to follow the changes in the motion (rotation) of an entire macromolecule, or of a part of the molecule, to which fluorescent chromophores are attached; it can also be used to indicate the environment of a **fluorescent probe** in, e.g., a membrane – an environment that results from less tightly packed structures permits greater movement of the probe molecule, thus leading to greater depolarization. See also **photon counter**, **time-resolved fluorescence polarization**.

fluorescence energy transfer a process of energy transfer between two **fluorophores**, which can occur when the emission spectrum of the first fluorophore overlaps the absorption spectrum of the second fluorophore. **Quenching** of the emission from the first compound occurs but the excitation energy is absorbed by the second compound, which then emits its own characteristic fluorescence. If the two fluorophores are attached at different positions to a macromolecule, observations

fluorescamine 4-phenylspiro[furan-2(3H),1'-(3'H)-isobenzofuran]-3,3'-dione; a nonfluorescent compound forming fluor-



fluorescence enhancement

of fluorescence energy transfer between them may be used to determine the distance between the two attachment positions.

fluorescence enhancement or **enhancement of fluorescence** a phenomenon occurring with a number of **fluorophors** whose fluorescence is markedly dependent on the polarity of the medium, in which fluorescence is greatly enhanced in a non-polar or rigid environment, but strongly quenched in an aqueous environment. Examples are 8-anilino-naphthalene sulfonate (ANS) when bound to an antibody or trapped in the hydrophobic environment within a protein; and ethidium when intercalated into double-helical regions of a nucleic acid.

fluorescence immunoassay or **fluorescent immunoassay** or **fluoroimmunoassay** *abbr.*: FIA; a method of assaying very small amounts of material that is similar in principle to **radioimmunoassay** but uses a fluorescent rather than a radioactive label for the antigen. Various techniques can be used for determination of the ratio of bound to free labelled antigen that do not necessitate a separation of the two; these may depend instead on **fluorescence energy transfer**, **fluorescence enhancement**, **fluorescence depolarization**, **fluorescence quenching**, or other phenomena.

fluorescence in situ hybridization *abbr.*: FISH; a procedure that involves the use of DNA probes to locate in a tissue section specific regions of DNA in the chromosomes. The probes are tagged with a selection of fluorescent molecules that are both excited by and emit light of different wavelengths and hence different colours. The structure and behaviour of individual chromosomes at all stages of the cell cycle can thereby be studied.

fluorescence microscopy a type of light microscopy by means of which fluorescent substances in a specimen may be detected and examined. Exciting light, commonly UV and of appropriate wavelengths, is isolated by means of a suitable filter placed between the light source and the condenser, and emitted radiation is selected by a different filter placed between the objective lens and the eye-piece. The technique is useful for visualizing certain components that are difficult to see by direct light, for localizing fluorescent substances bound to specific components of the specimen, and for determining the orientation of a specimen using the **fluorescence depolarization** technique.

fluorescence photobleaching recovery *an alternative name for fluorescence redistribution after photobleaching.*

fluorescence polarization or **polarization of fluorescence** the phenomenon that the light emitted from a fluorescent molecule is always partially polarized. This is true whether or not the exciting radiation is plane polarized. In consequence, when plane polarized radiation is used to excite fluorescence, **fluorescence depolarization** occurs.

fluorescence probe or **fluorescent probe** any small molecule that undergoes changes in one or more of its fluorescence properties as a result of noncovalent interaction with a protein or other macromolecular structure. Such probes are useful for studying the properties and behaviour of macromolecules. *See probe* (def. 1).

fluorescence quantum yield in any fluorescent system, the ratio of the number of photons emitted to the number of photons absorbed; the fraction of excited singlets that become de-excited by fluorescence. *See also quantum yield* (def. 1).

fluorescence quenching the reduction in the **fluorescence quantum yield** of a system due to de-excitation of the fluorescent molecules by environmental factors such as collision or complex formation with quenchers in the system, or by **fluorescence energy transfer** to them. The phenomenon is useful in studies of the binding of small molecules to proteins and other macromolecules.

fluorescence redistribution after photobleaching *abbr.*: FRAP; a technique, also termed **fluorescence photobleaching recovery**, for measuring two-dimensional lateral mobility of fluorescent particles in small ($\approx 10 \mu\text{m}^2$) regions of a cell surface. A small spot on the surface is photobleached by a brief

exposure to an intense, focused laser beam, and the subsequent recovery of fluorescence, due to replenishment of intact fluorophor in the bleached spot by lateral transport from the surrounding surface, is followed.

fluorescence spectrophotometry the measurement, by means of a **spectrofluorometer**, of the wavelengths and the intensities of the light emitted from a fluorescent sample that is excited by more or less monochromatic exciting radiation.

fluorescence spectrum the **emission spectrum** of a fluorescent system that is excited by radiation of a given wavelength. *Compare excitation spectrum.*

fluorescent antibody technique an **immunofluorescence** technique in which antibody conjugated with a fluorophor is used to locate corresponding antigen in tissue sections or smears.

fluorescent immunoassay a variant spelling of **fluorescence immunoassay**.

fluorescent probe a variant spelling of **fluorescence probe**.

fluorescent screen a screen coated with a substance, e.g. zinc sulfide or calcium tungstate, that fluoresces when irradiated with electromagnetic radiation of certain wavelengths or with particulate radiation.

fluoride a variant form of (sometimes before a vowel) **fluoro-**.

fluoridation the addition of up to approximately 1 ppm of fluoride ion to a drinking water supply with the object of reducing tooth decay in consumers of the water. —**fluoridate** *vb*.

fluorimeter a variant spelling of **fluorometer**.

fluorinate 1 to treat or react (a substance) with difluorine. 2 to introduce (one or more) fluorine atoms into (a molecule of an organic compound), whether by addition or by substitution. —**fluorinated** *adj.*; **fluorination** *n*.

fluoro- or (sometimes before a vowel) **fluor-** *comb. form* 1 denoting a fluorine in organic linkage. 2 or **fluori-** indicating fluorescence.

fluoroacetate the anion of fluoroacetic acid (= fluoroethanoic acid); a toxic substance found in the leaves of *Dichapetalum cymosum* and other plants that in animals is converted enzymically into **fluorocitrate**. Fluoroacetate and fluoroacetamide have been used as rodenticides.

fluorochrome any substance exhibiting fluorescence, especially one used to impart fluorescence to another molecule or to a specific structure in a biological specimen.

fluorocitrate the trianion of 1-fluoro-2-hydroxy-1,2,3-propanetricarboxylic acid; an extremely toxic substance formed in animal tissues from fluoroacetate. Fluorocitrate is a potent inhibitor of aconitase; it is believed that the fluorine atom interacts strongly with the Fe^{2+} ion at the active centre of the enzyme thereby inhibiting its activity. *See also lethal synthesis.*

fluorodeoxyuridine *see* **fluorouracil**.

fluorodeoxyuridylate *see* **fluorouracil**.

fluorodinitrobenzene *see* **Sanger's reagent**.

fluorogen the nonfluorescent precursor of a **fluorophor**.

fluorogenic 1 producing fluorescence. 2 of or pertaining to a **fluorogen**; e.g. a **fluorogenic** substrate is a nonfluorescent material that can be acted on by an enzyme to produce a fluorescent substance.

fluorogram 1 or (sometimes) **autofluorogram** an autographic image derived by **fluorography** (def. 1). 2 or **fluorograph** a photographic image produced by **fluorography** (def. 2).

fluorography 1 or **autofluorography** or **scintillation autography** or **scintillation autoradiography** an elaboration of autoradiography that shows a considerable increase in the efficiency of detection of low-energy-beta-particle-emitting nuclides, e.g. carbon-14, sulfur-35, and (especially) tritium, in preparations such as thin-layer chromatograms, gel electrophoretograms, tissue or whole-body sections, etc. The otherwise poor response of autoradiography in these situations results from internal absorption of most of the emitted radiation by the specimen in question. In fluorography this effect is diminished by impregnating the specimen (which preferably should be translucent and colourless) with fluor(s) prior to placing it in

fluoroimmunoassay

contact with an appropriately sensitive photographic emulsion or film, which then responds both to direct emission of beta particles from the specimen and to emission of photons of UV light (scintillations) generated by beta particles, exciting the fluor within the specimen. **2** photography of a fluorescent body, especially of an image on a fluorescent screen. —**fluorograph** *n.*, *vb.*; **fluorographic** *adj.*

fluoroimmunoassay *abbr.*: FIA; *a variant spelling of fluorescence immunoassay.*

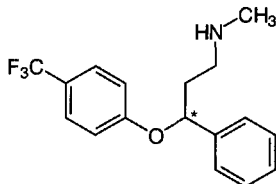
fluorometer or **fluorimeter** **1** a device for measuring the intensity and duration of fluorescence. **2** a device that may be attached to a **fluoroscope** to enable the position of a sought object to be determined with precision. —**fluorometric** or **fluorimetric** *adj.*; **fluorometry** or **fluorimetry** *n.*

fluorophor or **fluorophore** **1** a group of atoms in a molecule that renders the latter fluorescent. **2** a fluorescent substance.

fluoroscope **1** a screen coated with a substance that fluoresces when excited by X-rays; it is used to view the X-ray image of an object. **2** a device used to compare the fluorescence of a solution with that of a standard.

fluorouracil *symbol*: FUra (not FU); 5-fluorouracil; 5-fluoro-2,4(1*H*,3*H*)-pyrimidinedione; a synthetic analogue of uracil that is enzymically converted via its β -D-ribofuranoside, fluorouridine (*symbol*: FUr), to fluorodeoxyuridine (*symbol*: FdUrd) and then to fluorodeoxyuridylylate, which is a potent inhibitor of **thymidylate synthase** and, hence, of DNA synthesis. Thus fluorouracil is a **suicide inactivator** of thymidylate synthase. Fluorouracil and fluorodeoxyuridine are anticancer drugs, commonly used in combination therapy against fast-growing tumours.

fluoxetine ($\pm$)-*N*-methyl- γ -[4-(trifluoromethyl)phenoxy]benzenepropanamine; a serotonin uptake inhibitor, used as an antidepressant, particularly as the hydrochloride (*proprietary name*: Prozac).



* = racemic

flush end or **blunt end** an end of a duplex DNA fragment in which the two strands are coextensive, i. e. terminate at the same point. Such ends may be formed by scission of duplex DNA with some types of endonucleases. *Compare cohesive end.* —**flush-ended** *adj.*

flux **1** the rate of passage of energy or matter (usually in crossing a given area or passing through a given volume) under steady-state conditions. *See diffusion coefficient, luminous flux, radiant power.* **2** the number of particles passing through unit cross-sectional area of beam per unit time. **3** the rate of (chemical or physical) transformation of a substance within a living cell or tissue; the rate of translocation of a substance within a living cell or tissue, especially across a cell membrane. Such processes are usually measured in moles per second. *See also metabolic flux.* **4** the strength of a field of force orthogonal to unit area. *See magnetic flux.* **5** an alternative term for **flux density** (def. 2).

flux control coefficient *symbol*: C_E^J ; in a metabolic sequence at steady state, the response, δJ , of the original system flux, J , to a small change, δE , in either the concentration or one of the kinetic parameters of a specified enzyme, E ; it is given by:

$$C_E^J = \left(\frac{\partial J / J}{\partial E / E} \right) p, q, r, \dots$$

and similarly for every other enzyme in the system. This is the

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flux control coefficient of the enzyme E on the flux J , and is also called **sensitivity coefficient** or **control strength**. Similarly one can calculate a flux control coefficient for the substrate concentration, S , on the flux, J (this is also called **sensitivity** or **net sensitivity**) and a flux control coefficient of an effector concentration, I , on the flux, J (this is also called **control strength**).

flux density **1** (of energy or field strength) the flux per unit cross-sectional area; *see luminous flux density, magnetic flux density.* **2** or (sometimes) **flux** (of particulate radiation, including photons) the arithmetic product of the particle density and the mean velocity of the particles; **neutron flux (density).**

flux-ratio method a method for investigating the mechanisms of enzymic reactions in which simultaneous measurements are made of the rates of enzyme-catalysed isotopic transfers from one product to two substrates while the reaction system is in a steady state. It provides relatively unambiguous information about the order of binding of substrates.

Fm *symbol for fermium.*

fMet *symbol for a residue of the α -acylamino acid *N*-formyl-L-methionine.*

FMN *abbr. for flavin mononucleotide.*

FMN adenyllyltransferase EC 2.7.7.2; *other name*: FAD pyrophosphorylase; an enzyme of the pathway for biosynthesis of FAD. It catalyses the reaction of ATP with FMN to form FAD with release of pyrophosphate.

FMNH₂ *symbol for flavin mononucleotide (reduced).*

Fmoc *symbol for the fluorenylmethyloxycarbonyl group.*

FMOC *abbr. for the fluorenylmethyloxycarbonyl group.*

fmol *symbol for femtomole (10⁻¹⁵ mole).*

fms *symbol for either the retroviral oncogene, v-fms, which is carried by the McDonough strain of feline sarcoma virus, or its related protooncogene, c-fms, which encodes the M-CSF-1 receptor precursor (see colony stimulating factor).* The latter is a type I transmembrane protein of the immunoglobulin superfamily; database code KFMS_HUMAN, 972 amino acids (107.83 kDa). It possesses intrinsic protein tyrosine kinase activity within its cytoplasmic domain, which is activated by the binding of CSF-1 (*see colony stimulating factor*) to the extracellular domain. Its transforming counterpart, v-fms, encodes a constitutively active receptor kinase that can transform both fibroblasts and hematopoietic cells in a CSF-1-independent manner. Mutations in the c-fms gene as well as a critical alteration of the 3' coding sequences appear to be responsible for fully activating its latent transforming potential. In the v-fms protein product, 40 C-terminal amino acids of the normal CSF-1 receptor are replaced by 11 unrelated residues, suggesting that residues near the C terminus normally negatively regulate kinase activity in the absence of CSF-1.

f-number the numerical value of the relative aperture of a lens; it is the ratio of the focal length of a lens to its diameter.

foam **1** a dispersion in which a large proportion of gas by volume, in the form of gas bubbles usually >1 μ m in diameter, is dispersed in a liquid, solid, or gel. **2** to produce or cause to produce such a foam.

foam fractionation a method of separation in which a component of the bulk liquid is preferentially adsorbed at the liquid/vapour interface and is removed by foaming.

foaming agent a **surfactant** that when present in small amounts facilitates the formation of a foam and/or stabilizes it.

focal contact any of the areas on the surface of fibroblasts where actin-containing microfilaments terminate. The filaments are involved both in the attachment of the cells to the substrate and in cell movement.

fodrin or **brain spectrin** or **calpectin** a protein from bovine brain, immunologically related to **spectrin**, that is involved in secretion, interacts with calmodulin in a calcium-dependent manner, and thus is a candidate for the calcium-dependent movement of the cytoskeleton at the membrane. It contains 20 repeats of 106 amino-acid residues; example from mouse (partial sequence); database code FODA_MOUSE, 381 amino



acids (43.26 kDa).

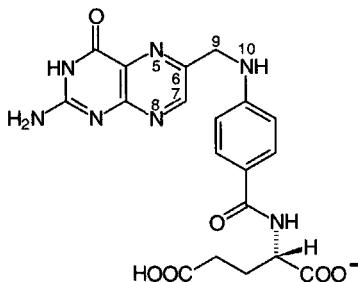
foetal a variant spelling of fetal (*see* fetus).

foeto+ a variant spelling of feto+.

foetus an etymologically misconceived variant spelling of fetus.

folacin a common name for folic acid.

folate 1 the recommended trivial name for pteroylglutamate; N-[4-[[[(2-amino-3,4-dihydro-4-oxopteridin-6-yl)methyl]-amino]benzoyl]-L-glutamate; the anion of **folic acid** (def. 1). 2 any member of the family of pteroylglutamates, irrespective of the level of reduction of the pteridine ring, of the nature of any one-carbon substituent, and of the number of glutamate residues. *See also* **folic acid** (def. 2).



folate coenzyme any of various interconvertible tetrahydrofolates that bear a one-carbon substituent and function as coenzymes in metabolic reactions involving the transfer of one-carbon units. The substituent may be 5-formimino, 5-formyl, 10-formyl, 5,10-methenyl, 5-methyl, or 5,10-methylene.

Folch-Lees protein the most abundant type of protein that is extractable from central-nervous-system myelin by chloroform:methanol (2:1 v/v); it is of at least 150 kDa and contains a high proportion of aliphatic amino acids. *Compare* **Wolffgram proteolipid protein**.

Folch method a method for the fractionation and isolation of lipids from either tissues or fluids. It involves extraction with various chloroform-methanol-water mixtures.

fold a description of protein conformations; there is no universally agreed criterion for proteins to be regarded as having the same fold but folds appear in the SCOP and CATH hierarchies (*see* Appendix E, Table 3).

foldback DNA an alternative term for hairpin DNA. *See* **hairpin**.

folding (in molecular biology) 1 the process whereby an unorganized polypeptide acquires a specific three-dimensional structure. 2 the detailed pattern of three-dimensional structure of a native protein molecule.

folding unit a locally ordered region within the hydrophobic core of a globular protein molecule, formed by three-dimensional interaction between two or three segments of secondary structure (α helix and/or β strand) that are near one another along the polypeptide chain. The most important types are: ($\alpha\alpha$), ($\alpha\beta$), ($\beta\beta\beta$), ($\beta\alpha\beta$). *See also* **domain**.

folic acid *abbr.*: (not recommended) FA or F; 1 the recommended trivial name for pteroylglutamic acid, the corresponding acid of **folate** (def. 1). 2 or **folacin** the name under which any **folate** (def. 2) is generally known in medicine, nutrition, etc. Folic acids are widely distributed members of the vitamin B complex, particularly plentiful in green leafy vegetables, liver, and yeast. Folates are essential for the synthesis of purines and pyrimidines, functioning as carriers of one-carbon units. A deficiency in humans results in megaloblastic anemia and in birth defects in children born to deficient mothers. Folic acids are also growth factors for certain microorganisms. *See also* **folinic acid**.

folic acid conjugate the corresponding acid of any **folate** (def. 2) containing two or more glutamyl residues.

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folic acid reductase an alternative name for **dihydrofolate reductase**.

Folin-Ciocalteu reagent a phosphomolybdotungstic acid reagent that gives a blue colour with phenols. The reagent has been used in the determination of tryptophan and tyrosine in proteins and peptides. *See also* **Lowry method**.

folinic acid 1 or (formerly) **citrovorum factor** (*abbr.*: CF) or (formerly) **leucovorin** or **5-formyltetrahydrofolic acid** 5-formyl-5,6,7,8-tetrahydropteroyl-L-glutamic acid; the common name for the corresponding acid of **folinate** (5-formyltetrahydrofolate), a folate coenzyme discovered to be a growth factor for the bacterium *Leuconostoc citrovorum* (hence its former synonyms). Folinic acid is an intermediate in the metabolic conversion of formiminoglutamate to glutamate, ammonium, and **active one-carbon**. The calcium salt, calcium folinate (or formerly calcium leucovorin), is useful therapeutically, e.g. as an antidote to doses of methotrexate or other folate antagonists. 2 name sometimes used for certain other folates, e.g. tetrahydrofolate, especially as growth factors for microorganisms.

Folin-Lowry method a sensitive method of protein assay in which protein reacts both with an alkaline copper tartrate solution and with the **Folin-Ciocalteu reagent** to give a blue colour, the absorbance of which is measured at 750 nm. *Compare* **Lowry method**.

Folin reaction a colorimetric reaction used for the determination of ammonia and amino-acid nitrogen in which a red colour is produced when an alkaline solution of amino acids is treated with 1,2-naphthoquinone-4-sulfonate. [After Otto Knute Olof Folin (1867-1934).]

Folin-Wu reagent a phosphomolybdic acid solution, containing a little tungstate, used in the determination of glucose. It gives a blue colour with the copper (I) oxide produced when the sample containing glucose is heated with an alkaline copper (II) solution. [After Otto Knute Olof Folin (1867-1934) and Hsien Wu (1893-1959).]

folliberin the recommended name for follicle-stimulating hormone releasing hormone (*abbr.*: FSH-RH or FRH). Luteinizing hormone releasing hormone (*abbr.*: LHRH) stimulates not only release of luteinizing hormone from the anterior pituitary but also release of follicle-stimulating hormone. This activity is commonly named **gonadotropin-releasing hormone**. There is, however, evidence that a specific FSH-RH may exist in addition.

follicle 1 any small cavity or recess. 2 a cavity that has a secretory or excretory function, e.g. the **Graafian follicle**.

follicle-stimulating hormone *abbr.*: FSH; *recommended name*: follitropin; one of two gonadotropic **glycoprotein hormones** secreted by the anterior pituitary; of 34 kDa (human), 33 kDa (ovine), or 29 kDa (porcine), it consists of α and β subunits, the α subunit being almost identical within a given species to that of other glycoprotein hormones and the β subunit conferring hormonal specificity; the carbohydrate content is 16% (human), 8-9% (ovine), or 7-8% (porcine). In the female it stimulates the growth of Graafian follicles in the ovaries, whereas in the male it stimulates the epithelium of the seminiferous tubules to increased spermatogenesis. Example, β subunit from *Bos taurus*: database code FSHB_BOVIN, 129 amino acids (14.71 kDa).

follicle-stimulating hormone releasing factor (*abbr.*: FRF or FSH-RF) or **follicle-stimulating hormone releasing hormone** (*abbr.*: FRH or FSH-RH) *see* **gonadotropin-releasing hormone**.

follicular 1 like, pertaining to, consisting of, or provided with a follicle or follicles. 2 affecting or involving follicles.

follicular phase *see* **estrous cycle**.

folliculate(d) having or consisting of follicles.

folipain a serine proteinase from pig ovarian fluid. It is a heterodimer, and cleaves Arg-Xaa bonds.

folliastatin an activin-binding glycoprotein that regulates biosynthesis and secretion of **follicle-stimulating hormone**. Alternative isoforms are formed by alternative splicing. Example

follitropin

(precursor) from human: database code FSA_HUMAN, 338 amino acids (37.33 kDa).

follitropin the recommended name for follicle-stimulating hormone.

foyl the trivial name for pteroylglutamyl, the acyl group derived from **follic acid** (def. 1). Use of the name is no longer recommended.

fomecin any of several antibiotic substances produced by certain strains of the basidiomycetous fungus *Fomes juniperinus*, especially fomecin A, 2,3,4-trihydroxy-6-(hydroxymethyl) benzaldehyde.

food any material that can be ingested by an organism and metabolized to provide energy and precursors for biosynthetic reactions.

footprinting a technique for identifying sites where there is protein bound to DNA. Protein-DNA complexes are resistant to the action of nucleases, so that undegraded DNA (the 'footprint') remains after treatment with a nuclease.

For symbol for the formyl group, H-CO- (alternative to HCO).

Forbes' disease an alternative name for type III glycogen disease. [After Gilbert Burnett Forbes (1915-), US physician.]

forbidden clone a hypothetical clone of immunologically competent cells with specificity for some autoantigen that, in accord with the clonal selection theory, has become suppressed in the fetus but may be reactivated in adult life and cause an autoimmune disease.

forbidden transition any transition between energy states in an atom or molecule that is forbidden by the selection rules of quantum mechanics (but can occur nevertheless!).

force symbol: *F* (in physics) a vector quantity, the external influence that changes or tends to change the state of rest or uniform motion of a body, or the shape of a fixed body. The SI derived unit of force is the **newton**.

formaldehyde dismutase EC 1.2.99.4; *systematic name*: formaldehyde:formaldehyde oxidoreductase. An enzyme that catalyses a dismutation of two molecules of formaldehyde to formate and methanol. The enzyme functions to detoxify formaldehyde thus rendering the organism resistant to formaldehyde. Its synthesis is induced by addition of formaldehyde to the medium. Example from *Pseudomonas putida* (strain F61) DNA: database code PSEFDM, 399 amino acids (42.93 kDa).

formaldehyde transketolase EC 2.2.1.3; *systematic name*: D-xylulose-5-phosphate:formaldehyde glycolaldehydetransferase; *other names*: dihydroxyacetone synthase; DHAS; glycerone synthase. An enzyme that catalyses a reaction between D-xylulose 5-phosphate and formaldehyde to form glyceraldehyde 3-phosphate and glycerone; thiamine diphosphate is a coenzyme. The enzyme is important in organisms (some fungi and many prokaryotes) that can utilize methanol as a carbon source in converting formaldehyde, produced by methanol oxidation, into carbohydrate, xylulose 5-phosphate acting as C₂ donor. Example from *Hansenula polymorpha* (a species of yeast): database code DAS_HANPO, 710 amino acids (78.75 kDa).

formalin or **formol** a 40% w/v aqueous solution of formaldehyde stabilized with approximately 12% v/v methanol. It is useful as a general disinfectant and for fixation and preservation of biological specimens.

formate **1** trivial and preferred name for methanoate, the anion, H-COO⁻, derived from formic acid (methanoic acid). **2** any mixture of formic acid and its anion. It is useful as a buffer over the pH range 2.6-4.8; p*K*_a (25 °C) = 3.75. **3** any salt or ester of formic acid.

formate-dihydrofolate ligase EC 6.3.4.17; *systematic name*: formate dihydrofolate ligase (ADP-forming); an enzyme that catalyses the formation of 10-formyldihydrofolate from ATP, formate, and dihydrofolate with release of ADP and orthophosphate.

formate-tetrahydrofolate ligase EC 6.3.4.3; *systematic name*: formate:tetrahydrofolate ligase (ADP forming); *other name*: formyltetrahydrofolate synthetase; an enzyme involved

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in one of the transformations between **folate coenzymes**. It catalyses the formation of 10-formyltetrahydrofolate from ATP, formate, and tetrahydrofolate with release of ADP and orthophosphate. It is one of three enzymes (*see also* **methylenetetrahydrofolate dehydrogenase (NADP⁺)** and **methenyltetrahydrofolate cyclohydrolase**) necessary for the biosynthesis of purines, thymidylate, methionine, histidine, pantothenate, and formyl tRNA-Met. It is one activity of the trifunctional enzyme **C1-tetrahydrofolate synthase**. *See* **methylenetetrahydrofolate dehydrogenase (NADP⁺)** for an example.

formation constant an alternative name for **stability constant**.

formation function *see* **Bjerrum formation function**.

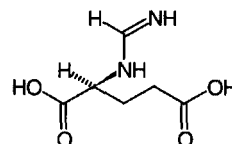
formazan **1** the compound NH₂-N=CH-N=NH. **2** any substitution derivative of this compound. Many are brightly coloured and may be reversibly oxidized into the corresponding colourless tetrazolium compounds (*see also* **Blue Tetrazolium**). Formazans prepared by coupling the phenylhydrazones of acyclic aldoses with diazonium compounds are useful in characterizing such sugars.

formed elements of the blood a general term for the **erythrocytes**, **leukocytes**, and **platelets**.

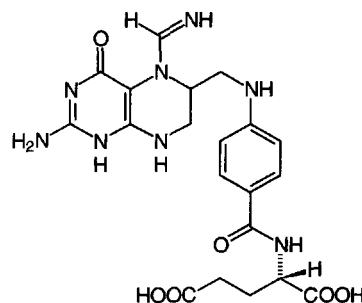
formimidoyl the acyl group, CH(=NH)-, derived from (the hypothetical) formimidic acid; the latter may also be regarded as the imino tautomer, CH(=NH)-OH, of formamide; hence also known as **formimino**.

formimino *see* **formimidoyl**.

formiminoglutamate *abbr.*: FIGLU; *N*-formimidoyl-L-glutamate; formamidino-L-glutarate; an intermediate in the catabolism of histidine. Its formimino (= formimidoyl) group is transferred to tetrahydrofolate to produce **5-formiminotetrahydrofolate**. In folate (or vitamin B₁₂) deficiency FIGLU accumulates in the tissues and its excretion in the urine increases. Estimation of the level of FIGLU in the urine is useful in assessing the extent of folate (or vitamin B₁₂) deficiency.



5-formiminotetrahydrofolate a derivative of tetrahydrofolate that is an intermediate in the metabolism of one-carbon units. It is formed from tetrahydrofolate and **formiminoglutamate**, and with **cyclodeaminase** yields 5,10-methenyltetrahydrofolate.



formin one of several developmental nuclear phosphoproteins found in several tissue types. They are formed by alternative splicing of the mouse *ld* (limb deformity) gene. Mutations in this locus result in defects of growth and patterning of the limb and kidney during embryonic development. Example



formol

from *Mus musculus*: database code FORM_MOUSE, 1468 amino acids (163.62 kDa).

formol (formerly) a proprietary name for formalin.

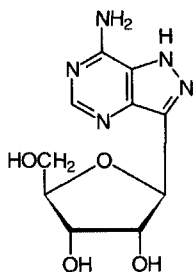
formolize or **formolise** to react the amino groups of amino acids, peptides, or proteins with formaldehyde.

formol titration the pH titration, with alkali, of amino acids, peptides, or proteins in the presence of formaldehyde. The formaldehyde reacts with the amino groups and lowers their apparent pK values, allowing identification of the region in the titration curve where they are being titrated.

formula (pl. **formulas** or **formulae**) **1** (in chemistry) a collection of symbols of atoms and numbers giving information about the composition and structure of a molecular chemical entity. **2** (in mathematics) a general relationship, principle, or rule expressed in symbols, frequently as an equation. **3** a recipe, or prescription, for the preparation of a medicine, food, etc. giving the constituents, their amounts, and the method to be used.

formula weight a former name for **relative formula mass**.

formycin either of two related **nucleoside antibiotics** synthesized by the actinomycete *Nocardia interforma* that display antitumour activity. **Formycin A**; 8-aza-9-deazaadenosine; 7-amino-3-(β-D-ribofuranosyl)-1H-pyrazolo[4,3-d]pyrimidine; an analogue of adenosine that inhibits *de novo* purine biosynthesis in tumour cells and is incorporated into RNA. It is deaminated in erythrocytes to **formycin B**, a competitive inhibitor of NAD⁺ in the action of NAD⁺ ADP-ribosyltransferase and an inhibitor of purine nucleoside phosphorylase.



formycin A

formyl symbol: For or HCO; the acyl group, H-CO-, derived from formic acid.

formylate to introduce (one or more) formyl groups into an organic compound. —**formylation** n.

N-formylmethionine symbol: fMet; codon: AUG (when used as start codon for initiation) or GUG (when present as the first codon; within genes GUG encodes valine); a modified amino-acid residue which is found at the N terminus of nascent polypeptide chains in prokaryotes and bacteriophages. It is also present in the mitochondria and chloroplasts of eukaryotes, supporting the hypothesis that they are derived from prokaryotes.

formylmethionine deformylase EC 3.5.1.31; *systematic name*: N-formyl-L-methionine amidohydrolase; *other name*: **polypeptide deformylase**; an enzyme, found in prokaryotes, that removes the formyl group from the N-terminal N-formylmethionine of many polypeptides produced by translation of mRNA; example from *Escherichia coli*: DEF_ECOLI, 169 amino acids (19.33 kDa). See also **post-translational modification**.

5-formyltetrahydrofolate see **folinic acid**.

5-formyltetrahydrofolate cyclo-ligase EC 6.3.3.2; *other name*: 5,10-methenyltetrahydrofolate synthetase; an enzyme that catalyses the reaction between ATP and 5-formyltetrahydrofolate to form 5,10-methylenetetrahydrofolate with release of ADP and orthophosphate.

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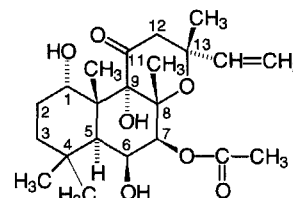
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formyltetrahydrofolate synthetase see **formate-tetrahydrofolate ligase**.

5-formyltetrahydrofolate an alternative name for **folinic acid**.

forskolin colforsin; 7β-acetoxy-8,13-epoxy-1α,6β,9α-trihydroxyl-14-en-11-one; a diterpene isolated from the roots of the Indian labiate plant *Coleus forskohlii*. It stimulates adenylate cyclase, has positive inotropic activity, and relaxes smooth muscle; at higher doses it is hypotensive and vasodilatory. The water-solubility of the active 7-O-hemisuccinyl-7-deacetyl- or (less potent) 7-deacetyl-6-(N-acetylglucyl)- derivatives is useful; the 1,9-dideoxy-derivative is inactive and can be used as a control.



Forssman antibody an antibody against **Forssman antigen**; it causes agglutination of sheep erythrocytes.

Forssman antigen or **Forssman hapten** abbr.: F antigen; a heterophil antigen, present on the tissue cells of many species, including horse, sheep, and mouse, but absent in human and rabbit. The glycolipid hapten (from horse) is a ceramide pentasaccharide, GalNAc(α1-3)GalNAc(β1-3)Gal(α1-4)Gal(β1-4)Glc(β1-1)Cer; when incorporated in liposomes the nonpolar ceramide end anchors the hapten in the bilayer leaving the polar pentasaccharide region accessible to Forssman antibodies.

Förster theory of fluorescence energy transfer a theory giving the rate of dipole-dipole energy transfer in fluorescence as:

$$k_T = 8.71 \times 10^{23} (r^{-6} K^2 J n^{-4} k_F)$$

and the efficiency of transfer as:

$$E = r^{-6} / (r^{-6} + R_0^{-6}),$$

where R_0 , the distance at which transfer efficiency is 50%, is given by:

$$R_0 = 9.71 \times 10^3 (JK^2 Q_0 n^{-4})^{1/6} \text{Å},$$

in which r is the distance between the centres of donor and acceptor chromophore, K^2 is the orientation factor for a dipole-dipole interaction, J is the spectral overlap integral, n is the refractive index of the medium between the donor and acceptor, k_F is the rate constant for fluorescence emission by the energy donor, and Q_0 is the quantum yield of fluorescence of the energy donor in the absence of the acceptor. For energy transfer to be efficient the fluorescence emission spectrum of the donor must overlap the absorption spectrum of the acceptor, i.e. the donor and acceptor must be in resonance, as measured by the spectral overlap integral J , which, in $\text{cm}^3 \text{M}^{-1}$, is:

$$J = (\int F(\lambda) \epsilon(\lambda) \lambda^4 d\lambda) / (\int F(\lambda) d\lambda),$$

where $F(\lambda)$ is the fluorescence intensity in arbitrary units of the energy donor at wavelength λ in cm, and $\epsilon(\lambda)$ is the extinction coefficient of the energy acceptor in $\text{cm}^{-1} \text{M}^{-1}$. The orientation factor, K^2 , is given by:

$$K^2 = (\cos\alpha - 3\cos\beta\cos\gamma)^2,$$

where α is the angle between the donor and acceptor transition moments and β and γ are the angles between the donor moment and the acceptor moment and the line joining the centres.



Förster transfer

tres of the donor and acceptor, respectively. [After T. Förster.]

Förster transfer the nonradiative, resonant energy transfer that occurs when electronic excitation of one chromophore in a molecule, or other structure, elicits fluorescence from a different chromophore located up to 5–7 nm distant. For example, excitation of a tyrosine residue in a protein molecule can lead to fluorescence from a tryptophan residue in the same protein, and excitation of a tryptophan residue can cause fluorescence of dyes attached to the protein surface or to an embedded coenzyme. Resonant energy transfer is important in photosynthesis; many of the chlorophyll molecules in a chloroplast transfer the energy they have absorbed to other chlorophyll molecules at reaction centres.

fortimycin any of several aminoglycoside antibiotics from the actinomycete *Micromonospora olivasterospora*.

FORTAN *abbr.* for formula translator; a high-level computer language much used in numerical applications and engineering and the development of structure refinement software in structural molecular biology.

forward mutation any gene mutation from wild-type to the mutant phenotype, as opposed to a **back mutation**.

forward scattering scattering of radiation in the direction of the beam of radiation, i.e. away from the radiation source.

FOS a gene family encoding nuclear transcription factors; *v-fos* is the oncogene of the FBJ and FBR murine osteosarcoma viruses and Fujinami (avian) sarcoma virus. *FOS* is rapidly (within 5 minutes) but transiently expressed following stimulation of cells by mitogens and also during the onset of differentiation and by wounding. The gene product, FOS, and JUN (from the related gene family *JUN*) form homo- and heterodimers that bind to DNA; the heterodimers bind to the AP-1 consensus site (see *JUN*); FOS homodimers bind to the TPA response element (TRE) if Glu¹⁶⁸ is substituted by Lys. The proteins are members of the **leucine zipper** and **helix-turn helix** superfamilies; five motifs; database code FOS_HUMAN, 380 amino acids (40.65 kDa).

fostomycin another name for **phosphonomycin**.

Fouchet's test a colour test for the presence of bilirubin in the urine. A green colour is produced when the sample is treated with iron (III) chloride solution and trichloroacetic acid.

Fourier analysis the fitting of the terms of a **Fourier series** to a set of periodic data.

Fourier series the expression of any ordinary, single-valued (univalent) periodic function:

$$y = f(x) \quad (1)$$

as the sum of an infinite number of sine waves of successively higher orders. Substituting $n\theta$ for x , to be appropriate to circular functions,

$$y = a_0 + a_1 \cos \theta + a_2 \cos 2\theta + a_3 \cos 3\theta + \dots + b_0 + b_1 \sin \theta + b_2 \sin 2\theta + b_3 \sin 3\theta + \dots \quad (2)$$

The coefficients a_n and b_n can be uniquely determined by equating equations (1) and (2) point by point for successive values of x and solving the simultaneous equations so found. [After Jean Baptiste Joseph Fourier (1768–1830), French mathematician.]

Fourier synthesis the process of computing the form of a function from the values of the coefficients of its **Fourier series**.

FP *abbr.* for flavoprotein.

F6P *abbr.* for fructose 6-phosphate.

F-pilus a type of sex pilus, one or two of which may be present in each cell of 'male' strains of *Escherichia coli*. Each pilus is approximately 8 nm in diameter and up to 20 μ m long. F-pili are consistently associated with the presence in the bacterium of the **F plasmid**, which directs their synthesis; they are believed to serve as conductors of nucleic acids in various circumstances, e.g. during bacterial conjugation.

F plasmid or **F factor** the prototype 'fertility factor' responsible for conjugation in the K-12 strains of *Escherichia coli* used in

the early studies of bacterial mating. The letter F refers specifically to this plasmid and thus should not be used to refer to other naturally occurring **conjugative plasmids**.

F' plasmid or **F' factor** an **F plasmid** derivative incorporating a segment of the bacterial chromosome. An organism in which an F' plasmid has arisen and that carries a chromosomal deletion corresponding to the segment incorporated by the plasmid is termed a **primary F'-containing strain**, and an organism into which an F' plasmid has been transferred is termed a **secondary F'-containing strain**.

FPS *see* **FES/FPS**.

Fr symbol for francium.

fraction **1** (*in chemistry*) any one of several portions of a mixture that can be separated by a fractional process, e.g. by fractional distillation or chromatography, and consisting either of a mixture or of a pure compound. **2** to separate or divide into portions; to **fractionate**.

fraction I–VI (of blood plasma) *see* **Cohn fractionation**.

fractional **1** of, relating to, or being a fraction. **2** of or denoting any process by which parts of a mixture are separated by exploiting differences in their physical properties, e.g. their boiling points, solubility, or chromatographic characteristics.

fractional centrifugation an alternative name for **differential centrifugation**.

fractional crystallization the separation of substances with nearly the same solubilities. It involves a series of recrystallizations at each stage of which the crystals will be enriched in one component and the mother liquor will be enriched in the other component.

fractional distillation the process of separating the components of a mixture by heating it and separately condensing portions of vapour at different temperatures, corresponding to the boiling points of the components of the mixture.

fractional precipitation the stepwise separation of substances, especially macromolecules, from a solution by exploiting differences in their solubilities in various conditions of pH, ionic strength, dielectric constant, etc. Changes in these conditions are effected by the addition of acid, alkali, salt (e.g. ammonium sulfate), ethanol, or other substances.

fractional sterilization an alternative name for **tyndallization**.

fractionate **1** to separate (a mixture) or cause (a mixture) to separate into portions or fractions, e.g. by distillation or crystallization. **2** to obtain (a part of a mixture) by a fractional process. —**fractionation** *n*.

fractionating column a vertical tube or hollow column frequently forming part of an apparatus for **fractional distillation** and usually filled with packing or intersected with perforated plates. An internal reflux takes place during distillation resulting in a gradual separation between the high-boiling-point fractions and the low-boiling-point fractions, the lowest-boiling-point fraction distilling over first.

fraction collector or (*sometimes*) **fraction cutter** an apparatus for the automatic collection of consecutive, commonly equalized, fractions of a flowing liquid, e.g. of the eluate from a chromatographic column. The size of the fractions may be predetermined by volume, mass, time, or number of drops.

fraction cutter **1** the device in a **fraction collector** that separates the flow of eluate into fractions of predetermined size, which then may be individually collected. **2** (*sometimes*) an alternative name for **fraction collector**.

fraction I protein a former name for RuBP carboxylase; *see* **ribulose biphosphate carboxylase**.

fragile X syndrome an X-linked inherited disorder, with mental retardation and connective tissue abnormalities, arising from a **trinucleotide repeat** of CGG.

fragment-antigen binding *see* **Fab fragment**.

fragmentation **1** (*in cell biology*) a the separation of a small piece (i.e. a fragment) of a chromosome from the rest. **b** the division of a cell nucleus by simple splitting. **2** (*in chemistry*) the breakdown of a radical into a diamagnetic molecule and a smaller radical; the breakdown of a radical ion in a mass



fragment-crystallizable

spectrometer to form an ion of lower molecular mass and a radical.

fragment-crystallizable *see* Fc fragment.

fragment ion (*in mass spectrometry*) any ion produced by the loss of one or more pieces from a parent **molecular ion** (or larger fragment ion).

fragmentography *see* molecular fragmentography.

fragmin a Ca^{2+} -sensitive protein, of M_r 43 000 and isolated from *Physarum polycephalum*, that regulates the length of actin filaments *in vitro*.

frame *short name for* reading frame.

frameshifting (*in ribosomal function*) *see* programmed ribosomal frameshifting.

frameshift mutagen any mutagen that is capable of causing frameshift mutations.

frameshift mutation or **phaseshift mutation** any mutation involving the addition or deletion of nucleotide residues in a DNA molecule that results in out-of-phase translation, at positions beyond the site of mutation, of the messenger RNA formed from it. Since the genetic code is read as unpunctuated base triplets, additions or deletions in the message that are not multiples of three bases result in a shift in the reading frame.

frameshift suppressor any of a class of external suppressor mutations that specifically suppress a frameshift mutation by changing (correcting) the phase of translation. Frameshift suppressors are genetic alterations outside the mutated cistron. Some give rise to altered tRNA molecules, **frameshift suppressor tRNAs**, that recognize a base quadruplet instead of a base triplet.

Franck, James (1882–1964) German-born US physicist notable for his work on quantum theory (with G. Hertz), on photochemistry (e.g. the Franck–Condon principle), and on the development of atomic energy (but in opposing the use of nuclear weapons); Nobel Laureate in Physics (1925) jointly with G. Hertz 'for their discovery of the laws governing the impact of an electron upon an atom'.

Franck–Condon principle the principle that the time taken for electronic transitions in molecules is very short in comparison with the period of vibration of the atomic nuclei, with the consequence that the nuclei do not appreciably alter their positions during the emission or absorption of a photon. [After J. Franck and Edward Uhler Condon (1902–74), US physicists.]

Frank, Il'ja Mikhailovich (1908–1990), Russian-born Soviet physicist noted for establishing the theoretical basis of the Cherenkov effect; Nobel Laureate in Physics (1958) jointly with P. A. Cherenkov and I. Y. Tamm 'for the discovery and interpretation of the Cherenkov effect'.

Franklin, Rosalind Elsie (1920–58) British biophysicist, who obtained X-ray data for DNA that were consistent with and provided insight that contributed to the proposal of a helical structure published by J. D. Watson and F. H. C. Crick.

FRAT *abbr. and proprietary name for* free radical assay technique (*see* spin immunoassay).

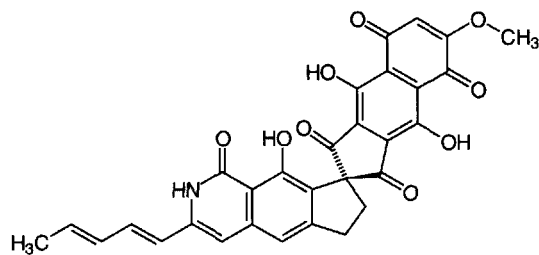
frataxin a 210-residue protein identified from a mutation found in **Friedreich's ataxia** (*abbr.*: FRDA). Some FRDA patients have point mutations in the gene encoding frataxin, but the majority are homozygous for an unstable GAA trinucleotide expansion in the first intron. The function of the protein is unknown.

fraudulent DNA a DNA molecule containing **fraudulent nucleotides**.

fraudulent nucleotide any (artificial) nucleotide that contains a non-naturally occurring base that is an analogue of a purine or pyrimidine found in natural DNA.

frayed end (of DNA) an end region of a double-stranded polynucleotide in which there is imperfect complementarity of the base pairs.

fredericamycin A an antibiotic produced by *Streptomyces griseus* that contains a *spiro* ring system and is active against Gram-positive bacteria and fungi. It is cytotoxic against some transplantable tumours.



fredericamycin A

free (*in chemistry*) not chemically combined (with another substance); not physically bound (to another substance).

free diffusion diffusion in a gas phase or across a boundary between two solutions (or between a solution and pure solvent), as distinct from diffusion across a membrane or some other porous barrier.

free-draining coil a model for polymer molecules in solution that are not well approximated by a solid particle. In this model the polymer is considered as a string of beads through which the solvent streams during viscous flow.

free electrophoresis *an alternative name for* moving boundary electrophoresis (def. 1).

free energy *an alternative name for* Gibbs energy.

free energy change *an alternative name for* Gibbs free energy change.

free fatty acid *abbr.*: FFA; a nonesterified fatty acid; the term is often used of such acids in the blood.

free-flow electrophoresis *an alternative name for* moving boundary electrophoresis (def. 1).

free radical *see* radical (def. 1).

free-radical scavenger *the former term for* radical scavenger.

free rotation sterically unhindered rotation of atoms or groups about a single covalent bond.

freeze-clamp to halt abruptly the metabolic processes in a piece of tissue, or other specimen, by clamping it into a thin, frozen wafer between two aluminium (or other metal) plates, previously cooled in liquid nitrogen. The frozen specimen then may be pulverized and treated with a protein precipitant, e.g. acetone at -70°C . The technique is useful in the determination of the concentrations of various metabolites that otherwise are rapidly consumed by enzymic action continuing during an investigation.

freeze-cleaving *an alternative term for* freeze-fracturing; *see also* freeze-etching.

freeze-dry to dry in a frozen state at very low pressure (i.e. in a high vacuum) so that ice, or other frozen solvent, sublimates rapidly, leaving a porous solid; to lyophilize. Used, e.g. in histochemistry, as frozen biological material can be preserved by removal of water under vacuum without damage to cell structures.

freeze-dryer any apparatus used for freeze-drying.

freeze-etching or **frozen replica method** a technique of preparing, for electron-microscopic examination, a surface that has been exposed by fracturing or cutting a deep-frozen specimen. The sample is rapidly frozen to about -150°C and placed in a vacuum, where it is cleaved or sliced with a cold knife – the process known as **freeze-fracturing**. Etching is then achieved by evaporation of some of the solvent from the surface in a vacuum at about -100°C to produce a textured field, in which nonvolatile macromolecules protrude from a frozen aqueous layer. A replica of the surface is then prepared by shadowing with carbon and/or a heavy metal while the sample is still in the vacuum. This replica is then lifted off, rinsed, and placed



freeze-fracturing

on an electron microscope grid for examination. *See also* **replica formation**.

freeze-fracturing (*in electron microscopy*) a method of visualizing the interior of cell membranes in which cells are frozen at the temperature of liquid nitrogen in the presence of antifreeze and then the frozen block is cracked with a knife blade. The fracture plane often passes through the hydrophobic middle of lipid bilayers, thereby exposing the interior of cell membranes. The resulting fracture faces are shadowed with platinum, the organic material is dissolved away and the replicas floated off for electron microscopy.

freeze-stop to stop a reaction rapidly by freezing the reaction mixture; to **freeze-clamp**.

freeze-thaw to disrupt cells in a suspension, or to disperse emulsions, by repeated freezing and thawing.

freezing 1 the change of a liquid into a solid by a lowering of temperature. 2 the stopping of a reaction or other process at a particular point, often by a lowering of temperature. 3 or **cryopreservation** the use of temperatures below 0 °C in order to preserve biochemicals and foodstuffs, or to maintain viable populations of (micro)organisms.

freezing-microtome a microtome in which the tissue sample is frozen and maintained in a frozen state with liquid carbon dioxide. It is useful for (rapidly) obtaining sections of unfixed soft tissue. *Compare* **cryomicrotome**.

freezing point the temperature at which **freezing** (def. 1) occurs.

French chalk *see* **talc**.

French pressure cell a device to disrupt chloroplasts and bacteria, yeasts, or other cells. A cell or chloroplast suspension is placed in a cylinder fitted with a piston and a narrow outlet controlled by a needle valve. After removal of air, the piston is forced into the cylinder with a hydraulic press to the desired operating pressure (<20 000 pound-force per square inch, 140 MPa). The valve is then opened. Extrusion through the narrow orifice together with the sudden release of pressure creates shear forces that break the suspended particles. *See also* **Hughes press**. [After Charles Stacy French (1907–), US plant physiologist.]

Freon the proprietary name for any of a group of fluorocarbon or chlorofluorocarbon compounds that are clear, colourless liquids, chemically inert, and of low toxicity. They are useful as refrigerants, aerosol propellants, cleaning fluids, and solvents. These substances have been shown to be important in the chemical removal of ozone by chlorine-catalysed cycles in the polar region during winter and spring. Nitric oxide and ice-clouds are formed in the polar stratosphere during winter. Heterogeneous chemical reactions take place on the surface of these particles, dramatically shifting the balance between inactive and active forms of chlorine and increasing the efficiency of chlorine in catalytic ozone removal. The use of Freon is being phased out in many countries; it is being replaced by substances such as tetrafluoroethane (ICI Trademark KLEA) which are claimed not to cause depletion of ozone.

frequency *symbol*: ν or f ; the number of complete cycles of any periodic process occurring in unit time. The SI derived unit of frequency is the **hertz**. The frequency of a wave is equal to its velocity divided by its wavelength.

frequency distribution a table, graph, or histogram constructed from a series of measurements to show the classes into which the measurements are divided and the numbers in each of these classes.

fresnel a non-SI unit of frequency of electromagnetic radiation, equal to 10^{12} Hz.

Fresnel diffraction the diffraction occurring when electromagnetic radiation encounters abrupt discontinuities in refractive index, as in a **Fresnel lens** or in the examination of some specimens in (electron) microscopy. [After Augustin Jean Fresnel (1788–1827), French physicist.]

Fresnel lens a lens whose surface consists of a concentric series of lens sections (or one convex surface and one stepped surface).

This arrangement allows a thin lens of short focal length and large diameter to be easily made, and finds application in searchlights, viewing devices, and overhead projectors.

fret a broad membrane-bound channel interconnecting the grana of **chloroplasts**.

Freund's adjuvant a widely used immunological **adjuvant** (def. 1), used to produce water-in-oil emulsions with aqueous antigens or antigen preparations. It consists of a mixture of mineral oil and emulsifier (**incomplete Freund's adjuvant**) in which killed mycobacteria have been dispersed (**complete Freund's adjuvant**). Omission of mycobacteria may modify the duration, intensity, and nature of the responses. [After Jules Freund (1891–1960).]

Frey-Wyssling particle a double-membrane particle (4–6 μm in diameter) representing a minor component of rubber latex. [After Albert Frey-Wyssling (1900–).]

FRF *abbr.* for follicle-stimulating-hormone-releasing factor; *see* **gonadotropin-releasing hormone**.

FRH *abbr.* for follicle-stimulating hormone releasing hormone; *see* **gonadotropin-releasing hormone**.

friable easily crumbled or pulverized.

friction the resistance to the relative motion of one body sliding, rolling, or flowing over another body with which it is in contact. —**frictional** *adj.*

frictional coefficient or **frictional constant** *symbol*: f ; the frictional force opposing the motion of a body in a fluid medium divided by the velocity of movement. It may be expressed in different ways. The **translational frictional coefficient**, f , for a rigid sphere of radius r moving at low velocity through a medium of viscosity η , is given by: $f = 6\pi\eta r$ (**Stokes' law of viscosity**). For particles of other shapes additional terms must be included. The **rotational frictional coefficient**, f_{rot} , is a parameter relating the velocity of rotation of a rigid sphere of volume V in a fluid medium of viscosity η . It is given by: $f_{\text{rot}} = 6\eta V$. *See also* **frictional ratio**.

frictional force the force exerted on a particle moving through a fluid medium as a result of friction. It is the product of the velocity, relative to the medium, of the particle and its **frictional coefficient**.

frictional ratio the ratio of the experimentally determined translational **frictional coefficient**, f , of a hydrated, anisotropic macromolecule to the calculated frictional coefficient, f_0 , of a nonhydrated spherical macromolecule of the same molecular mass. The ratio f/f_0 gives some indication of the asymmetry of the macromolecule.

Friedreich's ataxia an autosomal recessive inherited disease that involves the central and peripheral nervous systems and the heart. It is the most common hereditary ataxia, and is associated with defects in the gene encoding **frataxin**.

Friend cell a cell of a cultured cell line of leukemic mouse erythroblasts obtained by transformation with Friend leukemia virus. In such cells development is arrested before any globin synthesis occurs, but treatment with certain chemicals, e.g. dimethylsulfoxide, causes a resumption of development and hemoglobin synthesis.

Friend leukemia virus a strain of mouse type C oncovirus that specifically affects immature erythroid cells in mice to cause an erythroleukemia. [After Charlotte Friend (1921–).]

frit filter or **fritted filter** or **fritted glass filter** an alternative name for **sintered glass filter**.

Fromm's method a kinetic approach for studying the mechanism of action of enzyme systems that utilize three substrates. Such systems may be divided into two groups on the basis of their initial rate expressions: the first group is made up of sequential pathways of enzyme and substrate interaction alone, the second is the ping-pong type. The mechanisms in each category may further be separated either by initial rate measurements or from studies with competitive inhibitors. [After Herbert Jerome Fromm (1929–), US biochemist.]

frontal analysis any analytical technique by which the progress of the separation of components of a mixture, as by

frontal chromatography

frontal chromatography, moving boundary electrophoresis, or analytical ultracentrifugation, is followed, and the relative proportions of the components determined, by examination of the individual boundaries, or fronts, formed by each component with the solvent.

frontal chromatography a technique for chromatographic separation in which the sample (liquid or gas) is fed continuously into the chromatographic bed and only the component that emerges first is obtained in pure form.

fronting (*in chromatography*) the asymmetry of a peak such that, relative to the baseline, the front is less steep than the rear. In paper chromatography and thin-layer chromatography it refers to the distortion of a zone showing a diffuse region preceding the zone in the direction of flow.

frontside attack the mechanism of a chemical displacement reaction in which the configuration is retained.

frozen replica method an alternative name for freeze-etching.

FRS *abbr.* for Fellow of the Royal Society (of London).

FRSE *abbr.* for Fellow of the Royal Society of Edinburgh.

Fru symbol for a residue (or sometimes a molecule) of the ketohexose **fructose**. In the condensed system of symbolism of sugar residues, Fru signifies specifically a fructose residue with the common configuration and ring size, i.e. D-fructofuranose.

fructan any **glycan** composed only of D-fructose residues. Fructans occur commonly as food reserves in the roots, stems, leaves, and seeds of many plants, e.g. in tubers of dahlias and Jerusalem artichokes. There are two main types: (1) inulin-type, in which the fructofuranose residues are linked by $\beta 2 \rightarrow 1$ glycosidic linkages; and (2) levan-type, in which the linkages are $\beta 2 \rightarrow 6$, which were formerly referred to as **fructosans**.

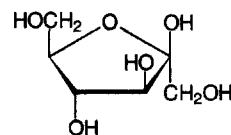
β -fructofuranosidase EC 3.2.1.26; *other names*: invertase; saccharase; an enzyme that catalyses the hydrolysis of terminal nonreducing β -D-fructofuranoside residues in β -D-fructofuranosides, e.g. sucrose. Examples: precursors of invertases types 1, 2 and 4 from yeast: database code INV1_YEAST, 532 amino acids (60.50 kDa); INV2_YEAST, 532 amino acids (60.57 kDa); INV4_YEAST, 532 amino acids (60.51 kDa).

fructokinase 1 the recommended name for the enzyme ATP:D-fructose 6-phosphotransferase; EC 2.7.1.4; this catalyses the phosphorylation of D-fructose by ATP to form D-fructose 6-phosphate and ADP. The enzyme occurs in plants and in schistosomes, and is highly specific for D-fructose. In bacteria it is important in fructose utilization. Structurally it is similar to other carbohydrate kinases. Example from *Zymomonas mobilis*: database code SCRK_ZYMMO, 301 amino acids (32.57 kDa). **2** an alternative name for **ketohehexokinase** (EC 2.7.1.3). This usage is misleading since ketohehexokinase yields D-fructose 1-phosphate.

fructosaccharase a trivial name for any **saccharase** that attacks the fructose-containing end of appropriate oligosaccharides, liberating fructose.

fructosan 1 (formerly) a trivial name for 2,6-anhydrofructofuranose. **2** an (older) incorrect name for **fructan**.

fructose symbol: Fru; the trivial name for the ketohexose arabino-2-hexulose; there are two enantiomers. D-(+)-fructose (symbol: D-Fru), commonly known as fructose, and formerly known as **fruit sugar**, is levorotatory, hence formerly known also as **levulose** or (*esp. UK*) **laevulose**; it is the sweetest of the sugars. Aqueous solutions of D-fructose at room temperature contain an equilibrium mixture of β -D-fructopyranose, β -D-fructofuranose, α -D-fructofuranose, and the open-chain form, whereas in crystals the sugar is present entirely as β -D-fructopyranose. Combined D-fructose is almost invariably in the furanose form, however. D-fructose is found free in a large number of fruits, in honey, and (as the sole sugar) in bull and human semen. In combined form it occurs at one residue per molecule of the disaccharide **sucrose** and the trisaccharides **gentianose**, **molezitose**, and **raffinose**, and as the sole component of the polysaccharide **fructans**.



β -D-fructofuranose

fructose-bisphosphatase EC 3.1.3.11; *systematic name*: D-fructose-1,6-bisphosphate 1-phosphohydrolase; *other names*: hexosediphosphatase; (formerly) fructose-1,6-diphosphatase. A phosphoric monoester hydrolase that hydrolyses D-fructose 1,6-bisphosphate to D-fructose 6-phosphate and orthophosphate. It is important in **gluconeogenesis** and in regulation of the **reductive pentose phosphate cycle** in chloroplasts. Example from rat: database code F16P_RAT, 362 amino acids (39.43 kDa); six motifs.

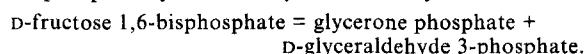
fructose-2,6-bisphosphatase an enzyme that hydrolyses fructose 2,6-bisphosphate to fructose 6-phosphate; it is the same protein as phosphofructokinase-2. *See* **phosphofructokinase** (def. 2).

fructose-1,6-bisphosphatase deficiency *see* **hereditary fructose-1,6-bisphosphatase deficiency**.

fructose 1,6-bisphosphate or (formerly) **fructose 1,6-diphosphate** *abbr.*: FBP; the D enantiomer is a metabolic intermediate in **glycolysis** and **gluconeogenesis**.

fructose 2,6-bisphosphate the D enantiomer is an important regulator of the **glycolytic** and **gluconeogenic pathways**. It inhibits fructose 1,6-bisphosphatase and activates phosphofructokinase, and its concentration is controlled by the **phosphofructokinase** (def. 2) and fructose 2,6-bisphosphatase activities of a 49 kDa protein.

fructose-bisphosphate aldolase EC 4.1.2.13; *systematic name*: D-fructose-1,6-bisphosphate D-glyceraldehyde-3-phosphate lyase; *other names*: aldolase; fructose-1,6-bisphosphate triosephosphate-lyase. An enzyme that catalyses the reaction:



It is a universally distributed enzyme that functions in the pathways for both the catabolism and biosynthesis of glucose. Example from *Drosophila melanogaster*: database code NRL_1FBAA, 360 amino acids (38.87 kDa); 3-D structure known.

fructose-2,6-bisphosphate 2-phosphatase EC 3.1.3.46; *other name*: fructose-2,6-bisphosphatase; an enzyme that catalyses the hydrolysis of D-fructose 2,6-bisphosphate to D-fructose 6-phosphate and orthophosphate, being part of the system for regulating the concentration of fructose 2,6-bisphosphate. This enzyme is the phosphorylated form of the protein. If dephosphorylated, it functions as a 2-kinase for fructose 6-phosphate. Example, human liver isoform: database code F26L_HUMAN, 471 amino acids (54.64 kDa). *See also* **phosphofructokinase** (def. 2).

fructose-1,6-diphosphatase a former name for **fructose-bisphosphatase**.

fructose 1,6-diphosphate a former name for **fructose 1,6-bisphosphate**.

fructose disease *see* **essential fructosuria**, **hereditary fructose intolerance**, **hereditary fructose-bisphosphatase deficiency**.

fructose intolerance an alternative name for **essential fructosuria**.

fructose 6-phosphate *abbr.*: F6P; the D enantiomer is an important intermediate in **glycolysis**, **gluconeogenesis**, and **fructose metabolism**.

fructosuria the presence of abnormal amounts of fructose in the urine. *See also* **essential fructosuria**.

Fru symbol for a fructofuranose residue.

fruit the structure that develops from the fertilized ovary of a



fruit fly

flowering plant and contains the ripening seed or seeds. It may be fleshy or dry; the term is often extended to include other associated parts such as a fleshy receptacle.

fruit fly a common name for species of flies belonging to the genus *Drosophila*.

fruit-ripening hormone *see ethylene*.

fruit sugar a former name for the D enantiomer of **fructose**.

FSF *abbr. for* fibrin-stabilizing factor (i.e. factor XIIIa); *see blood coagulation*.

FSH *abbr. for* follicle-stimulating hormone.

FSH-RF *abbr. for* follicle-stimulating hormone releasing factor; *see gonadotropin-releasing hormone*.

FSH-RH *abbr. for* follicle-stimulating hormone releasing hormone; *see gonadotropin-releasing hormone*.

F₁ sphere a submitochondrial structure containing the catalytic site for ATP synthesis.

F-strain a bacterial strain containing an **F** plasmid.

F-type pentose phosphate pathway *see pentose phosphate pathway*.

ftz the gene for Fushi Tazaru protein, a nuclear DNA-binding homeobox segmentation protein from *Drosophila melanogaster*. Database code HMFT_DROME, 413 amino acids (46.82 kDa).

FU *abbr. for* fluorouracil (use proscribed; the symbol FURa is recommended).

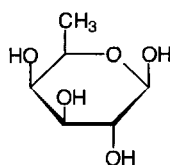
Fuc *symbol for* a residue (or sometimes a molecule) of the deoxyaldohexose **fucose**. In the condensed system of symbolism of sugar residues, Fuc signifies specifically a fucose residue with the common configuration and ring size, i.e. L-fucopyranose.

fucan a **glycan** composed only of fucose residues.

fucoidan or **fucoidin** any of a group of complex branched, sulfated heteropolysaccharides that are composed of residues of L-fucose (usually sulfated) together with small numbers of residues of various other monosaccharides. Fucoidans occur in the cell walls, intercellular matrix, and exuded mucilage of many (especially intertidal) species of marine algae, in which they appear to assist in water retention during periodic exposure.

fucosamine 2-amino-2,6-dideoxygalactose; an aminodeoxy-sugar, both enantiomers of which occur in lipopolysaccharides of *Salmonella* and other bacterial genera.

fucose *symbol:* Fuc; *the trivial name for* 6-deoxygalactose; there are two enantiomers. **D-Fucose** (*symbol:* D-Fuc), 6-deoxy-D-galactose, occurs in glycosides in various plants belonging to the Convolvulaceae. **L-Fucose** (*symbol:* L-Fuc), 6-deoxy-L-galactose, occurs in polysaccharides (fucans) in seaweeds, especially *Fucus* spp., and in the cell wall matrix of higher plants. In animals fucosyl residues occur in a number of glycoproteins and glycolipids; the presence of α -1,4-linked L-fucosyl residues at specific positions in the highly branched oligosaccharides of human erythrocyte surface antigens is controlled by the *H* and *Le* blood-group genes.



β -D-fucose

fucose-1-phosphate guanylyltransferase EC 2.7.7.30; *other name:* GDPfucosepyrophosphorylase; an enzyme that catalyses the formation of GDP-L-fucose from GTP and L-fucose 1-phosphate with release of pyrophosphate.

α -fucosidase any of several enzymes that hydrolyse α -fucosides. 1 α -fucosidase, EC 3.2.1.51; *systematic name:* α -L-fucoside fucohydrolase; it catalyses the hydrolysis of an

α -L-fucoside to an alcohol and L-fucose. Example from human: database code FUCO_HUMAN, 461 amino acids (53.10 kDa). 2 1,3- α -L-fucosidase, EC 3.2.1.111; *other names:* almond emulsin; fucosidase I; an enzyme that catalyses the hydrolysis of 1,3-linkages between α -L-fucose and *N*-acetylglucosamine residues in glycoproteins. 3 1,6- α -L-fucosidase, EC 3.2.1.127; an enzyme that catalyses the hydrolysis of 1,6-linkages between α -L-fucose and *N*-acetyl-D-glucosamine in glycopeptides such as immunoglobulin G glycopeptide and fucosyl-asialo-agalacto-fetuin.

fucosidosis an inherited human metabolic disorder in which there is an inability to remove α -L-fucosyl residues from various glycoproteins and glycolipids. It is subject to autosomal recessive inheritance.

FuDR or **FUDR** *abbr. for* fluorodeoxyuridine (use deprecated; the symbol FdUrd is recommended).

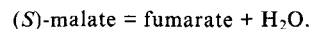
fugu toxin *see tetrodotoxin*.

full agonist an **agonist** that is able to elicit the maximal response of a tissue even when the tissue has spare receptors. In such a situation, several different agonists may be able to elicit the same maximal response, albeit at different receptor occupancies. Moreover, a full agonist in one tissue may be a **partial agonist** in another.

fuller's earth a naturally occurring, porous, absorptive aluminium-magnesium silicate clay. It is useful for decolorizing oils and other liquids and as a filtering medium. It is so-called because it was originally used to full (i.e. cleanse and thicken) wool and woollen cloth.

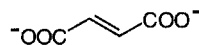
fully overlapping code *see overlapping code*.

fumarase EC 4.2.1.2; *recommended name:* fumarate hydratase. An enzyme of the **tricarboxylic-acid cycle** that catalyses the reaction:



There are two different enzymes in facultative bacteria (e.g. *Escherichia coli*). Example human mitochondrial enzyme, homotetramer: database code FUMH_HUMAN, 467 amino acids (50.20 kDa).

fumarate 1 *the trivial name for* *trans*-1,2-ethylenedicarboxylate, (*E*)-butenedioate, the dianion of fumaric acid; the diastereoisomer of **maleate** and a key intermediate in metabolism. In the tricarboxylic-acid cycle it is formed from succinate by **succinate dehydrogenase**, and is converted to malate by **fumarase**. 2 any mixture of fumaric acid and its mono- and dianions. 3 any salt or ester of fumaric acid.



fumarate hydratase *see fumarase*.

fumaroyl the bivalent *trans*-acyl group, $-\text{CO}-\text{CH}=\text{CH}-\text{CO}-$, derived from fumaric acid.

fumaryl the univalent *trans*-acyl group, $\text{HOOC}-\text{CH}=\text{CH}-\text{CO}-$, derived from fumaric acid.

fumed silica a preparation of pure silicon dioxide as an extremely fine, white powder with an external surface area of up to $400 \text{ m}^2 \text{ g}^{-1}$. It is useful for thickening and forming thixotropic gels. Fumed silica is used as a suspending and gelling agent in liquid-scintillation counting; a 3–5% concentration in dioxan, toluene, or xylene-based liquid scintillators will form an almost transparent, rigid gel, capable of holding an insoluble solid sample in suspension. *Proprietary name:* Cab-O-Sil.

fume hood a work space contained under a hood from which air is extracted to remove harmful vapours.

function 1 (*in chemistry*) a **functional group**. 2 (*in mathematics*) a either of two variables so related to each other that for any value of one variable there is a corresponding value of the other. b a variable or expression that can take a set of values



functional antagonism

each of which is associated with the magnitude of an independent variable or variables. **3** (*in physiology*) the normal vital activity of a cell, organ, or tissue within an organism. —**functional adj.**

functional antagonism **1** antagonism arising by the antagonist's producing an opposite effect to the substance being antagonized through the antagonist's action at a different receptor. **2** antagonism arising when the antagonist interferes with events that follow receptor activation by the substance being antagonized.

functional cloning a cloning strategy in which knowledge of a gene's product (function) is used to clone the gene. An example is the gene for the β chain of hemoglobin responsible for sickle-cell anemia. This is in contrast to **structural cloning** where the function of the gene is not known. A common procedure would be to construct a DNA oligonucleotide based on the amino-acid sequence of the peptide and use this as a probe to isolate the gene from a DNA library. In cases where one protein is dominant in a cell, such as hemoglobin in the reticulocyte, the preparation of a cDNA from the mRNA also provides a useful probe for gene isolation.

functional group or function (*in chemistry*) any **group** (def. 1) in a molecule of an organic compound that confers characteristic chemical and physical properties on the molecule and that may be considered to have replaced a hydrogen atom in a molecule of the corresponding hydrocarbon, e.g. the hydroxyl group in an alcohol.

fungicide any biological, chemical, or physical agent able to destroy fungi. —**fungicidal adj.**

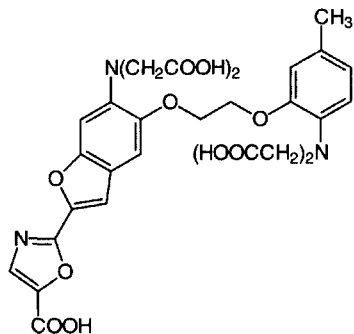
fungistat any drug, antibiotic, or other agent that inhibits the growth and multiplication of fungi. —**fungistatic adj.**

fungus (*pl. fungi*) any member of the taxonomic kingdom Fungi, a group of diverse and widespread unicellular, multicellular, or cenocytic eukaryotic organisms that do not contain chlorophyll. Fungi may be parasitic, saprophytic, or symbiotic. They reproduce through spores. *See also* **microfungus**. —**fungal adj.**

FUR *abbr. for* fluorouridine (use proscribed; the symbol FUrD is recommended).

FUra *symbol for* (a residue of) the synthetic pyrimidine base fluorouracil.

Fura 2 1-[2-(5-carboxyoxazol-2-yl)-6-aminobenzofuran-5-oxyl]-2-(2'-amino-5'-methylphenoxy)-ethane-*N,N,N'*-tetraacetic acid; a highly sensitive calcium ion chelator that exhibits a spectral shift on binding Ca^{2+} . Calcium ion concentration can be measured directly, independent of dye concentration, by measuring the ratio of fluorescence at 510 nm after excitation at 340 nm and then at 380 nm. It is frequently used as the acetoxymethyl ester (Fura 2/AM, Fura 2-pentakis(acetoxymethyl)ester), which is a membrane-permeable derivative that can enter cells, after which it is rapidly hydrolysed by cytoplasmic esterases to Fura 2, which remains trapped within the cell and can thus report the intracellular Ca^{2+} concentration. *See also* **Quin-2**.



furan the trivial name for oxa-2,4-cyclopentadiene, $\text{C}_4\text{H}_4\text{O}$. *See also* **furanose**.

furanoid furan-like, i.e. describing a molecular structure that consists of or contains a ring of four carbon atoms and one oxygen atom (as in furan).

furanose *symbol: f*; any monosaccharide or monosaccharide derivative whose molecule contains a furanoid ring. *Compare* **pyranose**, **septanose**.

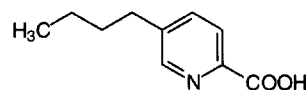
furanoside any glycoside in which the sugar moiety is in the furanose form.

FUrD *symbol for* (a residue of) the ribonucleoside fluorouridine (*see* **fluorouracil**).

furin the mammalian counterpart of **kexin**. Example from human: database code FUR1_HUMAN, 794 amino acids (86.58 kDa).

furyl either 2-furyl or 3-furyl, the two oxa-2,4-cyclopentadienyl groups derived from a molecule of **furan** by loss of a hydrogen atom from C-2 and C-3 respectively.

fusaric acid 5-butylpyridine-2-carboxylic acid; an antibiotic originally isolated from the fungus *Fusarium heterosporium* and obtained also from *Gibberella fujikuroi*. It inhibits **dopamine β -hydroxylase** and has hypotensive activity.

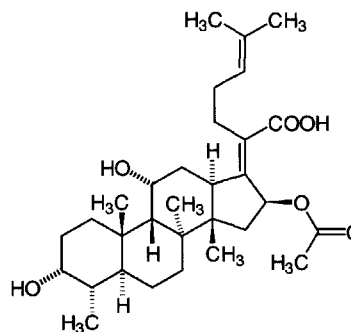


fused ring (*in chemistry*) any ring of atoms in a molecule that is joined to another ring of atoms in such a way that two atoms are common to both rings.

fused rocket immunoelectrophoresis a modification of **rocket immunoelectrophoresis** designed to follow the distribution of proteins among the fractions obtained in a separation experiment. Small volumes of fractions obtained in the separation experiment are placed in a row of sample wells in an antibody-free agarose gel and the samples are allowed to diffuse into the gel for 30–60 min. The proteins are then forced at right angles into an antibody-containing gel by electrophoresis; the resulting precipitation lines (rockets) show the elution profiles for each individual protein in the separation experiment and the distance between the precipitation line and the corresponding well is proportional to the amount of antigen precipitated.

fusel oil a colourless, oily liquid of variable composition obtained as a by-product in the distillation of alcoholic fermentation liquors. It contains chiefly isopentanol, 2-methyl-1-butanol, isobutanol, and propanol. It is considerably more toxic than ethanol. [From German *fuseln*, to bungle.]

fusidic acid $3\alpha,11\alpha,16\beta$ -trihydroxy- $4\alpha,8,14$ -trimethyl- 18 -nor- $5\alpha,8\alpha,9\beta,13\alpha,14\beta$ -cholesta- $17(20),24$ -dien-21-oic acid 16-acetate; a steroid antibiotic, structurally related to cephalosporin P (*see* **cephalosporin**) and isolated from the fermentation broth of



fusiform

Fusidium coccineum. Fusidic acid is active particularly against Gram-positive organisms, especially *Staphylococcus* spp. It prevents translocation during protein synthesis and inhibits the accumulation of ppGpp.

fusiform spindle-shaped; elongated and tapering at both ends.

fusion 1 the act or process of melting. 2 the act or process of melting together or uniting. 3 the process whereby two membranes are joined together, as in **cell fusion**. 4 the process whereby nuclei of light elements are united to form nuclei of heavier elements with the release of energy: **nuclear fusion**. 5 the state of being fused. 6 something produced by fusion. 7 (in molecular biology) the act, process, or result of artificially linking genes that code for two different proteins, with the aim of generating a **fusion protein**.

fusion protein an expression product resulting from the fusion of two genes. Such a protein may be produced, e.g., in recombinant DNA expression studies or, naturally, in certain viral **oncogenes** in which the oncogene is fused to **gag**. Their production sometimes results from the need to place a cloned eukaryotic gene under the control of a bacterial promoter for expression in a bacterial system; sequences of the bacterial system are then frequently expressed linked to the eukaryotic protein. Fusion proteins are used for the analysis of structure, purification, function, and expression of heterologous gene products.

fusogen any agent, or set of conditions, that gives rise to fusion of membranes, including cell membranes (and hence of cells). Killed Sendai virus was formerly much used for this purpose but now polyethylene glycol is commonly used as a

fusogen particularly in the preparation of hybridomas for the production of monoclonal antibodies. —**fusogenic** *adj.*

futile cycle any metabolic cycle that, if not controlled, acts as an ATPase and hence converts chemical energy (stored as ATP) to heat. For example, a coupling of phosphofructokinase, which converts fructose 6-phosphate to fructose 1,6-bisphosphate at the expense of ATP, with fructose-bisphosphatase, which hydrolyses fructose 1,6-bisphosphate to fructose 6-phosphate and (inorganic) phosphate, can form a cyclic process whose net effect is merely the hydrolysis of ATP to ADP and phosphate. A futile cycle may be an important control mechanism in some physiological conditions.

fuzzy coat or **fuzzy layer** the indistinct layer seen on electron microscopy outside the cell coat of various eukaryotic cells, e.g. cells of mammalian intestinal epithelium and a number of single-celled amoebae. It is composed principally of glycoproteins.

Fv fragment the N-terminal part of the **Fab fragment** of an immunoglobulin molecule, consisting of the variable portions of one light chain and one heavy chain.

FYN a gene family encoding nonreceptor tyrosine kinases of the Src family (see **src**) that are implicated in the control of cell growth. The gene product, p59^{lyn}, associates with the p85 subunit of **1-phosphatidylinositol 3-kinase**. The proteins undergo myristoylation and phosphorylation on serine and tyrosine residues; overexpression transforms fibroblasts. Example, database code KFYN_HUMAN, 536 amino acids (60.56 kDa). The alternative gene symbols are *SYN* (from *src/yes*-related novel gene) and *SLK*.

Gg

g symbol for 1 gram. 2 gaseous state. 3 gluon.

g 1 symbol for acceleration of free fall (formerly known as acceleration due to gravity). 2 *abbr. for gauche*.

g_n symbol for standard acceleration of free fall, a fundamental physical constant equal by definition to 9.806 65 m s⁻².

G symbol for 1 giga+ (SI prefix denoting 10⁹ times). 2 gauss (alternative to Gs). 3 a residue of the α-amino acid glycine (alternative to Gly). 4 a residue of the base guanine in a nucleic acid sequence. 5 a residue of the ribonucleoside guanosine (alternative to Guo). 6 a residue of the monosaccharide glucose (alternative to Glc). 7 ganglioside.

G symbol for 1 Gibbs energy. 2 gravitational constant. 3 weight (alternative to *W* or *P*). 4 electric conductance.

Ga symbol for gallium.

Ga₂ symbol for galabiose (alternative to GaOse₂).

Ga₃ symbol for galatriose (alternative to GaOse₃).

GA *abbr. (sometimes) for* 1 glyceric acid. 2 glutamic acid. 3 glucuronic acid.

GAA a codon in mRNA for L-glutamic acid.

GABA *abbr. for* γ-amino-*n*-butyric acid or γ-aminobutyrate.

GABAergic 1 describing a nerve or other cell that is acted upon by γ-aminobutyrate. 2 describing nerves that act by releasing γ-aminobutyrate.

GABA-gated channel *see* GABA receptor.

GABA-modulin a brain protein that down-regulates the high-affinity binding site for γ-amino-*n*-butyric acid (GABA) on synaptic membranes.

GABA receptor any of several membrane proteins that bind γ-aminobutyrate (GABA) and mediate its effects as an inhibitory neurotransmitter. Two main types have been distinguished: GABA_A receptors, known also as GABA-gated channels, which function as a Cl⁻ channel; and GABA_B receptors, which are G-protein-coupled receptors. **Baclofen** is a potent GABA mimetic that is selective for the GABA_B receptor; **muscimol** is its counterpart for the GABA_A receptor. The GABA_A receptor is multimeric, with α (types α1–α6), β (types β1–β4), γ (types γ1–γ3), δ, and ρ subunits. Examples (all precursors, human): α₁, database code GAA1_HUMAN, 456 amino acids (51.71 kDa); β₁, database code GAB1_HUMAN, 474 amino acids (54.18 kDa); γ₂, database code GAC2_HUMAN, 467 amino acids (54.12 kDa); ρ₁, database code GAR1_HUMAN, 473 amino acids (55.19 kDa). Subunits α and β have four putative transmembrane segments, with a domain distribution closely similar to that seen in the nicotinic acetylcholine receptor (nAChR) (*see* **cholinoceptor**), with some homology between specific regions of the two receptor types: for the α subunit with the M1 region of nAChR, and for the β subunit with regions M1–M3. Increased Cl⁻ ion flux brings about hyperpolarization of an excitatory neuron, which inhibits it from firing on receipt of a normal impulse. **Benzodiazepine** drugs interact with GABA_A receptors; co-expression studies with cloned subunit variants have indicated that the γ subunit confers benzodiazepine sensitivity. The mode of action of GABA_B receptors remains unknown. Indications variously suggest they may be coupled through G-proteins to stimulation of outward K⁺ current, inhibition of inward Ca²⁺ channels, inhibition of adenylate cyclase, and stimulation of the phosphatidylinositol–Ca²⁺ system. *See also* **chloride channel**.

gabase an informal name for aminobutyrate aminotransferase (EC 2.6.1.19).

GABA shunt *abbr. for* γ-aminobutyrate shunt.

GABA transaminase *see* 4-aminobutyrate transaminase.

GAC a codon in mRNA for L-aspartic acid.

G-actin the monomeric form of **actin**.

Gaddum equation a relationship, suggested to replace the **Hill–Langmuir equation**, that describes the situation when two

ligands, A and B, are in equilibrium with a common binding site. It is:

$$P_{AR} = [A]/K_A(1 + [B]/K_B) + [A],$$

where P_{AR} is the proportion of the binding sites occupied by ligand A; the square brackets indicate concentrations. *Compare* **Schild equation**. [After (Sir) John Henry Gaddum (1900–67), British pharmacologist.]

gadelaide the trivial name for (*E*)-eicos-9-enoate, CH₃–[CH₂]₉–CH=CH–[CH₂]₇–COO⁻ (*trans* isomer), the anion derived from gadelaideic acid, (*E*)-eicos-9-enoic acid. *Compare* **gadoleate**.

gadoleate the trivial name for (*Z*)-eicos-9-enoate, CH₃–[CH₂]₉–CH=CH–[CH₂]₇–COO⁻ (*cis* isomer), the anion derived from gadoleic acid, (*Z*)-eicos-9-enoic acid. *Compare* **gadelaide**.

gag a gene that encodes the glucosaminoglycan core protein for a **retrovirus**. It is named from group-specific antigen gene, or glucosaminoglycan gene. Example from avian erythroblastosis virus: database code AEVGAG, 962 amino acids (108.19 kDa). *See also* **arba**, **erbb**.

GAG 1 a codon in mRNA for L-glutamic acid. 2 *abbr. for* glucosaminoglycan.

Gaia a theory of the role of biota in the maintenance of a climatic homeostasis, proposed by James Lovelock [(1919–), British biochemist] in 1979. The original concept, and still the heart of the philosophy, is that all of the living organisms that inhabit the Earth can be regarded as a single vast organism capable of manipulating the atmosphere, geosphere, and hydrosphere to suit its needs. Lovelock named this organism Gaia, after the Greek goddess of the earth.

Gal symbol for a residue (or sometimes a molecule) of the aldohexose galactose. In the condensed system of symbolism of sugar residues, it signifies specifically a galactose residue with the common configuration and ring size, i.e. D-galactopyranose.

GAL4 a gene involved in the regulation of galactose utilization in yeast. The GAL4 protein (database code GAL4_YEAST, 881 amino acids (99.39 kDa)) has a zinc-finger motif for binding to its upstream activator sequence (UAS), which it then activates, and a site for binding the complex of the GAL80 protein – encoded by the *GAL80* gene – and galactose. When this is bound, activation by GAL4 protein is prevented.

Gala symbol for a residue (or sometimes a molecule) of the uronic acid galacturonic acid.

galabiose symbol: Ga₂ or GaOse₂; the trivial name for the disaccharide Gal(α1-4)Gal.

galactan any glycan composed solely of galactose residues. The galactans thus form a subgroup of plant hemicelluloses.

galactitol or **meso-galactitol** or (formerly) **dulcitol** the trivial name for the hexitol formally derived by reduction of the aldehyde group of either D- or L-galactose. It is optically compensated internally and therefore does not require D or L as a configurational descriptor; the prefix *meso-* may however be added for clarity. It occurs widely especially in seaweeds and higher plants.

galacto- prefix (in chemical nomenclature) indicating a particular configuration of a set of (usually) four contiguous CHOH groups, as in the acyclic form of D- or L-galactose. *See* **monosaccharide**.

galactoceramide *see* **galactosylceramide**.

galactocerebroside *see* **galactosylceramide**.

galactokinase EC 2.7.1.6; *systematic name*: ATP:D-galactose 1-phosphotransferase. An enzyme that catalyses the phosphorylation by ATP of D-galactose to D-galactose 1-phosphate, with release of ADP; this is the first step in the metabolism of

galactolipid

galactose. Example from *Escherichia coli*: database code GAL1_ECOLI, 381 amino acids (41.26 kDa).

galactolipid any **glycolipid** containing one or more residues of galactose and/or *N*-acetylgalactosamine. The term includes **galactosylceramide** and **galactosyldiacylglycerol**.

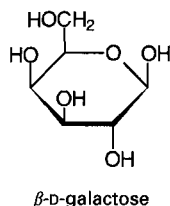
galactomannan any heteroglycan containing D-galactose and D-mannose residues; the structures have not been fully elucidated. They occur in seeds of various plants and trees, probably as storage polysaccharide. *See also guaran*.

galactoprotein A *abbr.*: Gap A; *an alternative name for fibronectin*.

galactosaemia a variant spelling (*esp. Brit*) of **galactosemia**.

galactosamine *symbol*: GalpN; *the trivial name for the amino-deoxysugar 2-amino-2-deoxygalactopyranose*; there are two enantiomers, D- and L-galactosamine. D-Galactosamine (*symbol*: D-GalpN), formerly known as **chondrosamine**, is a constituent (commonly as its *N*-acetyl derivative) of some glycolipids and of chondroitin sulfate, dermatan sulfate, and many bacterial glycosaminoglycans.

galactose *symbol*: Gal; *the trivial name for the aldohexose galacto-hexose*; there are two enantiomers, D- and L-galactose. D-Galactose (*symbol*: D-Gal), commonly known as galactose, and formerly known as **brain sugar** or **cerebrose**, rarely occurs free but is widely distributed in combined form in plants, animals, and microorganisms as a constituent of many oligo- and polysaccharides; it occurs also in **galactolipids** and as its glucoside in **lactose** and **melibiose**. Residues of both D-galactose and 3,6-anhydro-L-galactose (*symbol*: 3,6An-L-Gal) occur in the disaccharide **agarobiose** and in polysaccharides such as **agaropectin** and **agarose**.



galactose/glucose binding protein a signal transducing protein for bacterial chemotaxis. Example from *Escherichia coli*: database code NRL_2GBP, 309 amino acids (33.33 kDa); 3-D structure known.

galactosemia or (*esp. Brit.*) **galactosaemia** comparatively rare condition in humans in which there is a supranormal concentration of D-galactose in the blood and a specific inability to metabolize galactose. It is caused either by (1) a hereditary autosomal recessive trait in which there is a virtual absence of **UDPGlucose-hexose-1-phosphate uridylyltransferase** activity; or less frequently (2) an inherited autosomal recessive trait in which there is a deficiency in **galactokinase** activity.

galactose oxidase EC 1.1.3.9; a copper-containing enzyme that catalyses the oxidation by dioxygen of D-galactose to D-galacto-hexodialdase and H₂O₂. Example from the fungus *Cladobotryum dendroides*: database code NRL_1GOG, 639 amino acids (68.44 kDa); 3-D structure known.

galactose-1-phosphate uridylyltransferase *see* **UDPGlucose-hexose-1-phosphate uridylyltransferase**.

galactose-6-sulfate sulfatase *see* ***N*-acetylgalactosamine-6-sulfatase**.

galactosidase either of two types of enzyme: **α-galactosidase** EC 3.2.1.22; *systematic name*: α-D-galactoside galactohydrolase; *other name*: melibiase. An enzyme that hydrolytically removes terminal nonreducing α-D-galactose residues in α-galactosides, including galactose oligosaccharides, galactomannans, and galactolipids. Example from *Escherichia coli*: database code AGAL_ECOLI, 451 amino acids (50.60 kDa).

β-galactosidase EC 3.2.1.23; *systematic name*: β-D-galactoside

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galactohydrolase; *other name*: lactase. An enzyme that removes terminal β-D-galactose residues in β-galactosides by hydrolysis. Example, *lacZ* gene product of *E. coli*: homotetramer, similar to other glycosyl hydrolases, database code BGAL_ECOLI, 1023 amino acids (116.22 kDa).

galactoside any **glycoside** in which the sugar moiety is galactose.

galactoside acetyltransferase EC 2.3.1.18; *recommended name*: galactoside *O*-acetyltransferase; *systematic name*: acetyl-CoA:β-D-galactoside 6-acetyltransferase; *other name*: thiogalactoside transacetylase. An enzyme that catalyses the acetylation by acetyl-CoA of β-D-galactoside to 6-acetyl-β-D-galactoside with release of GDP. Such an enzyme is the *lacA* gene product in *Escherichia coli* (*see lac operon*). Its function remains unclear. Example from *E. coli*: database code THGA_ECOLI, 203 amino acids (22.80 kDa).

galactoside 3(4)-L-fucosyltransferase EC 2.4.1.65; *systematic name*: GDP-L-fucose:β-D-galactosyl-*N*-acetyl-D-glucosaminyl-R 4-L-fucosyltransferase; *other name*: Lewis transferase. An enzyme that catalyses the transfer by GDP-L-fucose of fucose to 1,3-β-D-galactosyl-*N*-acetyl-D-glucosaminyl-R to form 1,3-β-D-galactosyl-(α-1,4-L-fucosyl)-*N*-acetyl-D-glucosaminyl-R with release of GDP; this forms the blood group determinant Le^a of **Lewis antigen**. The enzyme also acts on the corresponding 1,4-galactosyl derivative, forming 1,3-L-fucosyl links. It is a type II membrane glycoprotein, and a product of the Lewis blood group locus: database code GFT_HUMAN, 361 amino acids (42.07 kDa).

β-galactoside permease any of several transport proteins in bacteria that facilitate the uptake of β-galactosides. An example is melibiose carrier protein from *Klebsiella pneumoniae*: database code MELB_KLEPN, 471 amino acids (52.27 kDa). This is responsible for melibiose transport; it is capable of using hydrogen and lithium cations as coupling cations for co-transport, depending on the particular sugar transported (e.g. H⁺-melibiose, Li⁺-lactose), but cannot recognize Na⁺ ions although it is an integral membrane protein of the sodium:galactoside symporter family. *See also lactose permease*.

galactosylceramidase EC 3.2.1.46; an enzyme that catalyses the hydrolysis of D-galactosyl-*N*-acylsphingosine to D-galactose and *N*-acylsphingosine. Example from human: database code JC2397, 669 amino acids (75.06 kDa).

galactosylceramide *the recommended term for any D-galactosyl-*N*-acylsphingosine, i.e. any cerebroside in which the sugar moiety is D-galactose (less correctly termed galactoceramide or galactocerebroside)*.

galactosylceramide lipidosis or **Krabbe's disease** or **globoid cell leukodystrophy** one of a number of **sphingolipidoses** or sphingolipid lysosomal storage diseases. A rapidly progressive, invariably fatal, hereditary (autosomal recessive) disease of human infants caused by deficient activity of galactocerebroside β-galactosidase (galactosylceramidase; EC 3.2.1.46). Clinical symptoms usually start between the ages of 3 and 6 months and the condition is characterized by severe mental and motor deterioration and the presence of numerous globoid cells containing galactosylceramide in the white matter of the brain.

galactosylceramide sulfotransferase EC 2.8.2.11; an enzyme involved in **sulfatide** biosynthesis that catalyses the reaction between a galactosylceramide and 3'-phosphoadenylyl-sulfate to form galactosylceramide sulfate and adenosine 3',5'-bisphosphate.

galactosyldiacylglycerol any of a class of lipids in which one or two galactose residues are in glycosidic linkage with a 1,2-diacylglycerol (loosely termed mono- and digalactosyldiacylglycerols), more specifically as, e.g., 1,2-diacyl-*sn*-glycerol 3-β-D-galactoside or (preferred, as glycosphingolipids are named this way) 1,2-diacyl-3-β-D-galactosyl-*sn*-glycerol. They are widely distributed in plants and are the main constituent lipids of chloroplasts, where they occur in the thylakoid membrane. They are also present in mitochondria.

β -1,3-galactosyl-O-glycosyl-glycoprotein

β -1,3-galactosyl-O-glycosyl-glycoprotein β -1,6-*N*-acetylglucosaminyltransferase EC 2.4.1.102; *other names*: O-glycosyl-oligosaccharide-glycoprotein; *N*-acetylglucosaminyltransferase I; β^6 -*N*-acetylglucosaminyltransferase; core debranching enzyme; an enzyme that catalyses the reaction between UDP-*N*-acetyl-D-glucosamine and β -D-galactosyl-1,3-*N*-acetyl-D-galactosaminyl-R to form β -D-galactosyl-1,3-(*N*-acetyl- β -D-glucosaminyl-1,6)-*N*-acetyl-D-galactosaminyl-R with release of UDP. It introduces branches into O-linked glycoprotein oligosaccharides. Example from human: database code G6NT_HUMAN, 428 amino acids (49.73 kDa).

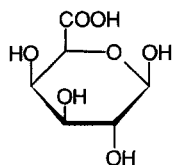
galactosyl transferase any enzyme belonging to the sub-class hexosyltransferases, EC 2.4.1, for which the glycosyl group is galactosyl; the acceptor is typically another carbohydrate molecule or lipid. *See also* B-transferase, lactose synthase, Lewis transferase.

galactowaldenase a former name for UDPglucose 4-epimerase.

galacturonan or (formerly) **polygalacturonide** any glycan composed solely of galacturonic-acid residues; a specific kind of glycuronan.

galacturonate 1 the anion of galacturonic acid. 2 any salt or ester of galacturonic acid.

galacturonic acid *symbol*: GaIA or (formerly) GaIU or GaIUA; the uronic acid formally derived from galactose by oxidation of the hydroxymethylene group at C-6 to a carboxyl group. There are two enantiomers, D- and L-galacturonic acid; D-galacturonic acid is a component of plant gums and bacterial cell walls.



β -D-galacturonic acid

galanin a biologically active neuropeptide with a wide distribution in the central neurons of several mammalian species. Human galanin is the 30-residue sequence

GWTLSAGYLLGPHAVGNHRSFSDKNGLTS;

porcine galanin had only 29 residues and differs at residues 16, 17 (I,D), 23 (H), 26 (Y), and 29 (A-NH₂). Galanin effects contraction of smooth muscle of the gastrointestinal and genitourinary tracts, and regulates somatotropin release; it may be involved in the control of adrenal secretion, and is also produced in the islets of Langerhans from stimulated intra-islet nerve terminals. It inhibits insulin release *in vivo* and *in vitro*, and inhibits formation of cyclic AMP in cells stimulated by **glucagon-related peptide**. A fragment, GWTLSAGYLLGPHAI, is an agonist at the hippocampal galanin receptor. Example from pig (precursor): database code GALA_PIG, 123 amino acids (13.07 kDa); the hormone is derived from amino acids 33–61, with C-terminal amidation; a galanin message associated peptide (GMAP) is derived from amino acids 65–123.

galaptin any of various proteinaceous, β -galactoside-specific, animal lectins of relatively low molecular mass, that are isolated from a number of developing and adult tissues and are believed to mediate intercellular adhesion in some developmental systems.

galatriose *symbol*: Ga₃ or GaOse₃; the trivial name for the trisaccharide Gal(α 1-4)Gal(α 1-4)Gal.

gall 1 an abnormal swelling or growth produced by a plant in response to the activities of various fungi, mites, and insects, especially members of the Cynipidae and Cecidomyiidae. Insect eggs and larvae are often found in galls. 2 a former name for **bile**.

gallate 1 the anion of gallic acid, 3,4,5-trihydroxybenzoic acid, a phenolic acid. 2 any salt or ester of gallic acid. The esters and polyesters are very widely distributed in angiosperms, usually as **gallotannins**.

gall bladder a hollow, muscular organ, found in many vertebrates, that receives dilute **bile** from the liver, concentrates and stores it, and discharges it into the duodenum.

gallotannic acid *see* **tannic acid**.

gallotannin or **hydrolysable tannin** any **tannin** consisting of a molecule of a polyhydric alcohol (usually glucose) esterified with several gallic-acid molecules, each of which may itself be esterified with a further gallic acid molecule, and so on.

Galp *symbol* for a galactopyranose residue.

GalpA *symbol* for a residue (or sometimes a molecule) of galactopyranuronic acid.

GalpN *symbol* for a residue (or sometimes a molecule) of the aminodeoxysugar **galactosamine**.

GalpNac *symbol* for a residue (or sometimes a molecule) of the acetamidodeoxysugar *N*-acetylgalactosamine.

GaIU or **GaIUA** former *symbol* for **galacturonic acid** (GaIA now recommended).

galvanic cell an alternative name for **cell** (def. 3).

gamete a mature reproductive cell whose nucleus (and often cytoplasm) fuses with that of another gamete of similar origin but of opposite sex to form a **zygote**, which develops into a new individual. Gametes are haploid and are differentiated into male (+) and female (–).

gamma *symbol*: γ (lower case) or Γ (upper case); the third letter of the Greek alphabet. For uses *see* **Appendix A**.

gamma chain or γ chain 1 the heavy chain of an IgG immunoglobulin molecule. 2 one of the two types of polypeptide chain in a human fetal hemoglobin (Hb F) molecule.

gamma globulin or γ -globulin any of a band of serum proteins that migrate most slowly on electrophoresis. In humans this normally constitutes quantitatively the largest fraction of the total globulins. Since the γ -globulins are predominantly antibodies, and are not synthesized in the liver as are virtually all the other serum proteins, the albumin/globulin ratio has traditionally been a useful indicator of liver function. The γ -globulins may be separated from the other serum globulins in pooled adult serum and may then be used for administration by intramuscular injection to provide passive immunity in premature infants or for those in contact with cases of infectious hepatitis. Such γ -globulin preparations tend to form aggregates spontaneously and these can lead to severe anaphylactic reactions when administered intravenously on account of their ability to aggregate platelets and to activate complement and generate C3a and C5a anaphylatoxins. In certain diseases such as Bruton's congenital agammaglobulinemia the production of immunoglobulin in affected males is grossly depressed. In patients with myeloma the γ -globulin band on electrophoresis of serum proteins is markedly more prominent, indicating the presence of a raised antibody against an unknown antigen. *See also* **globulin**.

gamma helix or γ helix a configuration of a polypeptide chain in which each amide group forms hydrogen bonds with the groups five residues away in each direction along the chain. It has 5.1 amino-acid residues per turn; the translation along the helical axis is 0.99 Å per residue and 5.03 Å per turn.

gamma-peptidase *see* **X-Pro dipeptidase**.

gamma radiation or γ -radiation electromagnetic radiation with wavelengths in the range 0.1–100 pm (frequencies 3–3000 EHz) emitted by some atomic nuclei during **transformation** (def. 3). Gamma rays have shorter wavelengths than X-rays and higher photon energies.

gammopathy any pathological disturbance of gamma-globulin metabolism. Gammopathies may be monoclonal or polyclonal and are associated with increased immunoglobulin synthesis.

gamone any of a group of biological agents secreted by gametes that act on gametes of the opposite sex to initiate the processes leading to fertilization. They may be subdivided into

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gynogamones, liberated from female gametes, and **androgamones**, liberated from male gametes.

ganciclovir 9-[(1,3-dihydroxy-2-propoxy)methyl]guanine; a nucleoside analogue of **acyclovir** that is more potent against cytomegalovirus. It is converted through ganciclovir mono- and diphosphate to ganciclovir triphosphate, which inhibits viral DNA synthesis by inhibition of viral DNA polymerase.

ganglion (*pl.* **ganglia**) a cluster of nerve cells and associated glial cells located outside the central nervous system.

ganglioside any ceramide oligosaccharide carrying, in addition to other sugar residues, one or more sialic residues. Gangliosides are so named because they were first isolated from ganglion cells (neurons) of the grey matter of the brain but are in fact widely distributed in vertebrate tissues. They are *N*-acetyl- (or *N*-glycoloyl-) neuraminosyl-(X)osylceramides, where (X) stands for the root name of the neutral oligosaccharide to which the sialosyl residue is attached; the acyl groups in the ceramide moiety are commonly stearyl or lignoceroyl. Gangliosides are soluble in water and do not pass through semipermeable membranes, so may be purified by dialysis. In the Svennerholm system of abbreviations gangliosides are designated G for ganglioside, plus subscript M, D, or T for mono-, di-, or trisialo, respectively, the subscript letter being followed by a subscript arabic numeral to indicate the sequence of migration in thin-layer chromatograms. The principal gangliosides of human brain are:

(1) monosialo gangliosides: G_{M1} , $\text{Gal}(\beta 1-3)\text{GalNAc}(\beta 1-4)(\text{NeuAc}(\alpha 2-3))\text{Gal}(\beta 1-4)\text{Glc}\beta 1\text{Cer}$ (G_{A1} denotes the asialo derivative of G_{M1}); G_{M2} , $\text{GalNAc}(\beta 1-4)(\text{NeuAc}(\alpha 2-3))\text{Gal}(\beta 1-4)\text{Glc}\beta 1\text{Cer}$ (G_{A2} denotes the asialo derivative of G_{M2}); G_{M3} , $\text{NeuAc}(\alpha 2-3)\text{Gal}(\beta 1-4)\text{Glc}\beta 1\text{Cer}$;

(2) disialo gangliosides: G_{D1a} , $\text{NeuAc}(\alpha 2-3)\text{Gal}(\beta 1-3)\text{GalNAc}(\beta 1-4)(\text{NeuAc}(\alpha 2-3))\text{Gal}(\beta 1-4)\text{Glc}\beta 1\text{Cer}$; G_{D1b} , $\text{Gal}(\beta 1-3)\text{GalNAc}(\beta 1-4)(\text{NeuAc}(\alpha 2-8)\text{NeuAc}(\alpha 2-3))\text{Gal}(\beta 1-4)\text{Glc}\beta 1\text{Cer}$; G_{D3} , $\text{NeuAc}(\alpha 2-8)\text{NeuAc}(\alpha 2-3)\text{Gal}(\beta 1-4)\text{Glc}\beta 1\text{Cer}$;

(3) trisialogangliosides: G_{T1a} , $\text{NeuAc}(\alpha 2-8)\text{NeuAc}(\alpha 2-3)\text{Gal}(\beta 1-3)\text{GalNAc}(\beta 1-4)(\text{NeuAc}(\alpha 2-3))\text{Gal}(\beta 1-4)\text{Glc}\beta 1\text{Cer}$; G_{T1b} , $\text{NeuAc}(\alpha 2-3)\text{Gal}(\beta 1-3)\text{GalNAc}(\beta 1-4)(\text{NeuAc}(\alpha 2-8)\text{NeuAc}(\alpha 2-3))\text{Gal}(\beta 1-4)\text{Glc}\beta 1\text{Cer}$.

ganglioside galactosyltransferase EC 2.4.1.62; *other name*: UDPgalactose–ceramide galactosyltransferase; an enzyme that catalyses the reaction between UDPgalactose and *N*-acetyl-D-galactosaminyl-(*N*-acetylneuraminyl)-D-galactosyl-D-glucosyl-*N*-acylsphingosine to form the **ganglioside** G_{M1} , with release of UDP.

gangliosidosis any of a group of inherited human diseases involving defects in **ganglioside** breakdown. They are characterized by: progressive mental and motor deterioration with fatal outcome; autosomal recessive inheritance; neuronal lipidosis secondary to storage of gangliosides and structurally related glycolipids, polysaccharides, or glycoproteins; and absence or severe deficiency of specific lysosomal glycohydrolases. The principal gangliosidoses are G_{M1} and G_{M2} gangliosidoses (the names indicating which ganglioside accumulates). G_{M1} gangliosidoses are due to a striking deficiency of β -galactosidase activity when tested using G_{M1} or galactose-containing glycoproteins as substrates. Type 1 or infantile G_{M1} gangliosidosis is a generalized gangliosidosis with bony involvement and onset of signs from birth or shortly after. Type 2 or juvenile G_{M1} gangliosidosis is of slower onset than type 1 and with milder bony abnormalities. There are three types of G_{M2} gangliosidosis. Type 1 G_{M2} gangliosidosis (*or* **Tay–Sachs disease**) is the commonest gangliosidosis, occurring especially in Ashkenazi Jews. It is caused by a severe deficiency of hexosaminidase A. Type 2 G_{M2} gangliosidosis (*or* **Sandhoff's disease**) occurs much less frequently than Tay–Sachs disease and is caused by severe deficiency of both hexosaminidases A and B; globoside accumulates in addition to G_{M2} in this disease. Type 3 or juvenile G_{M2} gangliosidosis is caused by either severe or partial deficiency of hexosaminidase A.

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gangliotetraose *symbol*: Gg_4 or $GgOse_4$; a *trivial name* for the tetrasaccharide $\text{Gal}(\beta 1-3)\text{GalNAc}(\beta 1-4)\text{Gal}(\beta 1-4)\text{Glc}$.

gangliotriaose *symbol*: Gg_3 or $GgOse_3$; a *trivial name* for the trisaccharide $\text{GalNAc}(\beta 1-4)\text{Gal}(\beta 1-4)\text{Glc}$.

GaOse₂ *symbol* for galabiose (alternative to Ga_2).

GaOse₃ *symbol* for galatriaose (alternative to Ga_3).

gap 1 a space introduced into the written form of a sequence of residues in a biopolymer to demonstrate homology with the sequence of a similar polymer from the same or another source. 2 a discontinuity in one of the two strands of duplex DNA resulting from enzymic elimination of one or more nucleotide residues. *Compare* **nick**.

GAP *abbr.* for 1 GnRH-associated peptide (*see* **gonadotropin-releasing hormone**). 2 *abbr.* for GTP-binding GTPase activating protein; a protein that activates the hydrolysis of GTP bound to certain monomeric proteins, especially p21^{ras} (Ras, *see* **RAS**), but also for other Ras-related proteins, e.g. Rho (*see* **Rho protein**) and Ran (*see* **RAN**). Since GTP-bound Ras is the active form, binding to GAP inactivates it. The protein has SH2 and SH3 domains (*see* **SH domains**) and **pleckstrin homology domains**. The *v-ras* product is not stimulated by GAP. Example from *Bos taurus*: database code GTPA_BOVIN, 1044 amino acids (115.76 kDa).

Gap A *abbr.* for galactoprotein A (i.e. plasma **fibronectin**).

GAPDH *abbr.* for glyceraldehyde-3-phosphate dehydrogenase.

gap gene a type of gene in *Drosophila*. There are at least six gap genes, which act early in embryogenesis to make out the coarsest subdivisions of the embryo. Mutations in these genes result in gaps in the segmentation pattern; the **Krüppel** and **hunchback** proteins are products of gap genes.

gap junction *or* **nexus** any specialized area of the plasma membranes of apposed vertebrate cells where the membranes are 2–4 nm apart and penetrated by a **connexon** that bridges the extracellular space and provides open means of communication between the cytoplasm of one cell and that of the other cell. *Compare* **tight junction**.

gap phase either of the two phases, designated G_1 and G_2 , in the cell-division cycle during **interphase** in growing eukaryotic cells during which there is no DNA synthesis.

gap repair the process of repair of a **gap** (def. 2) in one strand of a duplex DNA molecule.

garosamine 3-deoxy-4-C-methyl-3-(methylamino)-L-arabinose; a component of **gentamicins**.

gas a substance whose physical state is such that it always occupies the whole of the space in which it is contained, irrespective of the amount present. *See also* **gas laws**; **ideal gas**.

gas chromatography *abbr.*: GC; any form of chromatography in which the mobile phase is a gas. The term commonly refers to **gas–liquid (partition) chromatography** in which the stationary phase is a liquid, normally deposited on a solid support in a coiled column of glass or stainless steel, or the wall of a capillary column, but it also includes **gas–solid chromatography**, in which the stationary phase is a solid. The column is normally heated in an oven.

gas chromatography/mass spectrometry *abbr.*: GC/MS; an analytical technique that combines the separation process of **gas chromatography** with the highly selective detection technique of **mass spectrometry**.

gas constant *symbol*: R ; a fundamental physicochemical constant; $R = 8.314\,510\,(70)\text{ J K}^{-1}\text{ mol}^{-1}$.

gas exchange (*in physiology*) the exchange of gases between an organism and its environment. It includes oxygen uptake and carbon dioxide elimination in the respiration of all aerobic organisms, including plants, and carbon dioxide uptake and oxygen release in photosynthetic processes.

gas flow counter any gas-filled radiation detector in which the gas filling is renewed continuously. It may be operated in either the Geiger or the proportional regions of the voltage–current curve. Such devices are especially useful for detection and counting of alpha and soft-beta emitters, as no window is present to absorb the radiation.



gas laws either of two laws describing the effects of changes in pressure, p , and thermodynamic temperature, T , on the volume, V , of an ideal gas. **Boyle's law** (or **Mariotte's law**) states that at constant temperature the volume of a given mass of gas is inversely proportional to the pressure; i.e. $pV = \text{constant}$. **Charles's law** states that at constant pressure the volume of a given mass of gas is directly proportional to its thermodynamic temperature; i.e. $V/T = \text{constant}$. These two laws may be combined in the expression: $pV \propto T$, or, for one mole of gas: $pV = RT$, where R is the **gas constant**.

gas-liquid (partition) chromatography an analytical method based on the partition chromatographic separation of the components of a mixture through the use of a stationary liquid phase held in place on a solid support and a mobile gas phase flowing over the liquid phase in a controlled manner. Typically, the column is heated to temperatures in the range 50–300 °C and the technique is applicable to compounds that partition appreciably into the vapour phase in that range. Detection of the eluents is commonly by a **flame ionization detector**.

gas-solid chromatography *see* **gas chromatography**.

gastr+ a variant form of **gastr-** (before a vowel).

gastric of, pertaining to, or derived from the stomach.

gastric acid the hydrochloric acid that is secreted from parietal (or oxyntic) cells of the wall of the stomach and forms part of the **gastric juice**.

gastric inhibitory peptide *abbr.*: GIP; or **glucose-dependent insulinotropic peptide** or **glucose-dependent insulin-releasing peptide** a 5 kDa polypeptide hormone produced by the K cells of the proximal small intestine and released into the bloodstream in response to the presence of glucose in the intestine. It is structurally related to glucagon, secretin, and vasoactive intestinal polypeptide having the sequence (human) YAEGT-FISDYSIAMDKIHQQDFVNWLLAQKGKKNDWKHNITQ, porcine being similar but with Arg¹⁸ and Ser³⁴. The porcine hormone appears to be heterogeneous, the major component having 42 amino-acid residues and the main secondary component being two residues shorter at the N-terminus. Its two main physiological functions are the inhibition of gastric secretion and motility (**enterogastrone effect**), and the stimulation of insulin release when administered during hyperglycemia (**insulinotropic effect**); in the latter action GIP may be synonymous with **incretin**.

gastric juice the thin, clear, acidic fluid secreted by glands of the stomach mucosa in response to the ingestion of food. It contains mucus, intrinsic factor, hydrochloric acid (about 155 mM in humans) from the parietal cells, and **pepsinogen**, the precursor of pepsin proteolytic enzymes.

gastricsin EC 3.4.23.3; *other names*: pepsin C, pig parapepsin II; an enzyme of the pepsin family that appears to be present in the gastric juice of most vertebrates. It effects preferential cleavage at Tyr-[Xaa] bonds. It is a single-chain phosphoprotein formed by limited proteolysis of pepsinogen C with a more restricted specificity than pepsin A but high activity towards hemoglobin. Example, (precursor) from human: database code PEPC_HUMAN, 388 amino acids (42.38 kDa).

gastrin any of various members of a group of interrelated peptide hormones from the **G cells** of the pyloric antral mucosa of the stomach, and of the proximal duodenum, that strongly stimulate acid secretion by the stomach mucosa. There are three principal, biologically active forms: **big gastrin** (or G-34); **little gastrin**, (or G-17 or gastrin I); and **mini gastrin** (or G-14); these have 34, 17, and 14 amino-acid residues, respectively. The sequence of human big gastrin (gastrin I in parentheses, but with pE for the terminal Q) is pELGPQGPP-HLVADPSKK(QGPWLEEEEEAYGWMDF-NH₂), where pE indicates pyroglutamylation. Each has a counterpart, e.g. G-17S (or gastrin II), in which the tyrosyl residue is sulfated. There are other, larger immunoreactive forms, known as **big-big gastrins**, that elute from a gel column before G-34. The entire physiological activity is possessed by the C-terminal tetrapeptide amide sequence that ends in a phenylalaninamide residue.

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The C-terminal pentapeptide sequence of gastrin is the same as that of **cholecystokinin** and **cerulein**. Example, gastrin precursor, a member of the gastrin/cholecystokinin family, a precursor for gastrin (17 amino acids) and big gastrin (34 amino acids) from human: database code GAST_HUMAN, 101 amino acids (11.38 kDa). *See also* **gastrinoma**, **pentagastrin**, **Zollinger-Elison syndrome**.

gastrinoma a tumour of the gastrin-producing **G cells** of the pancreas, stomach, or duodenum. Gastrinomas are associated with high gastrin levels in the circulating blood, gastric hypersecretion, and fulminant gastroduodenal ulcerations (**Zollinger-Elison syndrome**).

gastrin receptor *see* **cholecystokinin receptor**.

gastrin-releasing protein a growth factor secreted by some small-cell lung cancers.

gastro+ or (*often before a vowel*) **gastr+** *comb. form* meaning stomach. [From Greek *gastēr*, belly.]

gastroenteric an alternative term for **gastrointestinal**.

gastrointestinal or **gastroenteric** of, or pertaining to, the stomach and intestines.

gastrointestinal hormone any member of a diverse group of peptides with hormone or hormone-like actions that occur in specific cells of the gastrointestinal tract, the pancreatic islets, and other locations. The best characterized include **cholecystokinin**, **gastric inhibitory polypeptide**, **gastrin**, **glucagon**, **insulin**, **motilin**, **neurotensin**, **pancreatic polypeptide**, **secretin**, **substance P**, and **vasoactive intestinal polypeptide**. *See also* **gut hormone**.

gastrula an early stage of development of an animal embryo, following the blastula, in which the cells are invaginating to form the rudiment of a gut cavity.

gas vacuole any membrane-bound, gas-containing, intracellular organelle. Gas vacuoles provide aquatic prokaryotes with buoyancy. *See also* **gas vesicle**.

gas vesicle the submicroscopic component of the gas vacuole in certain aquatic prokaryotes, especially cyanobacteria. It is constructed of molecules of a single protein (≈ 15 kDa), which form a monolayer enclosing a hollow space into which gases diffuse. Typically it is cylindrical with conical ends, 3–300 nm in diameter and 300–1000 nm in length.

GATA-1 or **ERYF1** or **GF-1** or **NF-E1** a transcriptional activator protein that serves as a general switch factor for erythroid development. It binds to DNA sites with the consensus sequence [A/T]GATA[A/G] within regulatory regions of globin genes and other genes expressed in erythroid cells. Other transcription factors with similar sequences are termed **GATA-type**. Example from human: database code GAT1_HUMAN, 413 amino acids (42.70 kDa).

GATase *see* **glutamine amidotransferase**.

gate (*in electronics*) 1 a logic circuit that gives an output signal only in response to a combination of two or more input signals. 2 a signal that allows or controls the passage of another signal.

gated channel a channel in the membrane of a cell (especially an excitable cell) that allows entry of ions into the cell only under specific conditions. *See also* **ligand-gated**, **voltage-gated ion channel**.

gating the opening and closing of a gated channel.

gating current an electric current arising from the movement of charged groups in the protein of the sodium-ion channels of excitable nerve membranes, during the passage of an action potential, prior to the opening of the channel.

g-atom *abbr.* for gram-atom or gram-atomic weight.

GAU a codon in mRNA for L-aspartic acid.

gauche *abbr.*: *g*; (*in chemical nomenclature*) a conformational descriptor indicating a synclinal conformation. *See* **conformation**.

Gaucher's disease an autosomal recessive inherited lysosomal storage disease showing several types, all of which involve some degree of **glucosylceramide** accumulation, especially in macrophages which are found in large numbers in spleen, bone marrow, and liver. The disease is due to a deficiency of



gauss

glucosylceramidase (EC 3.2.1.45). There is a juvenile form which is associated with mental retardation, and a milder form that is not manifested until adult life. *See also* **saposin**. [After Philippe Charles Ernest Gaucher (1854–1918), French physician.]

gauss *symbol*: G or Gs; the cgs unit of magnetic flux; it is equal to a magnetic flux density of one maxwell per cm², one line of magnetic induction per cm², or 10⁻⁴ tesla.

Gaussian distribution *an alternative name for* **normal distribution**.

gavage the administration of food or other material in liquid form through a tube directly into the stomach; forced feeding in this manner.

Gb₃ *symbol for* globotriaose (alternative to GbOse₃).

Gb₄ *symbol for* globotetraose (alternative to GbOse₄).

G-banding or **Giemsa banding**; a commonly used procedure in which **Giemsa's stain** is used to identify chromosomal bands in a karyotype.

GbOse₃ *symbol for* globotriaose (alternative to Gb₃).

GbOse₄ *symbol for* globotetraose (alternative to Gb₄).

GBP *abbr. for* glucose bisphosphate (formerly called glucose diphosphate).

G-1,6-BP *abbr. for* glucose 1,6-bisphosphate (formerly called glucose 1,6-diphosphate).

Gc *symbol for* the glycolyl group.

GC *abbr. for* gas chromatography.

GCA a codon in mRNA for L-alanine.

GC box a nucleotide sequence element common in eukaryotic promoters; consensus GGGCGC.

GCC a codon in mRNA for L-alanine.

G cell a gastrin-producing cell found in the antrum of the stomach and to a lesser extent in the mucosa of the duodenum, though the distribution varies in different species. In humans, G cells are small and pyriform while in dogs they are larger and triangular in shape. They can be identified by immunofluorescence using a gastrin-specific antibody.

GCG 1 a codon in mRNA for L-alanine. 2 a suite of computer programs widely used for nucleic acid and protein sequence analysis, structure prediction, and sequence alignment. It is available for **UNIX**, **VMS**, and **X Windows**, and is produced by **UWCGC**.

GC/MS *abbr. for* gas chromatography/mass spectrometry.

GCN4 a gene for a yeast transcription factor that binds, as a dimer, to the sequence TGA[C/G]TCA. The protein has a **leucine zipper**, and resembles **FOS** and **JUN**. Database code: GCN4_YEAST, 281 amino acids (31.28 kDa).

G-CSF *abbr. for* granulocyte colony stimulating factor.

GCU a codon in mRNA for L-alanine.

Gd *symbol for* gadolinium.

GDC *abbr. for* Graves' disease carrier protein; it is similar in structure and some other properties to **ADP/ATP carrier protein**.

G4 DNA *abbr. for* DNA containing a **G-quartet**.

GDNF *abbr. for* glial cell line-derived neurotrophic factor; a glycosylated disulfide-bonded homodimer that promotes survival and morphological differentiation of dopaminergic neurons. Human and rat GDNFs have 93% homology. The precursor, database code GDNF_HUMAN, 211 amino acids (23.72 kDa), yields a mature protein of 134 amino acids. *Compare* **brain-derived neurotrophic factor**.

GDP *abbr. for* 1 guanosine diphosphate. 2 glucose diphosphate (now called glucose bisphosphate).

G-1,6-DP *abbr. for* glucose 1,6-diphosphate (now called glucose 1,6-bisphosphate).

GDP-fucosepyrophosphorylase *see* **fucose-1-phosphate guanylyltransferase**.

GDPm or **GDPmannose** *abbr. for* guanosinediphosphomannose.

GDPmannose 6-dehydrogenase EC 1.1.1.132; an enzyme that catalyses the reaction between GDP-D-mannose, 2 NAD⁺ and H₂O to form GDP-D-mannuronate and 2 NADH. Example from *Pseudomonas aeruginosa*: ALGD_PSEAE, 436 amino acids (47.58 kDa).

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GDPmannose 3,5-epimerase EC 5.1.3.18; an enzyme that catalyses the interconversion of GDP-D-mannose and GDP-L-galactose.

GDS (*in clinical chemistry*) *abbr. for* guanine deaminase.

Ge *symbol for* germanium.

Geiger counter or **Geiger-Müller counter** a device for detecting and measuring ionizing radiation (alpha or beta particles or gamma-ray photons). It consists of a glass envelope containing low-pressure gas and an anode and a cathode, between which a large electric-potential difference is maintained. The ions produced in the gas by the incoming ionizing radiation are accelerated towards their appropriate electrodes, causing a momentary drop in the potential between the latter. The voltage pulses so produced may be counted with a suitable electronic circuit. [After Hans Wilhelm Geiger (1882–1945), German physicist, and Erwin Wilhelm Müller (1911–77), German-born US physicist.]

Geiger region or **Geiger-Müller region** the part of the curve of applied voltage against number of ions collected for a given incoming radiation by a **Geiger counter**, where all possible ion-pairs, both primary and secondary, are collected; i.e. the signal is relatively independent of the applied voltage. *Compare* **proportional region**.

gel 1 a colloidal system, with the semblance of a solid, in which a solid is dispersed in a liquid. A gel has a finite, usually rather small, yield stress. 2 to become, or cause to become, a gel.

gelatin a heterogeneous mixture of water-soluble proteins of high average *M_r* derived from collagen by partial hydrolysis in either acid (type A) or alkali (type B). Collagen obtained by boiling skins, ligaments, tendons, etc. with water is also termed gelatin as also is collagen denatured by heating for 15 minutes at 45 °C, used as a substrate for **gelatinase**. It is colourless, or slightly yellow, and transparent, and occurs as brittle sheets, flakes, or coarse powder; these swell in water to form gels below 35–40 °C, containing 5–10 times as much water as gelatin by weight. Nutritionally gelatin is an incomplete protein containing only small amounts of certain essential amino acids, especially tryptophan. It is widely used in foods and in the manufacture of adhesives, photographic films and plates, and light filters.

gelatinase EC 3.4.24.24; *recommended name*: gelatinase A; *other names*: 72 kDa gelatinase; matrix metalloproteinase 2; type IV collagenase. An enzyme that catalyses the cleavage of **gelatin** and collagen types IV, V, VII, and X; it cleaves the collagen-like substrate sequence Pro-Gln-Gly-[Ile-Ala-Gly-Gln; zinc and calcium are cofactors. It is a metallo (zinc) endopeptidase with **fibronectin** domains, and is produced by fibroblasts. Example from human (precursor): database code COG2_HUMAN, 660 amino acids (73.80 kDa); there is a large pre-pro region; the enzyme starts at amino acid 110.

gelation 1 the (usually reversible) act or process of forming a **gel** (def. 1) from a **sol**. *Compare* **solution**. 2 the solidification of a liquid by cold. —*gelate* *vb*.

gel chromatography *see* **liquid-gel chromatography**.

gel diffusion (*in immunology*) a technique in which antibodies and antigens are allowed to diffuse independently through a gel, often of agar, and form a precipitate within the gel where homologous antigens and antibodies meet in optimal proportions.

gel electrophoresis a type of **zone electrophoresis** in which the supporting medium is a gel of uniform concentration, commonly of polyacrylamide, agarose, or starch. *See* **polyacrylamide (gel) electrophoresis**. *Compare* **gradient (acrylamide) gel electrophoresis**.

gel filtration or **gel-filtration chromatography** *see* **gel-permeation chromatography**.

gel-filtration chromatography *see* **gel-permeation chromatography**.

gel-filtration high-performance liquid chromatography a technique of **gel-permeation chromatography** that uses a support material of high mechanical strength and chemical inertness

gel fluorography

thereby enabling separations to be effected under the conditions, and with the advantages of **high-pressure liquid chromatography**.

gel fluorography the application of **fluorography** (def. 2) to the detection of substances separated in gels, as by gel electrophoresis.

gellify to cause to become a **gel** (def. 1); to become gel-like; to **gel** (def. 2).

gelonin or **ribosome-inactivating protein** or **rRNA N-glycosidase** EC 3.2.2.22; *recommended name*: rRNA N-glycosidase; an extremely stable, potent toxin extracted from seeds of *Gelonium multiflorum* (a plant of the spurge family). It is a single-chain glycoprotein of 28–30 kDa containing terminal mannose residues. It acts as a powerful inhibitor of protein synthesis in cell-free systems but not in intact cells; it inactivates the 60S ribosomal subunit but has no effect on the 40S subunit. Its action is to hydrolyse one specific N-glycosidic bond of an adenosine on the 28S rRNA. In cell-free protein-synthesizing systems gelonin acts like the A chains of **abrin**, **ricin**, and **mod-eccin**, suggesting that it lacks the ability to bind to cell surfaces and enter cells. Example from *G. multiflorum*: RIPG_{GELMU}, 316 amino acids (35.42 kDa).

gel-permeation chromatography a method for the separation of substances in solution, *commonly known as gel filtration or gel-filtration chromatography*. The separation is based mainly on exclusion effects, such as differences in molecular size and/or shape or in charge, when the stationary phase is a swollen gel. The gel used is commonly a water-swollen insoluble carbohydrate polymer from which macromolecules are excluded and migrate without retention in the interstitial fluid. Substances of low or intermediate relative molecular mass penetrate into the gel particles to an extent that is, in most instances, determined by their molecular dimensions and the degree of cross-linking of the gel. The larger molecules therefore pass through the bed more rapidly than smaller molecules. The technique can be used to determine the relative molecular mass (M_r) of unknown materials, since, for a particular column, elution occurs at a volume determined by the equation

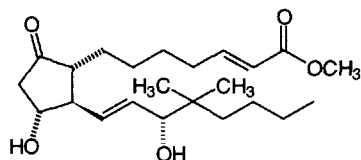
$$V_R/V_0 = 1 - \log(M_r/M_{r0})$$

where V_R is the elution volume of the unknown, V_0 is the volume of the mobile phase in the column, M_r is the M_r of the unknown, M_{r0} is the M_r of the largest totally permeating molecule, and p is a factor that is a property of the column. Thus each column must be calibrated using standard proteins, and a plot made of the elution volumes against $\log M_r$ of the standards, from which the unknown M_r can be determined.

gelsolin a heat-labile, monomeric protein of apparent M_r 90 000 isolated from macrophages, platelets, and plasma. It appears to promote the gel-sol transformation of **actin** and may thereby be important in the control of locomotion, secretion, and endocytosis in these and other cells; β -actin has a similar action. There are two forms of gelsolin, plasma and cytoplasmic, generated by alternative splicing. The cytoplasmic, calcium-regulated, actin-modulating form binds to the barbed ends of actin monomers or filaments, preventing monomer exchange; it also promotes nucleation and binds with high affinity to fibronectin. Example (cytoplasmic form) from mouse: database code GELS_MOUSE, 731 amino acids (80.79 kDa).

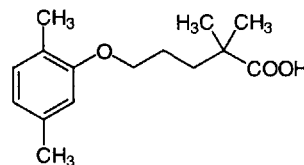
gem- prefix (in chemical nomenclature) denoting the presence of geminal substituents.

gemeprost 16,16-dimethyl-*trans*- Δ^2 -prostaglandin E_1 methyl



ester; 11,15-dihydroxy-16,16-dimethyl-9-oxoprostano-2,13-dien-1-oic acid methyl ester; an analogue of **prostaglandin E_1** with uterine stimulant activity.

gemfibrozil 2,2-dimethyl-5-(2,5-xylyloxy)valeric acid; an anti-hyperlipoproteinemic agent with actions similar to **clofibrate**.



geminal (in chemistry) describing two like groups or atoms attached to the same atom in a molecule, i.e. the geminal groups or atoms are separated by two bonds attached to the same atom. The presence of geminal substituents is denoted by the prefix *gem-* attached to the name of the compound. *Compare vicinal* (def. 2).

geminate occurring in pairs; doubled, twin.

+gen or (after a consonant) **+ogen** *comb. form* signifying producing or capable of producing (either directly or indirectly). In biological sciences it is used especially (1) with word-stems relating to physiological or pathological processes, states, conditions, etc., to denote a causative agent (e.g. antigen, lactogen, mutagen, estrogen); (2) with names of certain proteinases or blood components, to denote their inactive precursors or substances from which they can be derived by enzymic action (e.g. angiotensinogen, caseinogen, fibrinogen, trypsinogen) (*see also* **pro-** (def. 3), **zymogen**); (3) with names or name-roots of certain metabolites, to denote either a storage substance (e.g. glycogen, phosphagen) or a biosynthetic precursor (e.g. porphobilinogen); (4) with word-stems relating to physical properties (e.g. chromogen, luminogen). — **+genic** or **+genous** *adj. suffix*.

Genaminox KC the proprietary name for detergents similar to lauryldimethylamine oxide, but with different alkyl chain lengths (9–13).

Genapol the proprietary name for a series of nonionic polyoxyethylene detergents of the general formula $\text{CH}_3(\text{CH}_2)_x\text{O}(\text{CH}_2\text{CH}_2\text{O})_y\text{H}$; Genapol X-080 has $x = 12$, $y = 8$, CMC 0.06–0.15 mM; Genapol X-100 has $x = 12$, $y = 10$, aggregation number 88, CMC 0.15 mM. *Compare* **C₁₂E₈**, **Lubrol**, **Triton**, **Tween**.

GenBank a large database of protein and nucleic-acid sequences. *See* **Appendix E**.

gene in classical genetics, a statistical entity that correlates with a particular phenotypic characteristic; the functional unit of heredity. Before their biochemical nature was understood, genes were defined in terms of units of mutation and/or recombination. Discovery of the role of DNA in genetic processes, followed by elaboration of the **central dogma** of protein synthesis, led to enunciation of the 'one gene—one enzyme' hypothesis, i.e. that a gene consisted of DNA that coded for a protein that performed the functions associated with the phenotypic expression of the gene. In current molecular genetics, the concept requires modification in a number of ways. First, the genomic DNA that codes for a polypeptide is associated with regulatory sequences such as promoters. Second, the polypeptide resulting from translation of mRNA may be subsequently split to give several polypeptides with different functions (*see* **polyprotein**). Third, the RNA transcribed may give rise to several different proteins as a result of **alternative splicing**. Three types of genes are now distinguished: those that are both transcribed into mRNA and translated into polypeptides (**structural genes**); those that are only transcribed into RNA (e.g. rRNA, tRNA); and those that function as regulators of the expression of the other two types (**regulator genes**).

In diploid organisms a gene may occur in alternative forms



gene amplification

(**alleles**). The term gene is sometimes used interchangeably with **cistron**.

gene amplification the selective, repeated replication of a certain gene or genes without a proportional increase in other genes in the genome. It occurs, e.g., in the DNA puffs of *Rhynchosciara* and some other flies and in the amplification of the genes coding for ribosomal RNA (so-called ribosomal DNA) in amphibian oocytes.

gene bank an alternative name for **gene library**.

gene cloning see **molecular cloning**.

gene cluster or **gene complex** any group of two or more functionally related genes that are closely linked on a chromosome. The genes of a gene cluster are often structural genes coding for the enzymes that catalyse the various steps of a metabolic pathway.

gene complex an alternative name for **gene cluster**.

gene dosage the number of times a particular gene is present in the genome.

gene duplication the phenomenon, occurring in some higher organisms, in which there is duplication of the DNA sequences representing particular genes. This process is thought to have played a decisive part in the evolution of these organisms, through the occurrence of different mutations in the two duplicated genes.

gene expression the process by which the information carried by a gene or genes becomes manifest as the phenotype. It involves **transcription** of the gene into complementary RNA sequences and, for structural genes, subsequent **translation** of mRNA into polypeptide chains and their assembly into the ultimate protein products. Gene expression is tightly regulated by **promoters**, **enhancers**, and **transcription factors**.

gene frequency a measure of the proportion of an allele in a given population, equal to the number of loci at which a given allele occurs, divided by the total number of loci at which it could occur.

gene library or **cloned library** or **gene bank** or **shotgun collection** a random collection of DNA fragments cloned in a **vector** (def. 3), and which may include all the genetic information of a particular species. It may be prepared from a variety of sources, including an extract of mRNA, in contrast to a **genomic library** which is prepared from genomic DNA.

gene mutation any **mutation** (def. 1) occurring within a single gene.

gene pool the sum of the genetic information in the reproductive members of a population of sexually reproducing organisms.

gene product any of the types of RNA (transcription products) or any of the proteins or protein subunits (translation products) synthesized biochemically on the basis of the information encoded in a genome.

general acid-catalysis **homogeneous catalysis** in which the catalysts are various hydron donors (acids). Compare **specific acid-catalysis**.

general base-catalysis **homogeneous catalysis** in which the catalysts are various hydron acceptors (bases). Compare **specific base-catalysis**.

generally labelled describing the labelling of a molecule in such a way that a radionuclide may be present at any or all (but not necessarily all) possible positions. Compare **uniformly labelled**.

general transcription factor any of the proteins whose assembly around the **TATA box** is required for the initiation of transcription of most eukaryotic genes.

generate (in *mathematics*) to conceive a point, line, or surface to be moving in a specified way so as to form a line, surface, or solid, respectively.

generation 1 the act or process of producing or reproducing, naturally or artificially. 2 the phase in a life cycle that extends from one to the immediately successive reproduction. 3 any group comprising all those members of a population who are equally removed from a common ancestor or from coeval ancestors.

generation of diversity (in *immunology*) the process by which a large number of variable regions are generated in the immunoglobulins. The **stem cell** genome contains multiple variants of L-chain V (variable) and J genes and of the H-chain V, J, and D (diversity) genes. As the stem cells differentiate, the maturing lymphocyte constructs particular L and H genes of virtually unique structure by a recombination process that randomly selects one out of each set of gene segments and assembles them, together with a C (for the constant region) gene into a mature H or L gene. This, combined with recombinational inaccuracies, somatic point mutations, and the varied combinations of L and H chains found in immunoglobulins results in a repertoire of millions of lymphocytes each with H and L genes encoding unique molecules. See also **immunoglobulin**.

generation time the time between division of a cell and that of its daughter cells, averaged over a whole cell population.

gene redundancy the presence in a cell of many copies of a single gene. The multiple copies may be inherited or result from selective **gene duplication** during development.

gene-regulatory protein any protein that binds to a specific DNA sequence to alter the expression of a gene.

+genesis *comb. form* denoting beginning or origin; development; generation. See also **+gen.** — **+genetic** or **+genic** *adj. comb. form*.

gene splicing biochemical and/or chemical manipulations with the object of attaching one DNA molecule to another. The neatest method is to cleave the DNA to be inserted (foreign DNA) with a **restriction endonuclease** that yields single-stranded ends that are complementary to each other. The **cloning vector** into which the foreign DNA is to be inserted is treated with the same endonuclease so that the complementary ends of the two DNAs specifically associate under annealing conditions and are subsequently covalently joined (spliced) through the action of **DNA ligase**. If the endonuclease produces fragments with 'blunt' or **flush ends** then a similar procedure is adopted. If the foreign DNA and the cloning vector have no common restriction sites then terminal **deoxynucleotidyl transferase** may be used which adds nucleotides to the 3'-terminal OH group of a DNA chain. For this purpose tails of poly(dT) and poly(dA) may be used. The transformed cells are then grown and those containing the spliced DNA are selected (cloning). The preferred method of splicing is one that allows the foreign DNA to be removed easily by means of a restriction endonuclease, preferably the same one that was initially used in splicing. By this means large amounts of the foreign DNA can be synthesized and used for many different purposes. Alternatively the cloned cells containing the foreign DNA can be used to permit the expression of a foreign protein which is then recovered.

genetic or **genetical** of or pertaining to **genetics**, **genes**, or the origin of something. — **genetically** *adv.*

genetically engineered describing a cell, strain, or organism whose phenotype has been altered by manipulation of its genetic material. See **genetic engineering**.

genetic block a reduction in the activity of a particular enzyme in a metabolic pathway as a result of gene mutation. A genetic block is termed **complete** when the particular enzyme activity is absent, or **incomplete** (or **leaky**) when the enzyme formed is defective and of limited activity.

genetic carrier see **carrier** (def. 9).

genetic code all the regularities in nucleotide sequence according to which genetic information for the sequences of all the polypeptides synthesized by transcription and translation is encoded in DNA (or RNA in some viruses). A sequence of three nucleotide residues (a **codon**) is required to code for one amino-acid residue, and since there are four different kinds of base in nucleic acids (apart from minor constituents), 4³ i.e. 64 different nucleotide triplets can exist. But only 20 different amino-acid residues in polypeptides can be coded for; therefore the code is degenerate in that most amino acids can be

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genetic colonization

specified by more than one codon. In terms of the sequence of nucleotides in mRNA, i.e. the nucleotides complementary to those in the genetic material, the genetic code, reading from the 5' end to the 3' end of each codon, is given in the table. (The letter A symbolizes a nucleotide residue containing adenine; G, guanine; C, cytosine; and U, uracil; each amino-acid residue is represented by its usual three-letter symbol.)

UUU: Phe	UCU: Ser	UAU: Tyr	UGU: Cys
UUC: Phe	UCC: Ser	UAC: Tyr	UGC: Cys
UUA: Leu	UCA: Ser	UAA: Ter†	UGA: Ter†
UUG: Leu	UCG: Ser	UAG: Ter†	UGG: Trp
CUU: Leu	CCU: Pro	CAU: His	CGU: Arg
CUC: Leu	CCC: Pro	CAC: His	CGC: Arg
CUA: Leu	CCA: Pro	CAA: Gln	CGA: Arg
CUG: Leu	CCG: Pro	CAG: Gln	CGG: Arg
AUU: Ile	ACU: Thr	AAU: Asn	AGU: Ser
AUC: Ile	ACC: Thr	AAC: Asn	AGC: Ser
AUA: Ile	ACA: Thr	AAA: Lys	AGA: Arg
AUG: Met‡	ACG: Thr	AAG: Lys	AGG: Arg
GUU: Val	GCU: Ala	GAU: Asp	GGU: Gly
GUC: Val	GCC: Ala	GAC: Asp	GGC: Gly
GUA: Val	GCA: Ala	GAA: Glu	GGA: Gly
GUG: Val	GCG: Ala	GAG: Glu	GGG: Gly

† termination codon. UAA was once called **ochre codon**; UAG, **amber codon**; UGA, **opal codon**.

‡ initiation codon. The methionine codon AUG is the most common starting point of translation of a genetic message but GUG sometimes serves.

The genetic code is almost universal but differences have been found in the DNA of mitochondria from a number of organisms. For example, in human mitochondria UGA codes for Trp and not for termination; AUA codes for Met and not for Ile; AGA and AGG are termination codons and do not code for Arg; AUA and possibly AUU act as initiation codons as well as AUG.

genetic colonization the introduction by a parasite of genetic information into a host such as to cause the host to synthesize products that only the parasite can use. This occurs in some plants, e.g., in cases of parasitism by *Agrobacterium* spp.

genetic complementation the phenomenon whereby the introduction of two chromosomes, each from different heterozygote cells and carrying different mutations in the same genetic region, results in the restoration of normal biochemical function, when this is deficient in either heterozygote.

genetic cross 1 any mating of two organisms that results in a **genetic recombinant**. 2 the offspring of two (or more) parent organisms produced by mating or other means.

genetic defect any alteration in the genetic information carried in an individual organism, the effect of which is to produce a phenotype that is at a disadvantage in competing with a normal organism of the same species or in meeting the challenges of its environment. *See also* **inborn error of metabolism**, **molecular disease**.

genetic disease any disease ultimately caused by a genetically determined biochemical defect. In the strict sense a genetic disease is only slightly influenced by environmental factors.

genetic drift 1 any change with time in gene frequency in a population. It may be either directed, **steady drift**, or undirected, **random drift**. 2 irregular, random fluctuations in gene frequency in a (relatively) small population caused by statistical effects.

genetic engineering the genetic manipulations used to produce individuals having a new combination of inherited properties. Such manipulation may be of two kinds: (1) cellular manipulation, involving the culturing of (haploid) cells and

the hybridization of somatic cells (*see* **cell fusion**); and (2) molecular manipulation, involving the construction of artificial recombinant DNA molecules, their insertion into a **vector** (def. 3), and their establishment in a host cell or organism. The latter approach has been called **recombinant DNA technology**.

genetic information the information carried in a sequence of nucleotides in a molecule of DNA or RNA.

genetic locus *see* **locus**.

genetic map any diagram showing the linear order of, and the relative distances between, mutable sites on a chromosome as deduced from genetic recombination experiments.

genetic mapping any method that may be used to determine the positions of, and the relative distances between, genes of a linkage group or of sites within a gene (fine-scale mapping). In classical genetics, mapping relied on a study of recombination frequency and was measured in **morgans**. In molecular biology, mapping relies on the sequencing of DNA in a chromosome. Current work is aimed at mapping entire genomes. *See also* **genetic map**, **map unit**.

genetic marker 1 any genetically controlled phenotypic difference that can be used in analysing the genetic make-up of an organism. 2 any gene difference in either one or both alleles used in the detection of genetic recombination events.

genetic material the molecular carrier of primary genetic information that serves as a template for its own replication and provides the structural and regulatory information for the processes of protein synthesis and cell development. It consists of double-stranded DNA in higher organisms, bacteria, and most bacteriophages, single-stranded DNA in some bacteriophages, and RNA in the RNA viruses.

genetic polymorphism the regular and simultaneous existence in the same population of two or more discontinuous variants or genotypes, in frequencies that cannot be due to recurrent mutations, e.g. the different human blood groups.

genetic profiling or **DNA fingerprinting**, a technique for providing profiles of DNA fragments (resulting from digestion with restriction enzymes) that characterize a genome. The human genome is scattered with minisatellites, regions of DNA consisting of tandem repeats of short base sequence which can show extensive variation in the number of repeats, leading to multi-allelic variation and high degrees of heterozygosity. This extreme level of variation renders minisatellite or VNTR (**variable number tandem repeats**) loci invaluable as genetic markers. The DNA is digested with a suitable restriction endonuclease and the fragments are electrophoresed through agarose gel and transferred by blotting to a nylon filter. Hybridization is with a minisatellite probe labelled with ³²P. The use of the polymerase chain reaction (**PCR**) to amplify hypervariable loci, including very short tandem-repetitive microsatellites, has greatly increased the sensitivity of DNA typing systems and the ability to type degraded human DNA. The method can be used to identify an individual's DNA and also to trace the parentage of an individual. The method is not confined to humans but has been applied to a wide range of species.

genetic recombinant any organism produced by **genetic recombination**.

genetic recombination the processes by which a new genotype is formed by reassortment of genes resulting in gene combinations different from those that were present in the parents. In eukaryotes genetic recombination can occur by chromosome assortment, intrachromosomal recombination, or nonreciprocal interchromosomal recombination. Intrachromosomal recombination occurs by **crossing over**. In bacteria it may occur by genetic transformation, conjugation, transduction, or F-duction. *See also* **Holliday junction**.

genetics 1 the branch of biology concerned with heredity and variation in organisms. It embraces the phenomenology and physiology of heredity; the nature of genetic information; the storage of genetic material; the replication, mutation, transmission, and recombination of genetic material; and the way it



genetic screening

is translated into systems that control development and metabolism, and determine the reappearance of parental characters among progeny. **2** the genetic features and constitution of any single organism, species, or group of organisms. *See also* **reverse genetics**.

genetic screening the testing of individuals or groups of individuals for some specific gene product or resulting metabolite for the purpose of identifying genetic defects or diseases.

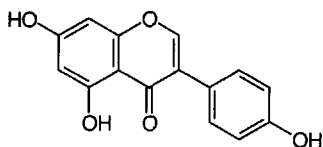
genetic transformation *see* **transformation** (def. 1).

genetic variance that part of the phenotypic variance caused by the varying genotypes of the individuals within a population.

genic of, or relating to, a gene or genes.

genin **1** the class name for the noncarbohydrate residue of steroid glycosides. It is used as a suffix in forming trivial names of the naturally occurring glycoside precursors; e.g. digitogenin (from digitonin), sapogenin (from saponin). **2** a former name for spirostan, 16,22,22,26-diepoxycholestane.

genistein 4,5,7-trihydroxyisoflavone; 5,7-dihydroxy-3-(4-hydroxyphenyl)-4H-1-benzopyran-4-one; an inhibitor of tyrosine-specific protein kinases and other protein kinases. It is a competitive inhibitor at the ATP binding site, which is a conserved structural feature in protein kinases.



genome the whole of the genetic information of an organism. It is contained as DNA in eukaryotes and prokaryotes, and as either DNA or RNA in viruses. A given organism has only one genome regardless of whether the organism is haploid, diploid, or polyploid. The term was originally used to denote one haploid set of chromosomes in a eukaryote organism. — **genomic** *adj.*

genome mutation any **mutation** (def. 1) occurring by alteration of the number of chromosomes in the genome.

genome-regulatory protein any site-specific protein involved in the regulation of transcription, such as a polymerase, repressor, or activator.

genomic blotting an alternative name for **Southern blotting**.

genomic DNA all of the DNA comprising one genome.

genomic library a set of clones containing DNA fragments that *in toto* embrace an entire genome. It is prepared by firstly randomly cutting the genomic DNA to give fragments of varying length. The average fragment length is controllable by discrimination in the choice of restriction enzymes used for cutting, and is selected to optimize the length such that fragments will contain entire open reading frames without excessive lengths of additional DNA at either end. Secondly the DNA fragments thus obtained are ligated into a suitable **vector** (def. 3), which is cloned in a host of choice.

genophore the structural equivalent of a genetic **linkage group** in any organism. The term comprehends the linear chromosome of the (enveloped) eukaryon of all animal and plant cells, the circular chromosome of the (nonenveloped) prokaryon of cells of bacteria, the duplex DNA of mitochondria, chloroplasts, kinetoplasts, plasmids, and episomes, and the nucleic acid, whether DNA or RNA, single-stranded or double-stranded, of viruses.

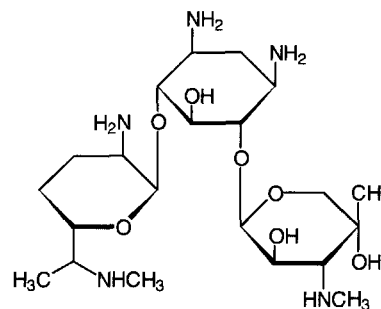
genothèque a term proposed as an alternative to **gene library**. [From *gene* (ultimately from Greek *genos*, descent + *thēkē*, case, box).]

genotoxic exerting an effect by damaging or otherwise interfering with the action of a gene.

genotype **1** the total genetic constitution of an organism. It comprises the genetic information carried both in the chromo-

somes and extrachromosomally. *Compare* **phenotype**. — **genotypic** *adj.*

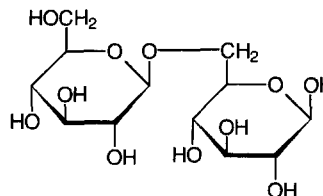
gentamicin or **gentamycin** any of a group of aminoglycoside antibiotics produced by fermentation of some *Micromonospora* spp. They are inhibitors of protein synthesis, and are particularly effective in the treatment of *Pseudomonas aeruginosa* infections.



gentamicin C₁

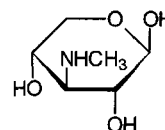
gentianose the trisaccharide, *O*-β-D-glucopyranosyl-(1→6)-*O*-α-D-glucopyranosyl-(1→2)-β-D-fructofuranoside. It is found in the rhizomes of many species of *Gentiana*.

gentiobiose or (formerly) **amygdalose** the disaccharide, *O*-β-D-glucopyranosyl-(1→6)-D-glucose. It is a component of some glycosides, e.g. **amygdalin**.



β-gentiobiose

gentosamine 3-deoxy-3-methylamino-D-xylose; a component of gentamicin A, one of the structural variants of **gentamicin**.



genus (*pl. genera*) (*in taxonomy*) a category used in biological classification consisting of species with a common phylogenetic origin that appear to be clearly differentiated from other species.

+**geny** *comb. form* denoting origin or mode of development. — +**genic** *adj.*

geometric isomer or **geometrical isomer** an alternative term for **cis-trans isomer**.

geometric isomerism or **geometrical isomerism** an alternative term for **cis-trans isomerism**.

geometric mean or **geometric average** for *n* positive numbers, the *n*th root of the product of the numbers. Thus for a set of positive numbers $x_1, x_2, x_3, \dots, x_n$, the geometric mean is $(x_1 \cdot x_2 \cdot x_3 \dots x_n)^{1/n}$. *Compare* **mean**.

geotaxis the movement of an organism or a cell under the influence of a gravitational field. In **positive geotaxis** the organ-



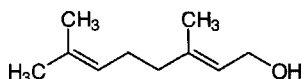
geotropism

ism moves in the same direction as the gravitational field, while in **negative geotaxis** the movement is in a direction opposite to the gravitational field. — **geotactic adj.**

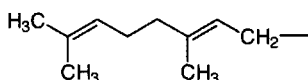
gephryin a cytoplasmic protein that associates with microtubules and is required for clustering of glycine receptors in spinal cord. It has a role in molybdoenzyme function in non-neural tissue. Example, GEPH_RAT, 736 amino acids (79.83kDa).

g-equiv *abbr.* for gram-equivalent or gram-equivalent weight.

geraniol (2*E*)-3,7-dimethyl-2,6-octadien-1-ol, a monoterpene alcohol with a sweet rose odour, present in oil of rose and other essential oils.

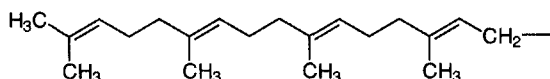


geranyl the (2*E*)-3,7-dimethyl-2,6-octadien-1-yl group. The 1-diphosphate derivative is an intermediate in the synthesis of cholesterol and in the formation of **geranylgeranyl** diphosphate.



geranyl-diphosphate synthase *see* **dimethylallyltranstransferase**.

geranylgeranyl the (2*E*,6*E*,10*E*)-3,7,11,15-tetramethyl-2,6,10,14-hexadecatetraen-1-yl group. The 1-diphosphate derivative is a substrate for enzymes introducing the geranylgeranyl group into proteins. *See also* **prenylation**.



geranylgeranyl pyrophosphate synthetase *see* **farnesyltranstransferase**.

geranyltranstransferase EC 2.5.1.10; *systematic name*: geranyl-diphosphate:isopentenyl-diphosphate geranyl transtransferase; *other names*: farnesyl-diphosphate synthase; farnesyl pyrophosphate synthetase (*abbr.*: FPP synthetase). An enzyme of the pathway synthesizing **prenyl** groups, **sesquiterpenes**, and **cholesterol**. It catalyses the reaction between geranyl diphosphate and isopentenyl diphosphate to form *trans,trans*-farnesyl diphosphate with release of pyrophosphate. Example from yeast: database code FPPS_YEAST, 352 amino acids (40.48 kDa).

Gerbich system *abbr.*: Ge; a blood-group system whose antigenic determinants are located on the extracellular, glycosylated domain of **glycophorins** C and D (GPC and GPD). The polypeptide sequences of these give rise to several Ge phenotypes: on GPD the antigenic site for Ge2 resides between residues 1 and 17, and that for Ge3 resides between residues 19 and 34; on GPC, the antigenic site for Ge4 is at the N terminus, and that for Ge3 between residues 40 and 55.

GERL *abbr.* for Golgi, endoplasmic reticulum, lysosomes; a term formerly used to describe a specialized region of smooth **endoplasmic reticulum** with high levels of acid phosphatase and other acid hydrolase activities. It is thought to be a region where acid hydrolases are concentrated and packaged into lysosomes. It may also be concerned in packaging other materials, including epinephrine granules in the adrenal medulla, tyrosinase and other materials in premelanosomes in mouse melanomas, peroxidase, etc. in secretory granules in

cells of parotid and submaxillary glands, and un-iodinated thyroglobulin in thyroid epithelial cells.

germ cell any cell of the **germ line**, including **gametes**.

germ-free describing any group of animals or plants that is bred and maintained in such conditions that they contain no microorganisms, either normal symbionts or abnormal, infective ones. *Compare* **gnatobiotic**.

germin a pepsin-resistant water-soluble glycoprotein of the walls of germinated plant embryos that may alter the properties of cell walls during germinative growth. Its synthesis coincides with the onset of growth in germination. The protein has oxalate oxidase activity. Example from *Triticum aestivum*: germin GF-2.8 precursor, homopentamer; database code GER2_WHEAT, 224 amino acids (23.58 kDa).

germ line any line of cells that gives rise to gametes and is continuous through the generations.

Gerstmann–Straussler syndrome a transmissible **spongiform encephalopathy** of humans, usually familial, characterized by widespread degeneration of the nervous system. *See also* **prion**. [After Joseph Gerstmann (1887–1969), Austrian neurologist.]

GF-1 *see* **GATA-1**.

GFR *abbr.* for (renal) glomerular filtration rate.

Gg₃ *symbol* for gangliotriaose (alternative to GgOse₃).

Gg₄ *symbol* for gangliotetraose (alternative to GgOse₄).

GGA a codon in mRNA for a glycine residue.

GGC a codon in mRNA for a glycine residue.

GGG a codon in mRNA for a glycine residue.

GgOse₃ *symbol* for gangliotriaose (alternative to Gg₃).

GgOse₄ *symbol* for gangliotetraose (alternative to Gg₄).

GGU a codon in mRNA for a glycine residue.

GH *abbr.* for growth hormone (i.e. **somatotropin**).

ghilanten *see* **antistasin**.

ghost an empty red-blood-cell membrane, usually obtained by lysis of red cells by the controlled reduction in the osmotic pressure of the suspending medium followed by restoration of normal osmotic conditions causing the membranes to reseal. Ghosts are extensively used in studies of the properties of membranes.

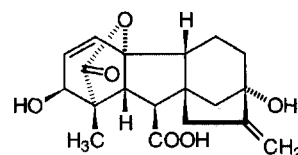
GHRF *abbr.* for growth hormone releasing factor (*see* **GHRH**).

GHRH *abbr.* for growth hormone releasing hormone; *other names*: growth hormone releasing factor (*abbr.*: GHRF) or somatoliberin; any of a family of peptide hormones that act on the anterior pituitary to stimulate the secretion of growth hormone and exert a trophic effect on the gland. Example from *Bos taurus*: database code SLIB_BOVIN, 44 amino acids (5.109 kDa). It is synthesized as a precursor; example from *Oncorhynchus nerka*: database code PACA_ONCNE, 173 amino acids (19.7 kDa). *See also* **GHRH receptor**.

GHRH receptor *abbr.* for growth hormone releasing hormone receptor; *other name*: growth hormone releasing factor receptor (*abbr.*: GRFR); any of a family of integral membrane proteins that bind GHRH and mediate its intracellular effects. These receptors are seven-transmembrane-helix receptors coupled to a G-protein that stimulates adenylate cyclase. A defect in this receptor (Asp⁶⁰ → Gly⁶⁰) is associated with hypoplasia of the anterior pituitary in mice. Example (precursor) from *Mus musculus*: database code GRFR_MOUSE, 423 amino acids (47.04 kDa).

GH-RIH *abbr.* for growth hormone release inhibiting factor (i.e. **somatostatin**).

gibberellin any of a widely distributed, multimembered group



gibberellin acid

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Gibbs–Duhem equation

of highly modified C₂₀ (and C₁₉) terpene plant hormones, the first of which was isolated from plants infected with the fungus *Gibberella fujikuroi*. The most important is gibberellin A₃, also known as **gibberellic acid**. Gibberellins accelerate shoot growth, and can actively stimulate RNA synthesis in dwarf mutant plants, making them grow like normal plants. Other physiological effects, e.g. overcoming the need for vernalization in long-day plants, have also been recorded.

Gibbs–Duhem equation an equation relating the chemical potential and concentration of a chemical species in a mixture at constant temperature and pressure. If n_i moles of species i , having chemical potential μ_i , are present in a mixture, then

$$\sum_i n_i d\mu_i = \sum_i x_i d\mu_i = 0$$

where $x_i = n_i / \sum n_i$ is the mole fraction and $d\mu_i$ is the change in chemical potential for dn_i , a small change in the amount of species i present, in moles. [After Josiah Willard Gibbs (1839–1903), US physical chemist, and Pierre Duhem (1861–1916), French physical chemist.]

Gibbs energy or **Gibbs free energy** or **Gibbs function** *symbol*: G ; a thermodynamic property of a system; it is defined as $H - TS$, where H is the enthalpy, S the entropy, and T the temperature. For a molecule in solution, Gibbs energy is concentration-dependent. The Gibbs free energy (G_i) of the i th component is equal to the standard free energy of the component (G_i°), the free energy at unit activity) plus a term to adjust for concentration. Thus

$$G_i = G_i^\circ + RT \ln a_i,$$

where R is the gas constant, and a_i the activity of the i th component. The molar free energy is referred to as the **chemical potential**. *See also* **Gibbs free energy change**.

Gibbs energy diagram or **Gibbs energy profile** for a chemical reaction, a diagram showing the relative standard Gibbs energies of reactants, transition states, reaction intermediates, and products in the same sequence as they occur in the reaction.

Gibbs energy of activation or **Gibbs free energy of activation** *symbol*: $\Delta^\ddagger G^\circ$; the standard Gibbs energy difference between the transition state of a chemical reaction (either elementary or stepwise) and reactants.

Gibbs free energy change the difference in Gibbs free energy (G) at any instant in time between the reactants and products of any reaction. Thus for a reaction $A + B \rightleftharpoons C + D$, at any particular concentration of reactants and products, the free energy change, ΔG , for that state is calculated as

$$G_C + G_D - G_A - G_B.$$

Since, for each substance,

$$G_X = G_X^\circ + RT \ln a_X$$

(*see* **Gibbs energy**), this leads to

$$\Delta G = \Delta G^\circ + RT \ln \{a_C a_D / a_A a_B\};$$

$$\Delta G^\circ = \Delta G^\circ_C + \Delta G^\circ_D - \Delta G^\circ_A - \Delta G^\circ_B,$$

and is the **standard free energy change**. All chemical reactions proceed in the direction of negative free energy change until equilibrium is reached, at which point ΔG is zero. Since at equilibrium $a_C a_D / a_A a_B$ is the equilibrium constant K , it follows that the standard free energy change $\Delta G^\circ = -RT \ln K^\circ$ where K° is the standard equilibrium constant.

Gibbs–Helmholtz equation an equation that may be expressed as

$$\Delta G - \Delta H = T[\partial \Delta G / \partial T]_P$$

where ΔG is the change in Gibbs energy during a reaction, ΔH the increase in enthalpy, P the pressure, and T the temperature.

Giemsa's stain a stain used for staining chromosomes, in which it reveals bands known as **G-bands** (from the name). These are rich in A–T nucleotide pairs. Other compounds, e.g. the antibiotic olivomycin, stain G–C-rich bands known as **R-bands** (because they give the 'reverse' of the G-bands). [After Gustav Giemsa (1867–1948), German parasitologist.]

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

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GIF *abbr.* for growth hormone-inhibitory factor (i.e. **somatostatin**).

giga+ *symbol*: G; 1 the SI prefix denoting 10⁹ times. **2** (*in computing*) prefix used to indicate a multiple of 2³⁰, e.g. gigabyte.

gigantism a condition arising from the excessive release of **somatotropin** leading to an individual of excessive size. *Compare* **acromegaly**.

Gilbert, Walter (1932–), US theoretical physicist and molecular biologist; Nobel Laureate in Chemistry (1980) jointly with F. Sanger 'for their contributions concerning the determination of base sequences in nucleic acids' [prize shared with P. Berg].

Gilman, Alfred Goodman (1941–), US pharmacologist; Nobel Laureate in Physiology or Medicine (1994) jointly with M. Rodbell 'for their discovery of G-proteins and the role of these proteins in signal transduction in cells'.

GIP *abbr.* for gastric inhibitory peptide.

GIPL *abbr.* for glycoinositolphospholipid (*see* **GPI anchor**).

GIP receptor any of the integral membrane proteins that bind **gastric inhibitory peptide** (GIP) and mediate its intracellular effects. They are G-protein-coupled, seven-transmembrane-helix receptors that activate adenylate cyclase. Example (precursor) from *Rattus norvegicus*: database code GIPR_RAT, 455 amino acids (52.26 kDa); residues 1–18 are the signal peptide, 19–455 the receptor.

Gla *symbol* for a residue of the α -amino acid L-4-carboxyglutamic acid.

Glanzmann thrombasthenia *see* **von Willebrand factor**.

glass electrode an electrode made of glass whose potential, when immersed in a solution, is a function of the hydrogen-ion activity, i.e. pH, of the solution. A common form of glass electrode consists of a thin glass bulb inside which is mounted a reference electrode, often a silver/silver chloride electrode, immersed in a solution of constant pH and containing the ion to which the inner reference electrode is reversible. A number of special glasses have been made in an effort to extend the range of pH over which the electrode responds linearly, to reduce errors caused by other ions in solution, to extend its useful temperature range, etc.

Glc *symbol* for a residue (or sometimes a molecule) of the aldohexose **glucose**. In the condensed system of symbolism of sugar residues, Glc signifies specifically a glucose residue with the common configuration and ring size, i.e. D-glucopyranose. (*Note*: Glu is the symbol for a glutamic residue.)

GLC or **glc** *abbr.* for gas-liquid chromatography.

GlcA **1** *symbol* for a residue (or sometimes a molecule) of the uronic acid glucuronic acid. **2** *former symbol* for a residue (or sometimes a molecule) of the aldonic acid **gluconic acid** (no symbol now recommended).

GlcF *symbol* for a glucofuranose residue.

GlcN *symbol* for a residue (or sometimes a molecule) of the aminodeoxysugar **glucosamine**.

GlcNAc *symbol* for a residue (or sometimes a molecule) of the N-acetylated aminodeoxysugar **N-acetylglucosamine**.

GlcP *symbol* for a glucopyranose residue.

GlcU or **GlcUA** *former symbol* for glucuronic acid (GlcA now recommended).

GLI *abbr.* for glucagon-like immunoreactant (or glucagon-like immunoreactivity).

glia or **neuroglia** the special connective tissue of the central nervous system constituting some 40% of the total volume of the brain and spinal cord. It has supportive and nutritive functions and is made up of **glial cells** – oligodendrocytes, astrocytes, ependymal cells, and microglial cells.

gliadin or (*formerly*) **gliadine** any of a group of proline-rich proteins (**prolamins**) found in the seeds of the cereals wheat and rye. All gliadins are mixtures of simple proteins of similar composition and properties. They are differentiated from **glutenin**, with which they are associated in gluten, by their solubility in 70% aqueous ethanol. With water, gliadins form a sticky mass, which binds flour into a dough. They constitute



glial cells

the major seed storage proteins in wheat. Gliadins fall into two groups: α/β gliadins (*also* prolamin or gliadin AI–V) are in five homology classes; there are more than 100 copies of the gene per haploid genome; γ -gliadins are in three homology classes (classes BI–III). For both groups, sequence divergence between the respective classes is due to single base substitutions and to duplications or deletions within or near direct repeats. These proteins have nine motifs. Examples (precursors): α/β , database code GDA1_WHEAT, 262 amino acids (30.37 kDa); γ , database code GDB1_WHEAT, 304 amino acids (34.21 kDa). [French *gliadine*; from Greek *glia*, glue.]

glial cells *see* glia.

glial derived nexin a glycoprotein serine protease inhibitor, similar to **serpin**, that promotes neurite extension. Example (precursor) from human: database code GDN_HUMAN, 398 amino acids (43.95 kDa). *See also* **nexin**.

glial fibrillary acidic protein *abbr.*: GFAP; a class III intermediate filament and a cell-specific marker that, during the development of the central nervous system, distinguishes astrocytes from other glial cells. It is similar to other intermediate filament proteins. Example from human astrocyte: database code GFAP_HUMAN, 432 amino acids (49.82 kDa).

glibenclamide *see* glyburide.

glicentin a peptide of originally obtained as a component of porcine enteroglucagon and known at first as **porcine gut GLI-I**. It reacts fully with antisera to the N-terminal moiety of glucagon from pancreas. The known amino-acid sequence includes the entire sequences of **glicentin-related pancreatic peptide** (at residues 1–30) and pancreatic **glucagon** (at residues 33–61), and is identical with the N-terminal moiety of the sequence of proglucagon. It was initially thought to be composed of 100 amino-acid residues, hence the name.

glicentin-related pancreatic peptide or **glicentin-related pancreatic polypeptide** *abbr.*: GRPP; originally, any polypeptide material present in extracts of porcine pancreas and in the secretory granules of the pancreatic A cell that reacted with certain antisera to glicentin but not with any antisera to glucagon. The term is now usually applied to a specific component of such material that is a 30-residue peptide identical in sequence to the N-terminal 30-residue sequence of **proglucagon**, is secreted by the pancreas concomitantly with glucagon, and thus is probably a cleavage product of proglucagon.

GLI-I *see* glicentin.

gliostatin *see* platelet-derived endothelial cell growth factor.

GLIP *abbr.* for glucagon-like insulinotropic peptide; a gastrointestinal peptide that potentiates release of insulin. It is released during a meal and after oral glucose administration. It is GLP-1^{7–37}. *See* **glucagon**.

Gln *symbol* for a residue of the α -amino acid L-glutamine (alternative to Q or Z).

globin the colourless and basic protein moiety (and biosynthetic precursor) of **hemoglobin** and **myoglobins**. In normal human adult hemoglobin, the globin component comprises two nonidentical pairs of polypeptide chains, whereas in myoglobin there is only one polypeptide chain. Pharmaceutical globin is prepared by adding ox hemoglobin to acetone–HCl; this causes the heme groups to separate and remain in solution while the globin precipitates. It is used in the preparation of globin–insulin, which is absorbed more slowly than soluble insulin and has a prolonged action.

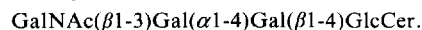
globo+ *prefix* for an oligosaccharide, signifying the core structure GalNAc(β 1-3)Gal(α 1-4)Gal(β 1-4)Glc (I), present in **globoside** from which the term originated. Also, by derivation it signifies Gal(α 1-4)Gal(β 1-4)Glc, residues 2–4 of I. *Compare* **isoglobo+**.

globoside or **cytolipin K** *symbol*: GbOse₄Cer or Gb₄Cer; globotetraosylceramide; ceramide globotetraoside; the major neutral glycosphingolipid in normal kidney and erythrocytes. The name originated from the discovery of this lipid on red cells, and led to the use of **globo+** as a root signifying this core

carbohydrate structure. *Compare* **globotriaosylceramide**. *See also* **globotetraosylceramide**, **P system**.

globotetraose *symbol*: GbOse₄ or Gb₄; the tetrasaccharide GalNAc(β 1-3)Gal(α 1-4)Gal(β 1-4)Glc. *See also* **P system**, **globo+**.

globotetraosylceramide *symbol*: GbOse₄Cer or Gb₄Cer; a glycolipid having the structure



It is the P determinant in the **P system** of blood groups. *See also* **globo+**.

globotriaose *symbol*: GbOse₃ or Gb₃; the trisaccharide Gal(α 1-4)Gal(β 1-4)Glc. *See also* **globo+**.

globotriaosylceramide *symbol*: GbOse₃Cer or Gb₃Cer; a glycolipid having the structure Gal(α 1-4)Gal(β 1-4)Glc-Cer. It is the major neutral glycosphingolipid in leukocytes, platelets, liver, spleen, and most other non-neural tissues. It is the P^k determinant in the **P system** of blood groups. *See also* **globo+**.

globular protein any protein whose polypeptide chain(s) are folded so as to give the whole molecule a rounded shape.

globulin any simple, globular protein that is insoluble or sparingly soluble in water but soluble in dilute salt solutions, and can be precipitated from solution by half-saturating the solution with neutral ammonium sulfate. Globulins may be distinguished by, e.g., electrophoresis into α -, β -, and γ -globulins, or by ultracentrifugation into 7S globulins, 19S globulins, etc. *See also* **gamma globulin**, **immunoglobulin**.

glomerular filtrate the filtrate, free of cells and major plasma proteins, that passes from the blood through a renal glomerulus into the lumen of a nephron in the kidney.

glomerular filtration rate *abbr.*: GFR; the rate at which the **glomerular filtrate** is produced in the kidney(s). It is a measure of renal efficiency and is usually expressed in terms of the renal clearance of some substance, e.g. inulin, that is freely filterable by the glomerulus and is neither reabsorbed nor excreted by the renal tubules.

glomerulus (*pl.* **glomeruli**) 1 a knot of blood capillaries projecting into the capsular end of a nephron and from which the **glomerular filtrate** is formed. *See also* **Malpighian body**. 2 any compact tuft or tangled mass of processes, nerve fibres, or blood vessels. —**glomerular** *adj.*

glp a regulon of *Escherichia coli* encoding proteins required for the utilization of glycerol 3-phosphate and its precursors. Operons for enzymes of these pathways are controlled negatively by *glp* repressor and positively by the cyclic AMP–cyclic AMP receptor protein complex.

Glp *symbol* for a residue of the α -imino acid L-5-oxoproline (often known as pyroglutamic acid or pyrrolidinecarboxylic acid).

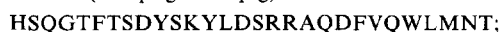
GLP *abbr.* for glucagon-like peptide (*see* **glucagon-related polypeptide**).

Glu *symbol* for a residue of the α -amino acid L-glutamic acid (alternative to E). (Note: Glc is the symbol for a glucose residue.)

gluc+ a variant (sometimes, before a vowel) of **gluco+**.

glucaemia a variant spelling (*esp.* UK) of **glucemia**; *see* **glycemia**.

glucagon a 3.5 kDa, single-chain, 29 residue hormone synthesized in the A cells of pancreatic islets and mammalian (except human) gastric mucosa. Formerly sometimes known as hyperglycemic-glycogenolytic factor (*abbr.*: HGF), or hyperglycemic factor. The amino-acid sequence is highly conserved, being identical in human and all other mammalian glucagons studied so far (except guinea-pig):



there are two motifs, characteristic of the glucagon family. It is secreted in response to falling blood glucose levels and has an opposing action to that of **insulin** by increasing the glucose concentration in the blood. The effects of glucagon are mediated in responsive tissues by either (1) stimulation of adenylated in responsive tissues by either (1) stimulation of adenylated



ate cyclase or (2) activation of inositolphospholipid-specific phospholipase C with increase in intracellular Ca^{2+} concentration. Its effects include the stimulation of glycogenolysis and gluconeogenesis in liver, increased lipolysis in adipose tissue, and increased insulin secretion. Three peptides (glucagon itself and glucagon-like peptides 1 and 2 (GLP-1 and -2)) derive from the precursor. These peptides have $\approx 50\%$ similarity in their primary structure. The sequence for GLP-1 is the same in all mammalian species studied but its biological role is unknown. While GLP-1 is the major form in the pancreas, a smaller fragment, GLP-1⁷⁻³⁶-amide, predominates in the intestine. This is a potent stimulator of insulin secretion and is a potent inhibitor of gastric secretion. Receptors for this peptide have been shown to be present in rat insulinoma cell lines and in brain. Exendin⁹⁻³⁹ is a GLP-1 receptor antagonist. Intracerebroventricularly administered GLP-1 powerfully inhibits feeding in the starving rat. This effect is blocked by exendin⁹⁻³⁹. Example, human (precursor): database code GLUC_HUMAN, 180 amino acids (20.89 kDa). *See also enteroglucagon, glicentin, glucagon receptor, proglucagon.*

glucagon-37 *see enteroglucagon.*

glucagon-like immunoreactant *abbr.*: GLI; any member of a group of peptides found in extracts of mammalian gastrointestinal tract that react with N-terminal-specific anti-glucagon antibodies. A particular component of such material may be denoted by means of a suffixed roman numeral, e.g. GLI-I, and the relative molecular mass of the component in question may be indicated by a superscript, e.g. ⁸⁰⁰⁰GLI. The principal molecular species recorded include one of ≈ 12 kDa (¹²⁰⁰⁰GLI), a second of ≈ 8 kDa (⁸⁰⁰⁰GLI), and a third of ≈ 4.5 kDa (⁴⁵⁰⁰GLI), the first two being the predominant forms. All have glucagon-like biological activities; e.g. they bind to glucagon receptors of liver plasma membrane, activate hepatic adenylate cyclase, and cause the release of insulin. *See also glicentin, proglucagon.*

glucagon-like insulinotropic peptide *see GLIP.*

glucagon-like peptide receptor any of the membrane proteins that bind glucagon-like peptides (*abbr.*: GLPs) and mediate their intracellular effects. A receptor for GLP-1 has been characterized. It is a seven-transmembrane-segment protein that is coupled to a G-protein activating adenylate cyclase. Example from human: database code GLPR_HUMAN, 463 amino acids (53.06 kDa).

glucagonoma a neoplasm (benign or malignant), originating usually from the A cells of pancreatic islets, that autonomously secretes glucagon or glucagon-like immunoreactant, frequently with associated diabetes mellitus.

glucagon receptor any of several membrane proteins that bind glucagon and mediate its intracellular effects. Activation of the liver glucagon receptor gives rise to both elevation of intracellular cyclic AMP and phosphoinositide hydrolysis with elevation of intracellular Ca^{2+} . A single clone has been isolated that, when expressed in transfected COS-7 cells, codes for a receptor that on activation can both increase cyclic AMP and intracellular Ca^{2+} , suggesting that two types of receptor may not be necessary for these two effects. This gene codes for a protein of the seven-transmembrane-segment type; database code GLR_RAT, 485 amino acids (55.04 kDa). The gene product is related to the receptors for calcitonin, parathyroid hormone, secretin, vasoactive intestinal peptide, and glucagon-like peptide I. The K_d of the expressed receptor has been found to be 20 nM (COS-7 cells) or 37 nM (BHK cells).

glucagon-related polypeptide or **glucagon-like peptide** any member of a diverse group of peptides found, e.g., in gastrointestinal tissue and blood that react with anti-glucagon antibodies and presumably contain part or all of the amino-acid sequence of glucagon. They include: **enteroglucagon**, **glicentin**, **glucagon-like immunoreactant**, and **proglucagon**.

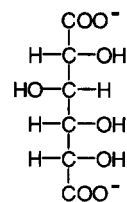
glucan any glycan composed solely of glucose residues. *See also dextran.*

glucan 1,4- α -glucosidase EC 3.2.1.3; *other names*: glu-

coamylase; amyloglucosidase; α -amylase; lysosomal α -glucosidase; acid maltase; exo-1,4- α -glucosidase. An enzyme that hydrolyses terminal 1,4- α -D-glucosidic bonds (at the nonreducing end) of polysaccharides, producing β -D-glucose with inversion of the configuration. It is less active towards oligosaccharides. Some of these enzymes (including the example) have a raw-starch binding domain. The enzyme from *Aspergillus niger* is used commercially in production of glucose. Example (precursor) from *Neurospora crassa*: database code S36364, 626 amino acids (66.50 kDa); this is a good example of a protein with a **kex2** site. *See also glucosidase.*

α -glucan phosphorylase a phosphorylase with affinity for α -glucans. Example (precursor) from potato chloroplasts: database code PHSL_SOLTU, 966 amino acids (109.38 kDa); this is the L-isomer.

glucarate 1 the dianion of glucaric acid, the **aldaric acid** derived from either glucose or gulose. There are two enantiomers L- and D-glucarate; D-glucarate, *formerly known as saccharate*, is derived from either D-glucose or L-gulose (and vice versa for L-glucarate). 2 any salt or ester of glucaric acid.



D-glucarate

glucemia or (*esp. UK*) **glucaemia** an alternative term for glycemia.

glucitol *see sorbitol.*

gluco+ or (*sometimes before a vowel*) **gluc+** *comb. form* indicating glucose. *See also glyco+* (def. 2).

gluco- *prefix (in chemical nomenclature)* indicating a particular configuration of a set of (usually) four contiguous CHOH groups as in the acyclic forms of D- or L-glucose. *See monosaccharide.*

glucoamylase another name for glucan 1,4- α -glucosidase.

β -glucocerebrosidase *see glucosylceramidase.*

glucoconjugate a former term for glycoconjugate.

glucocorticoid or **glucocorticosteroid** any naturally occurring or synthetic hormonal **corticosteroid** that acts primarily on carbohydrate and protein metabolism, e.g. by promoting hepatic glycogen deposition, and that has an anti-inflammatory effect. Glucocorticoids are produced in the middle layer (zona fasciculata) of the adrenal cortex, and include **cortisol**, **corticosterone**, and **cortisone**. In the rat, corticosterone is the major product; in humans and most other mammalian species, cortisol is the major product – in humans it is secreted at a rate of ≈ 8 –25 mg per day, compared with corticosterone at 1–4 mg per day. Glucocorticoid secretion is enhanced during stress, including hypoglycemia, hypotension, trauma (including surgery), and illness. The main physiological actions of glucocorticoids include stimulation of liver glucose output (by both glycogenolysis and gluconeogenesis), inhibition of glucose utilization by tissues, mobilization of amino acids by breakdown of muscle protein, and mobilization of fatty acids from adipose tissue. Anti-inflammatory actions include stabilization of lysosomal membranes, preventing the release of degradative enzymes, depression of phagocytosis and suppression of **T-cell** activity, suppressing the synthesis of **interleukin-1**, and stimulating the synthesis of **lipocortin**. The actions of glucocorticoids are mediated in large part through **glucocorticoid receptor** molecules, which regulate gene transcription. *Compare mineralocorticoid.*

glucocorticoid receptor a mammalian transcription factor in-



glucocorticoid response element

involved in the regulation of eukaryotic gene expression and affecting cellular proliferation and differentiation in target tissues. It is a cytosolic, soluble glucocorticoid-binding protein of $M_r \approx 90\,000$, probably existing in a complex with heat-shock protein hsp90, the total complex mass being about 300 kDa. Binding of steroid induces DNA-binding function and translocation to the nucleus where the receptor-steroid complex interacts with DNA at a **glucocorticoid response element** to regulate gene transcription. Cloning and other studies have indicated the presence of a glucocorticoid-binding site in the C-terminal region (beyond residue 527 in the human protein), with a DNA-binding region in mid-molecule with two **zinc fingers** (residues 421–481). There is also a modulating N-terminal domain. There are two forms (α , β), which differ only in their C-terminal sequences; they are probably generated by alternative splicing of a single gene. The human receptor cDNA sequence predicts a more abundant 777-residue α form, and a 742-residue β form, which have identical sequences up to residue 727. The glucocorticoid receptor is similar to other steroid/thyroid/retinoic nuclear hormone receptors. Example, α form (human): database code GCRA_HUMAN, 777 amino acids (85.56 kDa).

glucocorticoid response element *abbr.*: GRE; a specific regulatory DNA sequence involved in mediating the effects of glucocorticoids on gene transcription through interaction with the activated glucocorticoid receptor. As yet it is not well characterized, but it is likely that there are a number of GREs, probably *trans-acting*. A consensus sequence for one such GRE is the almost palindromic structure GGTACAnnnTGTCT.

glucocortin a protein, M_r 17 000, induced by glucocorticoids in adipocytes, liver, fibroblasts, and thymus, and common to all glucocorticoid target cells examined.

glucogenesis the production of glucose in an organism, especially from noncarbohydrate precursors.

glucogenic describing a substance, some or all the carbon atoms of which can be used to produce glucose in an organism. In particular, a glucogenic amino acid is any amino acid that gives rise to increased urinary glucose excretion when fed to a diabetic animal.

glucokinase **1** the recommended name for the enzyme ATP: D-glucose 6-phosphotransferase (EC 2.7.1.2), which catalyses the phosphorylation by ATP of D-glucose to D-glucose 6-phosphate with release of ADP. The enzyme occurs in invertebrates and microorganisms. Unlike **hexokinase**, which will catalyse the same reaction, it is highly specific for D-glucose. **2** a name frequently applied to an isoenzyme of hexokinase (EC 2.7.1.1) that is characteristic of mammalian liver.

glucon or (formerly) **glucone** any **glycon** in which the sugar involved is glucone.

gluconate **1** the anion of **gluconic acid**. **2** any salt or ester of gluconic acid.

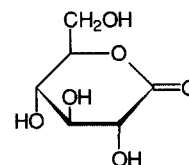
gluconeogenesis the metabolic formation of carbohydrates from non-carbohydrate organic precursors in liver. Formerly the term was restricted to the metabolic formation of glucose from noncarbohydrate precursors. Physiologically, an important source of precursor molecules is amino acid released from muscle protein during fasting. This is under hormonal control, especially by glucocorticoids. Alanine is an important precursor, and there is extensive conversion in muscle of other amino acids to alanine, which is then transported to the liver, where it undergoes transamination to form pyruvate. *See also* **gluconeogenic pathway**.

gluconeogenic pathway the pathway in liver by which non-carbohydrate precursors are converted to glucose 6-phosphate and thence to glucose or other carbohydrates. Precursor molecules must first be converted to pyruvate, thence successively to oxaloacetate, phosphoenolpyruvate, 2-phosphoglycerate, 3-phosphoglycerate, 1,3-bisphosphoglycerate, glyceraldehyde 3-phosphate plus glyceraldehyde phosphate, fructose 1,6-bisphosphate, fructose 6-phosphate, glucose 6-phosphate, and glucose (or other sugars). The enzymes involved are in many cases the same as those involved in the corresponding reverse step in the

glycolytic pathway. The enzymes specific to the gluconeogenic pathway are pyruvate carboxylase, phosphoenolpyruvate carboxykinase, fructose 1,6-bisphosphatase and glucose 6-phosphatase (see individual entries for each enzyme). The pathway is hormonally regulated, especially by glucagon, which increases cyclic AMP levels, thereby stimulating protein kinase activity. Control is effected by phosphorylation of liver pyruvate kinase, which inactivates the enzyme and thus restricts glycolysis, and phosphorylation of phosphofructokinase-2, which converts its action to that of a phosphatase which hydrolyses the 2-phosphate of fructose 2,6-bisphosphate, thereby inactivating phosphofructokinase.

gluconic acid the aldonic acid derived from glucose.

gluconolactone the trivial name for the γ -lactone (D-glucono-1,4-lactone) or δ -lactone (D-glucono-1,5-lactone) of D-gluconic acid.



D-glucono-1,5-lactone

glucophore the pharmacophore, or common structural feature, postulated to be present in molecules of all substances, whether chemically related or not, that possess a sweet taste, and thus to confer sweetness on them. The essential components of the glucophore are considered to be a polarized bond A–H (e.g. O–H in a sugar or N–H in a sweet amino acid), an electronegative atom B (typically =O), and a polarizable hydrophobic group X (such as H or Cl). The approximate distances between them are: A–B, 260 pm; A–X, 350 pm; B–X, 550 pm. The groups A–H and B are considered to bind respectively to similar B and A–H groups in the receptor, and the group X to bind to the receptor by charge-transfer or dispersion bonding.

glucoprotein a former name for **glycoprotein**.

glucosaccharase the trivial name for any **saccharase** that attacks the glucose end of appropriate oligosaccharides, liberating glucose.

glucosamine *symbol*: GlcN; the trivial name for the amino-deoxysugar 2-amino-2-deoxyglucopyranose; there are two enantiomers. D-Glucosamine (*symbol*: D-GlcN), formerly known as **chitosamine**, occurs in combined form in chitin, in mucoproteins, and in mucopolysaccharides, and is one of the most abundant natural monosaccharides.

glucosamine N-acetyltransferase EC 2.3.1.3; *other name*: glucosamine acetylase; an enzyme forming N-acetyl-D-glucosamine from acetyl-CoA and D-glucosamine while releasing CoA.

glucosamine-6-phosphate isomerase EC 5.3.1.10; *systematic name*: D-glucosamine-6-phosphate ketol-isomerase (deaminating). An enzyme that catalyses the hydrolysis of D-glucosamine 6-phosphate to D-fructose 6-phosphate and NH_3 . The amino-acid sequences of bacterial and fungal types are similar. Example from *Escherichia coli*: database code NAGB_ECOLI, 266 amino acids (29.74 kDa); 3-D structure known.

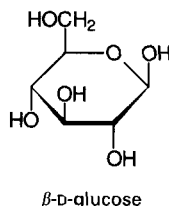
glucosaminic acid the common biochemical name for 2-amino-2-deoxygluconic acid.

glucosan **1** an obsolete name for **glucan**. **2** an old trivial name (established by usage but not recommended) for 1,6-anhydroglucopyranose.

glucose *symbol*: Glc; the trivial name for the aldohexose glucohexose; there are two enantiomers, D- and L-glucose. D-(+)-Glucose (*symbol*: D-Glu), commonly known as glucone, and formerly known as **grape sugar** or **corn sugar**, is dextrorotatory, hence also formerly (and sometimes still) known as **dextrose**; it is 0.74 times as sweet as sucrose. Aged aqueous



solutions of D-glucose contain an equilibrium mixture of α -D-glucopyranose, β -D-glucopyranose, and the open-chain form (see also **mutarotation**), whereas in crystals α -D-glucopyranose monohydrate is the stable form below 50 °C. Combined D-glucose may have either the α or the β configuration, but is invariably in the pyranose form. D-Glucose is an important source of energy for living organisms. It is found free in fruits and other parts of plants, in honey, and in animals, especially in the blood (≈ 5 mM in human blood). In combined form it occurs in many homo- and hetero-oligosaccharides and polysaccharides, especially in the animal storage polysaccharide glycogen and in the plant storage polysaccharides cellulose and starch.



glucose–alanine cycle or **alanine cycle** a metabolic cycle in which alanine is formed peripherally, e.g. in muscle, by transamination of glucose-derived pyruvate, and is transported to the liver where its carbon skeleton is reconverted to glucose.

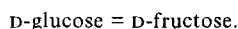
glucose-assimilation coefficient *symbol: K*; see **glucose-tolerance test**.

glucose-dependent insulinotropic peptide or **glucose-dependent insulin-releasing peptide** see **gastric inhibitory peptide**.

glucose effect the ability of glucose in the growth medium to inhibit the synthesis of certain enzymes in bacteria growing on the medium. See also **catabolite repression**.

glucose fatty-acid cycle a concept to explain how changes in plasma levels of glucose and fatty acids influence one another so as to provide a homeostatic control mechanism. Its essential features are: (1) the release, for oxidation, of fatty acids from muscle or adipose tissue acylglycerols restricts glucose metabolism in muscle; and (2) glucose uptake by muscle or adipose tissue restricts the release of fatty acids from acylglycerols in these tissues. The cycle thus provides a primitive, hormone-independent mechanism tending to maintain a constant plasma glucose level in animals that feed intermittently. Insulin modifies this mechanism by enhancing glucose uptake by muscle and adipose tissue, inhibiting fatty-acid release by adipose tissue, and increasing fatty-acid esterification in both muscle and adipose tissue.

glucose isomerase a bacterial enzyme preparation that catalyses the reaction:



It is used on a large scale in industry to convert glucose-containing materials to the much sweeter mixture of glucose plus fructose. The enzymes are usually immobilized on a variety of supports. The product is known as **high-fructose corn syrup** and is widely used as a sweetening agent. No enzyme for which this is the main function has been isolated. EC 5.3.1.18 (glucose isomerase) is now a deleted entry.

glucose oxidase EC 1.1.3.4; *other names*: glucose oxyhydrase; notatin; *systematic name*: β -D-glucose:oxygen 1-oxidoreductase; a flavoprotein (FAD) enzyme that catalyses the oxidation by dioxygen of β -D-glucose to D-glucono-1,5-lactone and H_2O_2 . It has application in the experimental determination of glucose concentration (e.g. in medicine) and has some industrial uses. Example from *Aspergillus niger*: database code NRL_1GAL, 581 amino acids (63.00 kDa); 3-D structure known.

glucose-1-phosphatase EC 3.1.3.10; *systematic name*: D-glucose-1-phosphate phosphohydrolase. An enzyme that catalyses the hydrolysis of D-glucose 1-phosphate to D-glucose and orthophosphate. Example (precursor) from *Escherichia coli*: database code AGP_ECOLI, 413 amino acids (45.63 kDa).

glucose-6-phosphatase EC 3.1.3.9; *systematic name*: D-glucose-6-phosphate phosphohydrolase. An enzyme that catalyses the hydrolysis of D-glucose 6-phosphate to D-glucose and orthophosphate. It has rather wide specificity. It is the final enzyme in the **gluconeogenic pathway**. Example from mouse: database code U00445, 357 amino acids (40.43 kDa).

glucose-6-phosphatase deficiency see **glycogen storage disease**.

glucose-6-phosphate dehydrogenase *abbr.*: G6PDH or (in *clinical chemistry*) GPD; EC 1.1.1.49; *recommended name*: glucose-6-phosphate 1-dehydrogenase; *systematic name*: D-glucose-6-phosphate:NADP⁺ 1-oxidoreductase; *other name (obsolete)*: Zwischenferment. An enzyme that catalyses the conversion of D-glucose 6-phosphate and NADP⁺ to D-glucono-1,5-lactone 6-phosphate and NADPH. This is an early step in the **pentose phosphate pathway** and the **Entner–Doudoroff pathway**, and thus in **heterolactic fermentation**. The enzyme has the greatest known variability of all enzymes in humans. Molecular variants that result in deficient activity are associated with a drug-induced hemolytic anemia, **favism**. This arises because the enzyme is involved in maintaining reduced glutathione in red cells. Example from *Drosophila melanogaster*: database code G6PD_DROME, 523 amino acids (60.05 kDa). See also **glucose-6-phosphate dehydrogenase deficiency**.

glucose-6-phosphate dehydrogenase deficiency a hereditary X-linked deficiency of **glucose-6-phosphate dehydrogenase** that affects mostly males. It is associated with a hemolytic anemia, but also confers some protection against malaria, since the parasite is unable successfully to pass through the schizont stage in red cells lacking this enzyme activity; hence there is a particularly high incidence in tropical Africa and the Middle and Far East. Lack of the enzyme in red cells results in failure to reduce NADP⁺ to NADPH in adequate amounts (the **pentose phosphate pathway** playing a major role in this in the red cell), as a result of which oxidized glutathione cannot be reduced by **glutathione reductase**. Consequently, there can be hemolysis, as hemoglobin is not maintained in the ferrous state and the peroxides produced as a result of drug action promote red-cell destruction. The antimalarial drug **primaquine** and related compounds generate oxidative stress, and during World War II their administration led to the discovery of the condition in many service personnel. The multi-allelic nature of the gene has led to many investigations of its distribution. See also **favism**.

glucose-6-phosphate isomerase *abbr.* (in *clinical chemistry*): GPI; EC 5.3.1.9; *systematic name*: D-glucose-6-phosphate ketol-isomerase; *other names*: phosphoglucose isomerase; phosphohexose isomerase; phosphohexomutase; oxoisomerase; hexosephosphate isomerase; phosphosaccharomutase; phosphoglucoisomerase; phosphohexoisomerase. An enzyme of the glycolytic pathway that catalyses the interconversion of glucose 6-phosphate and fructose 6-phosphate. Example from *Escherichia coli*, homodimer: database code G6PI_ECOLI, 549 amino acids (61.45 kDa).

glucose-tolerance curve the curve relating blood glucose concentration to time that is obtained in a **glucose-tolerance test**, either oral or intravenous.

glucose-tolerance factor *abbr.*: GTF; a water-soluble, relatively stable organic chromium complex of $M_r \approx 500$ that is believed to be required in the diet for normal glucose tolerance in animals, including humans. It is a coordination complex between chromium (III) ions, nicotinate, glycine, glutamate, and cysteine or glutathione.

glucose-tolerance test any procedure designed to assess the response of an individual to a loading dose of glucose, widely used in the diagnosis of diabetes mellitus. The standard **oral**

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glucose tolerance test is performed in the morning following at least 3 days' unrestricted diet, physical activity, and freedom from medication. The subject should have fasted for 12 hours before the test and remain seated and non-smoking during the test. After collection of the fasting blood sample, 75 g (or 1.75 g per kg of body weight for children) glucose is ingested in 5 minutes in 200–500 mL water (formerly loads of 50 g and 100 g glucose were used). Blood samples are then collected at 30-minute intervals for 2 hours, and urine samples are collected at 0, 1, and 2 hours. All samples are assayed for glucose. For capillary blood samples, a maximum glucose level below 10 mmol L⁻¹ (180 mg/100 mL) and a 2-hour value less than 7.5 mmol L⁻¹ (135 mg/100 mL) are the criteria for normality. If venous blood is used the corresponding values are 8.9 mmol L⁻¹ (160 mg/100 mL) and 6.1 mmol L⁻¹ (110 mg/100 mL). The urine samples should be glucose free. The **intravenous glucose tolerance test** may be used when abnormal carbohydrate absorption from the gut is suspected. A common procedure for this test is to take a fasting blood sample and further samples at 10-minute intervals for 1 hour following intravenous injection of 50 mL of 50% w/v glucose solution over a period of 2 minutes. Glucose is assayed in the blood samples and the logarithms of the values are plotted against time. The slope of the resulting straight line, multiplied by 100, is the **glucose-assimilation coefficient, K**, which represents the percentage fall in blood glucose per minute. Values of *K* in excess of 1.1 per cent per minute are taken as normal.

glucose transporter any sugar-transport protein for which glucose is a substrate. Apart from certain ABC types (e.g. in *Escherichia coli*) many glucose transporters belong to the 12-helix superfamily of integral transmembrane proteins. They resemble one another and also the typically sugar-H⁺ symport proteins of bacteria even though the mammalian examples are usually facilitated diffusion proteins. Examples from human; type 1 (erythrocyte and brain): database code GTR1_HUMAN, 492 residues (54.15 kDa); type 4 (insulin responsive, muscle and brown fat): database code GTR4_HUMAN, 509 amino acids (54.72 kDa). The latter is a candidate for the factor responsible for certain post-receptor defects in non-insulin-dependent diabetes mellitus. *See also* **sugar transporter**.

glucosidase any **glycosidase** that hydrolyses *O*-glucosyl-, *N*-glucosyl-, or *S*-glucosyl-compounds.

α -Glucosidase EC 3.2.1.20; *systematic name*: α -D-glucoside glucohydrolase; *other names*: maltase; glucoinvertase; glucosidase; maltase-glucoamylase. An enzyme that hydrolyses maltose and terminal (nonreducing end) 1,4- α -D-glucosidic bonds of oligosaccharides to yield α -D-glucose with retention of configuration. Maltases from some sources (especially intestine) act also on polysaccharides and may slowly hydrolyse (1→6)- α -D-glucosidic linkages as well. Most active with oligosaccharides, the enzyme catalyses the final step in the degradation of starch and is widely distributed in plants. Example (precursor) from the yeast *Candida tsukubaensis*: database code AGLU_CANTS, 1070 amino acids (118.50 kDa). *See also* **glucan 1,4- α -glucosidase**.

β -Glucosidase EC 3.2.1.21; *systematic name*: β -D-glucoside glucohydrolase; *other names*: cellobiase; gentiobiase. An enzyme that hydrolyses terminal (nonreducing end) 1,4- β -D-glucosidic bonds to yield β -D-glucose with retention of configuration. Example (precursor) from the yeast *Kluyveromyces marxianus*: database code bglsKLUMA, 845 amino acids (93.81 kDa).

glucoside or (formerly) **glycoside** any **glycoside** (def. 1) in which the sugar moiety is a glucose residue.

glucosuria the presence of glucose in the urine. It is sometimes loosely referred to by the older term **glycosuria**, which now strictly implies the presence of any type of sugar in urine. Glucosuria occurs in **diabetes mellitus** and is diagnostically important. It may also occur because of renal defects, as in **Fanconi syndrome**. *See also* **fructosuria**.

glucosyl any glucosyl group formed by detaching the anomeric

hydroxyl group, on C-1, from a cyclic form (pyranose or furanose) of glucose or a glucose derivative.

glucosylated hemoglobin a misnomer for **glycated hemoglobin**.

glycosylation **glycosylation** with one or more glucosyl groups. —**glucosylated** *adj.*

glucosylceramidase EC 3.2.1.45; *other names*: β -glucocerebrosidase; acid β -glucosidase; *systematic name*: D-glucosyl-*N*-acylsphingosine glucohydrolase; a lysosomal enzyme that catalyses the hydrolysis of D-glucosyl-*N*-acylsphingosine to D-glucose and *N*-acylsphingosine. A deficiency of this enzyme is the cause of **Gaucher's disease**, in which there is an accumulation of glucocerebrosides. Example (precursor) from human: database code GLCM_HUMAN 536 amino acids (59.63 kDa).

glucosylceramide the recommended term for any D-glucosyl-*N*-acylsphingosine, i.e. any **cerebroside** in which the sugar moiety is D-glucose.

glucosyloxy any of the glucosyloxy groups formed by detaching the hydrogen atom from the anomeric hydroxyl group, on C-1, from a cyclic form (pyranose or furanose) of glucose.

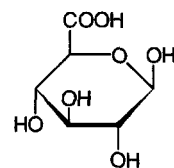
glucosyltransferase any hexosyltransferase enzyme (EC 2.4.1) for which the glucosyl group transferred is glucosyl. The acceptor may be orthophosphate (as for e.g. **phosphorylase**), but more often is another carbohydrate molecule (as for e.g. **glycogen synthase**) and sometimes a lipid (as in the synthesis of 3-D-glucosyl-1,2-diacylglycerol). *See also* **B-transferase**, **lactose synthase**, **Lewis transferase**.

glucuronate 1 the anion of glucuronic acid. 2 any salt or ester of glucuronic acid.

glucuronate pathway or **glucuronate oxidation pathway** the metabolic pathway by which D-glucuronate is converted, via L-gulonate, L-xylulose, xylitol, and D-xylulose, to D-xylulose 5-phosphate, an intermediate in the pentose phosphate pathway. In **essential pentosuria** there is defective conversion of L-xylulose to xylitol. In a side branch of the pathway L-ascorbate is formed, via L-gulonolactone, from L-gluconate. Primates and the guinea pig lack the ability to convert L-gulonolactone to L-ascorbate (*see* **ascorbic acid**).

glucuronate reductase EC 1.1.1.19; *systematic name*: L-gulonate:NADP⁺ 6-oxidoreductase. An enzyme that reduces the aldehyde group at C-1 of D-glucuronate thus forming L-gulonate. *See* **glucuronate pathway**.

glucuronic acid *abbr.* (sometimes): GA; *symbol*: GlcA (or (formerly) GlcU or GlcUA); the uronic acid formally derived from glucose by oxidation of the hydroxymethylene group at C-6 to a carboxyl group. There are two enantiomers; D-glucuronic acid is widely distributed in plants and animals, where it usually occurs as glucuronides.



β -D-glucuronic acid

β -glucuronidase *abbr.* (in clinical chemistry): GRS; EC 3.2.1.31, an enzyme that catalyses the hydrolysis of a β -D-glucuronoside to D-glucuronate and the corresponding hydroxy compound, which may be an alcohol, a phenol, or a saccharide. Deficiency of β -glucuronidase occurs in human type VII mucopolysaccharidosis. It is a homotetramer, and a lysosomal enzyme. Example (precursor) from human: database code BGLR_HUMAN, 651 amino acids (74.63 kDa).

glucuronide former name for **glucuronoside**.

glucuronoside or (formerly) **glucuronide** any compound formed by combination in (α or β) glycosidic linkage of a hydroxy compound with the anomeric carbon atom of glucose.

glucuronosyl

curonate. The hydroxy compound may be a saccharide, an alcohol, or a phenol. β -D-Glucuronosides are formed in so-called **detoxication** reactions.

glucuronosyl the glycosyl group formed by removal of the anomeric hydroxyl group from glucuronic acid or glucuronate.

glucuronosyltransferase *another name for* **UDPglucuronosyltransferase**.

glucuronoyl the univalent acyl group derived from glucuronic acid.

glued *see* **dynactin**.

Glu(NH₂) (formerly) *symbol for* a residue of the α -amino acid L-glutamine (Gln or Q now recommended).

Glu-NH₂ *symbol for* a residue of the α -amino acid L-isoglutamine.

Glu(OEt) *symbol for* a residue of γ -ethyl glutamate; *O*⁵-ethyl hydrogen glutamate.

Glu-OEt *symbol for* a residue of α -ethyl glutamate; *O*¹-ethyl hydrogen glutamate.

GluR receptor *see* **glutamate receptor**.

GLUT *an abbr. commonly used for* **glucose transporter**.

glutaconyl-CoA decarboxylase EC 4.1.1.70; an enzyme that catalyses the decarboxylation of glutaconyl-CoA (pent-2-enoyl-CoA) to but-2-enoyl-CoA, releasing CO₂. Biotin is a coenzyme. The enzyme is dependent on Na⁺, and functions as a sodium pump in the bacterial membrane.

glutamate 1 (*in biochemistry*) glutamate(1-); glutamic-acid monoanion; hydrogen glutamate. In theory, the term denotes any ion or mixture of ions formed from **glutamic acid** and having a net charge of -1, although the species ⁻OOC-[CH₂]₂-CH(NH₃⁺)-COO⁻ generally predominates in practice. **2** (*in chemistry*) the *systematic name for* glutamate(2-); glutamic-acid dianion. **3** any salt containing an anion of glutamic acid. **4** any ester of glutamic acid.

glutamate N-acetyltransferase EC 2.3.1.35; *other names*: ornithine acetyltransferase; ornithine transacetylase; an enzyme of the pathway in bacteria for the formation of ornithine from glutamate. It catalyses a reaction between N²-acetyl-L-ornithine and L-glutamate to form L-ornithine and N-acetyl-L-glutamate (*abbr.*: AGA). In bacteria, AGA is then converted to N-acetylglutamate 5-semialdehyde, which undergoes transamination to N²-acetylornithine. Example from *Bacillus stearothermophilus*: database code ARGJ_BACST, 410 amino acids (43.33 kDa); this is a bifunctional enzyme (*other name*: AGS/OCTase) containing activities for this and also **amino-acid N-acetyl transferase**.

glutamate-ammonia ligase EC 6.3.1.2; *systematic name*: L-glutamate:ammonia ligase (ADP-forming); *other name*: glutamine synthetase. An enzyme that catalyses a reaction between ATP, L-glutamate, and NH₃ to form ADP, orthophosphate, and L-glutamine. This is the first step in the fixation of ammonia in plants, the ATP-dependent incorporation of NH₄⁺ into the amide position of glutamine. The second step involves L-glutamate synthase (ferredoxin) and the combined effect of the two enzymes is to produce L-glutamate from 2-oxoglutarate and ammonium (NH₄⁺), a major NH₄⁺ assimilatory route. Glutamate-ammonia ligase has a molecular mass of 33 kDa and its eight monomers are arranged in two parallel tetramers. Example from *Escherichia coli*: 12 subunits in two hexagons, database code GLNA_ECOLI, 468 amino acids (51.71 kDa). The activity is controlled by adenylation in the presence of ATP by [glutamate-ammonia-ligase] **adenyl transferase** to adenylylate a tyrosine residue; the fully substituted enzyme is inactive; transcription is regulated by **nitrogen regulatory protein**.

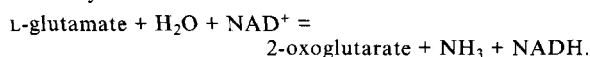
[glutamate-ammonia-ligase] **adenyl transferase** EC 2.7.7.42; *other name*: glutamine-synthetase adenylyltransferase; an enzyme that catalyses the adenylation of [L-glutamate:ammonia ligase (ADP-forming)] by ATP to form adenylyl-[L-glutamate:ammonia ligase (ADP-forming)] with release of pyrophosphate. The adenylation regulates enzyme activity (*see* **glutamate-ammonia ligase**). Example from *Escher-*

ichia coli: database code GLNE_ECOLI, 945 amino acids (108.15 kDa).

glutamate/aspartate transporter any membrane protein that transports glutamate and aspartate. Two types are known. (1) An H⁺-symport protein of the bacterial (inner) membrane; it is probably a 12-helix integral membrane protein. Although Na⁺ is not involved, there is a sequence resemblance to the second type (below). Example from *Escherichia coli*: database code GLTP_ECOLI, 437 amino acids (47.10 kDa). (2) The sodium-dependent transporter of brain; it is similar to sodium-dicarboxylate symporters. Putatively it is a glycosylated integral membrane protein of brain (high levels in Purkinje cell layer). Example from rat: database code GLT1_RAT, 573 amino acids (62.03 kDa).

glutamate-cysteine ligase EC 6.3.2.2; *other name*: γ -glutamylcysteine synthetase; an enzyme of the pathway for the formation of glutathione. It catalyses the formation of γ -L-glutamyl-L-cysteine from ATP, L-glutamate, and L-cysteine with release of ADP and orthophosphate. Example from *Escherichia coli*: database code GSH1_ECOLI, 518 amino acids (58.18 kDa).

glutamate dehydrogenase EC 1.4.1.2; *systematic name*: L-glutamate:NAD⁺ oxidoreductase (deaminating). An enzyme that catalyses the reaction:



This oxidative removal of the amino group from glutamate avoids the need for an oxo-acid acceptor, and in **ureotelic** animals is a major route for the removal of nitrogen as NH₄⁺, which is then converted to urea in the **ornithine-urea cycle**. Example from *Neurospora crassa*: database code DHE2_NEUCR, 1026 amino acids (115.52 kDa). Example from yeast, homotetramer: database code DHE2_YEAST, 1092 amino acids (124.19 kDa). *See also* **glutamate dehydrogenase (NADP⁺)**.

glutamate dehydrogenase (NADP⁺) EC 1.4.1.4; *systematic name*: L-glutamate:NADP⁺ oxidoreductase (deaminating). An enzyme that catalyses the oxidation by NADP⁺ of L-glutamate to form 2-oxoglutarate, NH₃, and NADPH. Two forms exist. One is a homotetramer. Example from *Neurospora crassa*: database code DHE2_NEUCR, 1026 amino acids (115.52 kDa). The other form, found in bacteria and plants, mainly proceeds in the direction of glutamate synthesis, as an NH₃-assimilatory enzyme. It is the product of the *am* gene, and is a homohexamer. Isoforms of this enzyme provided the first demonstration of **interallelic complementation**. Example from *Neurospora crassa*: database code DHE4_NEUCR, 453 amino acids (48.67 kDa).

glutamate-oxaloacetate transaminase *see* **aminotransferase**.

glutamate-pyruvate transaminase *see* **aminotransferase**.

glutamate receptor *abbr.*: GluR; any of a number of membrane proteins that bind glutamate and mediate its effects in neurotransmission. A major subdivision within glutamate receptors is between those (ionotropic) that act as ligand-gated ion channels and those (metabotropic) that are coupled to G-proteins. **Ionotropic GluR receptors** are further subdivided into three classes, named after dicarboxylic amino acids that act as agonists: (1) N-methyl-D-aspartate (NMDA) receptors; they play a key role in memory acquisition, learning, and neurological disorders and are the target for **conantokin-G**; (2) α -amino-3-hydroxy-5-methyl-4-isoxalone (**AMPA**) receptors, formerly designated quisqualate receptors; and (3) kainate receptors. Most of the fast excitatory synaptic transmission in the vertebrate central nervous system is mediated by GluR channels. There are many molecular subtypes of GluR in these categories; however, in all ionotropic receptors an ion channel is an intrinsic property of the receptor structure, generally with a broad ionic selectivity to monovalent ions; NMDA channels are also five to ten times more permeable to Ca²⁺ ions than to Na⁺ or K⁺ ions. The sequences of the subunits of these recep-



glutamate 1-semialdehyde

tors reveal four putative transmembrane segments; these subunits have been designated α , β , γ , δ , ϵ , and ζ . The members of the α subfamily constitute AMPA-selective GluR channels; the β and γ subfamilies include the subunits of the kainate-selective GluR channel; NMDA receptor channels are composed of $\alpha\zeta$ heteromers. Example of an ionotropic receptor (precursor) from rat: database code JN0339, 885 amino acids (99.34 kDa). **Metabotropic GluR receptors (mGluR)** that have been cloned include mGluR₁, of which two alternatively spliced versions exist (1199 and 906 amino acids), and mGluR₂ (872 amino acids); both of these types have seven transmembrane segments. The activity of this receptor is mediated by a G-protein that activates a phosphatidylinositol- Ca^{2+} second messenger system. Example, receptor 1 (precursor) from rat, transmembrane glycoprotein, seven motifs: database code MGR1_RAT, 1199 amino acids (133.08 kDa) of which amino acids 21–1199 comprise the receptor.

glutamate 1-semialdehyde glutamic α -semialdehyde; the L enantiomer, L-glutamate 1-semialdehyde, (S)-4-amino-5-oxopentanoate, is the precursor of 5-aminolevulinate (see **glutamate-1-semialdehyde 2,1-aminomutase**). It is synthesized in plants from glutamate bound to a specific tRNA; after reduction of the 1-carboxyl group, the tRNA is removed to yield glutamate 1-semialdehyde. *Compare* **glutamate 5-semialdehyde**.

glutamate 5-semialdehyde glutamic γ -semialdehyde; the L enantiomer, L-glutamate 5-semialdehyde, (S)-2-amino-5-oxopentanoate, is an intermediate in the synthesis of proline from ornithine. In plants and bacteria it is formed in a pathway in which L-glutamate is first converted to N-acetyl-L-glutamate, which is reduced to N-acetyl-L-glutamate 1-semialdehyde, from which the acetyl group is then removed. In animals it is formed from ornithine by transamination of the δ -amino group. *See* **proline**. *Compare* **glutamate 1-semialdehyde**.

glutamate-1-semialdehyde 2,1-aminomutase EC 5.4.3.8; *other name*: glutamate-1-semialdehyde aminotransferase. The enzyme synthesizing 5-aminolevulinate in plants; it catalyses the conversion of (S)-4-amino-5-oxopentanoate (i.e. **glutamate 1-semialdehyde**) to 5-aminolevulinate. Example from *Escherichia coli*: database code GSA_ECOLI, 426 amino acids (45.42 kDa).

glutamate synthase (ferredoxin) EC 1.4.7.1; *systematic name*: L-glutamate:ferredoxin oxidoreductase (transaminating). An enzyme that catalyses the second step in the fixation of ammonia in plants, i.e. the transfer of the NH_2 group from the amide group of glutamine to 2-oxoglutarate to form L-glutamate. The reaction is between L-glutamine, 2-oxoglutarate, and two molecules of reduced ferredoxin; the products are two molecules of L-glutamate and two of oxidized ferredoxin. In plants the combination of the action of **glutamate-ammonia ligase** and this enzyme results in the production of one molecule of glutamate from one of 2-oxoglutarate and one of NH_4^+ . *See also* **glutamate synthase (NADH)**, **glutamate synthase (NADPH)**.

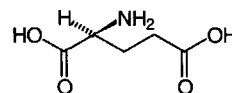
glutamate synthase (NADH) EC 1.4.1.14; *systematic name*: L-glutamate:NAD⁺ oxidoreductase (transaminating). An enzyme that catalyses a reaction between L-glutamine, 2-oxoglutarate, and NADH to form two molecules of L-glutamate and NAD⁺. FMN is a cofactor. Example from alfalfa: database code GLSN_MEDSA, 2194 amino acids (240.10 kDa).

glutamate synthase (NADPH) EC 1.4.1.13; *systematic name*: L-glutamate:NADP⁺ oxidoreductase (transaminating). An enzyme that catalyses the formation of two molecules of L-glutamate and one of NADP⁺ from L-glutamine, 2-oxoglutarate, and NADPH. It is an iron-sulfur flavoprotein. It brings together nitrogen and carbohydrate metabolism in bacteria, since in association with **glutamate-ammonia ligase** it functions as the second step in a major NH_3 -assimilatory pathway. It exists as an aggregate of four heterodimers of large and small subunits; glutamine binds to the large subunit and transfers NH_2 from its amido group to 2-oxoglutarate, which apparently binds to the small subunit. Example from *Escherichia coli*: small subunit, database code GLTD_ECOLI, 470 amino

acids (52.09 kDa); large subunit, database code GLTB_ECOLI, 1514 amino acids (166.04 kDa).

glutamic the general name for a residue of glutamic acid, whether or not the γ -carboxyl group is protonated.

glutamic acid *abbr.* (sometimes): GA; the trivial name for α -aminoglutaric acid; 2-aminopentanedioic acid; $\text{HOOC}-[\text{CH}_2]_2-\text{CH}(\text{NH}_2)-\text{COOH}$; a chiral α -amino acid. L-Glutamic acid (symbol: E or Glu), (S)-2-aminopentanedioic acid, is a coded α -amino acid found in peptide linkage in proteins; codon: GAA or GAG. In mammals it is a nonessential dietary amino acid and is glucogenic. Residues of D-glutamic acid (symbol: D-Glu or DGLu), (R)-2-aminopentanedioic acid, are found in the cell-wall material of various bacterial species and (as poly-D-glutamic acid) in the capsular substances of *Bacillus anthracis*, *B. subtilis*, and other bacteria. One residue of D-glutamic acid per molecule occurs in the antibiotic bacitracin A (see **bacitracin**) and also in **glutathione**.



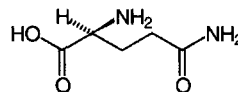
L-glutamic acid

glutamic amide 1 glutamic 1-amide (i.e. **isoglutamine**). 2 glutamic 5-amide (i.e. **glutamine**).

glutaminase EC 3.5.1.2; *other and systematic name*: L-glutamine amidohydrolase; an enzyme that catalyses the hydrolysis of L-glutamine to L-glutamate and NH_3 . Example (kidney isoform) from *Rattus norvegicus*: database code GLSK_RAT, 674 amino acids (73.94 kDa).

glutamate 1 the glutamine anion; the anion, $\text{H}_2\text{N}-\text{CO}-[\text{CH}_2]_2-\text{CH}(\text{NH}_2)-\text{COO}^-$, derived from glutamine. 2 any salt containing glutamine anion. 3 any ester of glutamine.

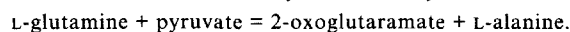
glutamine the trivial name for the γ -amide of glutamic acid; glutamic 5-amide; α -aminoglutaric acid; 2-amino-4-carbamoylbutanoic acid; $\text{H}_2\text{N}-\text{CO}-[\text{CH}_2]_2-\text{CH}(\text{NH}_2)-\text{COOH}$; a chiral α -amino acid. L-Glutamine (symbol: Q or Gln or (formerly) Glu(NH_2)), (S)-2-amino-4-carbamoylbutanoic acid, is a coded α -amino acid found in peptide linkage in proteins; codon: CAA or CAG. In mammals it is a nonessential α -amino acid and is glucogenic. D-Glutamine (symbol: D-Gln or DGLn), (R)-2-amino-4-carbamoylbutanoic acid, is not known to occur naturally. *Compare* **isoglutamine**.



L-glutamine

glutamine amidotransferase *abbr.*: GATase; a *nonsystematic name* for carbon-nitrogen ligases that employ glutamine as the amido-N-donor; i.e. members of the sub-subclass EC 6.3.5. These enzymes transfer the NH_2 group from glutamine to another reactant, e.g. **GMP synthase**. The enzyme activity is also an activity associated with **CAD protein** in mammals. *See also* **glutamate synthase (ferredoxin)**, **glutamate synthase (NADH)**, **glutamate synthase (NADPH)**.

glutamine-pyruvate aminotransferase EC 2.6.1.15; *other names*: glutaminase II; glutamine-oxo-acid transaminase; glutamine transaminase L; an enzyme that catalyses the reaction:



Pyridoxal-phosphate is a cofactor. *See also* **transamination**.

glutamine synthetase see **glutamate-ammonia ligase**.

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